



(12) **United States Patent**
Campbell et al.

(10) **Patent No.:** **US 12,403,419 B1**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **FILTER ELEMENTS AND ASSEMBLIES**

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(71) Applicant: **Donaldson Company, Inc.**,
Bloomington, MN (US)

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(BE)

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(73) Assignee: **Donaldson Company, Inc.**,
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Extended European Search Report for EP25177106.9 mailed Jun.
30, 2025.

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(Continued)

(21) Appl. No.: **19/210,888**

Primary Examiner — Robert Clemente

(22) Filed: **May 16, 2025**

(74) *Attorney, Agent, or Firm* — Merchant & Gould P.C.

Related U.S. Application Data

(60) Provisional application No. 63/768,562, filed on Mar.
7, 2025, provisional application No. 63/715,456, filed
(Continued)

(51) **Int. Cl.**
B01D 46/24 (2006.01)

(52) **U.S. Cl.**
CPC **B01D 46/2414** (2013.01); **B01D 2265/021**
(2013.01); **B01D 2265/06** (2013.01); **B01D**
2271/027 (2013.01); **B01D 2279/60** (2013.01)

(58) **Field of Classification Search**
CPC B01D 2271/027; B01D 2265/021; B01D
2265/06; B01D 46/2414; B01D 2279/60;
B01D 2201/347

See application file for complete search history.

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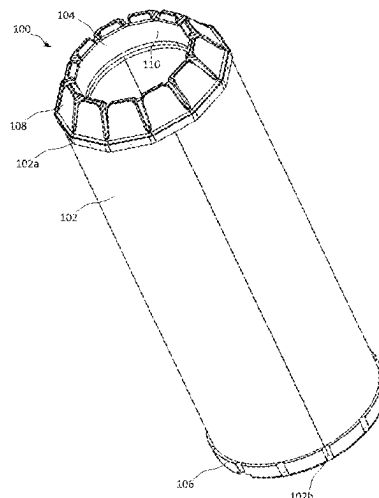
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(57) **ABSTRACT**

A filter cartridge can include a media pack including media extending between a first end and a second end along a longitudinal axis; an end cap secured to the media and including: a seal member presenting a radially inward or outward facing seal surface, the seal member being formed from a first material; and a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material harder and/or stiffer than the first material. At least a portion of the base part is covered by and/or embedded within the first material, and the plurality of flexible extensions includes an exposed portion that is uncovered by and/or embedded within the first material and defines an axial end of the filter cartridge and is coincident with a radial innermost portion of the end cap. The seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

20 Claims, 96 Drawing Sheets



Related U.S. Application Data

on Nov. 1, 2024, provisional application No. 63/649, 173, filed on May 17, 2024.

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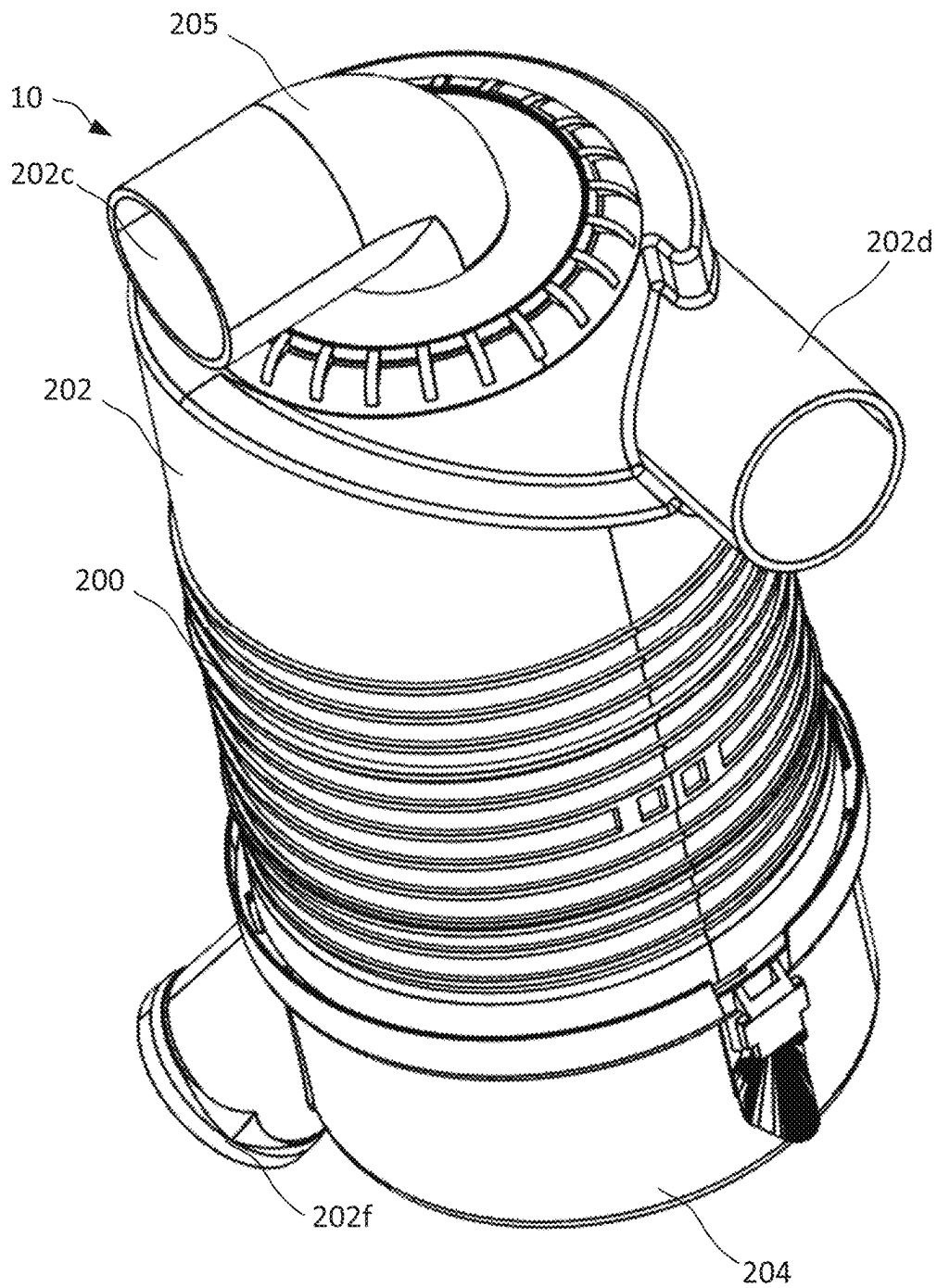


FIG. 1

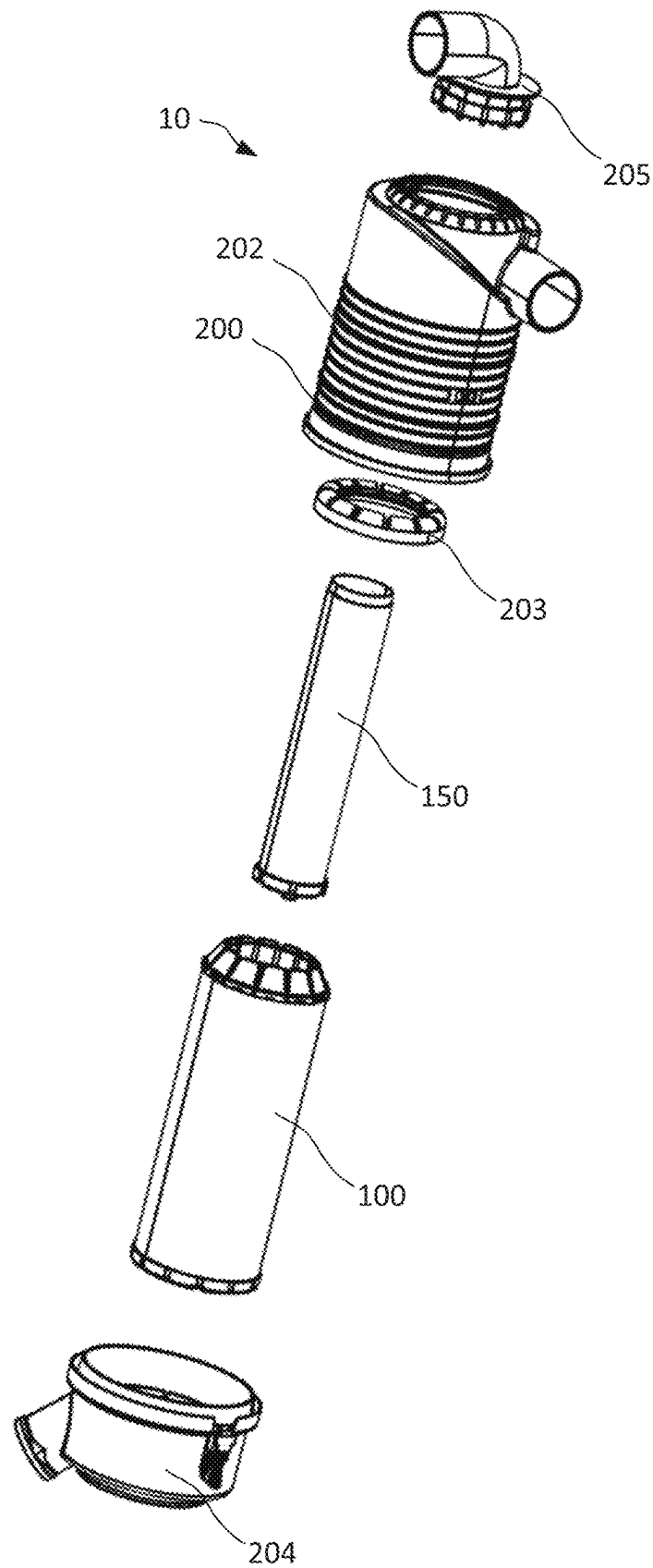


FIG. 2

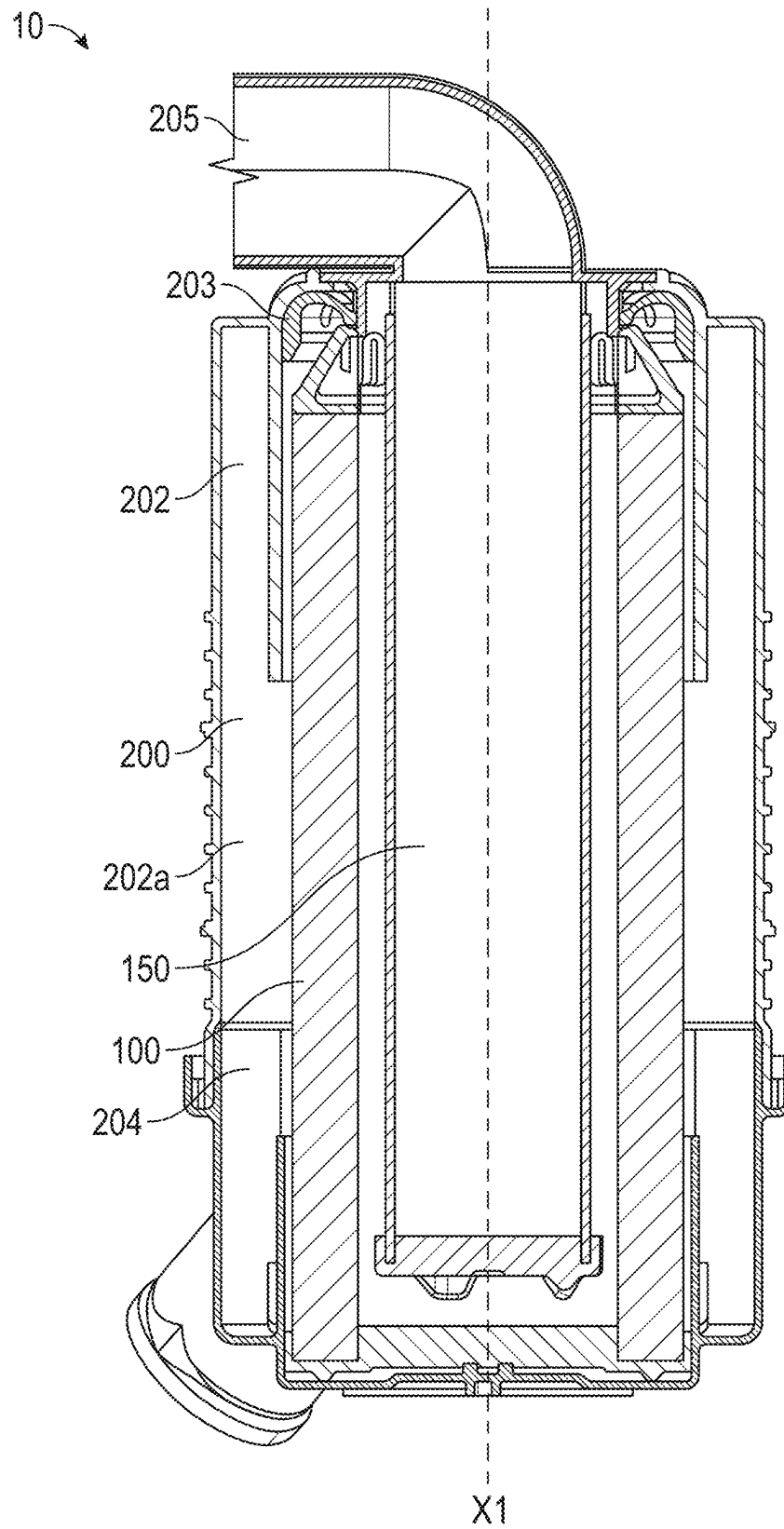


FIG. 3

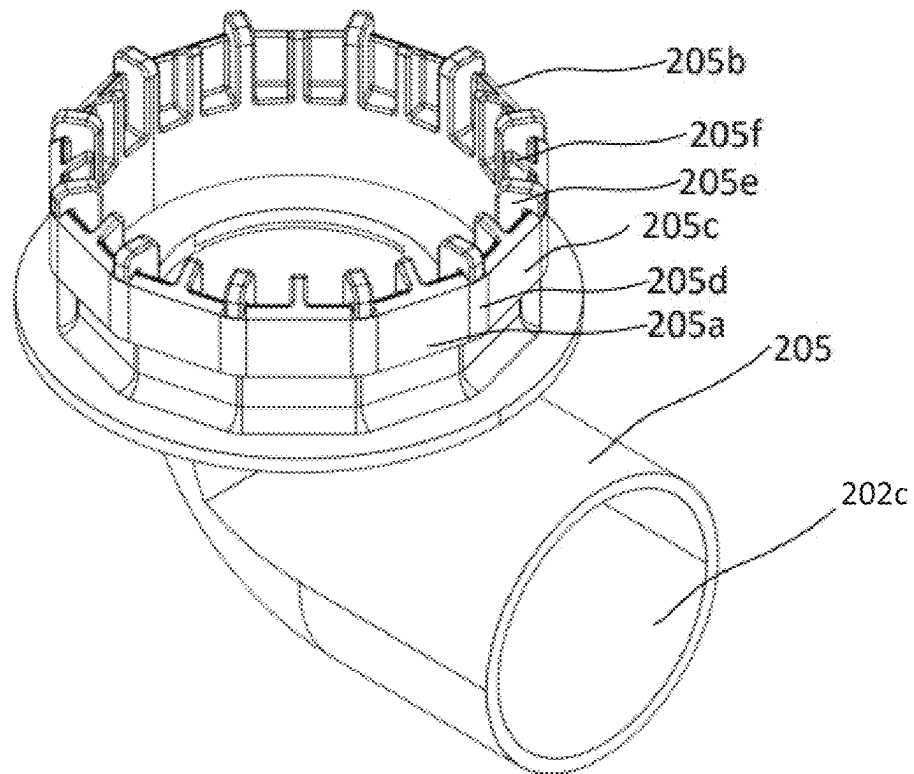


FIG. 4

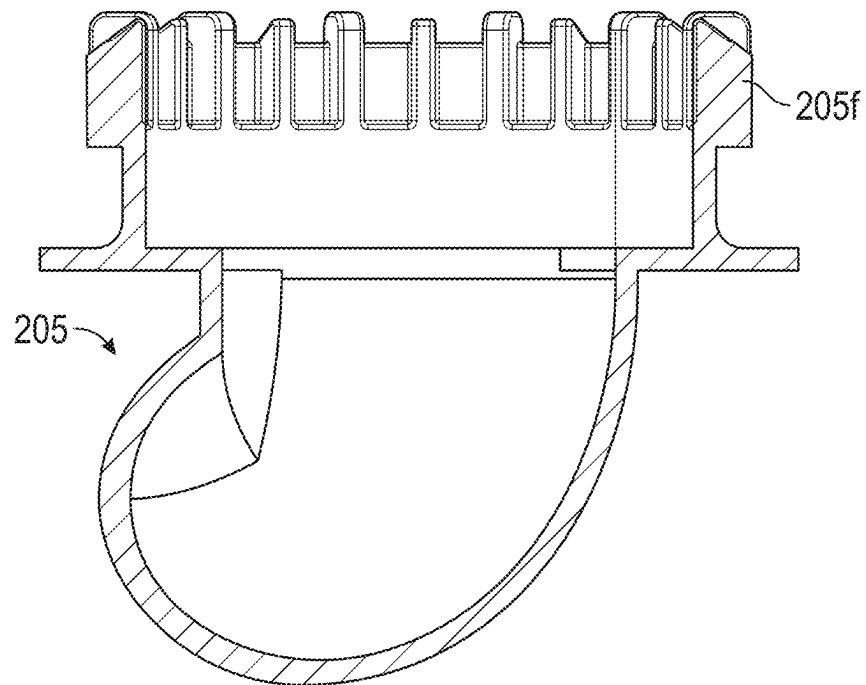


FIG. 5

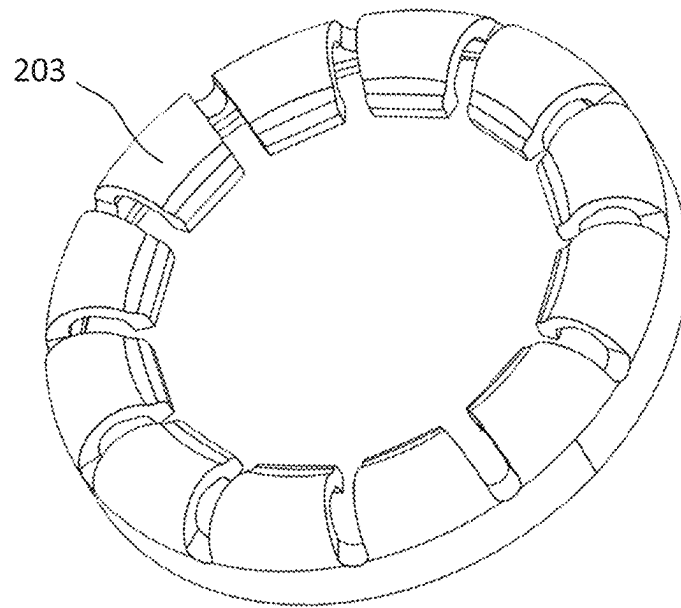


FIG. 6

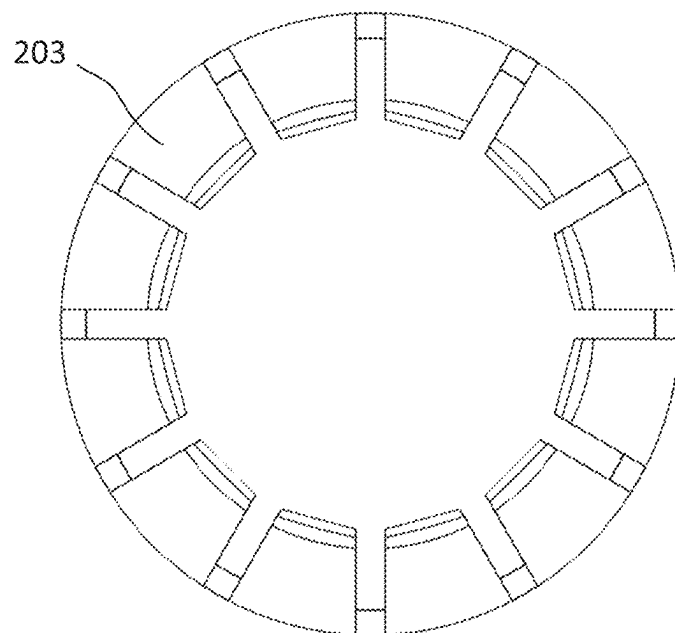


FIG. 7

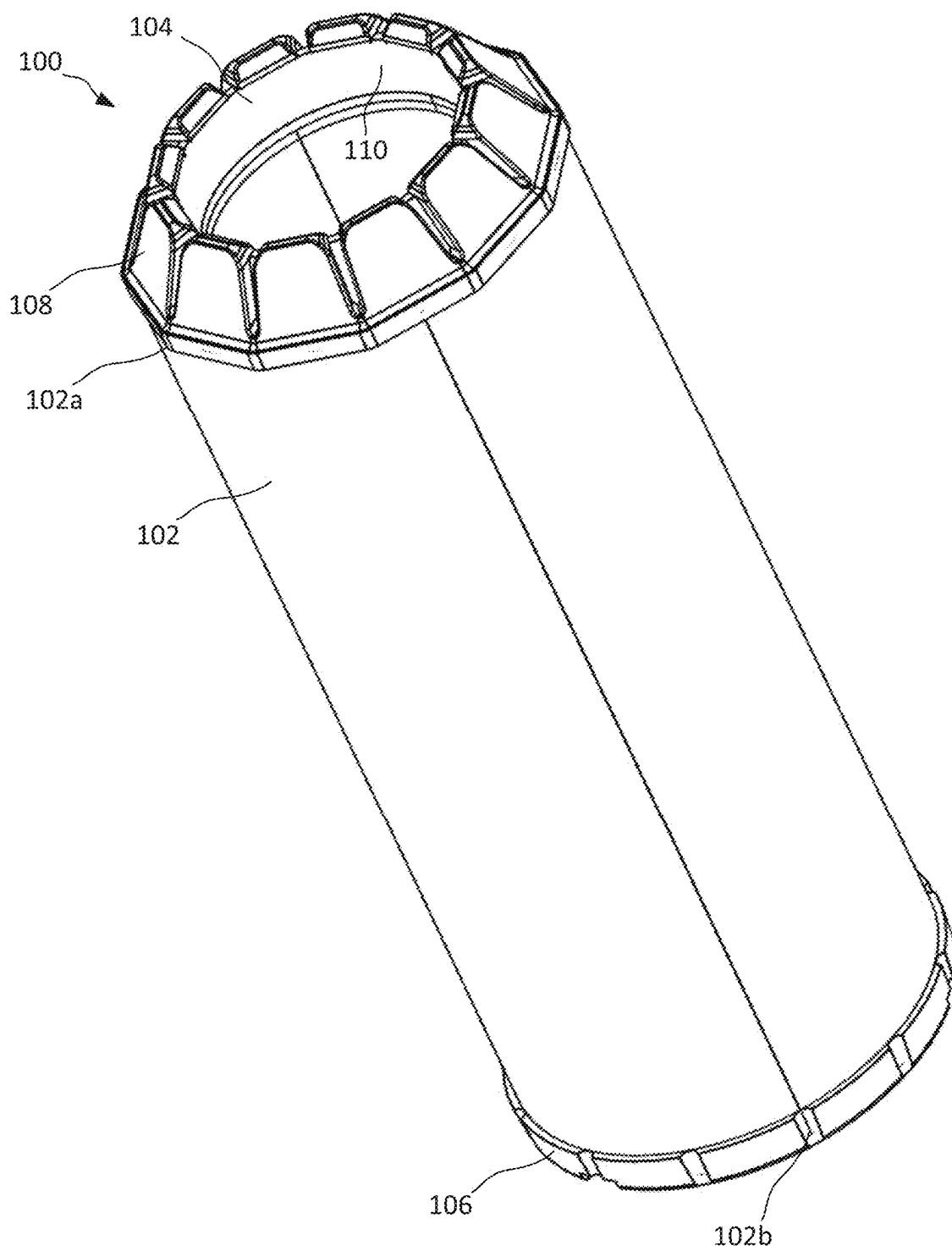


FIG. 8

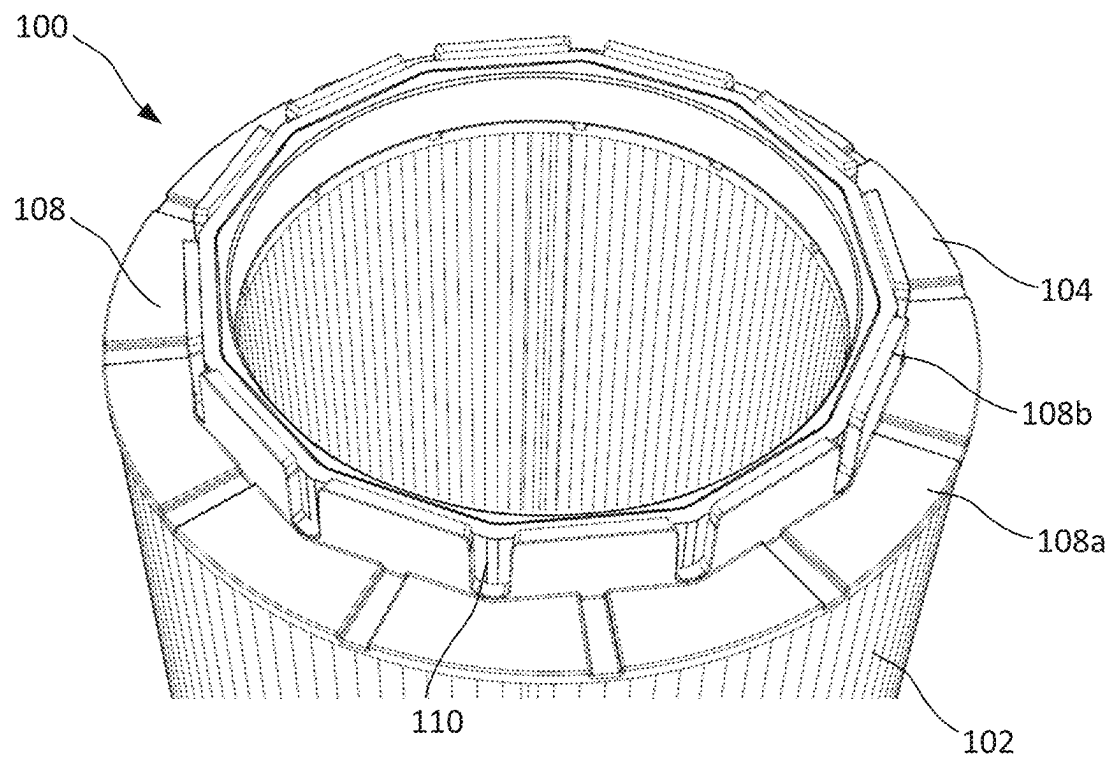


FIG. 8A

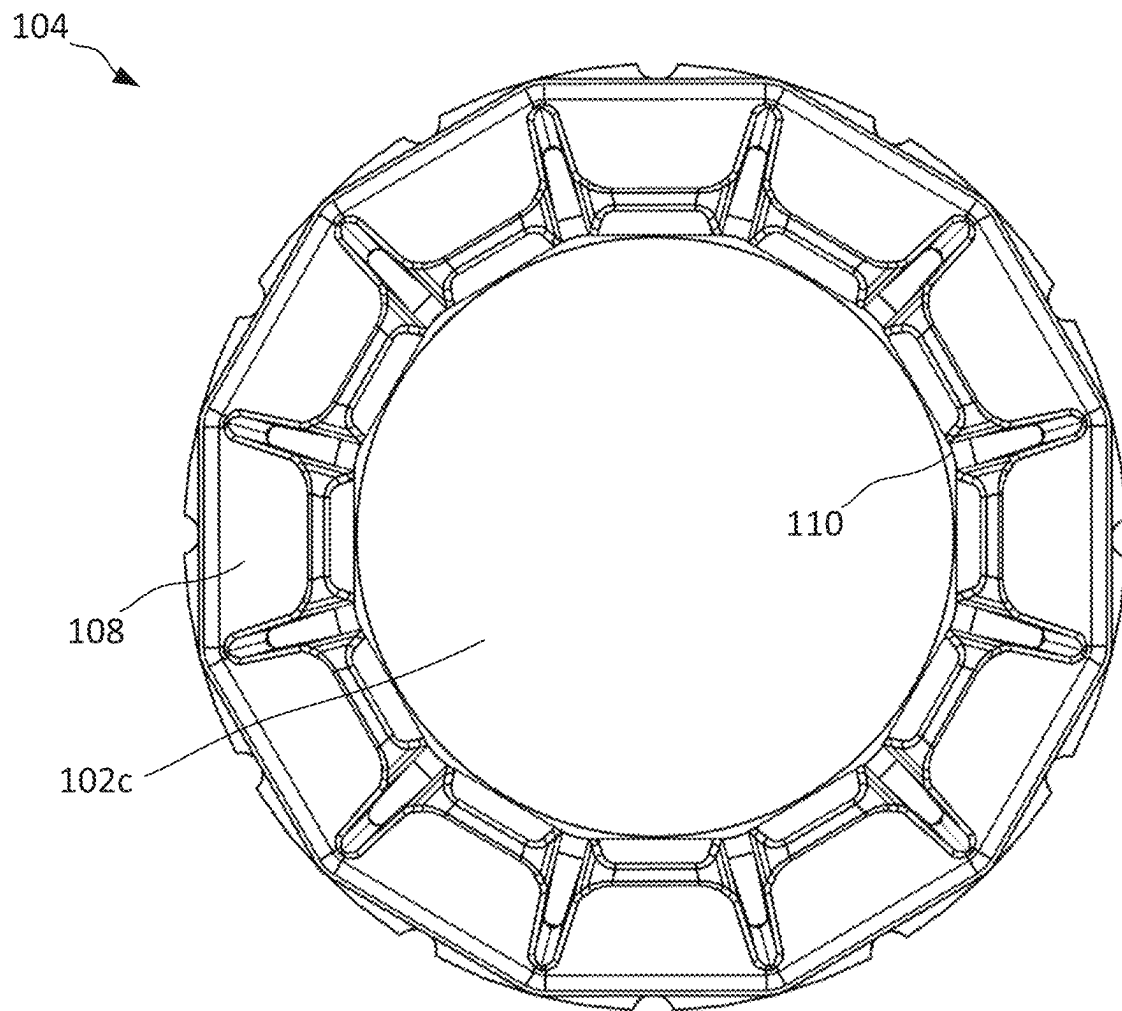


FIG. 9

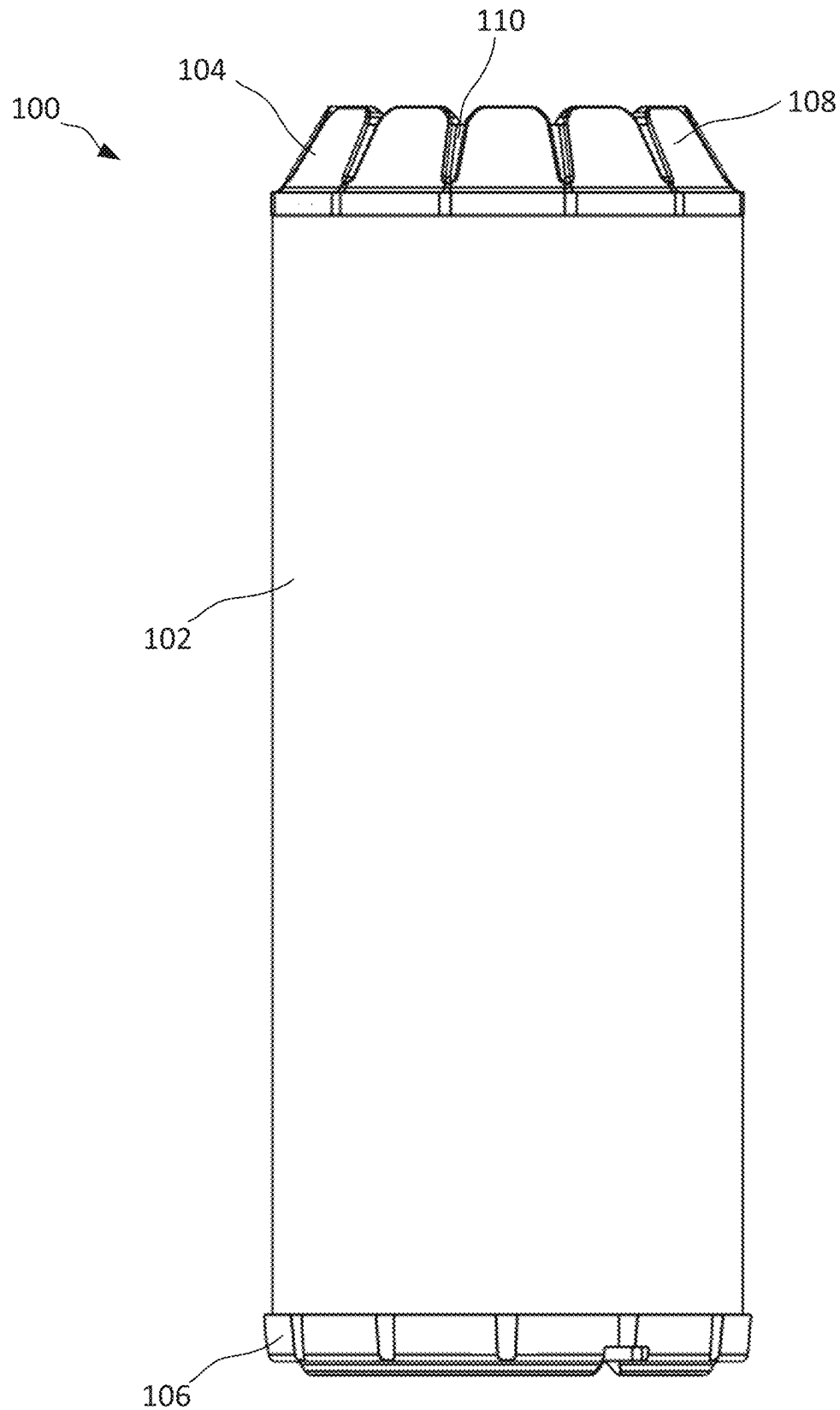


FIG. 10

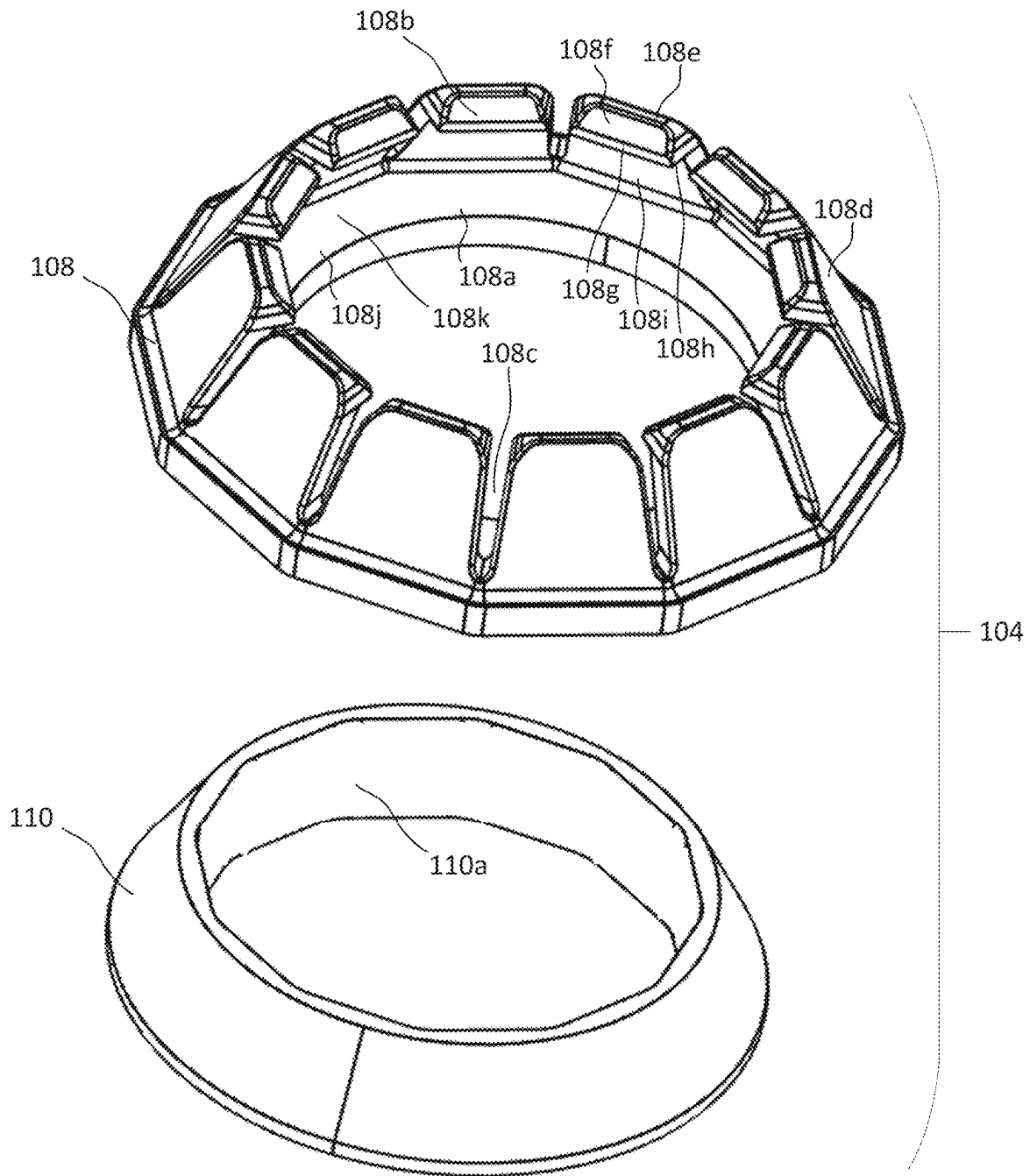


FIG. 11

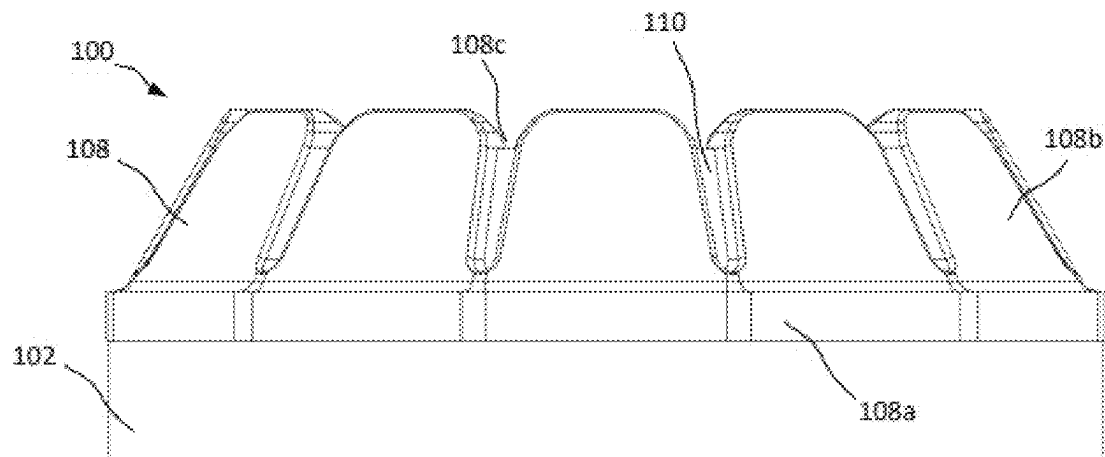


FIG. 12

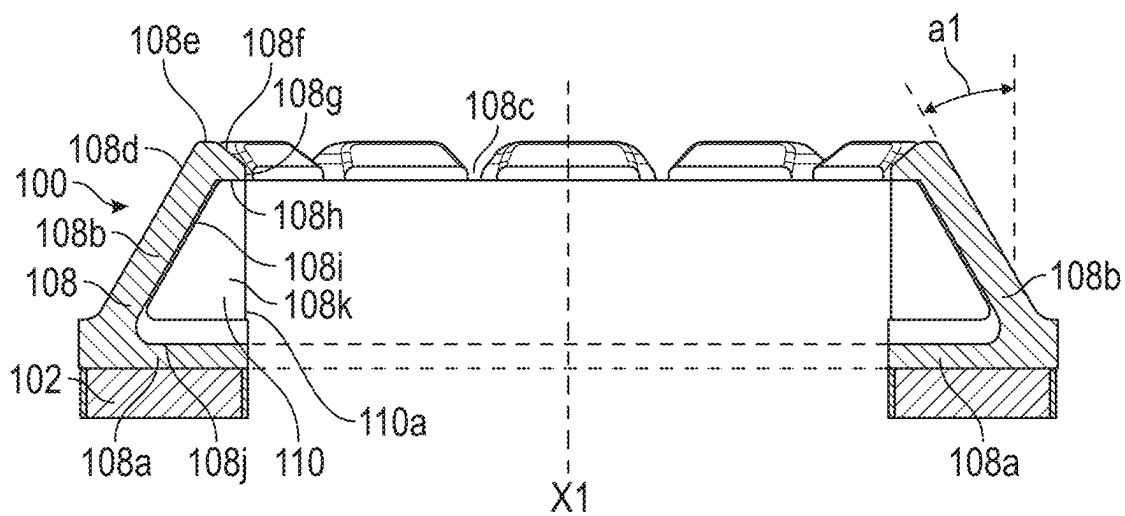


FIG. 13

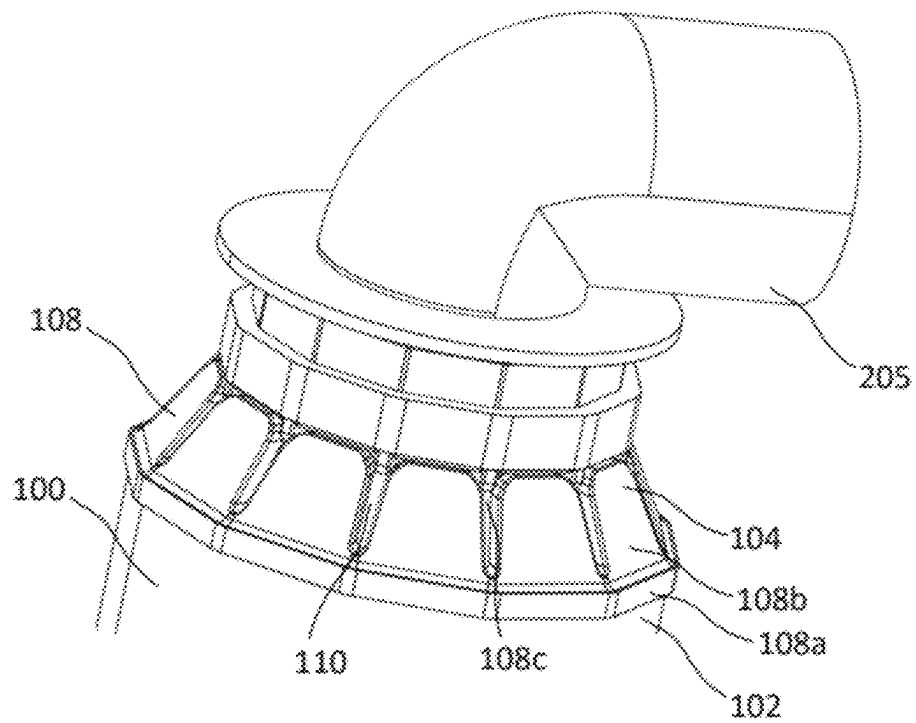


FIG. 14

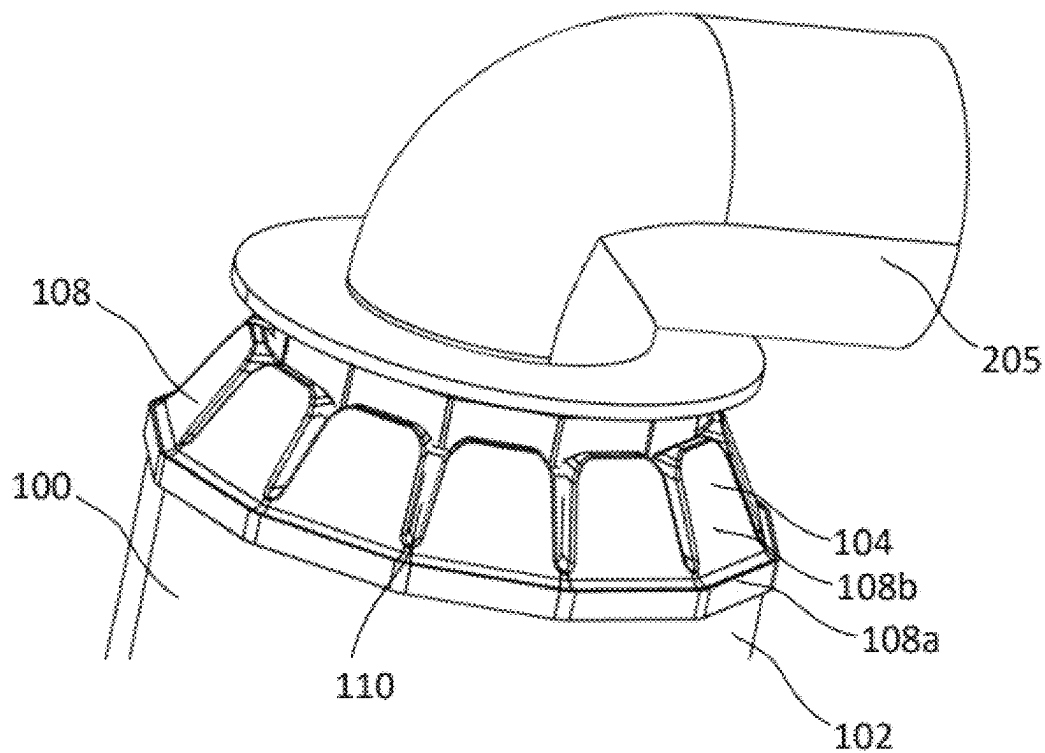


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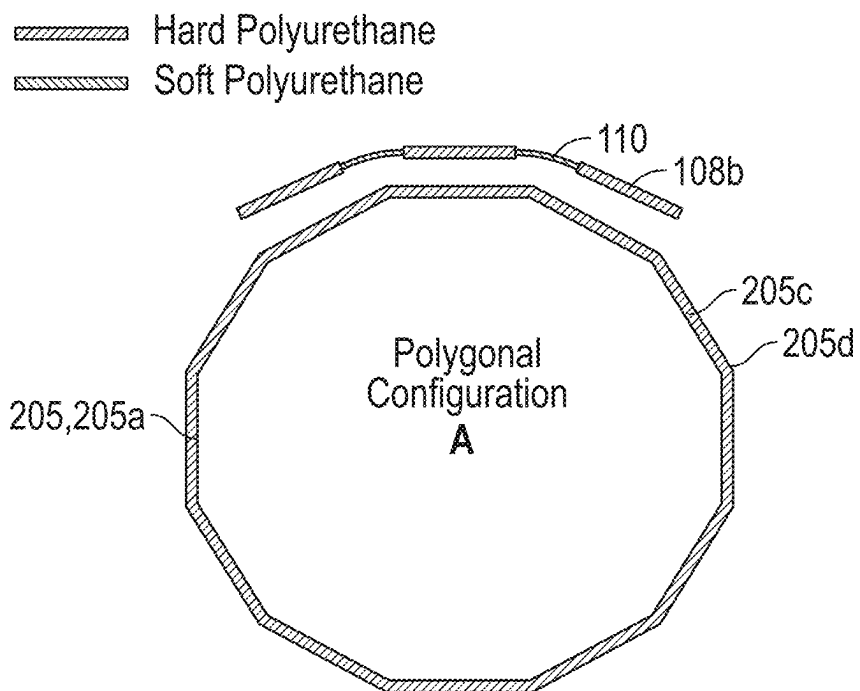


FIG. 16

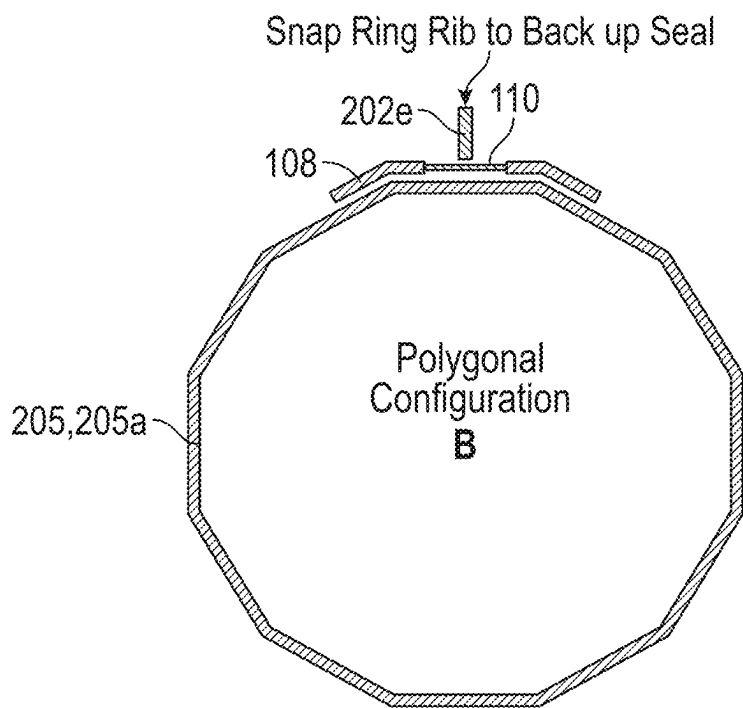


FIG. 17

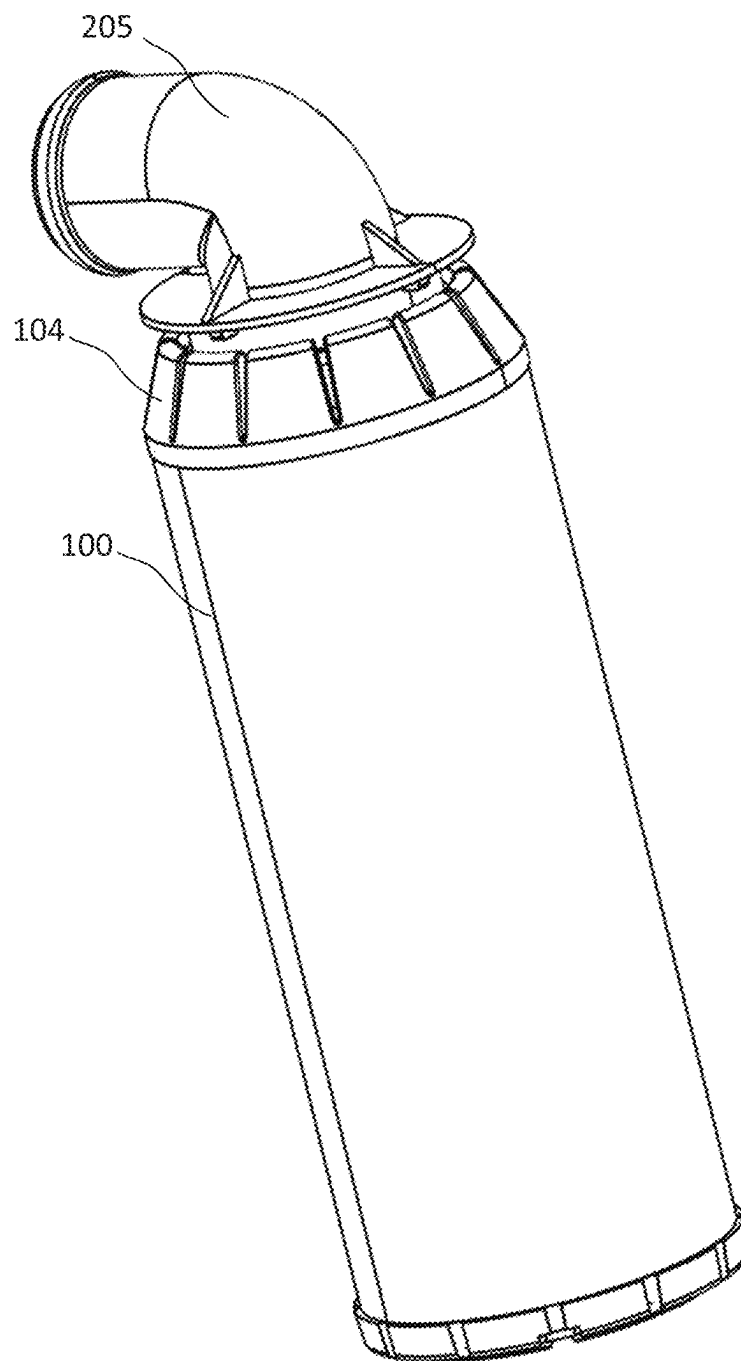


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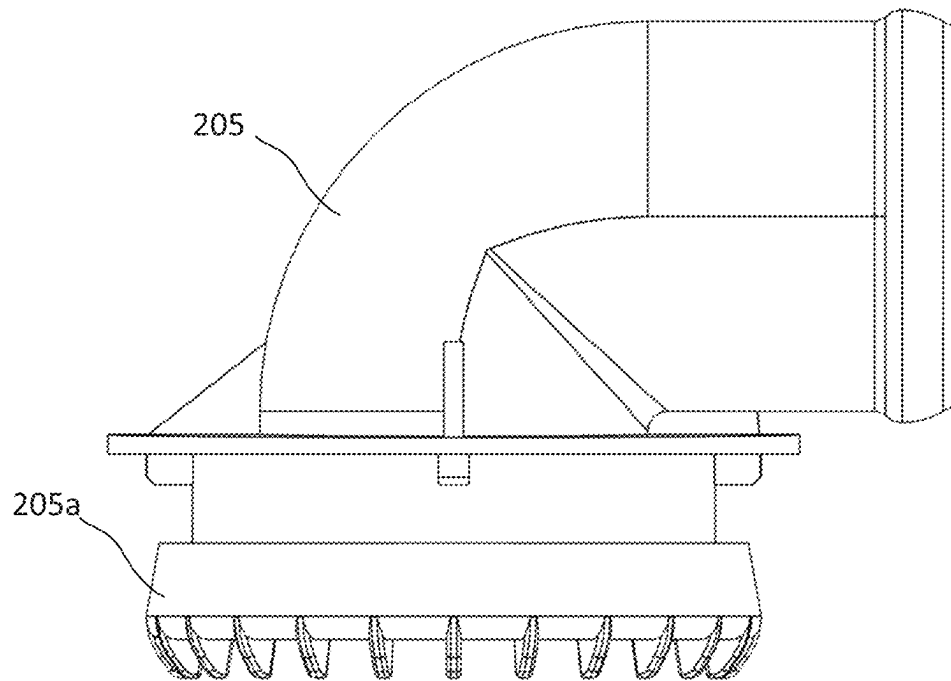


FIG. 19

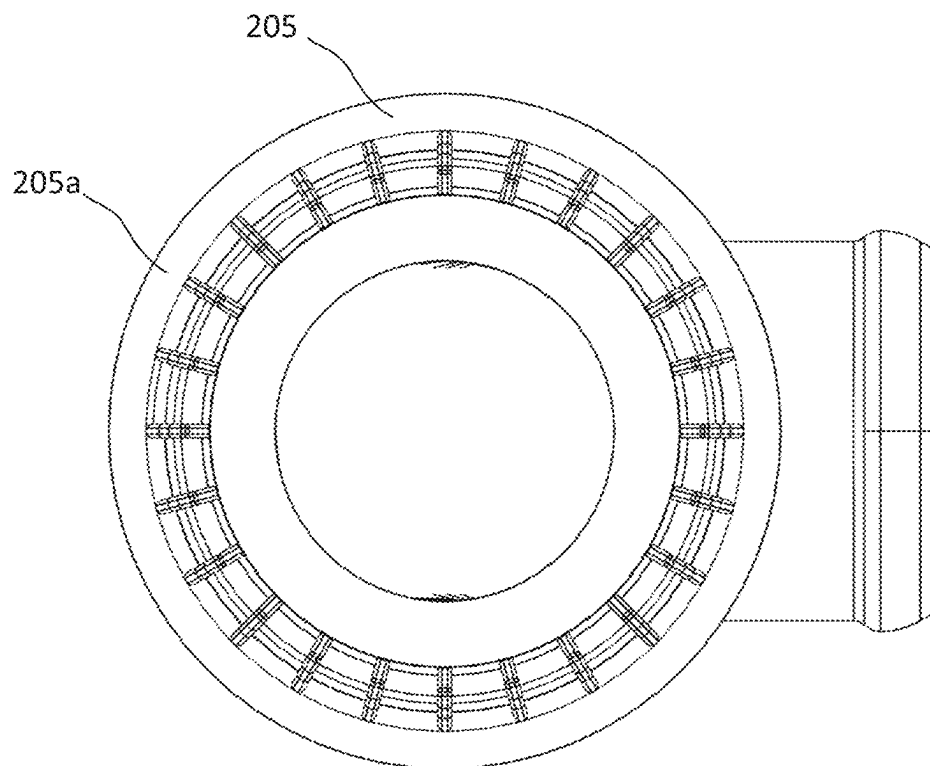


FIG. 20

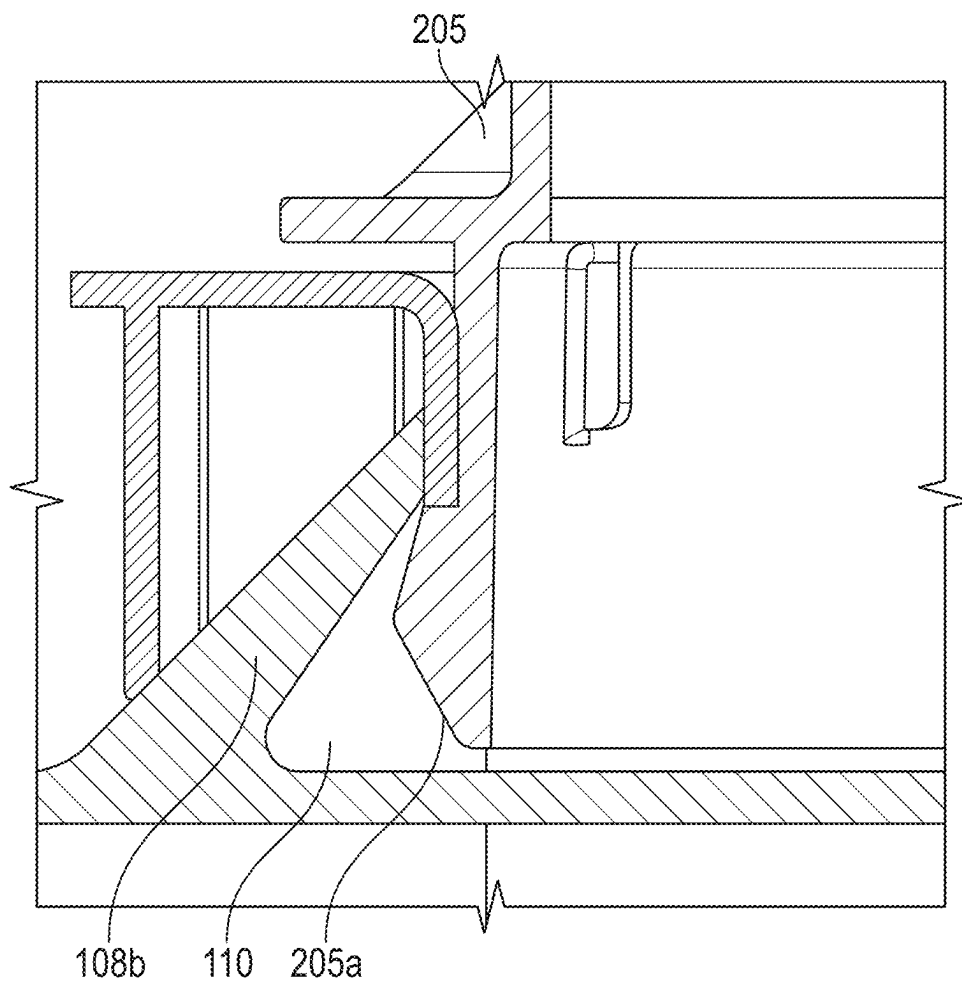


FIG. 21

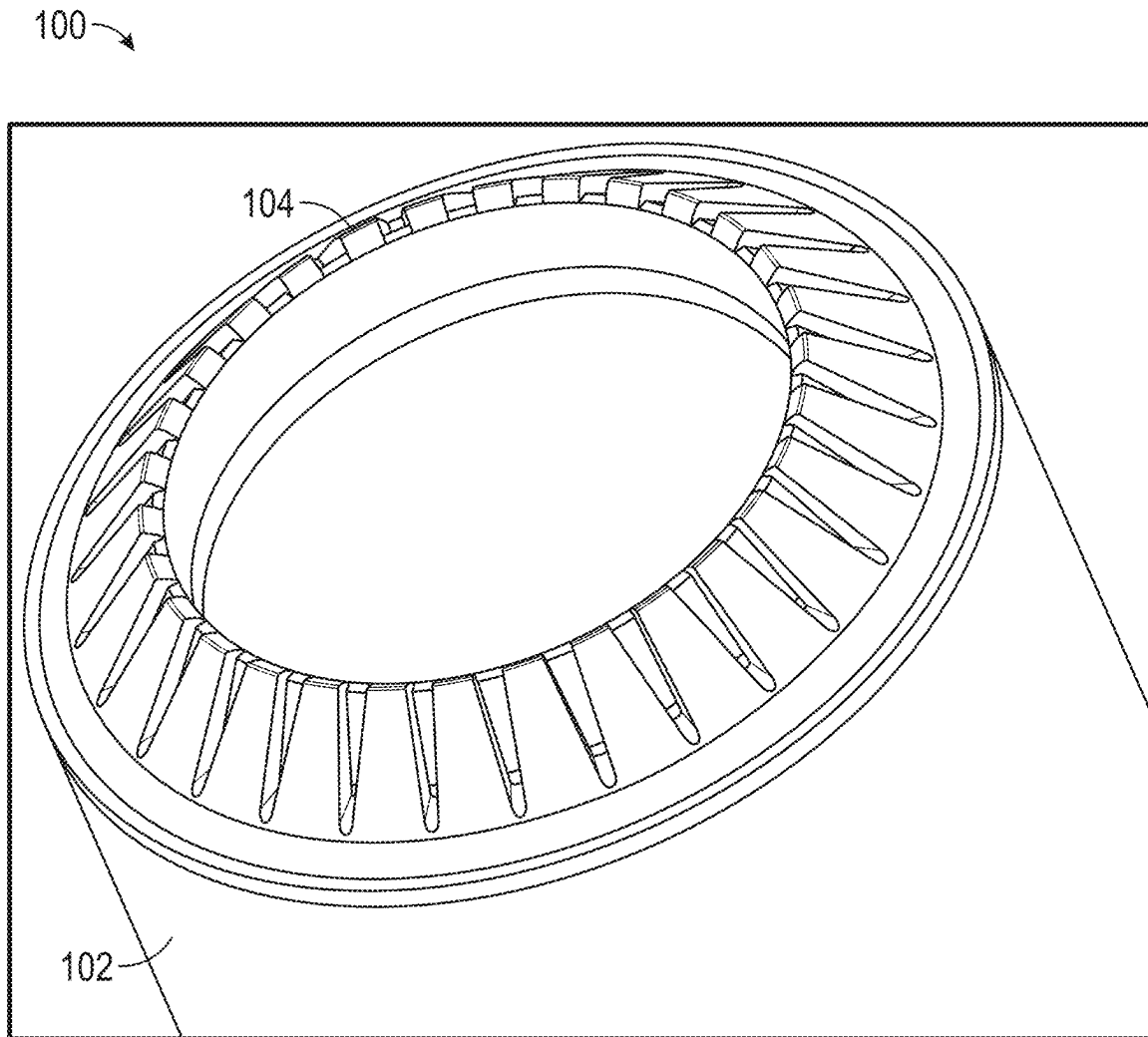
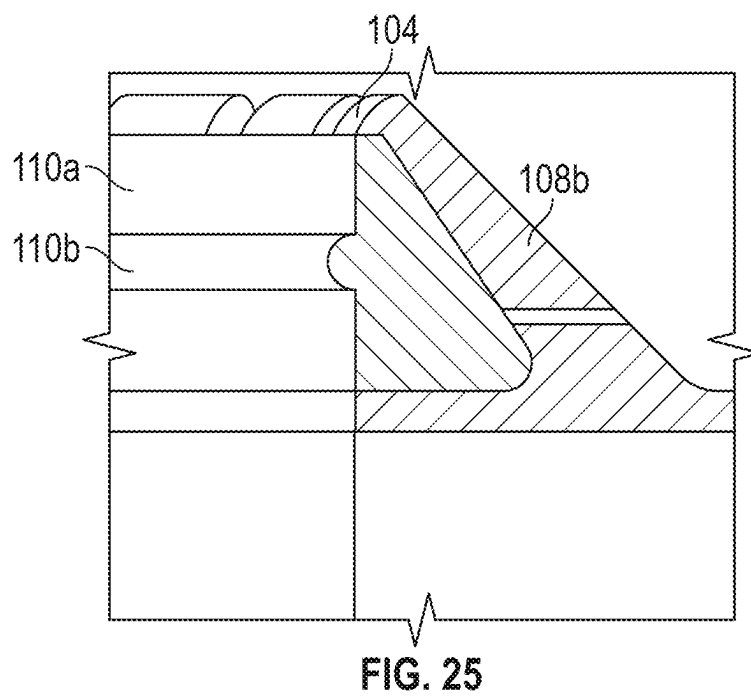
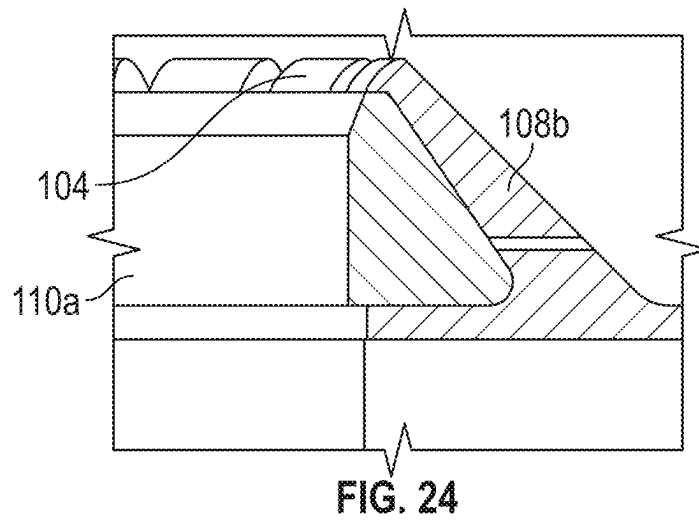
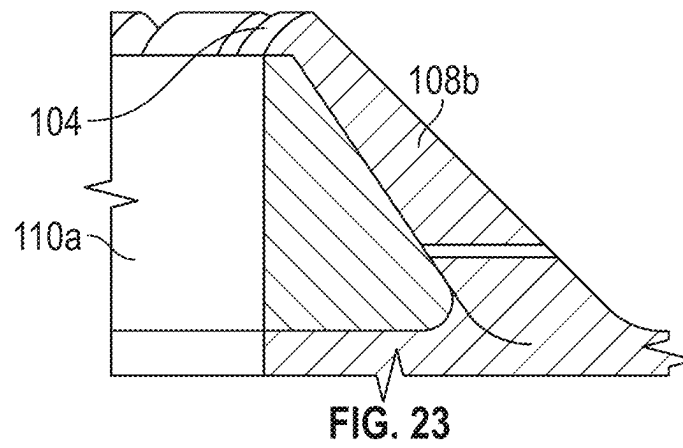


FIG. 22



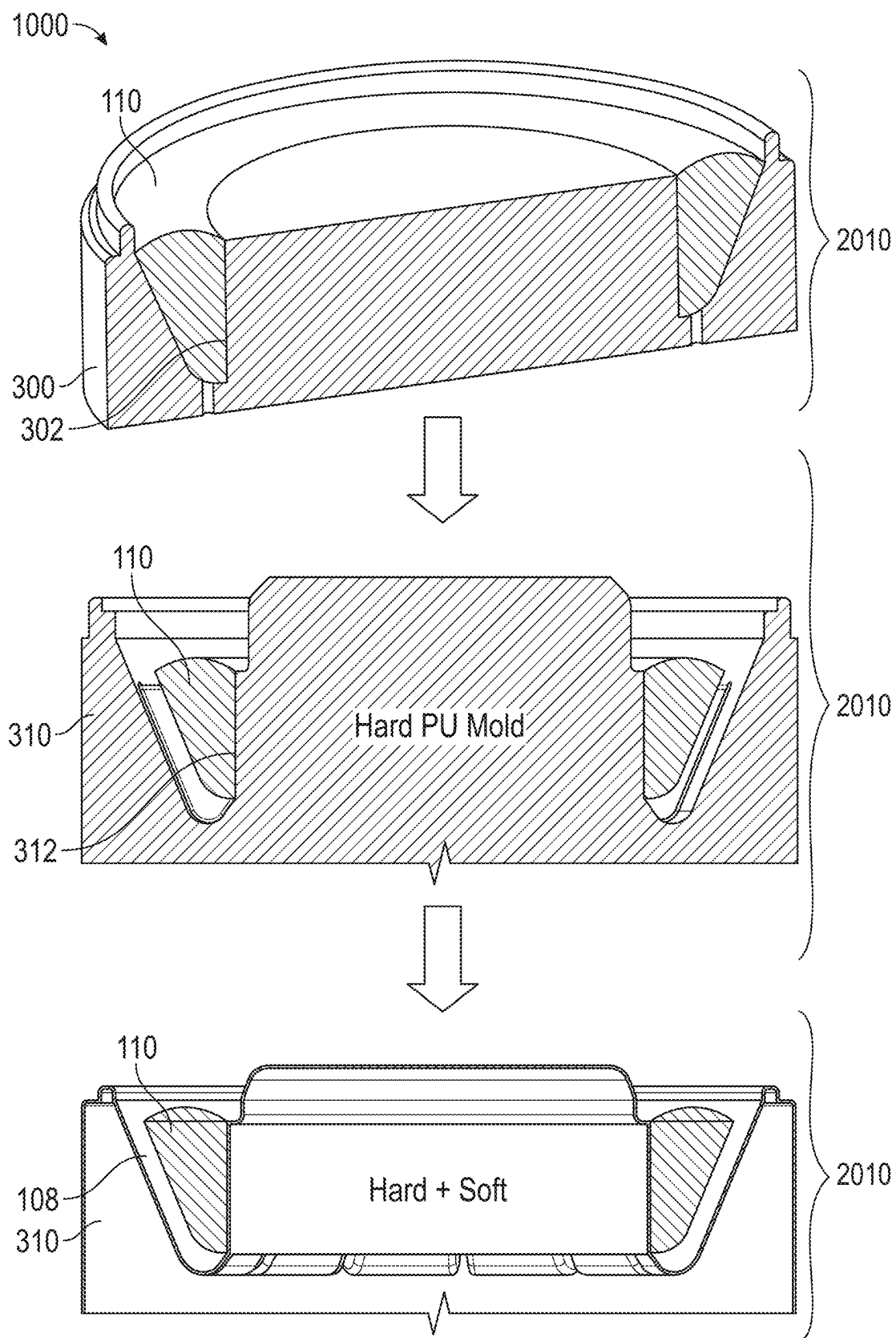


FIG. 26

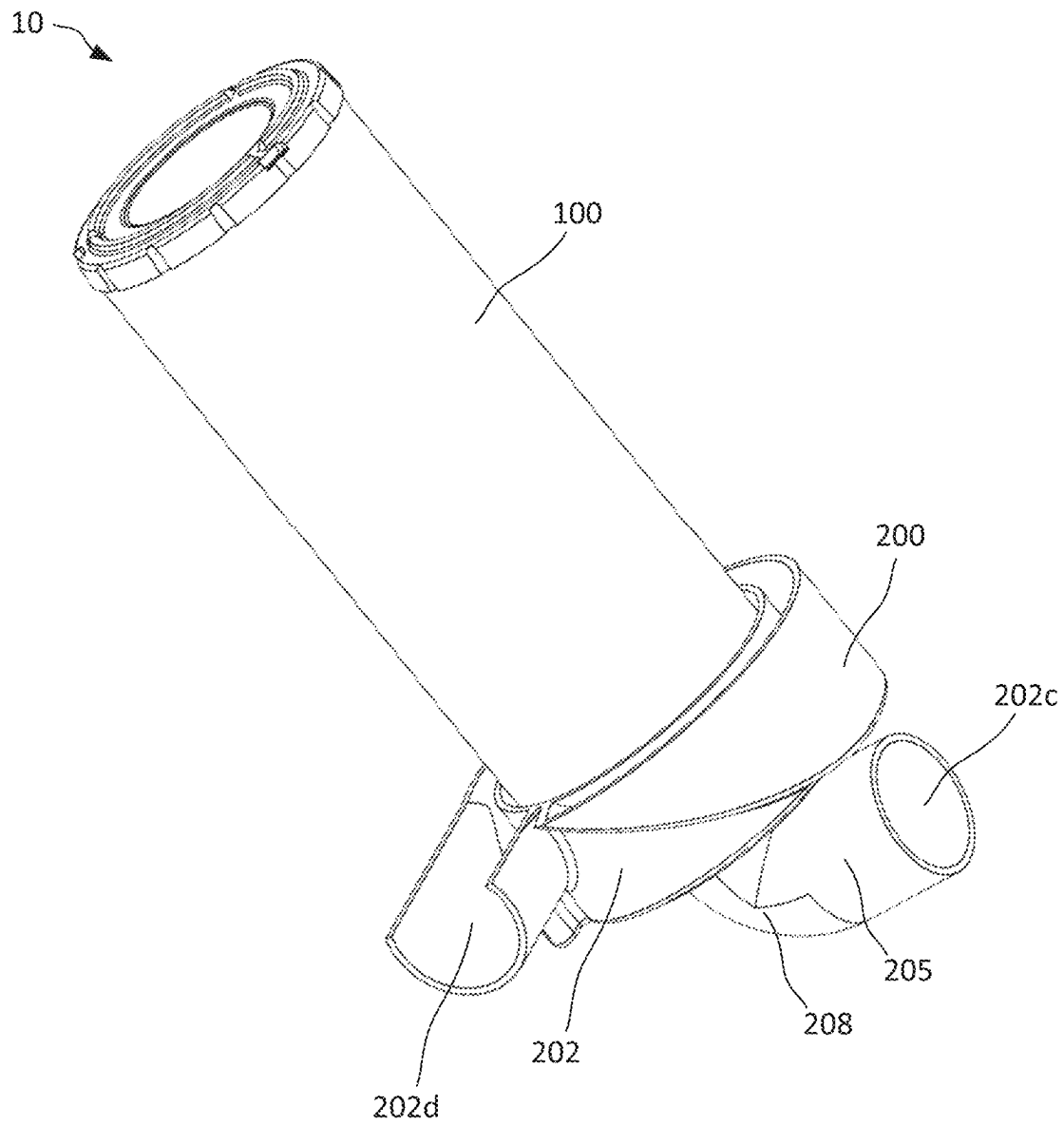


FIG. 27

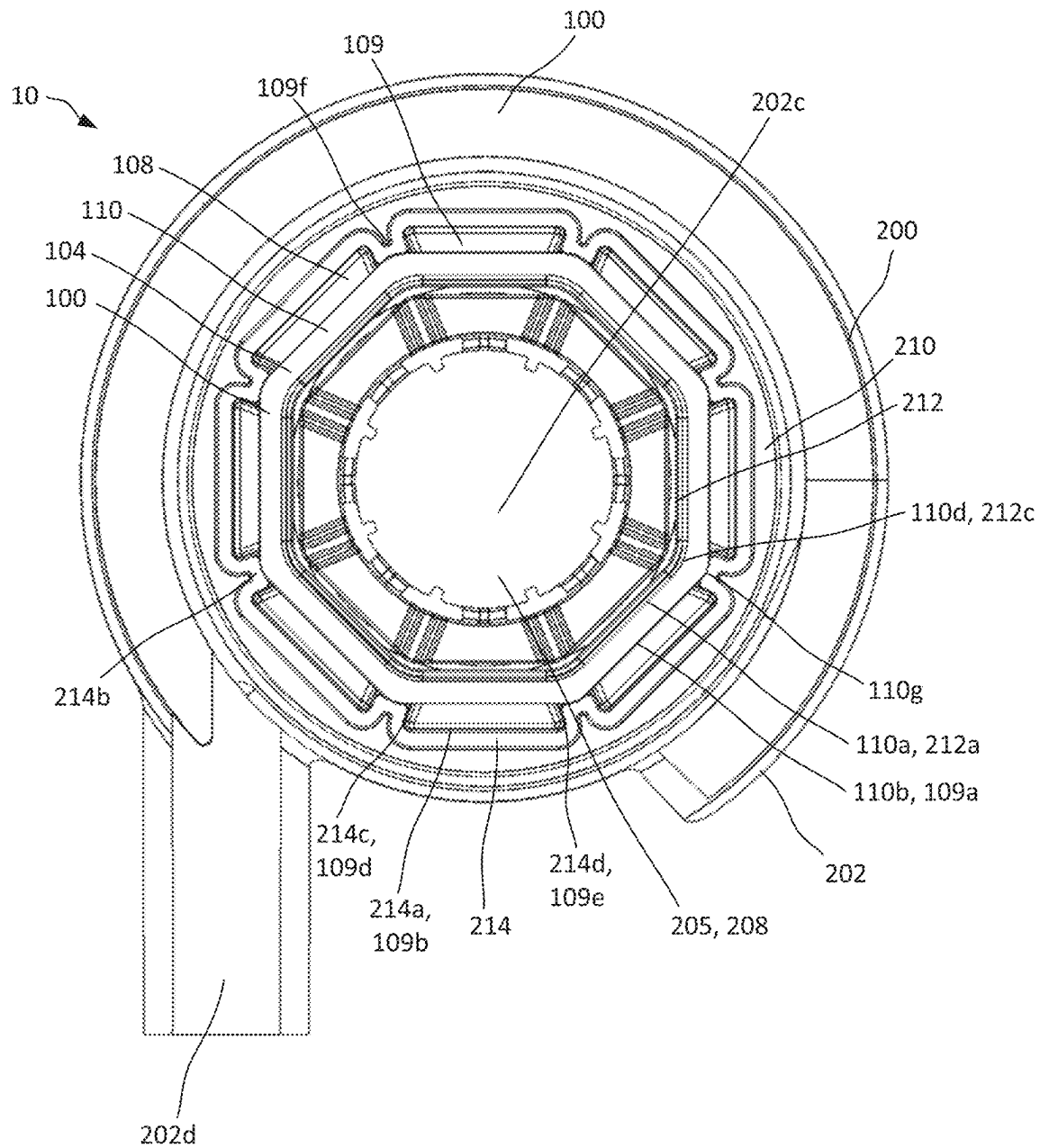


FIG. 28

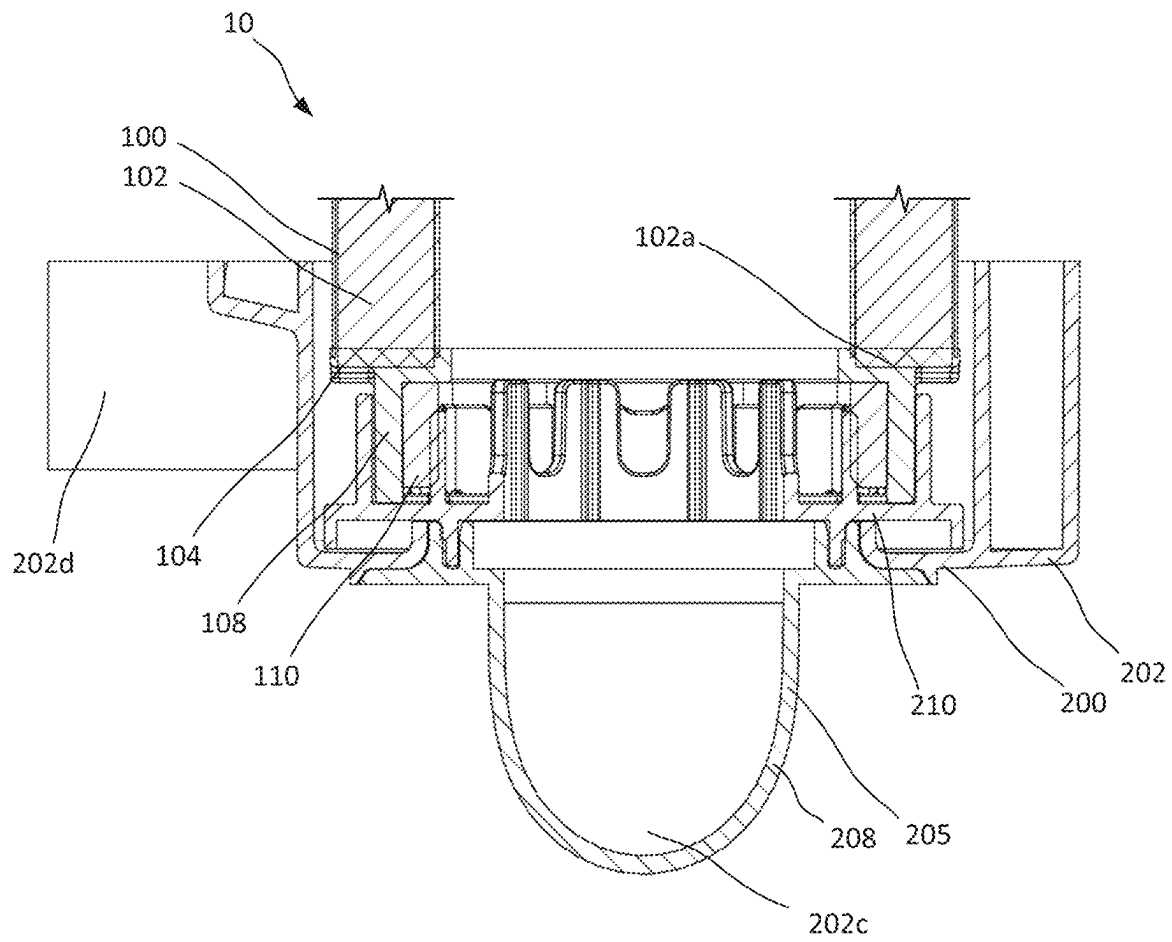


FIG. 29

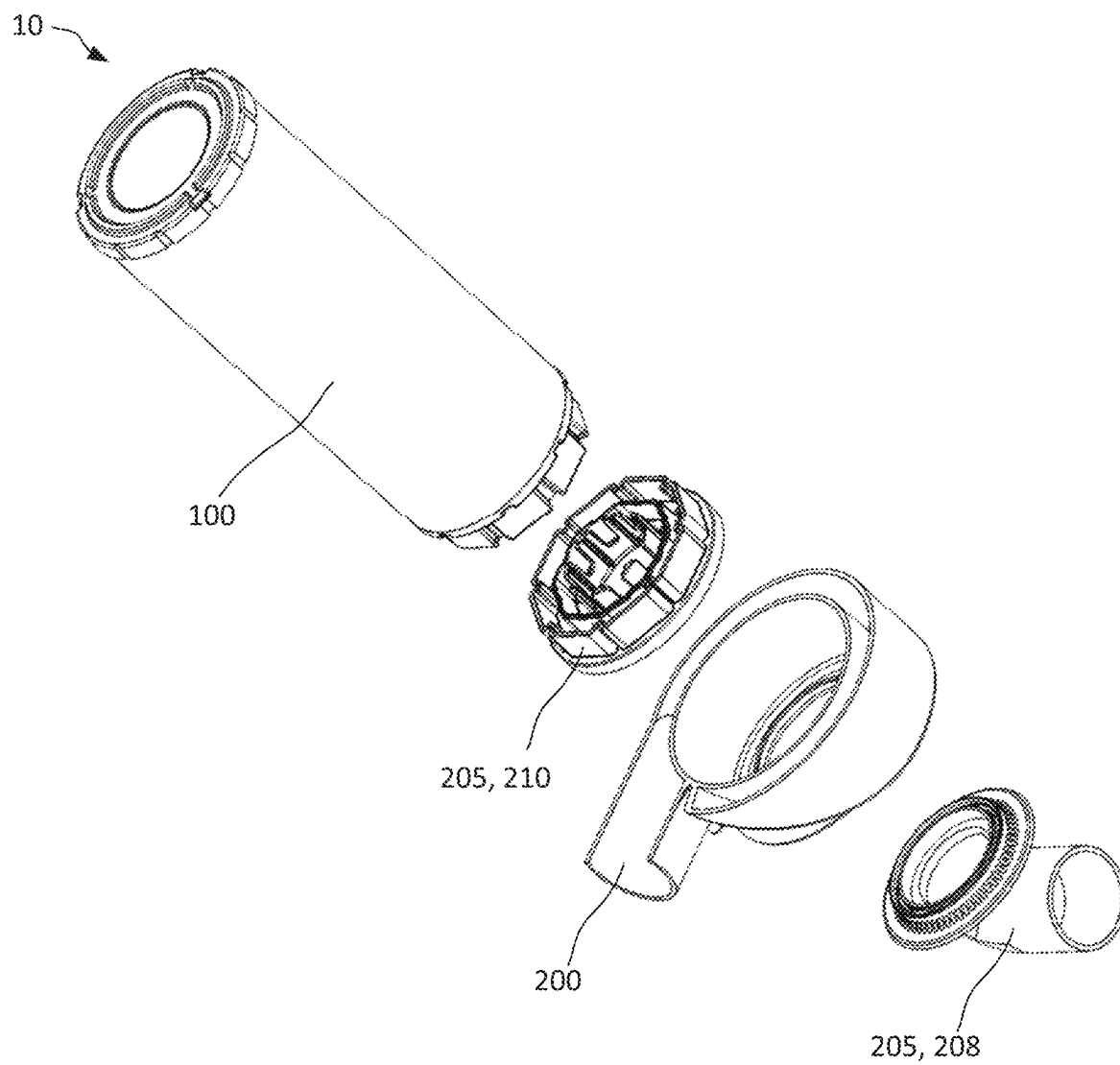


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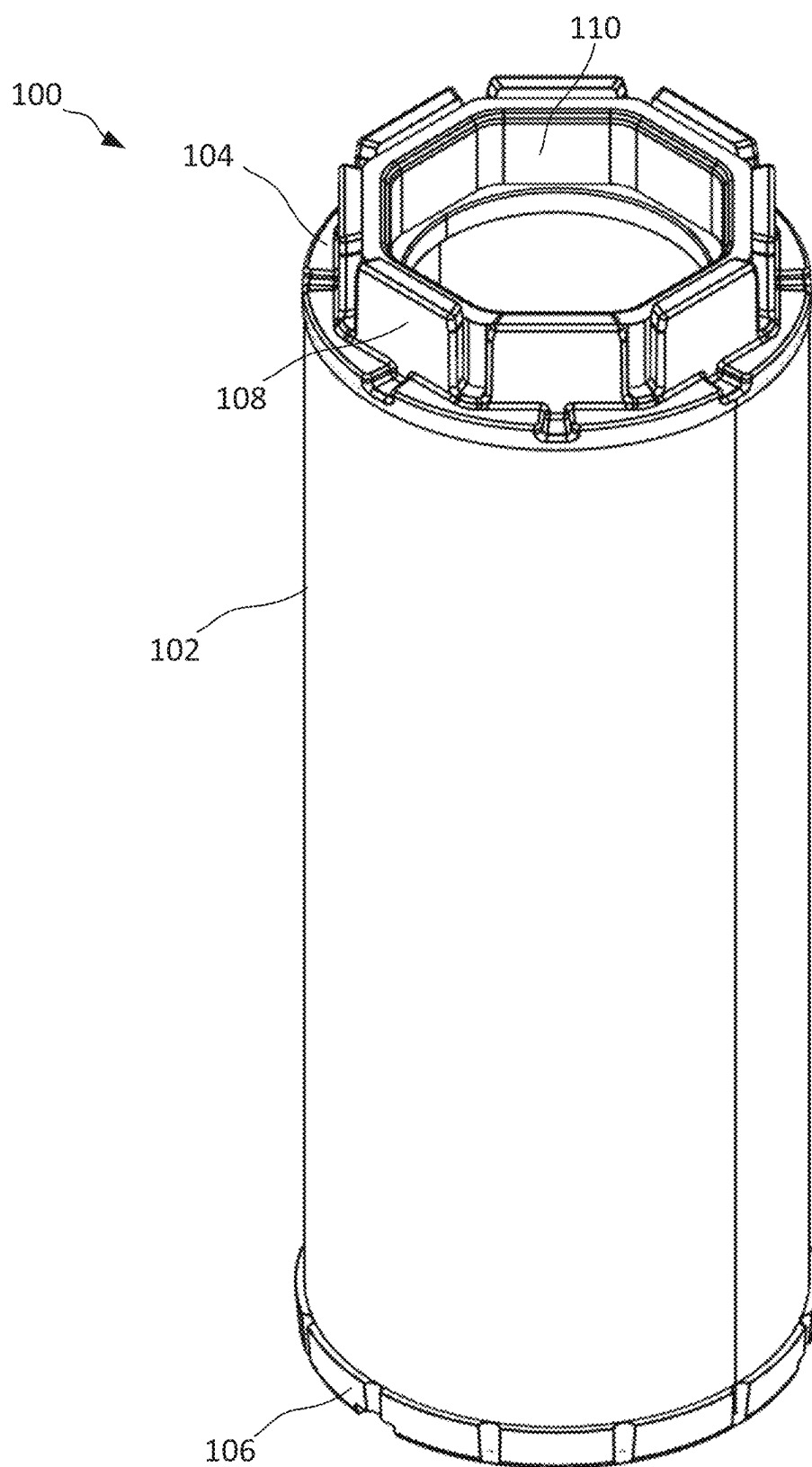


FIG. 31

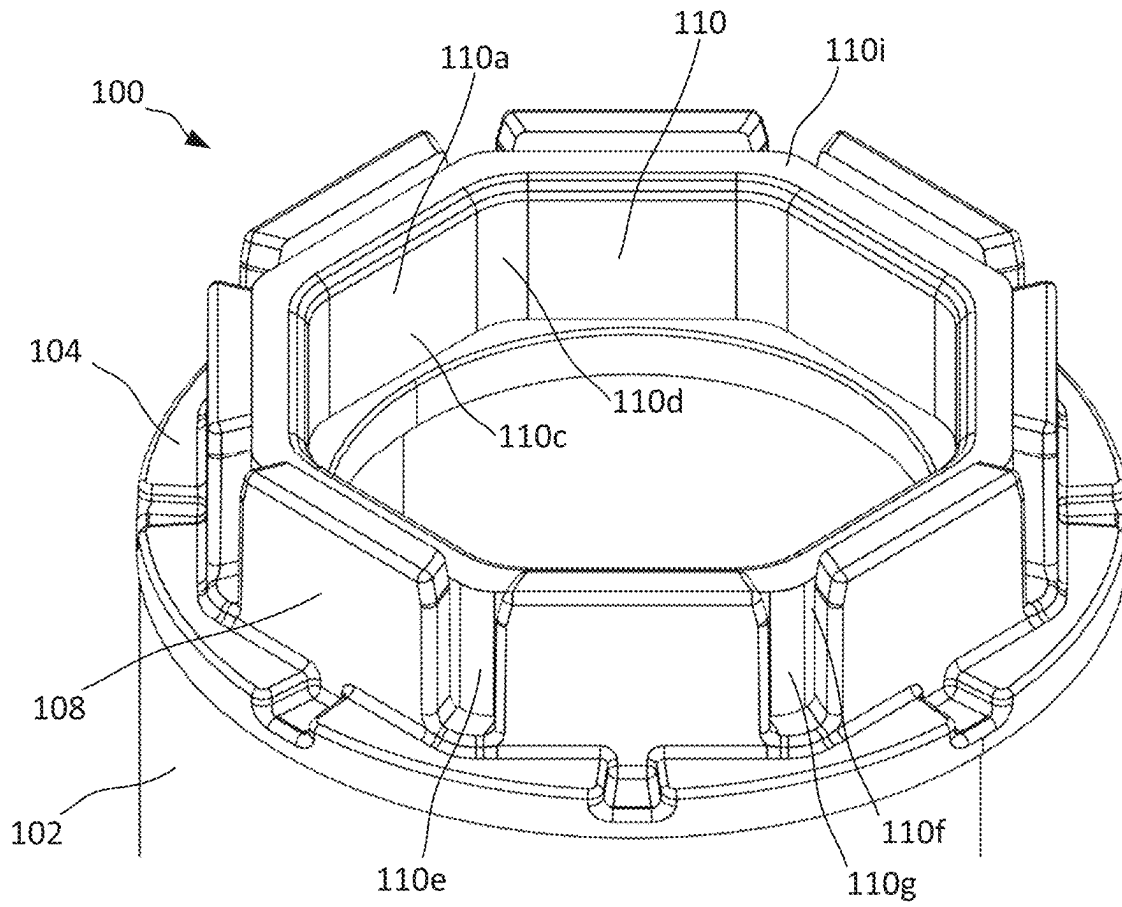


FIG. 32

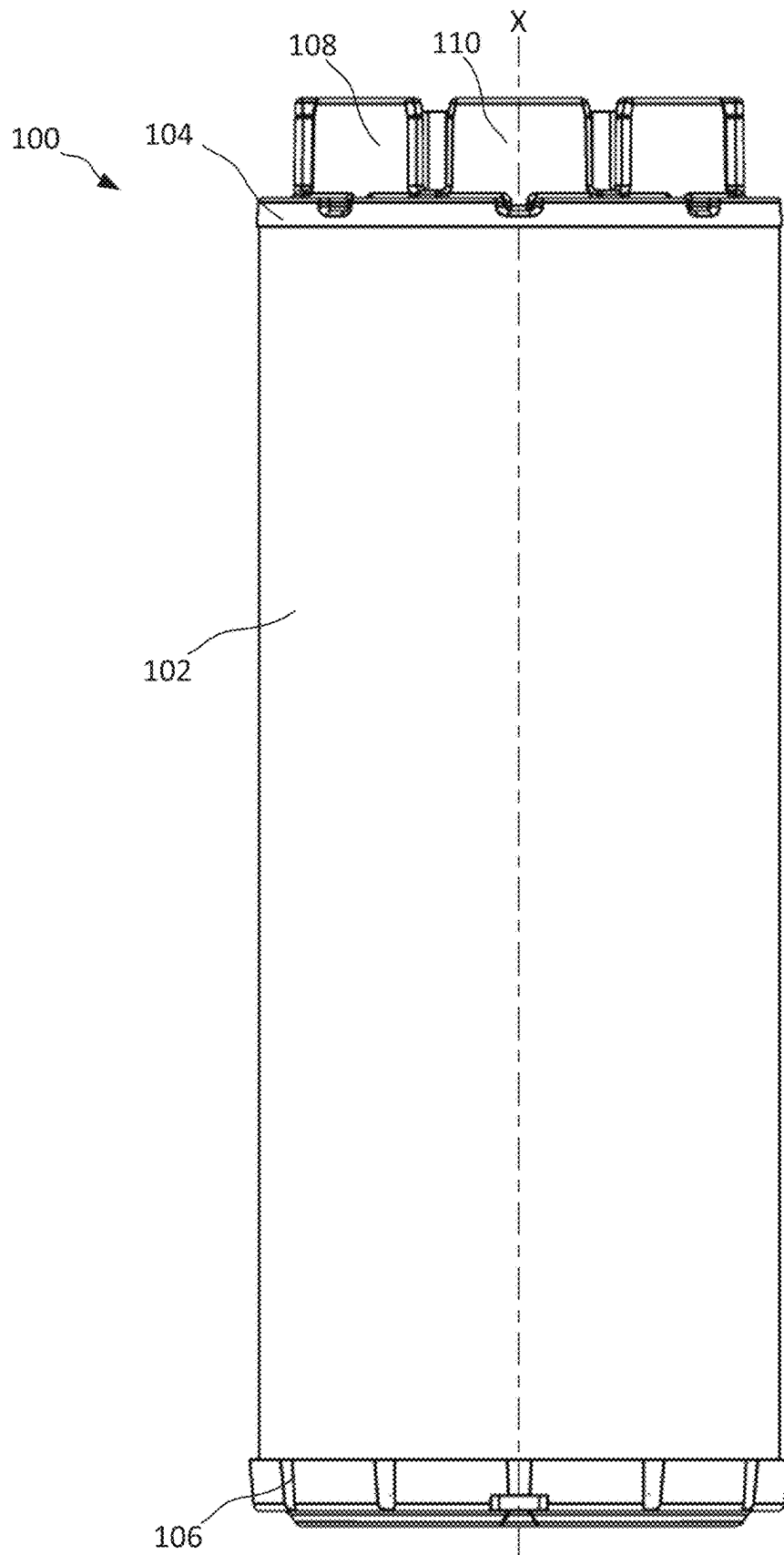


FIG. 33

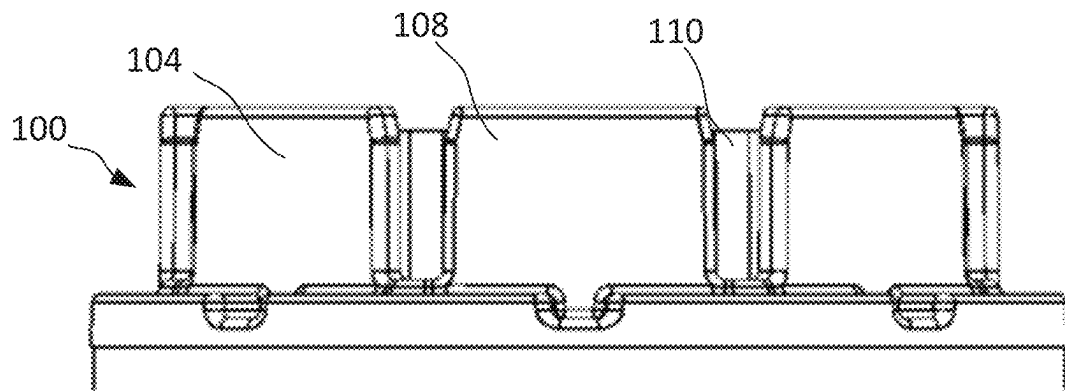


FIG. 34

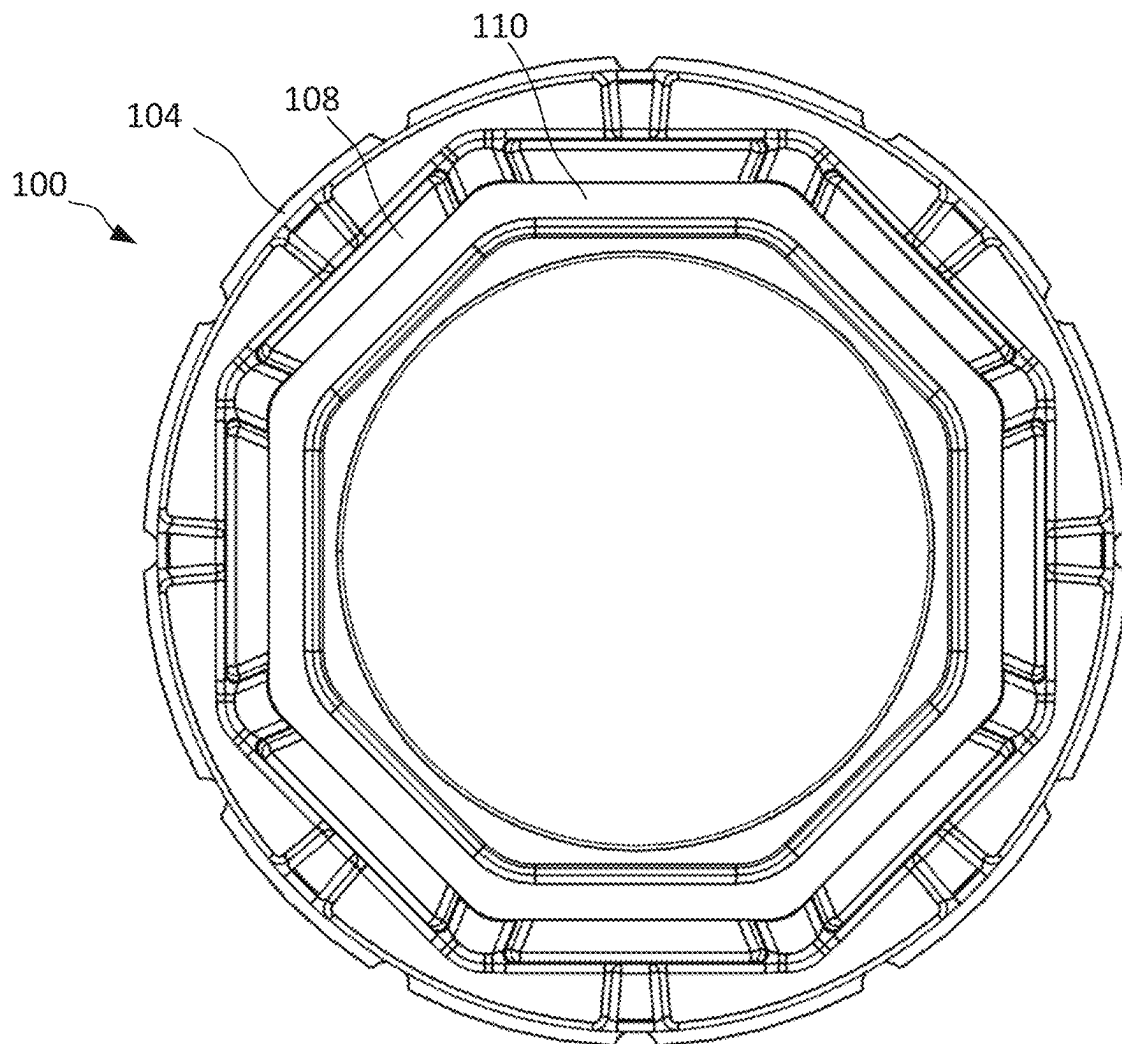


FIG. 35

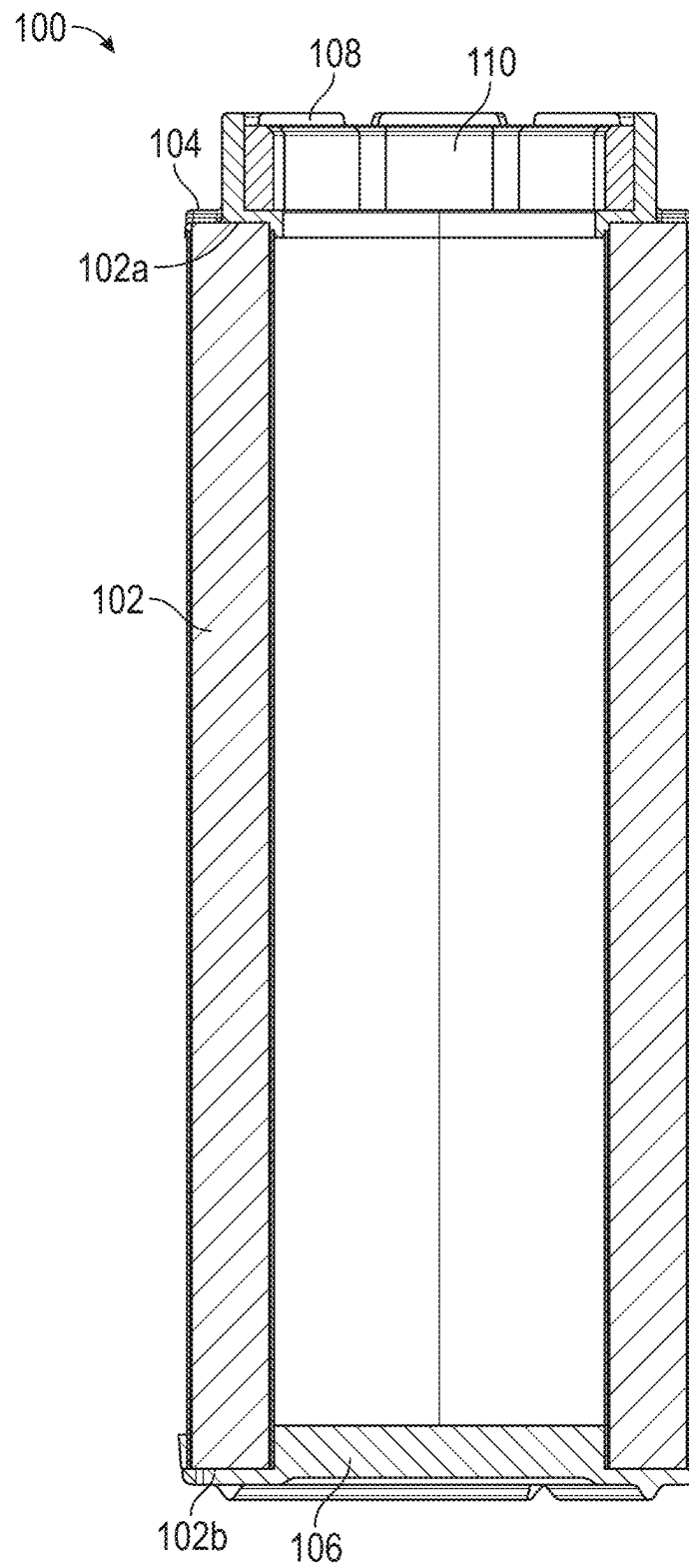


FIG. 36

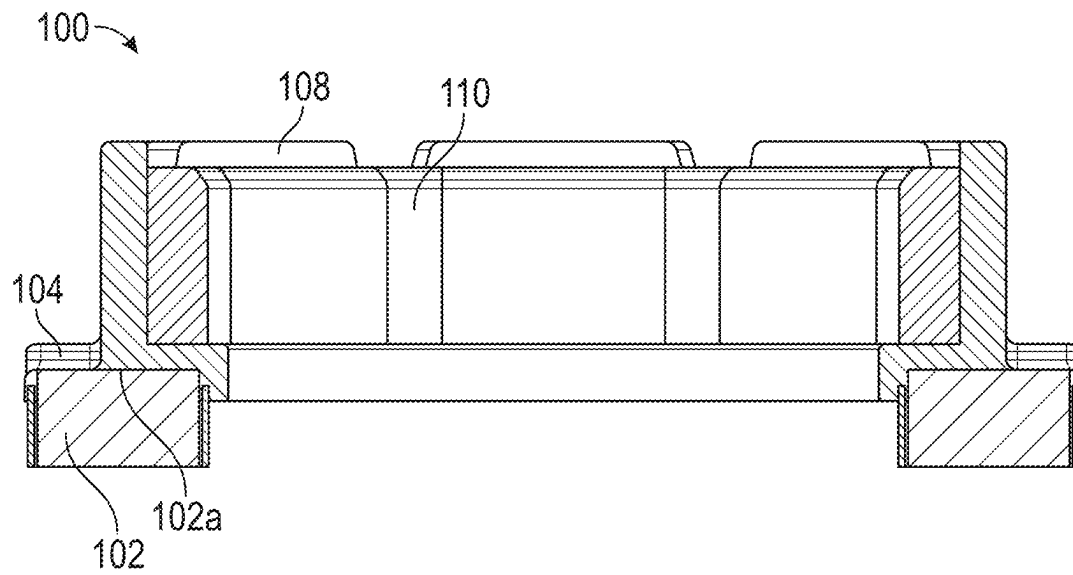


FIG. 37

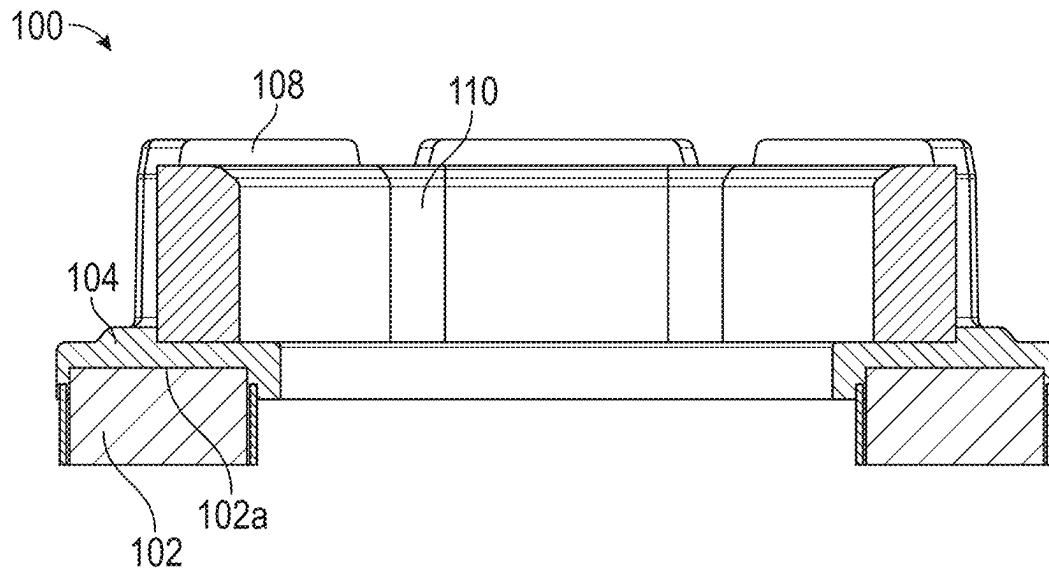


FIG. 38

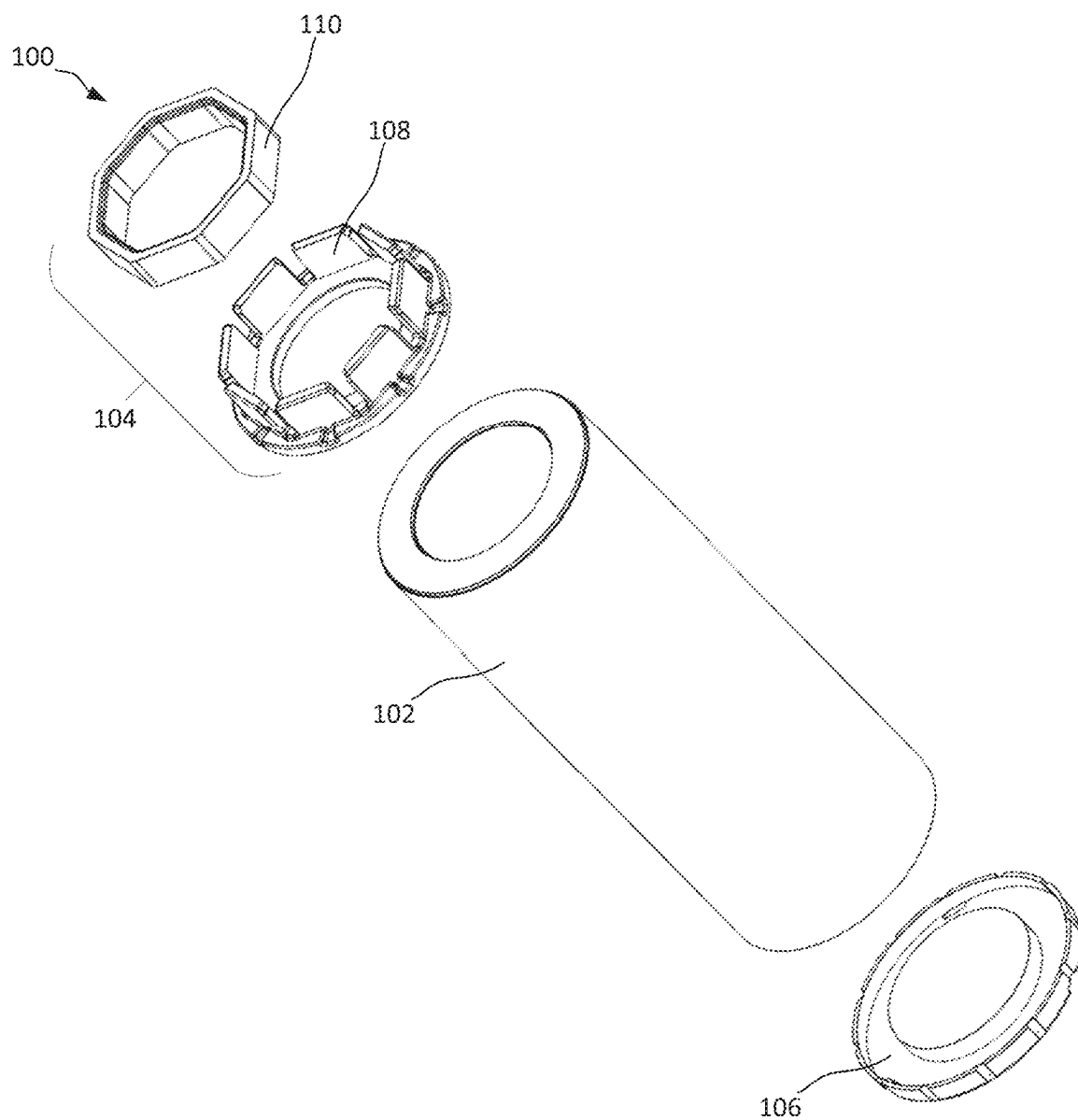


FIG. 39

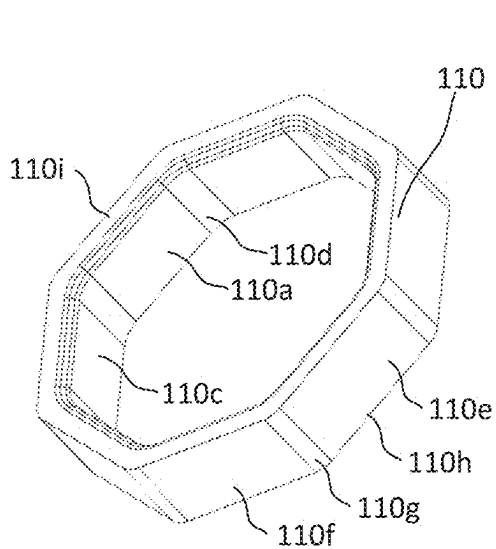


FIG. 40

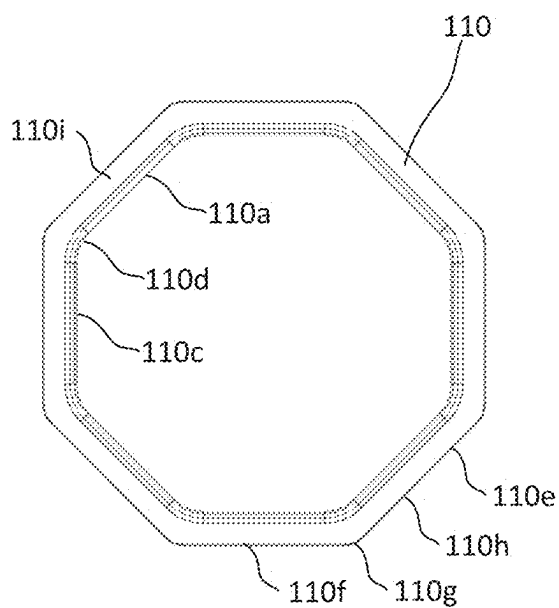


FIG. 41

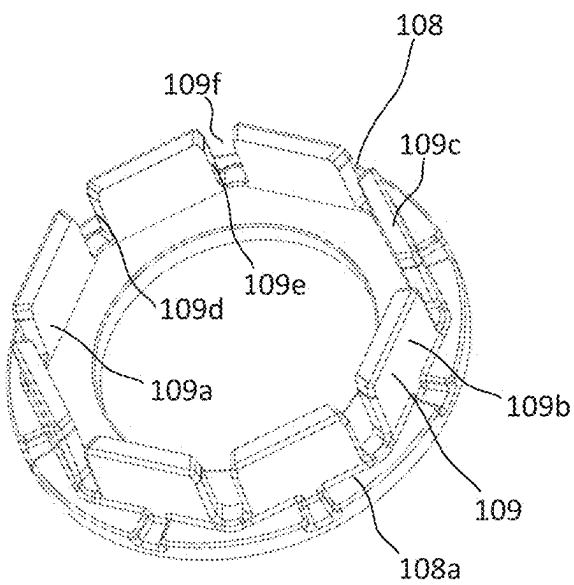


FIG. 42

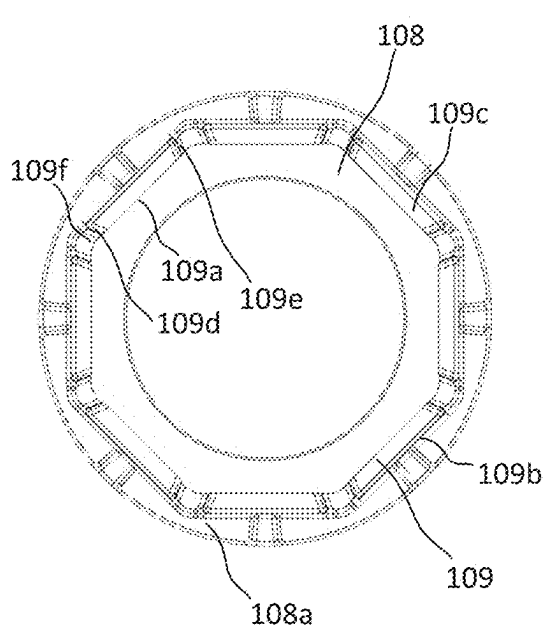
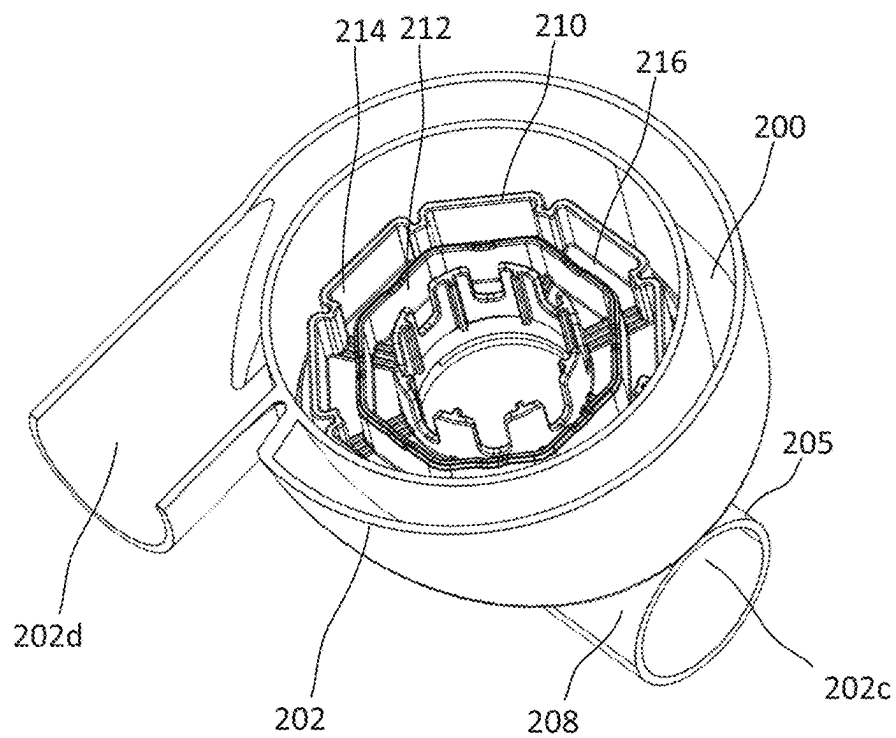
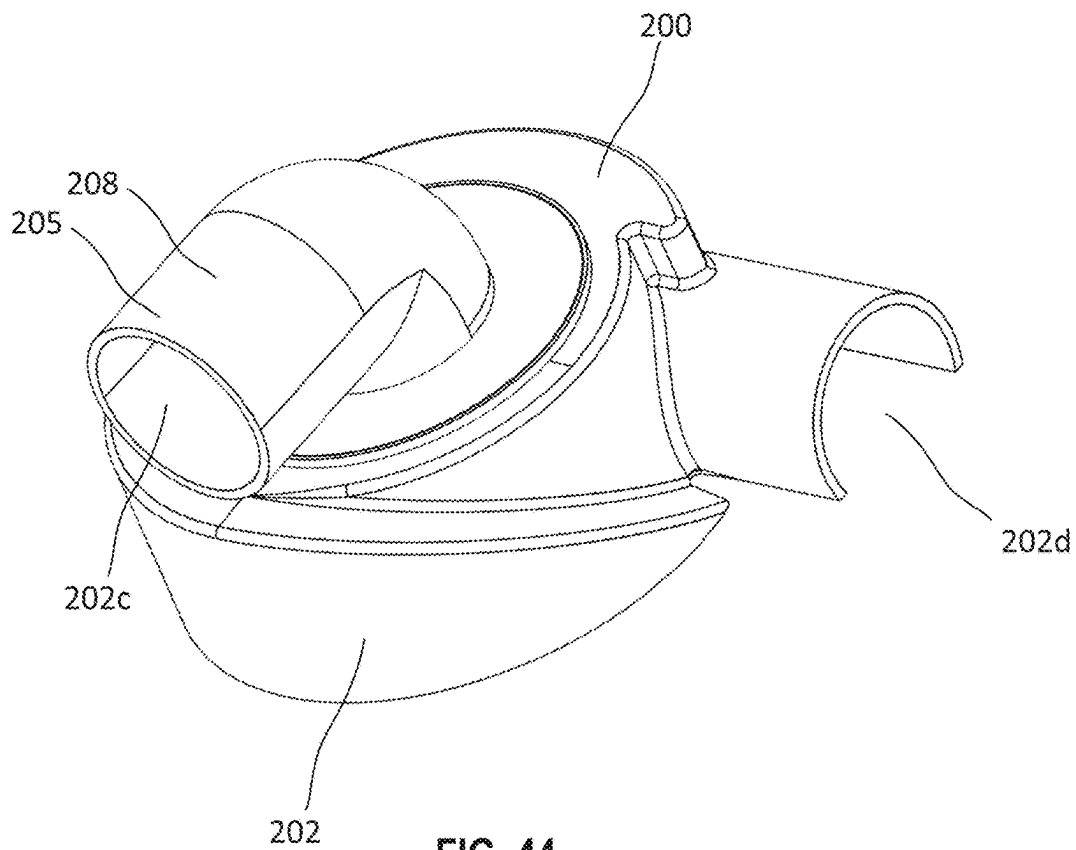


FIG. 43



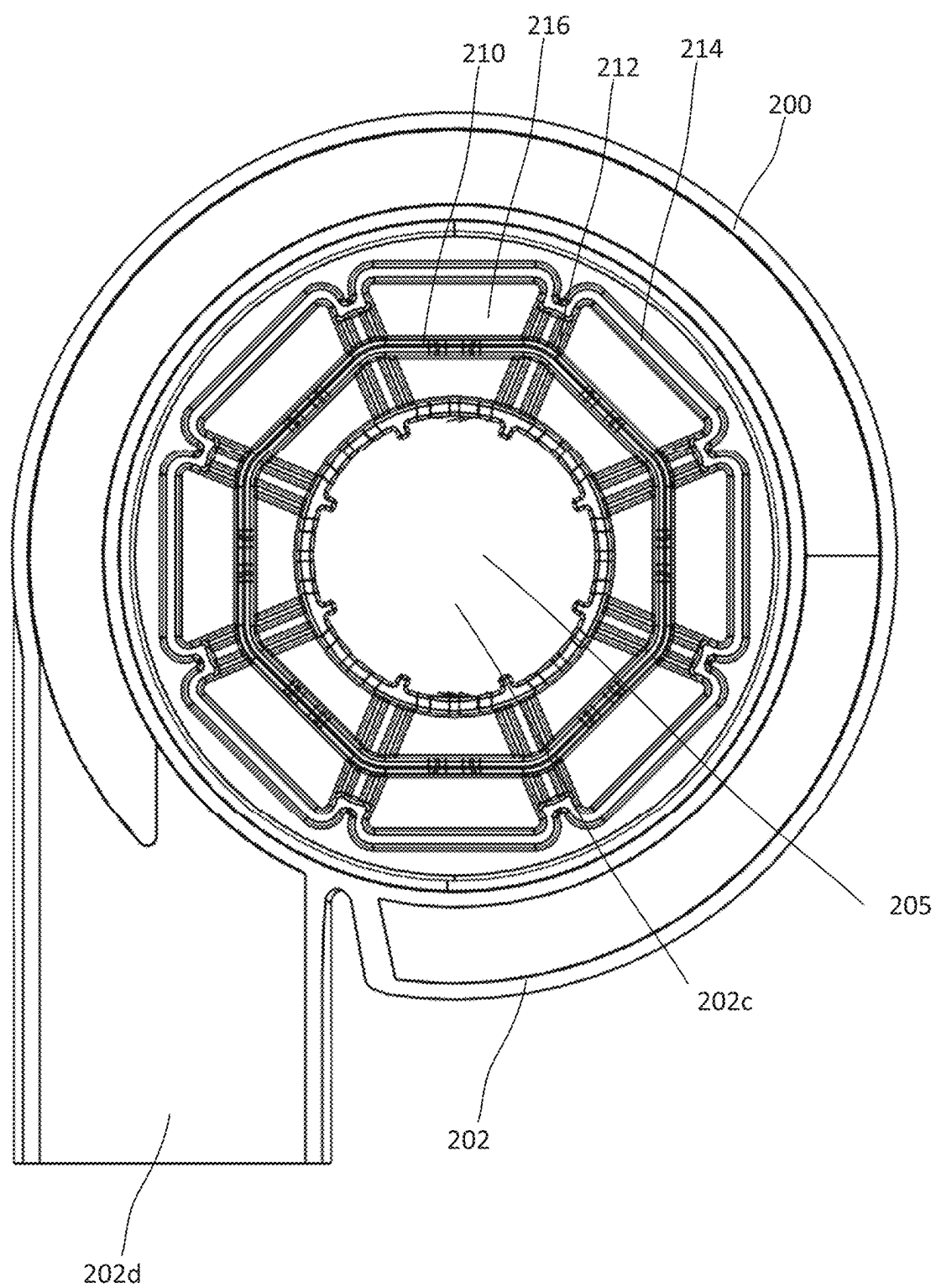


FIG. 46

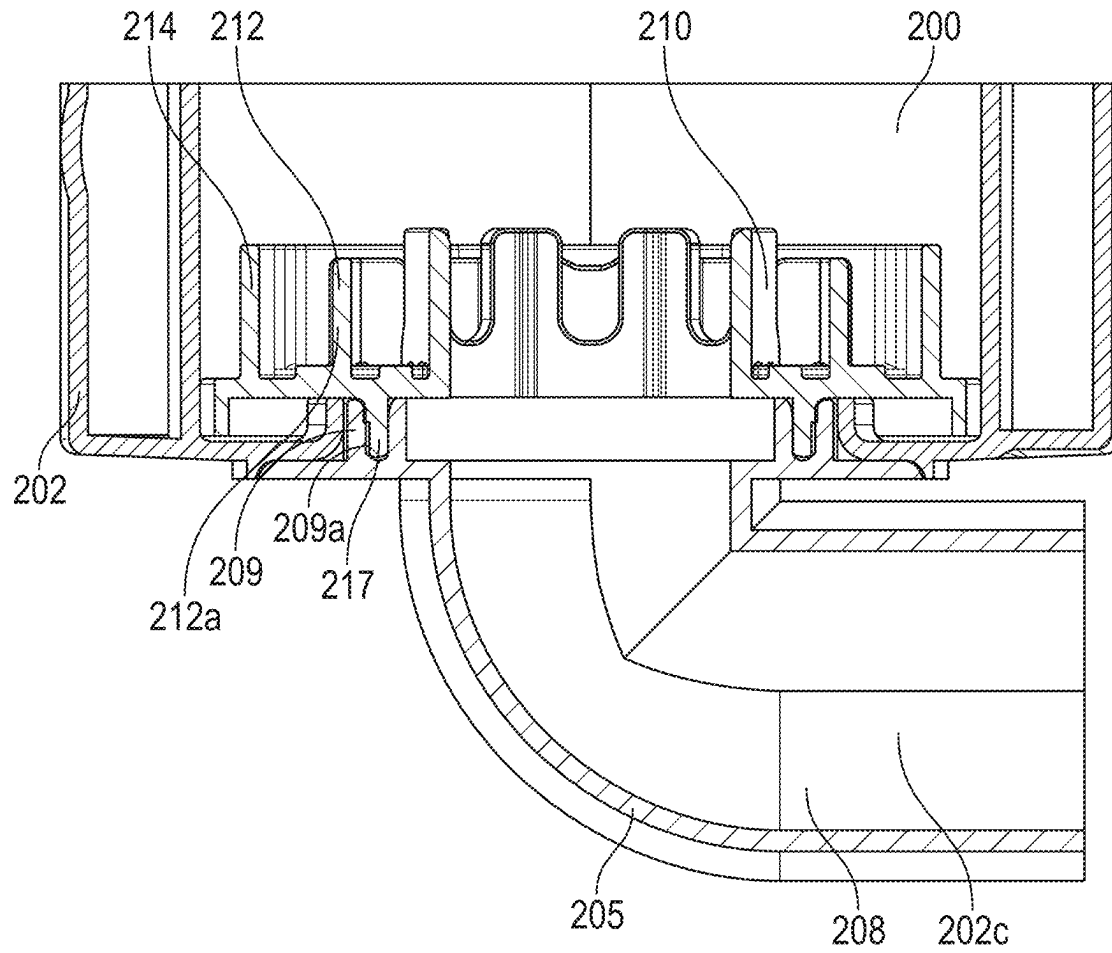


FIG. 47

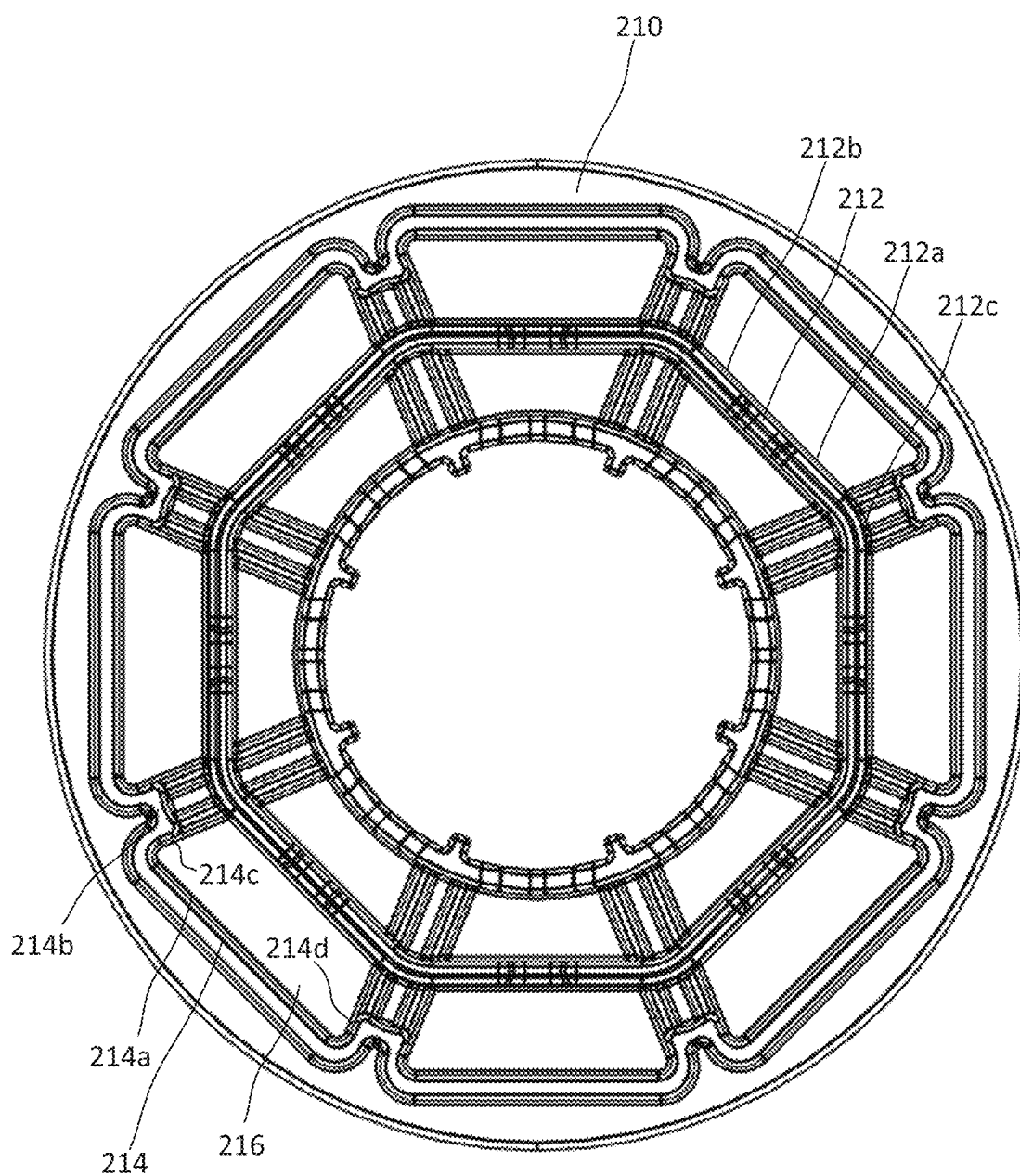


FIG. 48

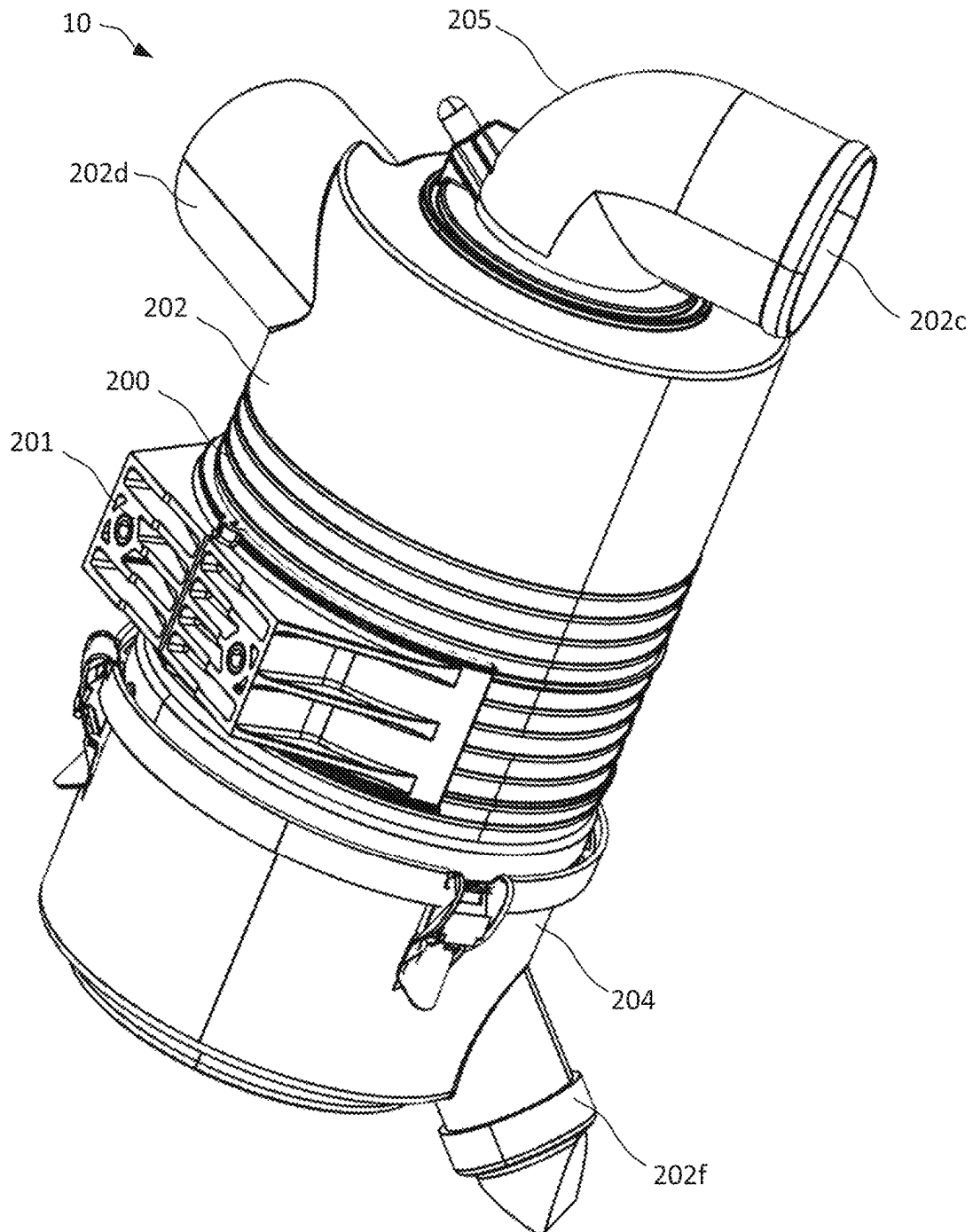


FIG. 49

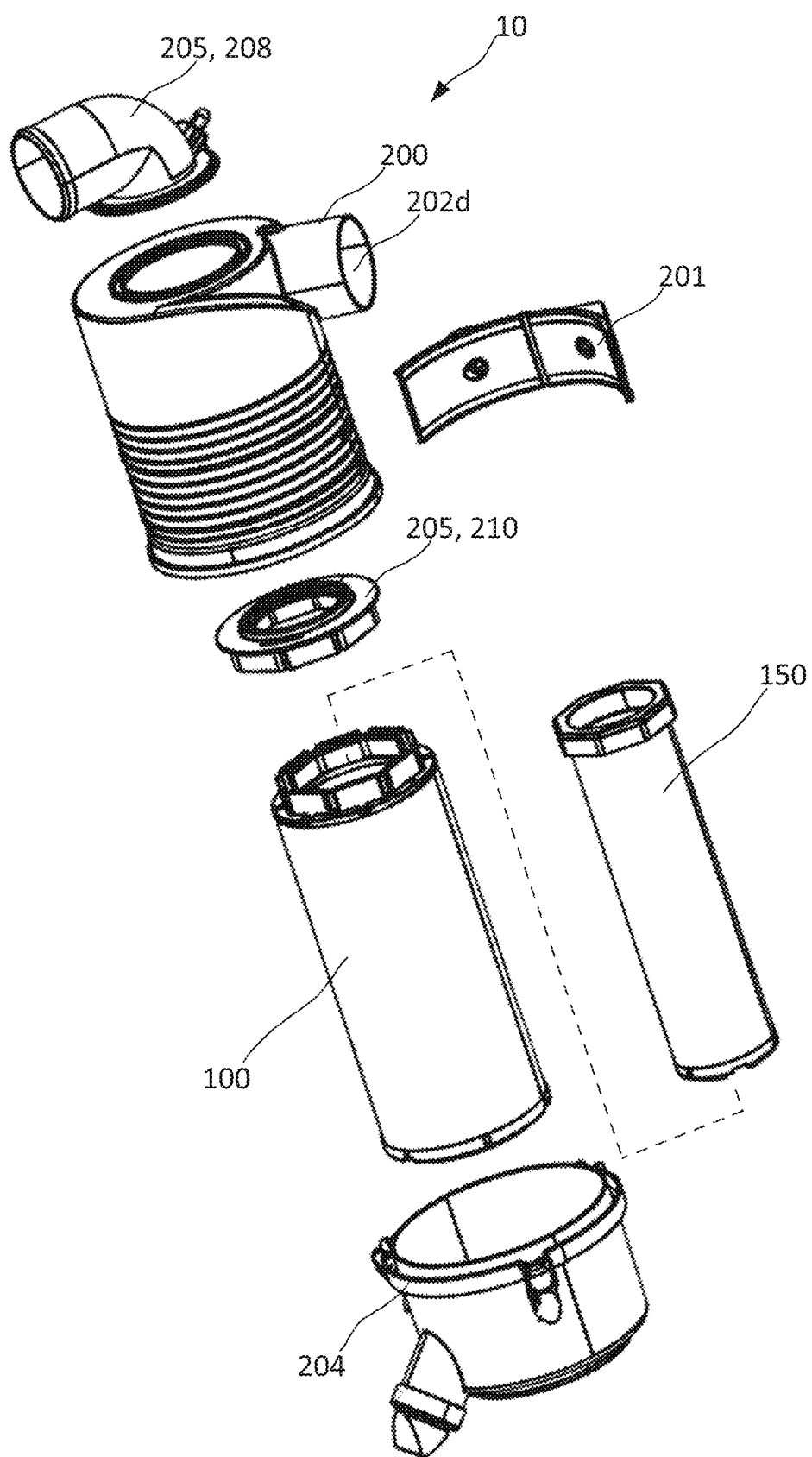


FIG. 50

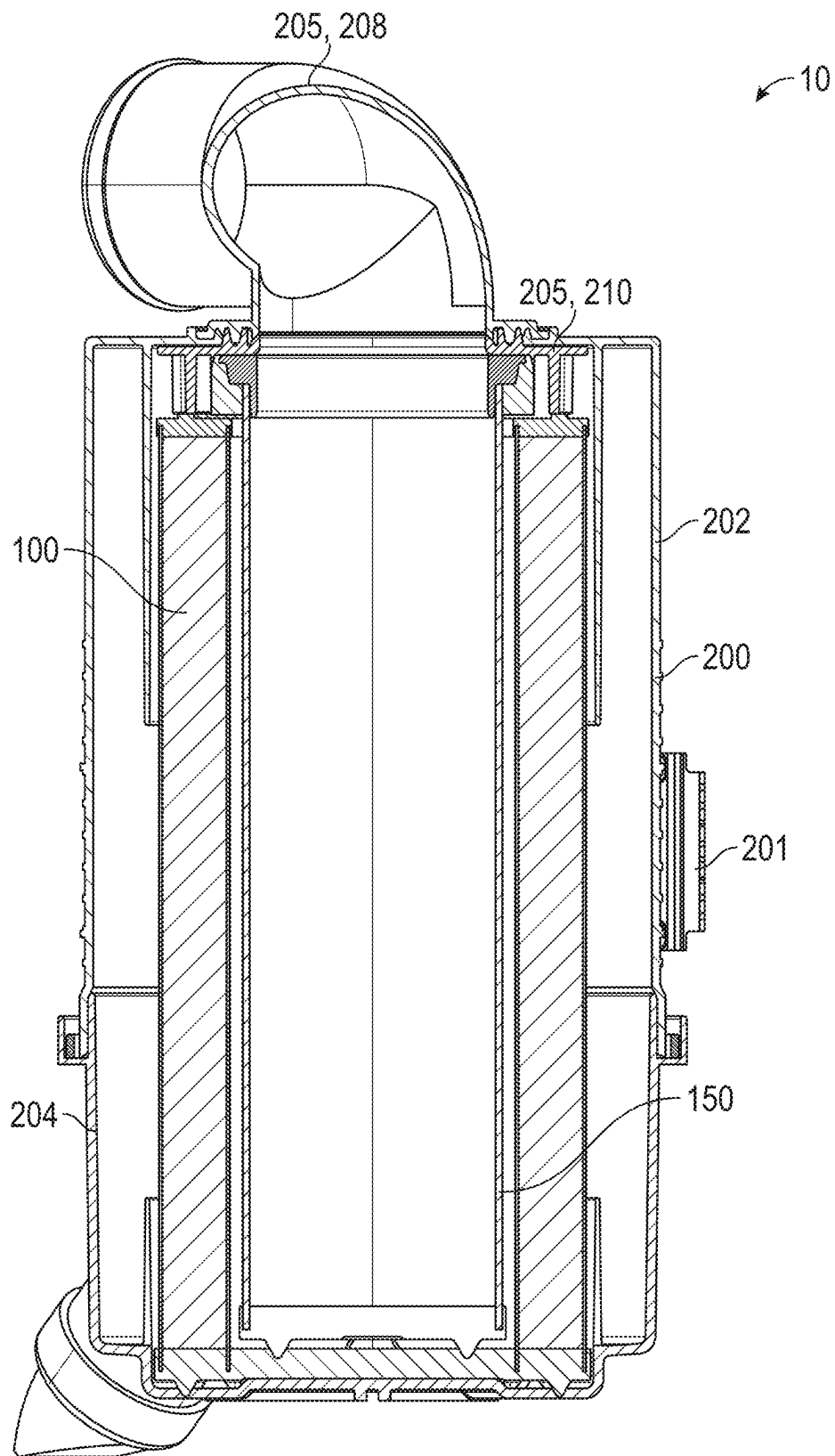


FIG. 51

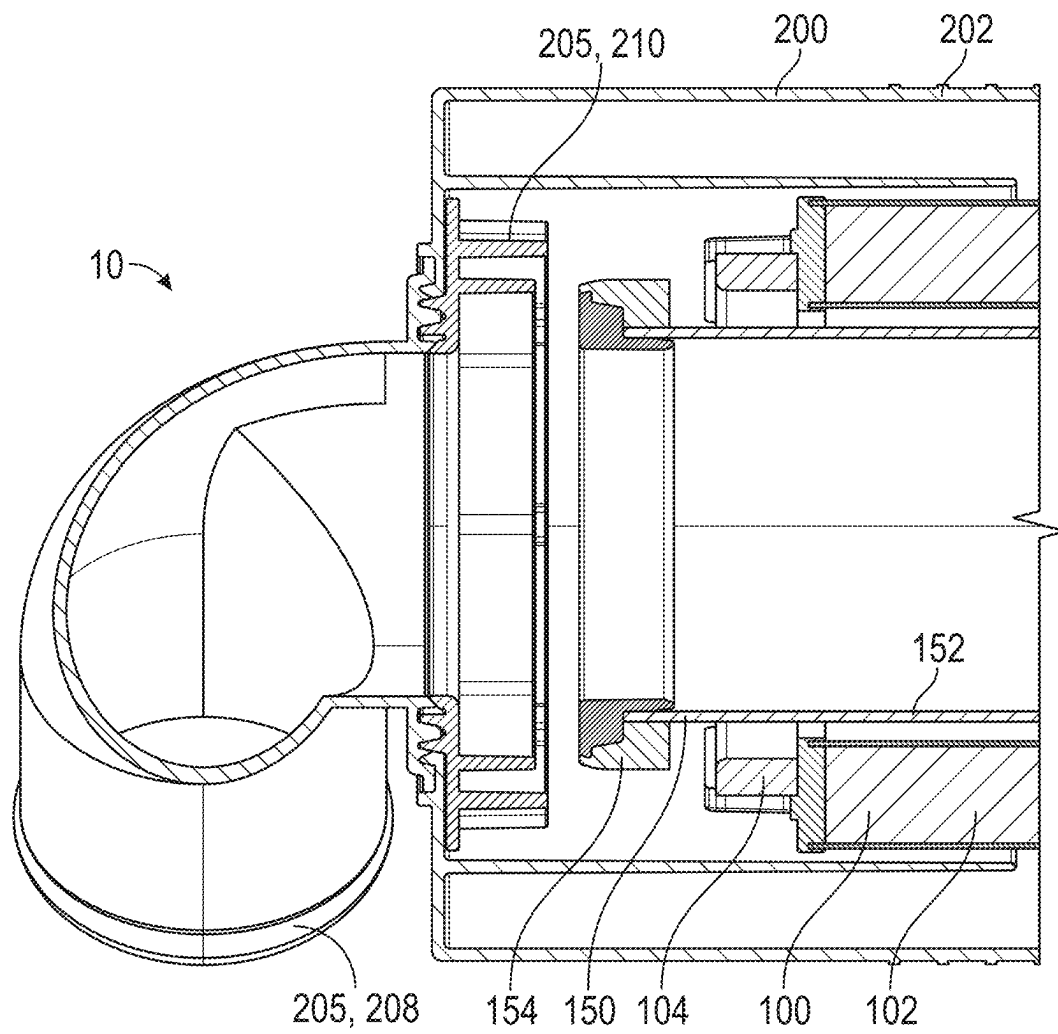


FIG. 51A

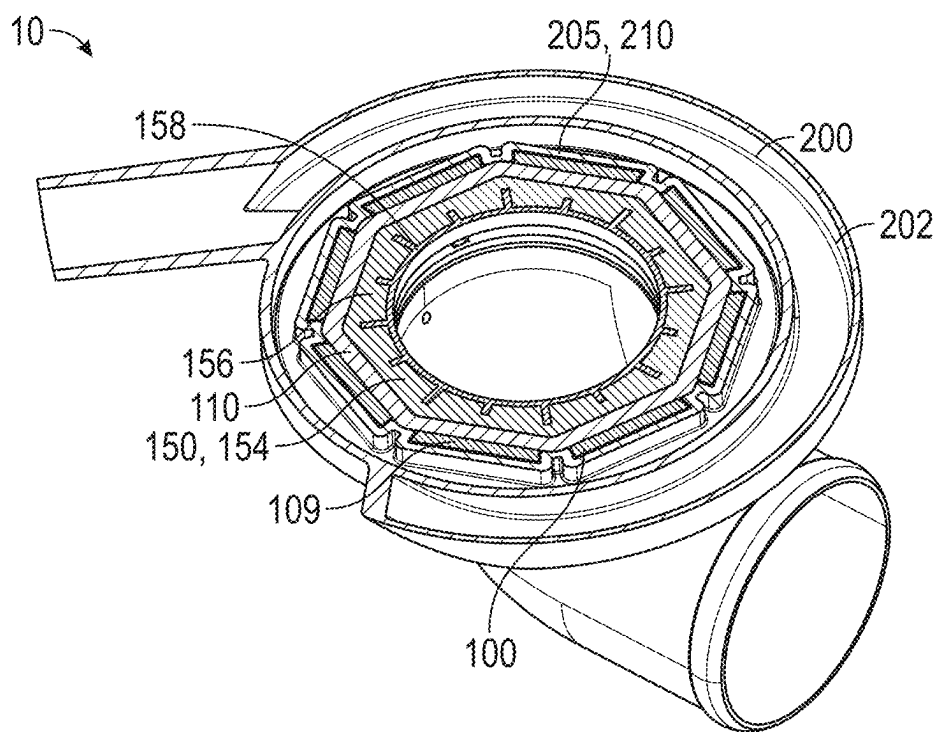


FIG. 51B

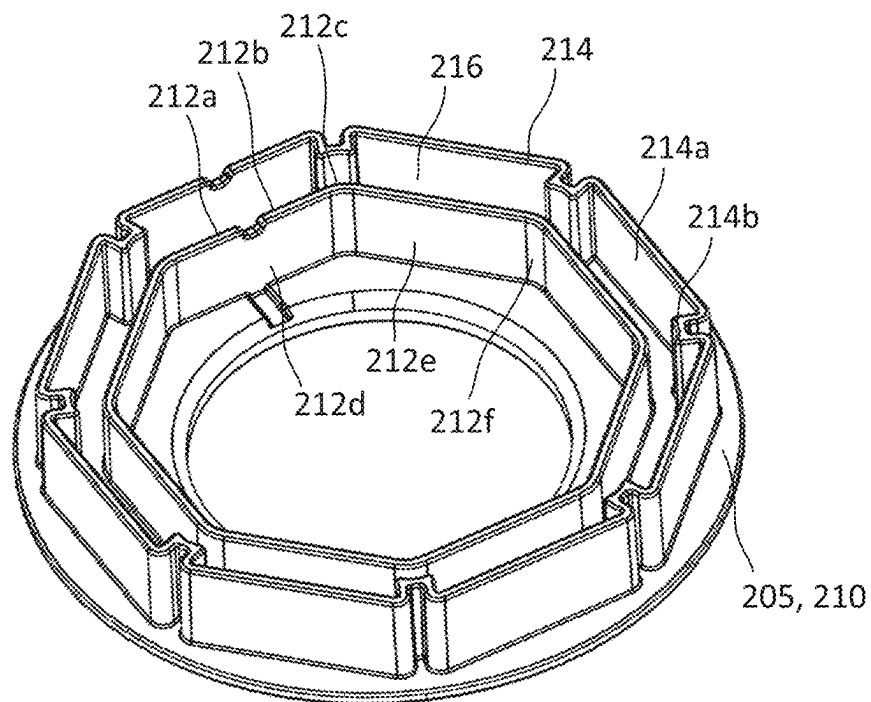


FIG. 52

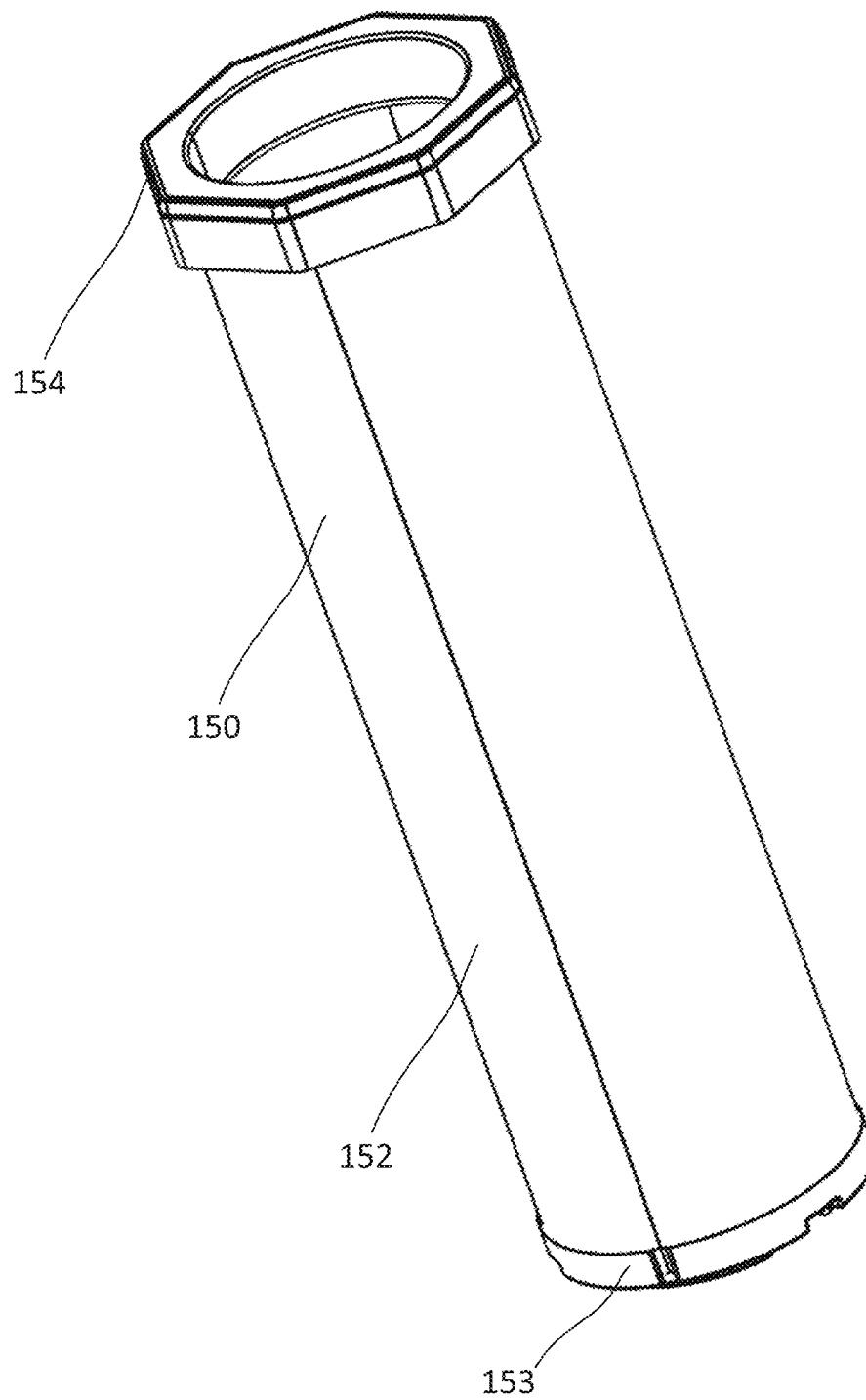


FIG. 53

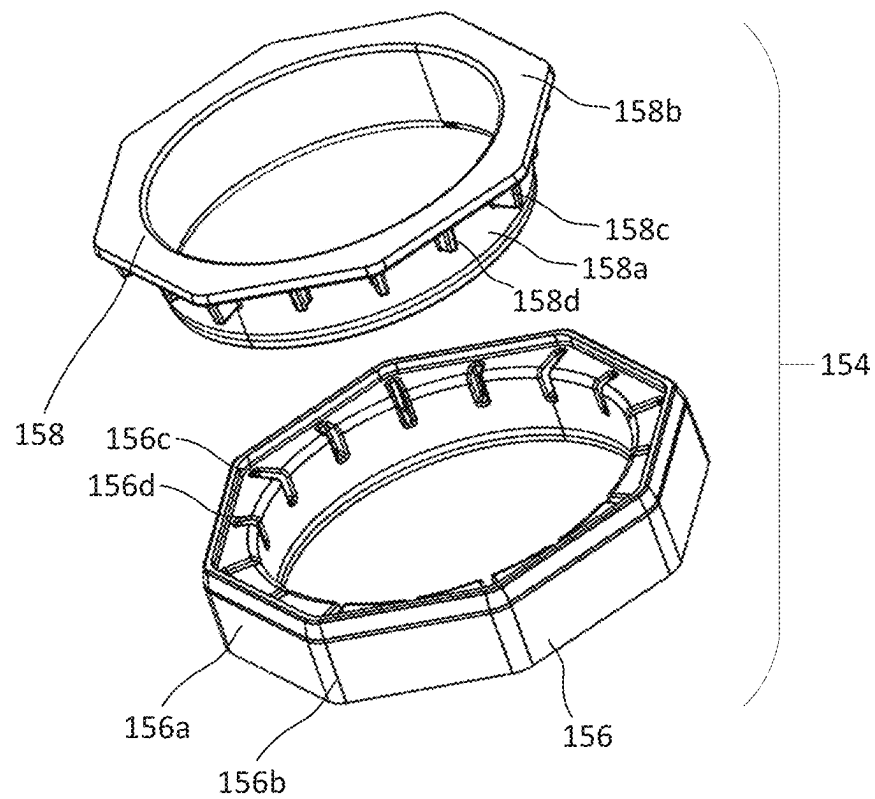


FIG. 54

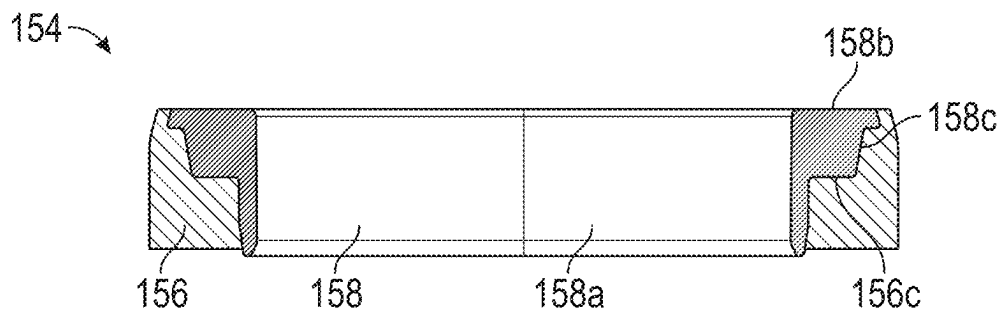


FIG. 55A

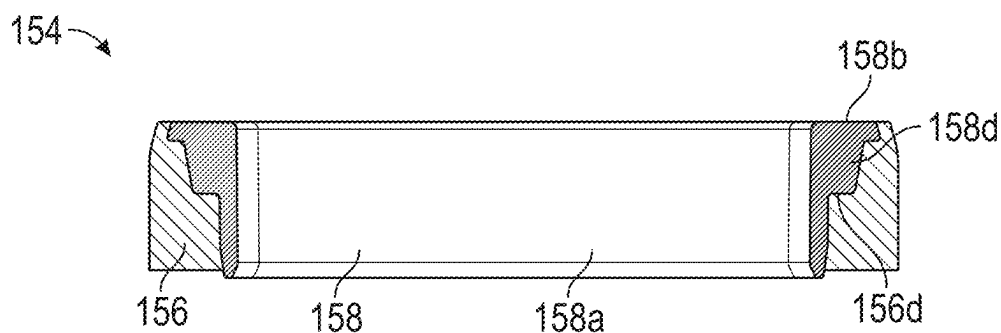


FIG. 55B

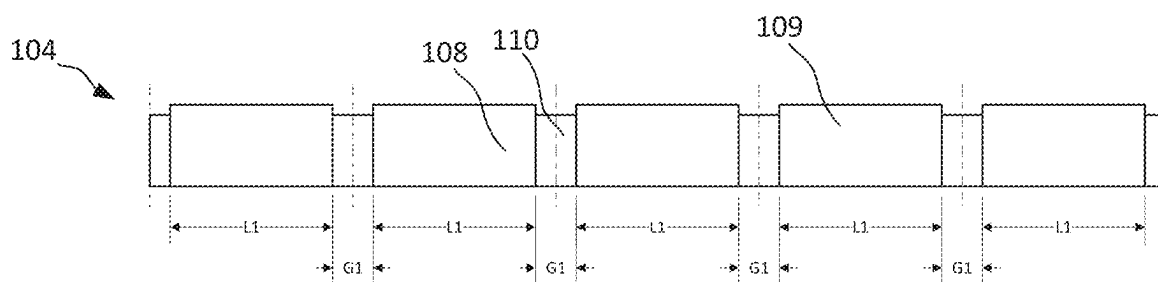


FIG. 56A

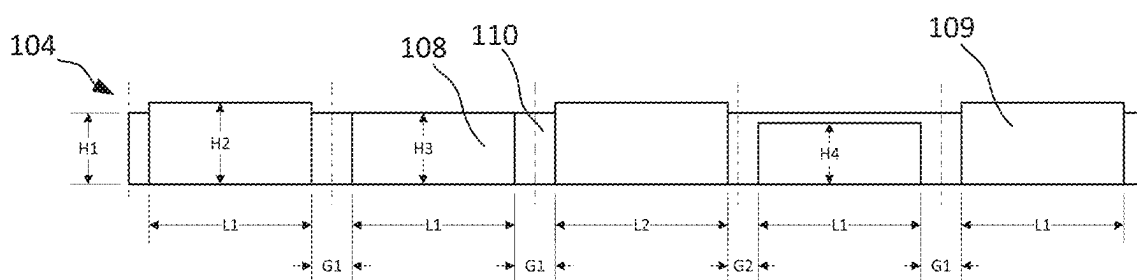


FIG. 56B

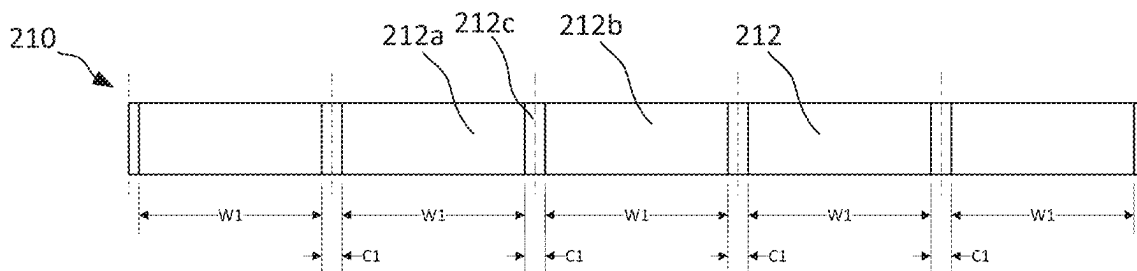


FIG. 56C

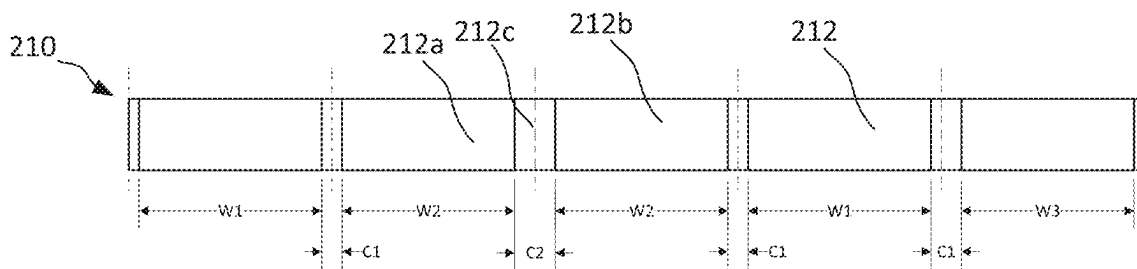


FIG. 56D

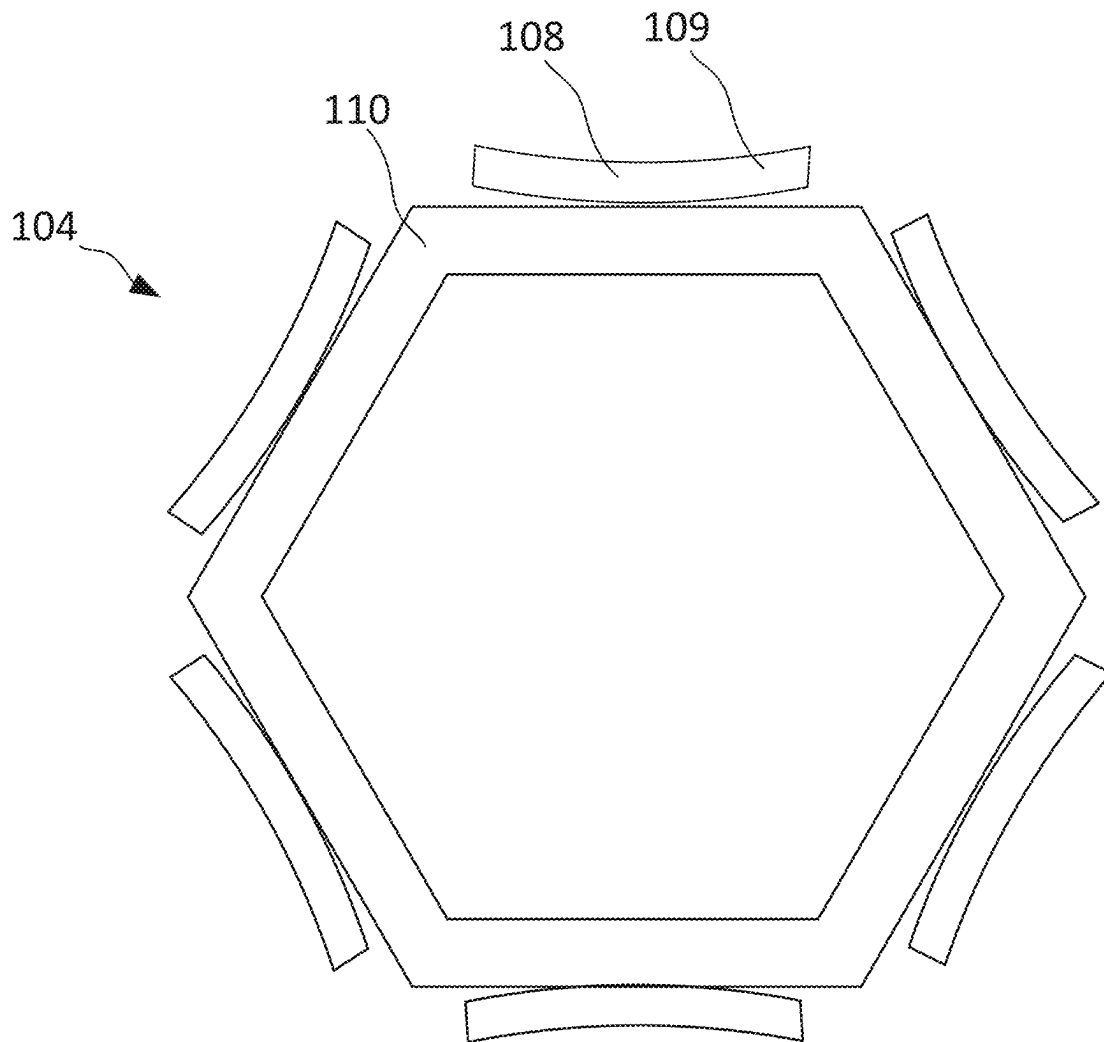


FIG. 57

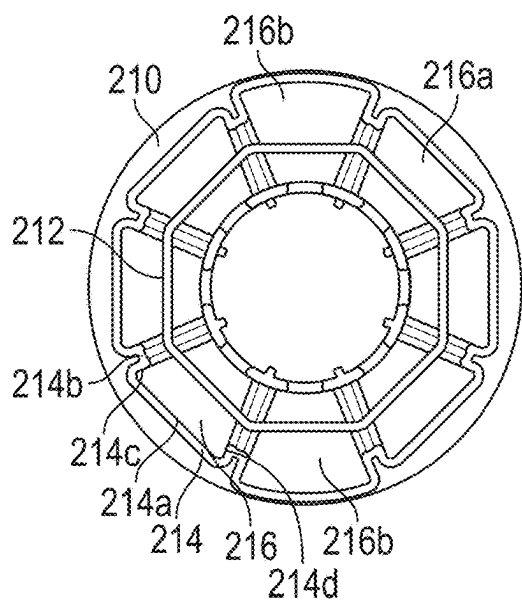


FIG. 58A

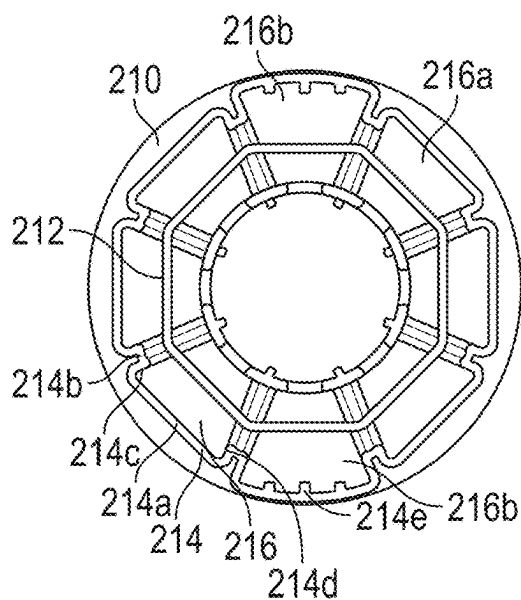


FIG. 58B

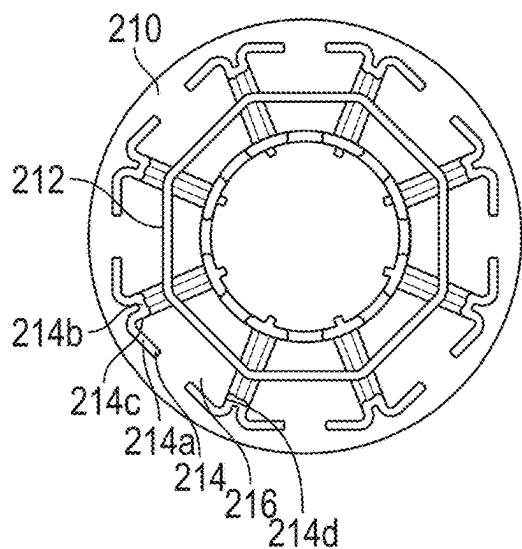


FIG. 58C

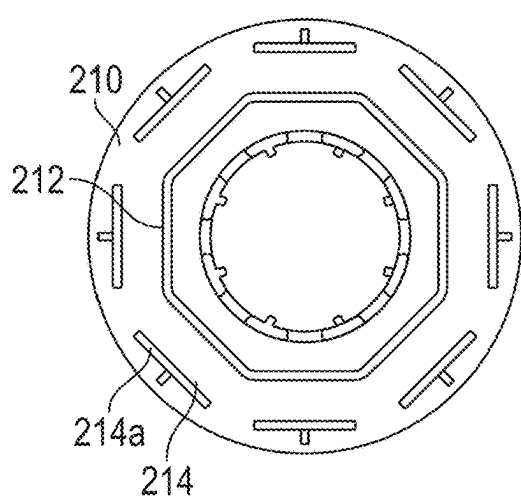


FIG. 58D

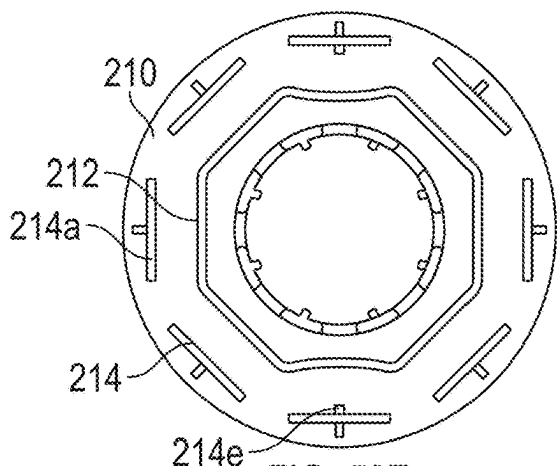


FIG. 58E

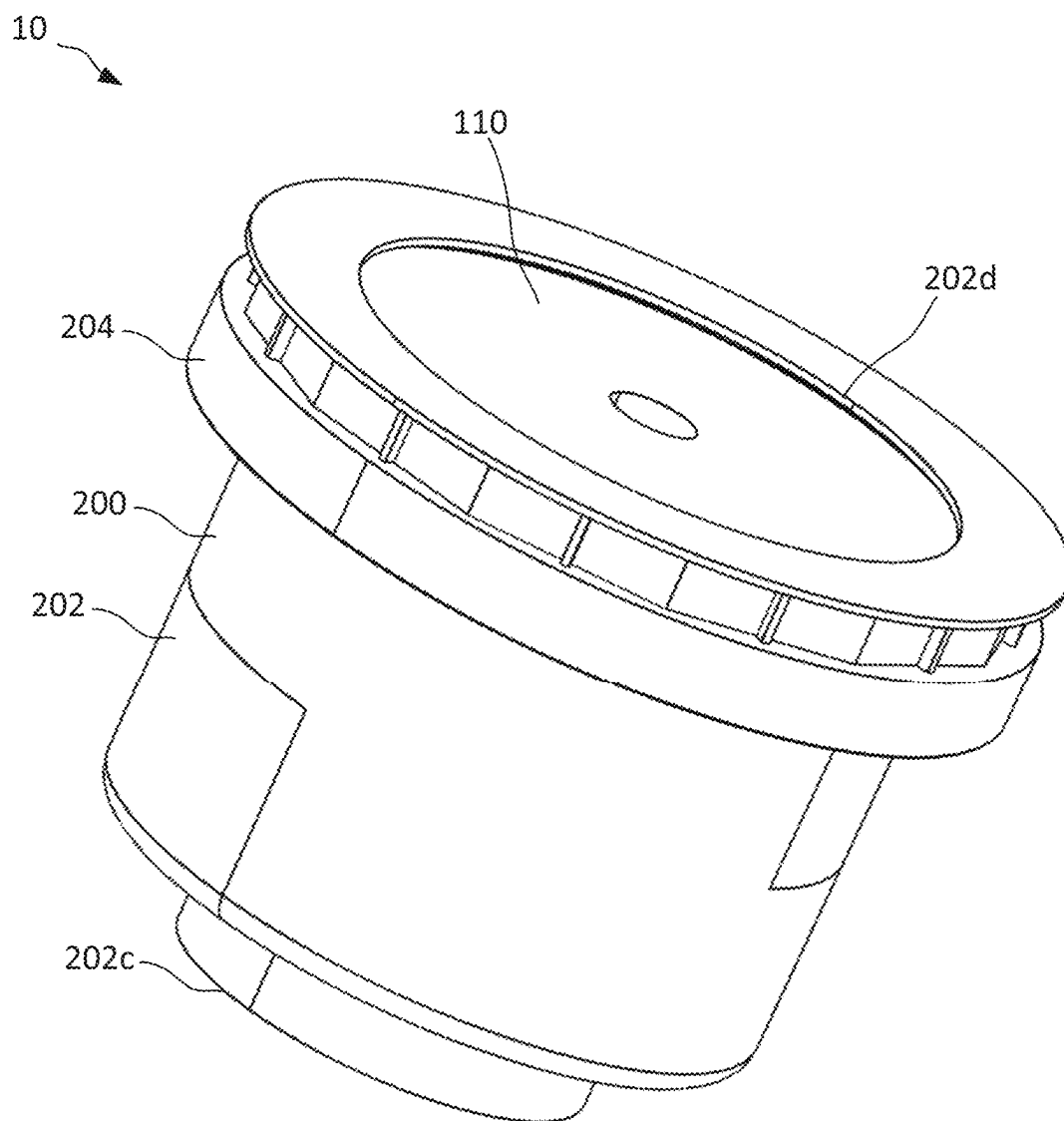


FIG. 59

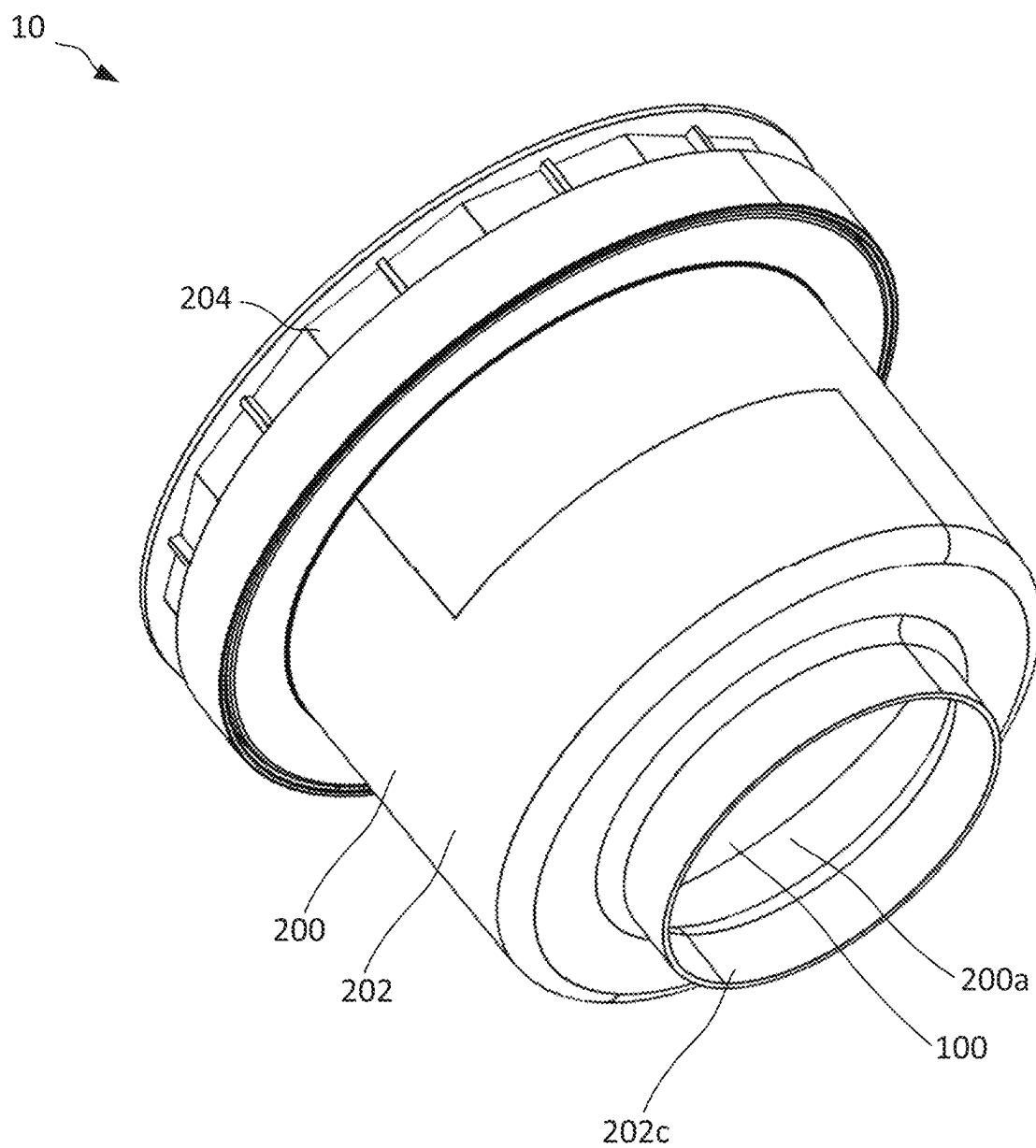


FIG. 60

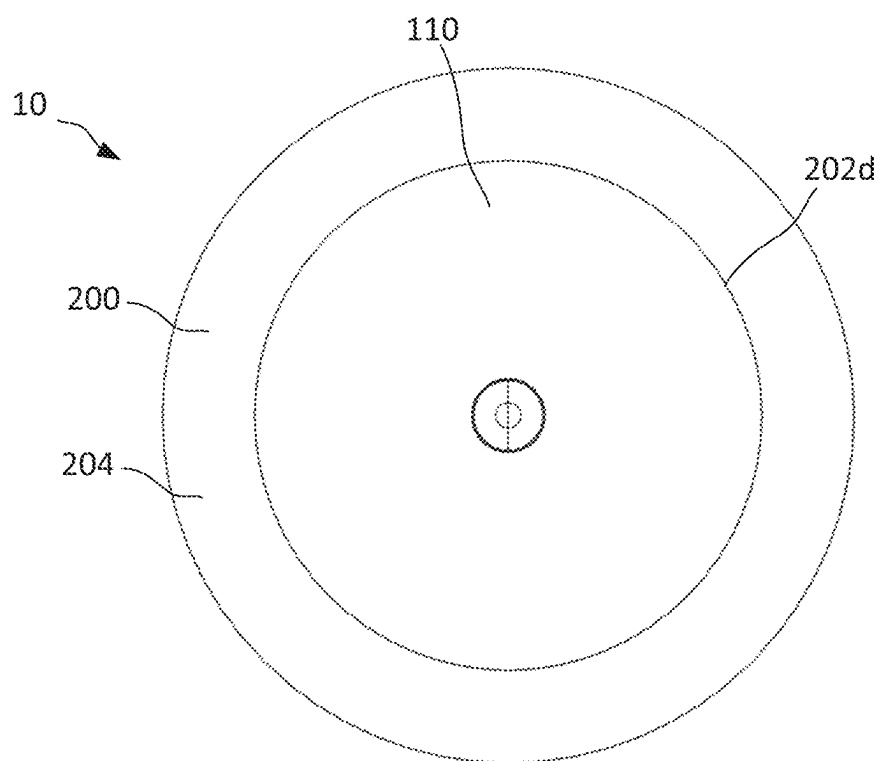


FIG. 61

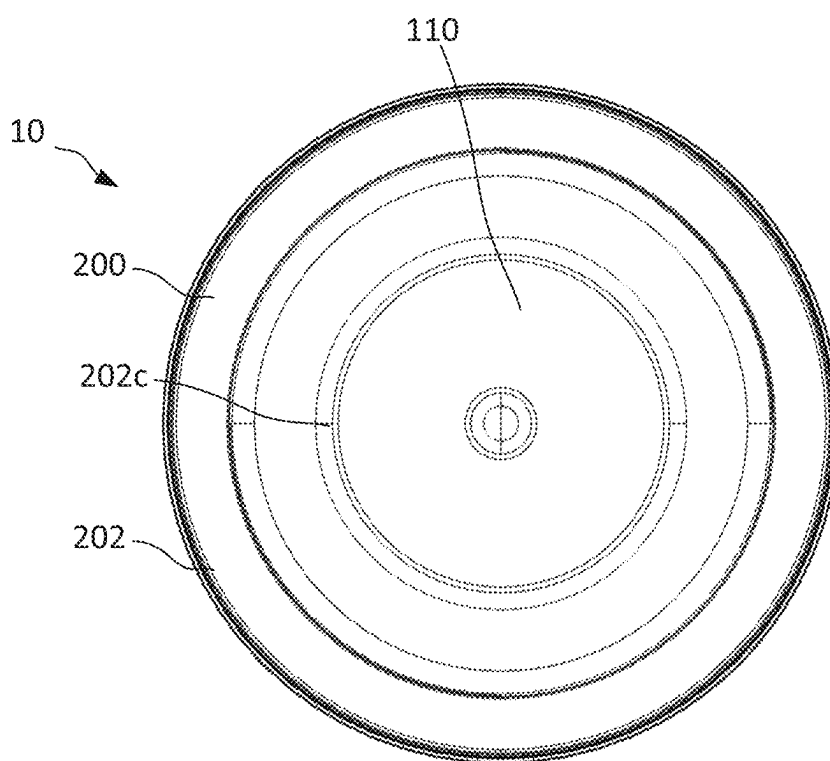


FIG. 62

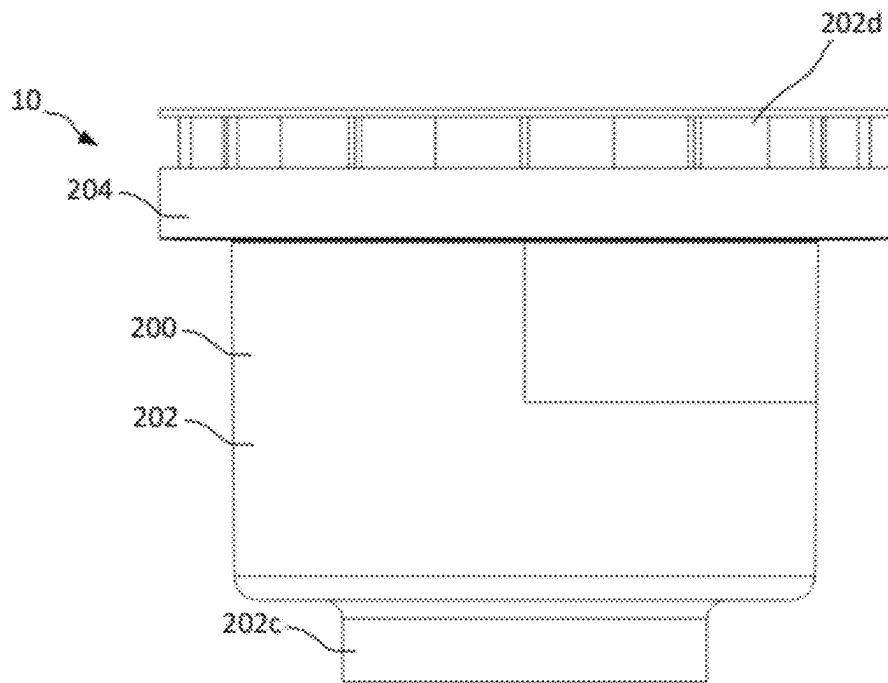


FIG. 63

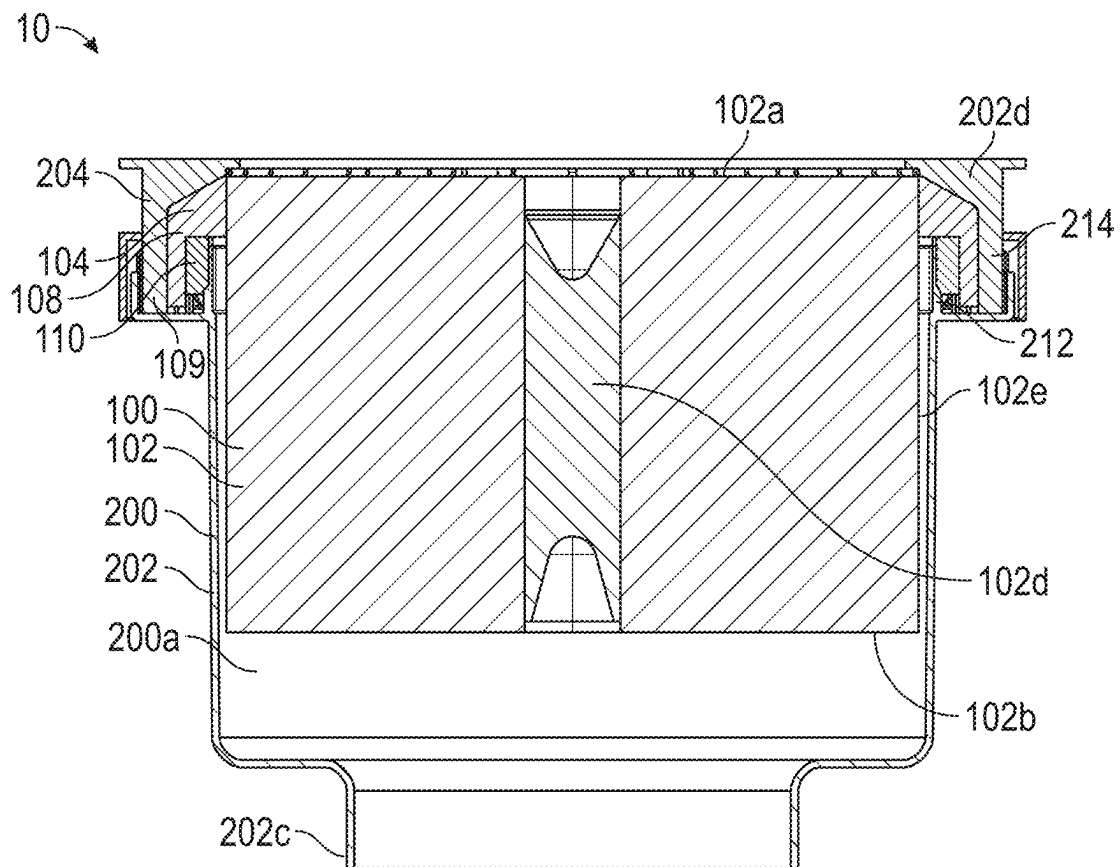


FIG. 64

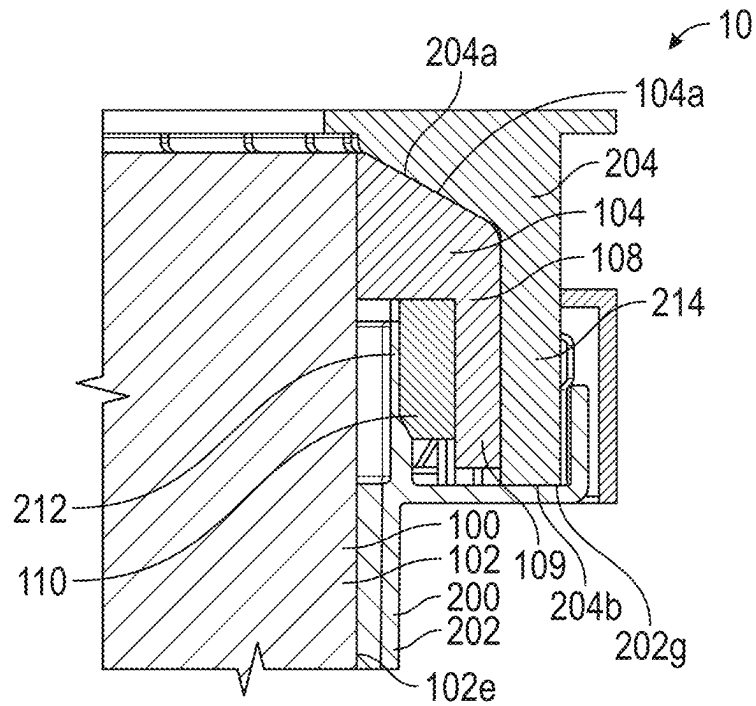


FIG. 64A

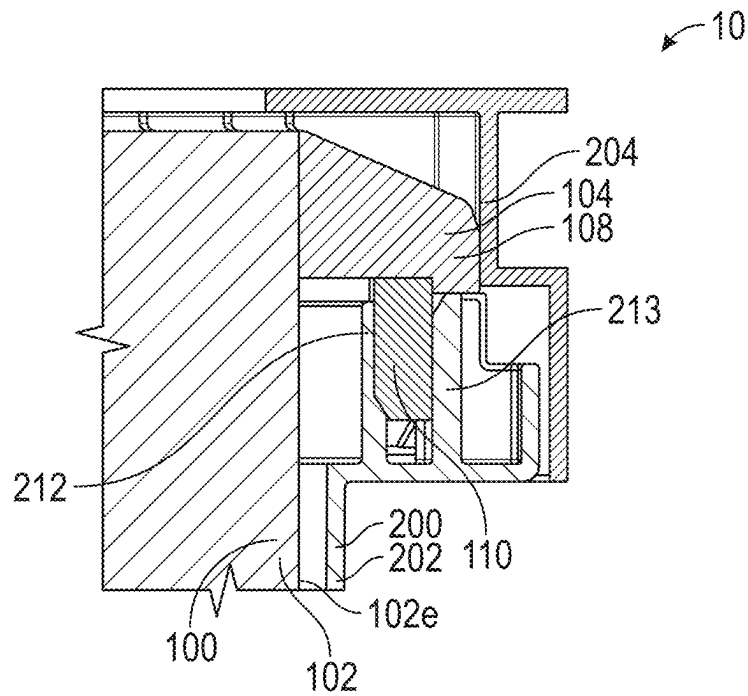


FIG. 64B

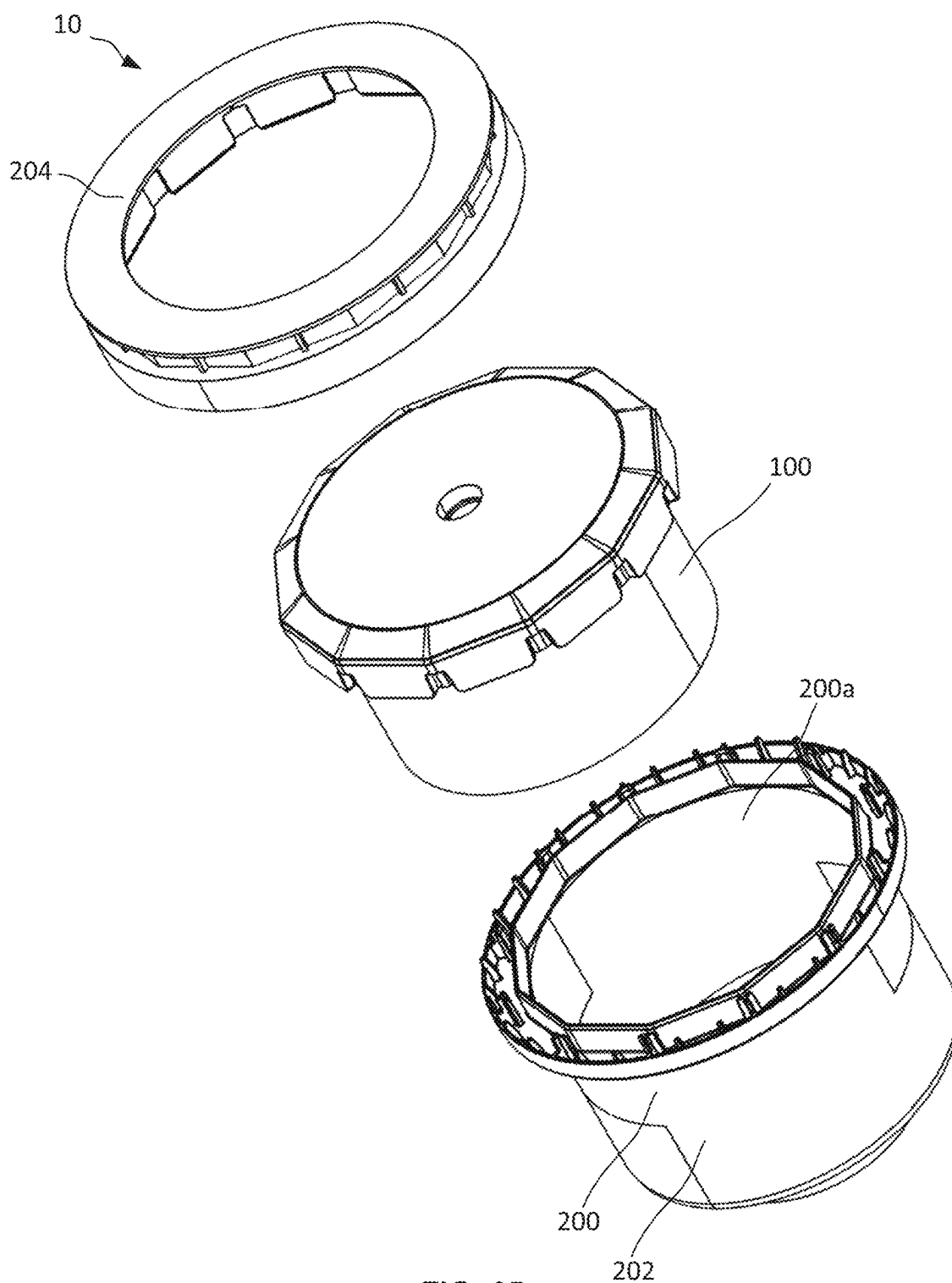


FIG. 65

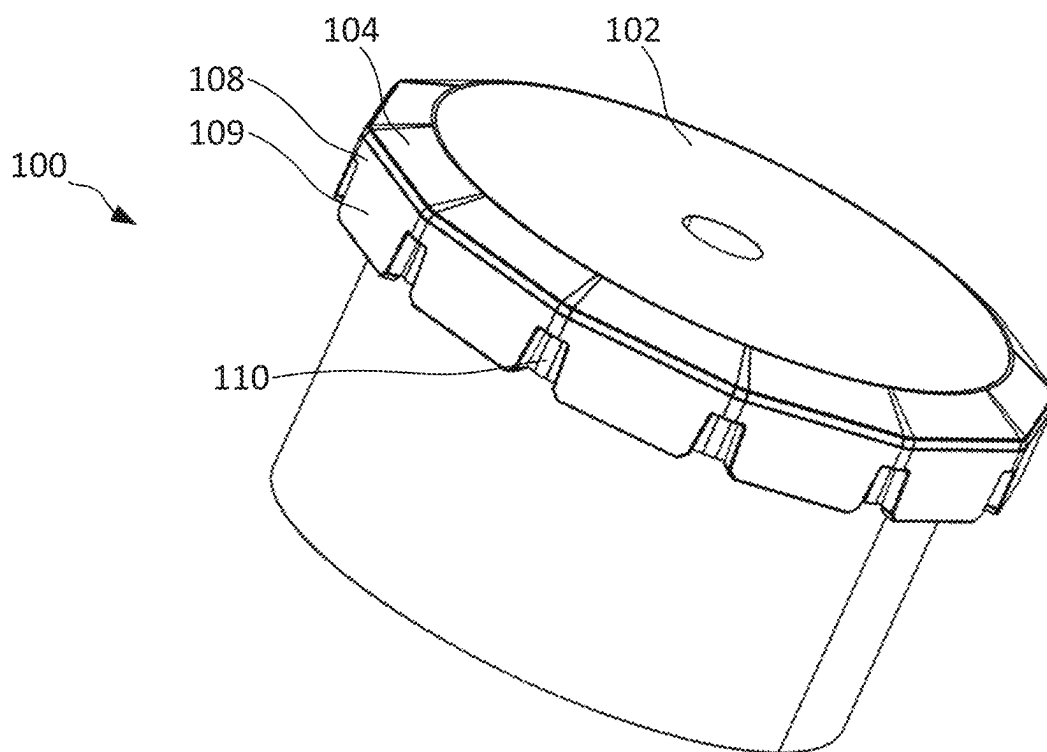


FIG. 66

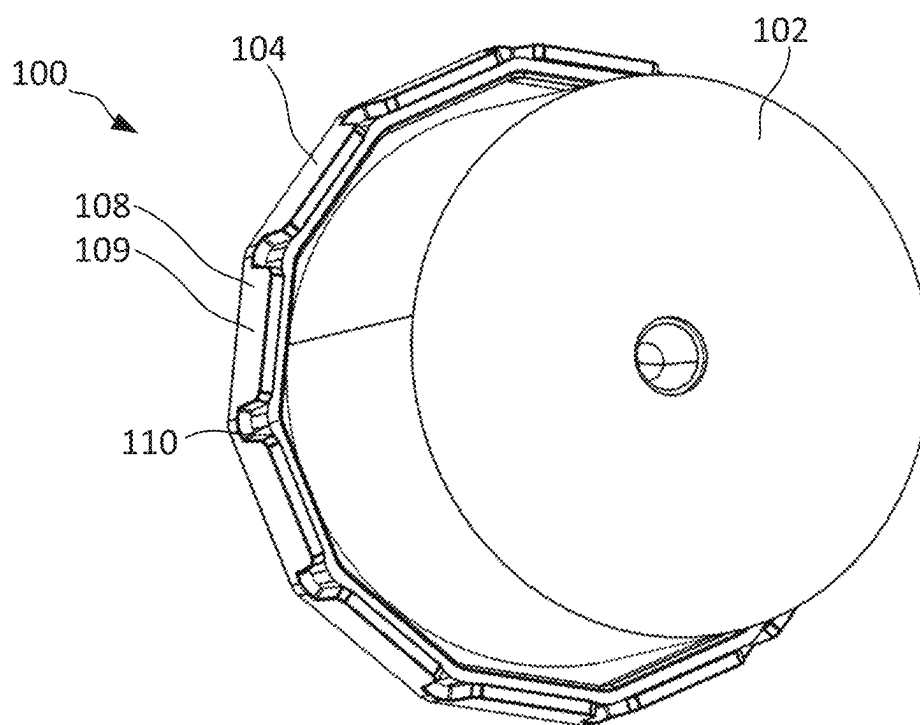


FIG. 67

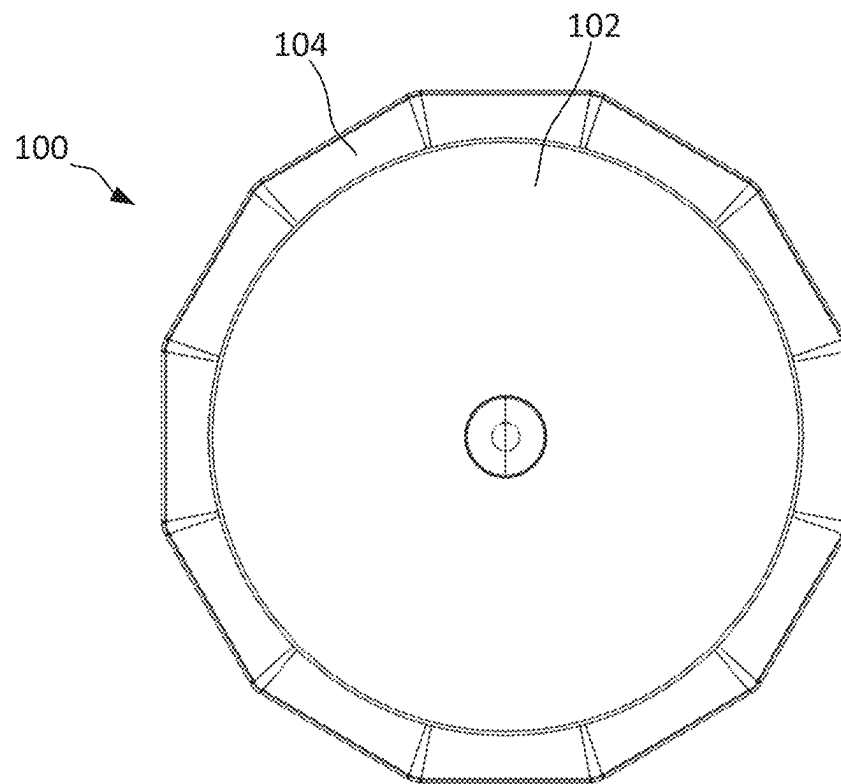


FIG. 68

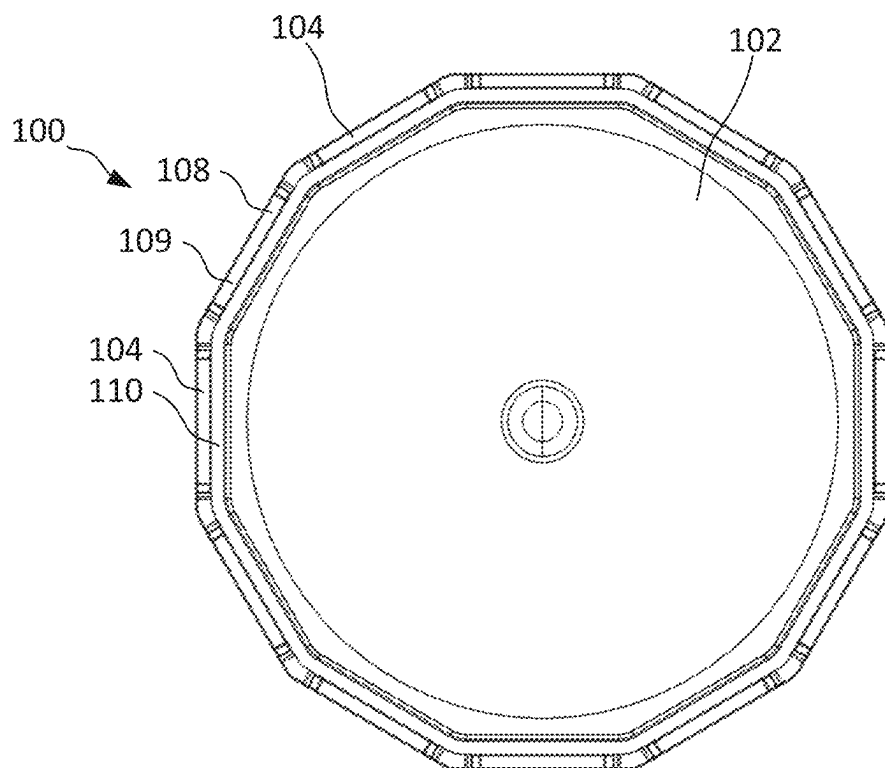


FIG. 69

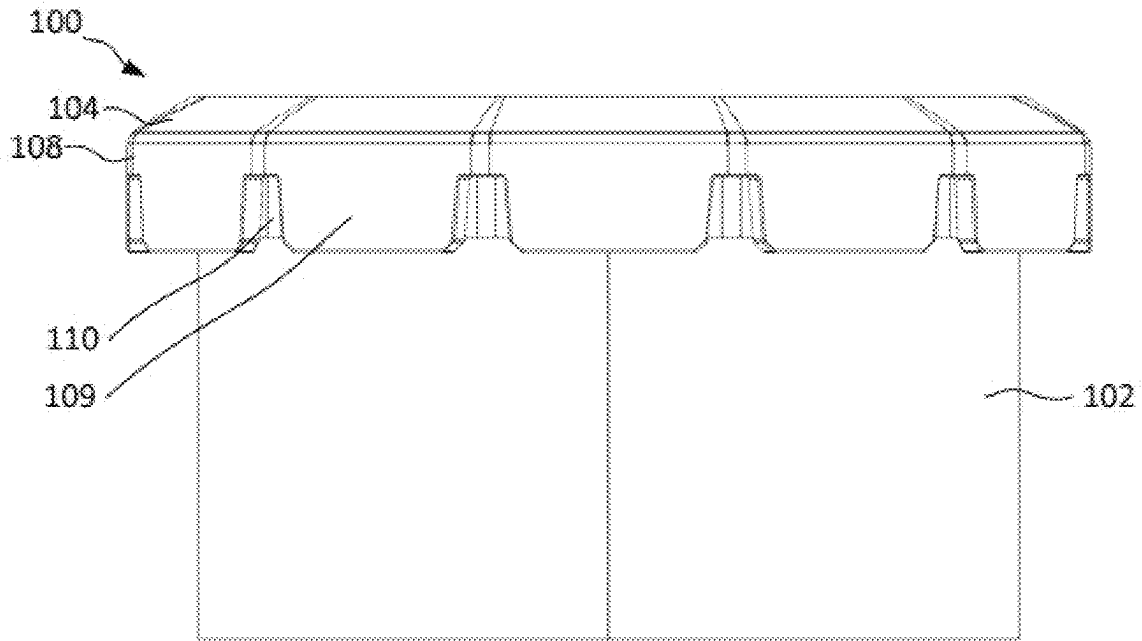


FIG. 70

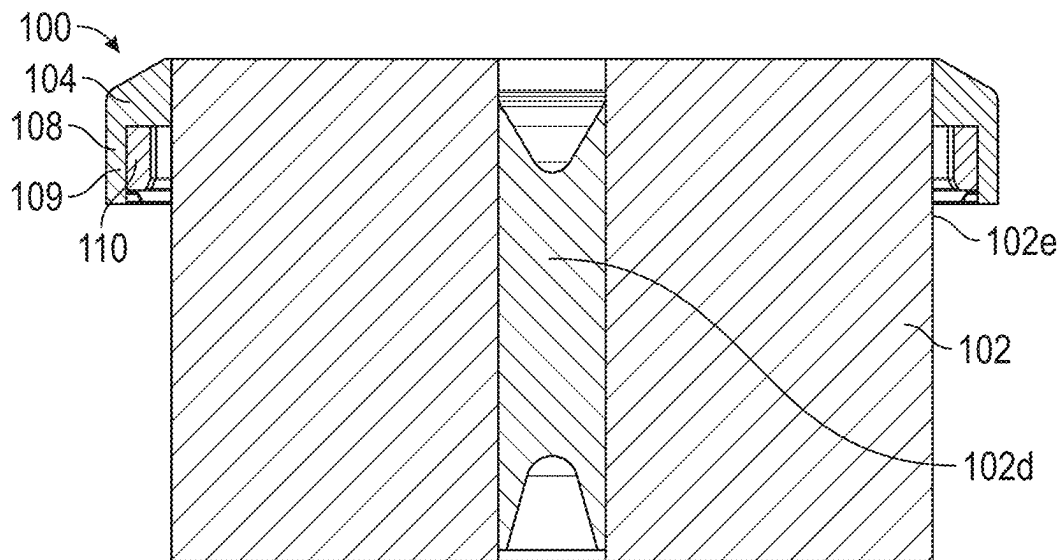


FIG. 71

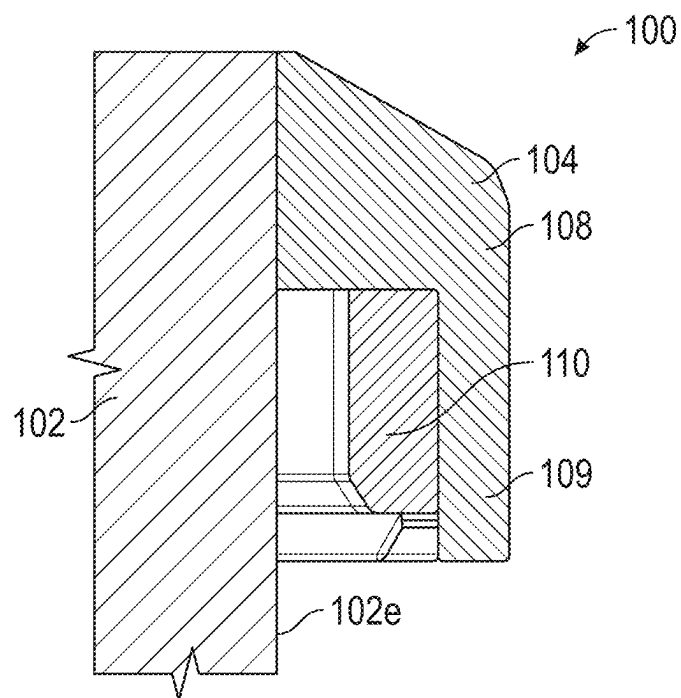


FIG. 72

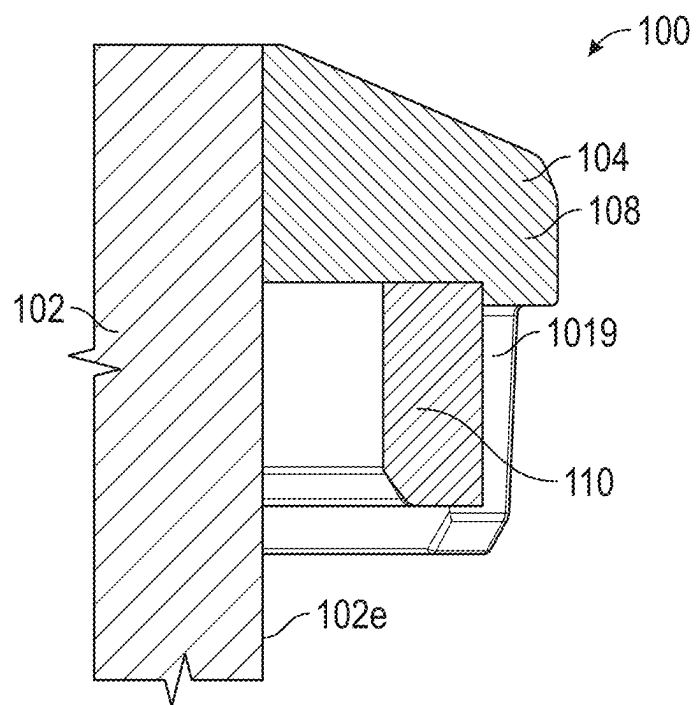


FIG. 73

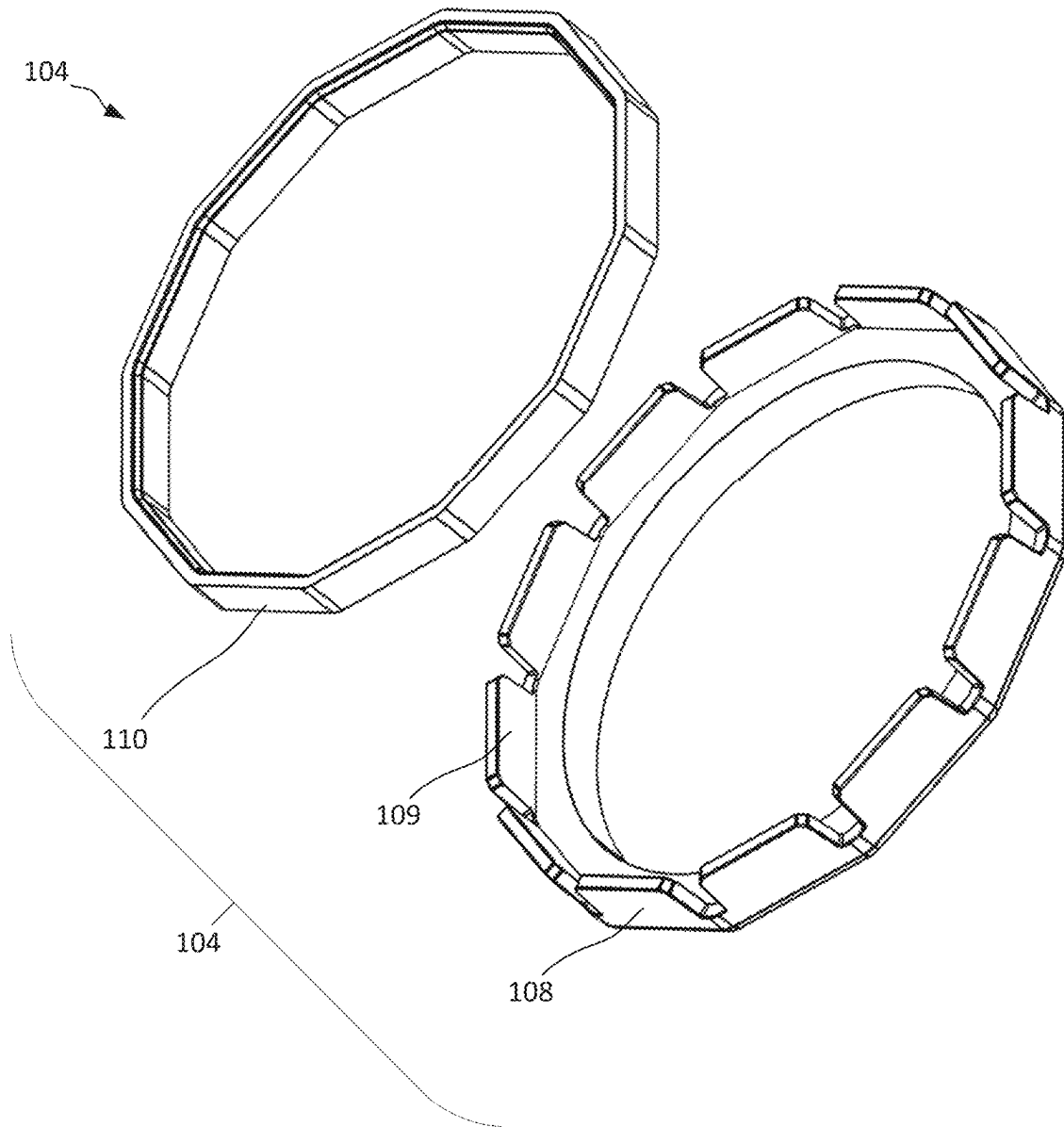


FIG. 74

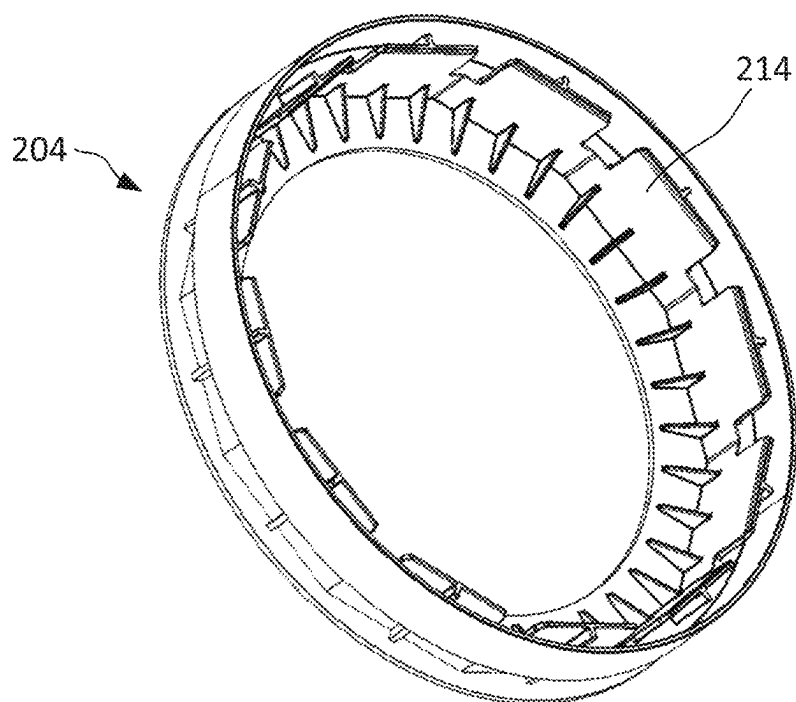


FIG. 75

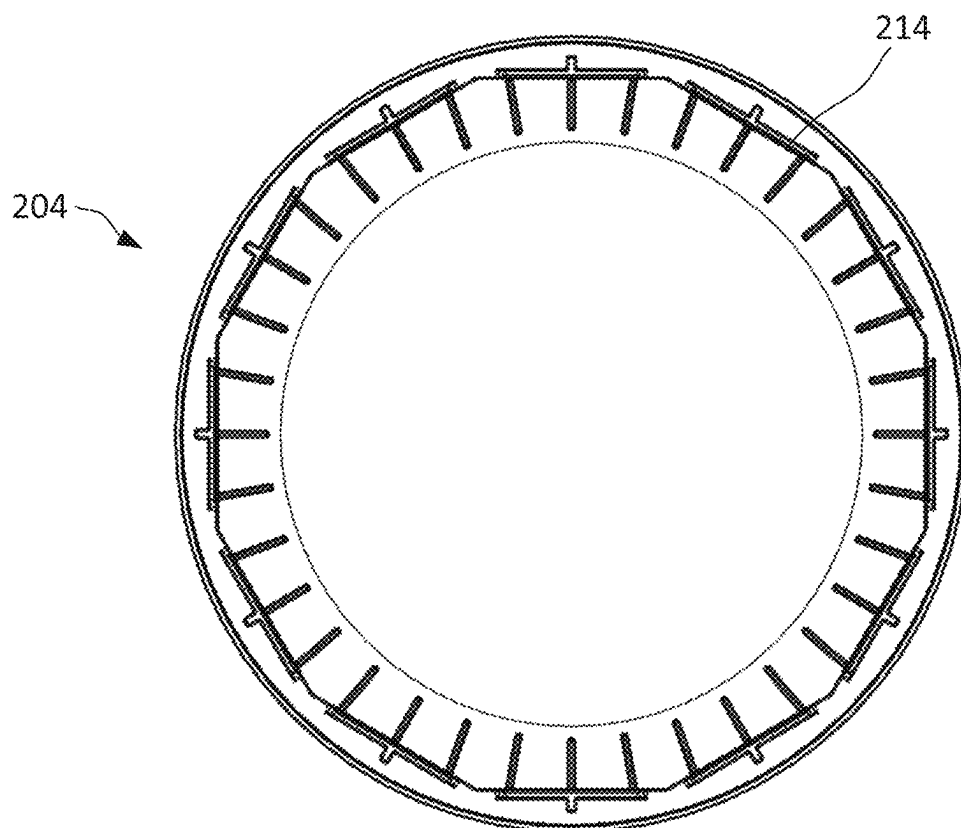


FIG. 76

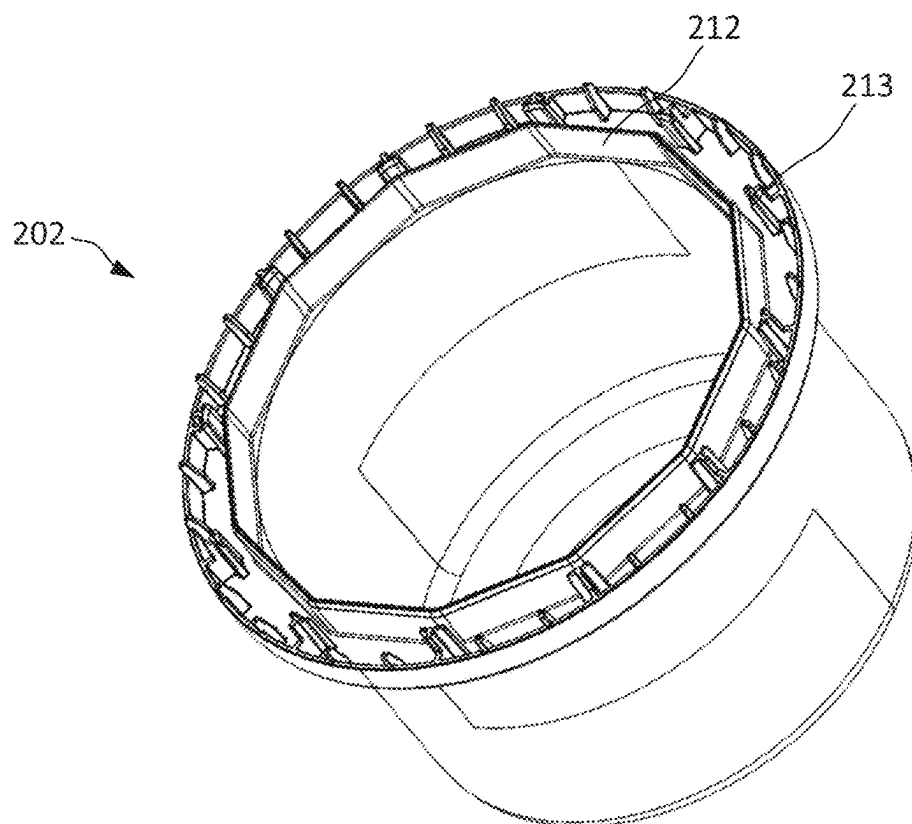


FIG. 77

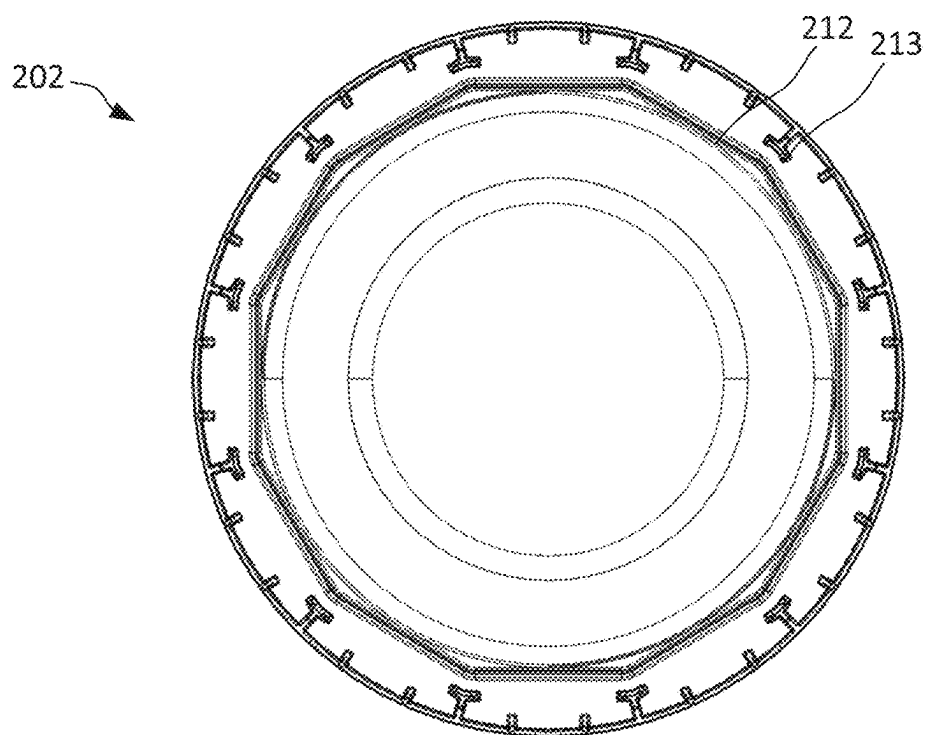


FIG. 78

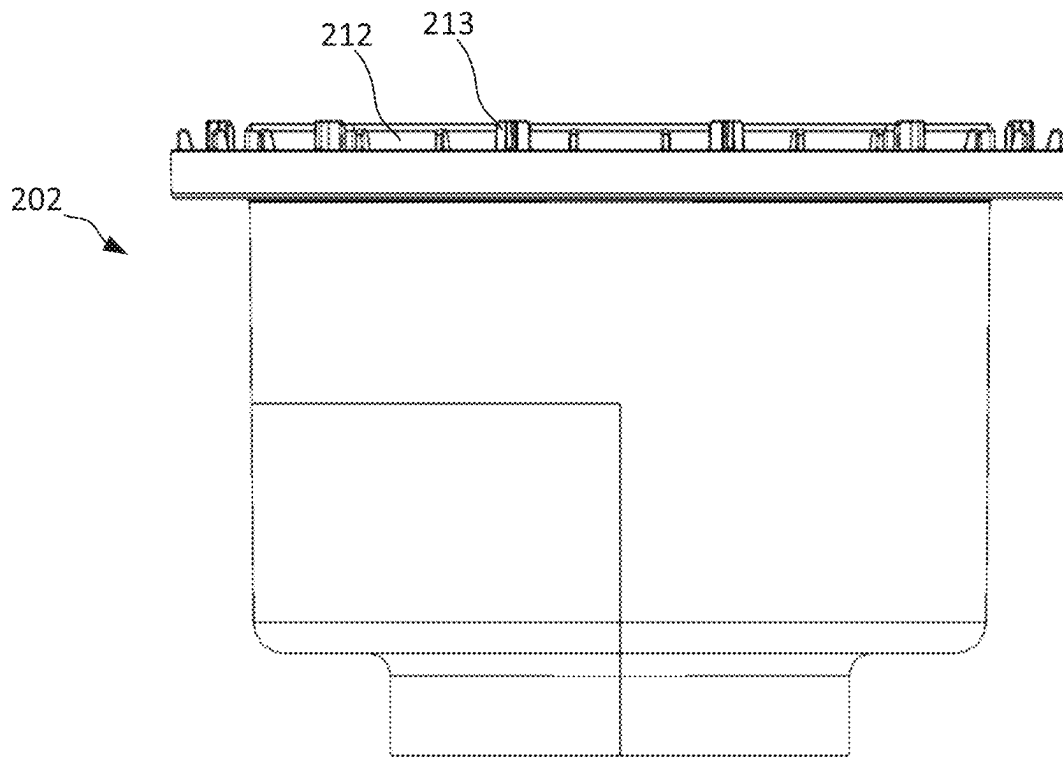


FIG. 79

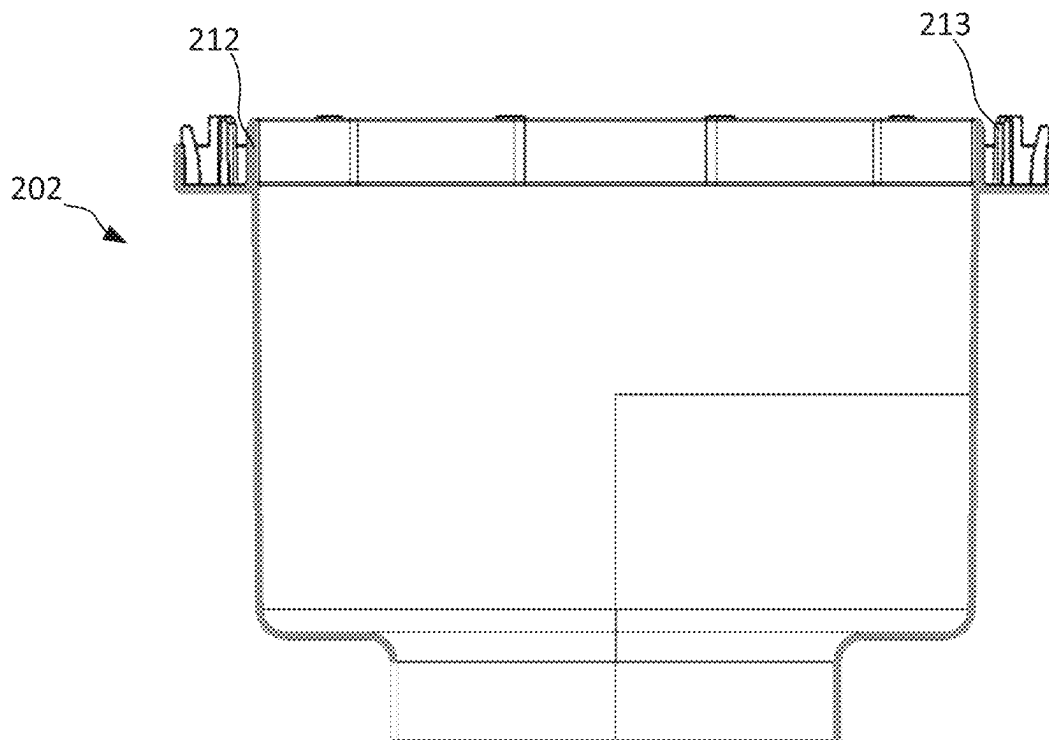


FIG. 80

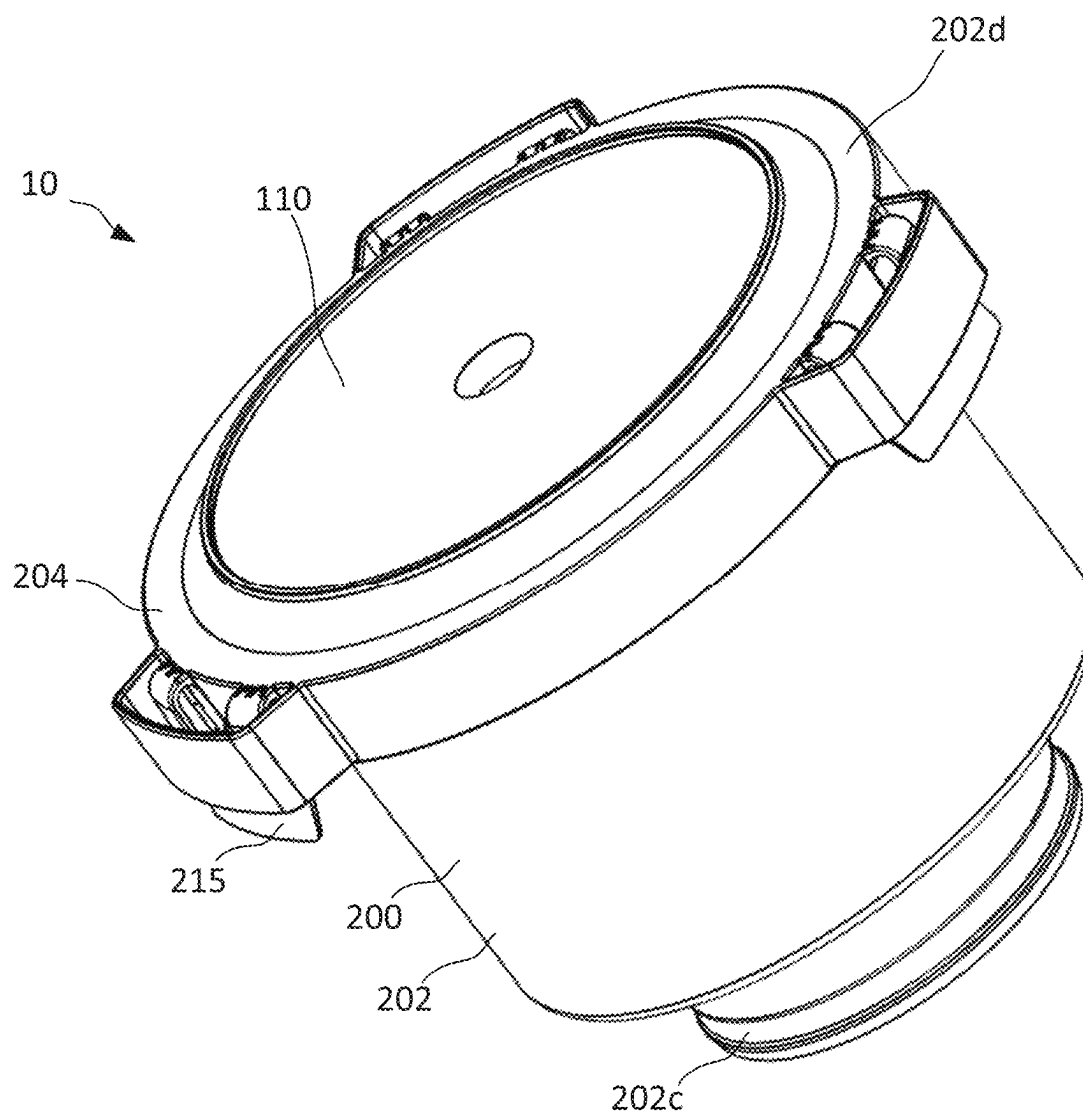


FIG. 81

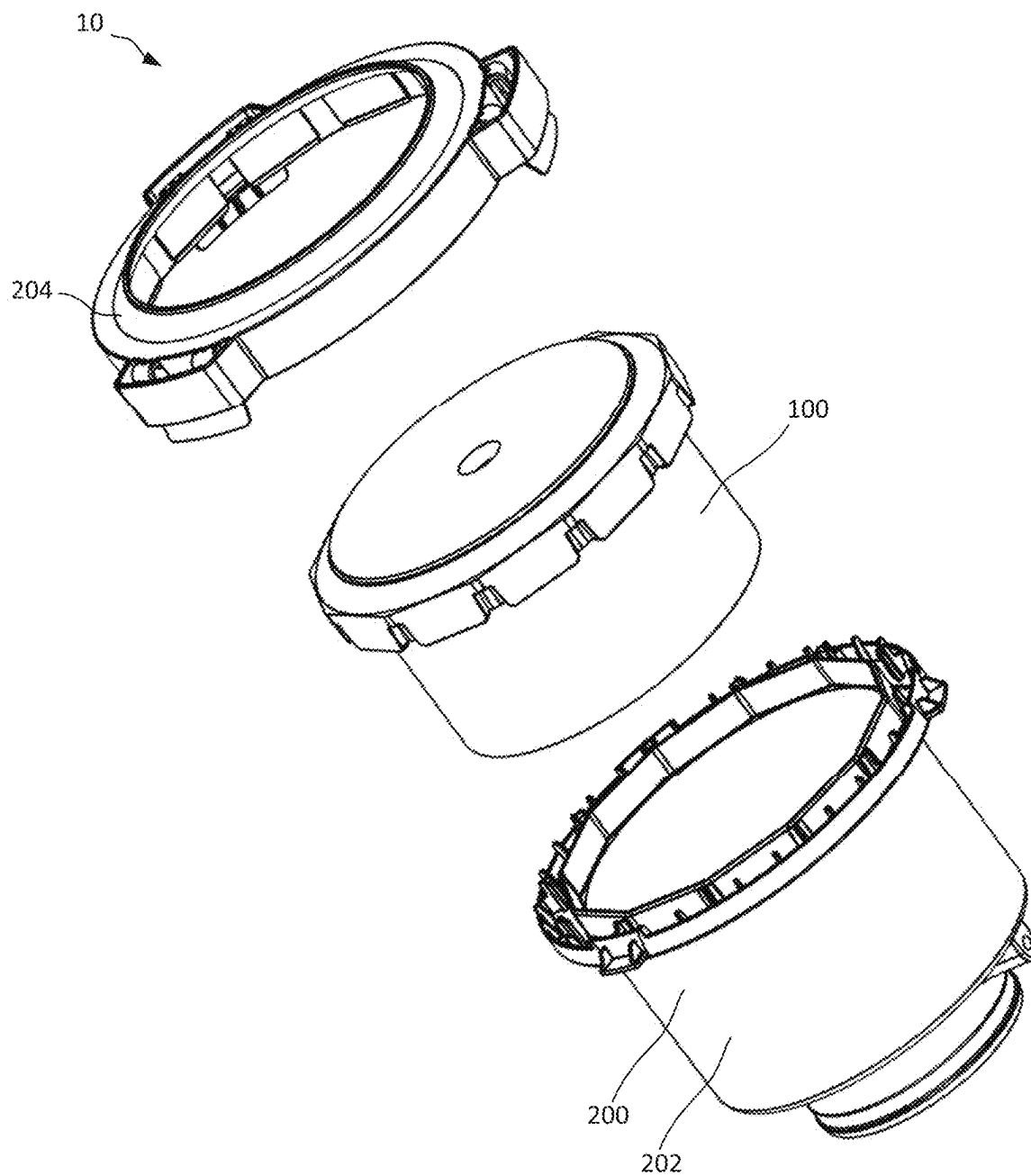


FIG. 82

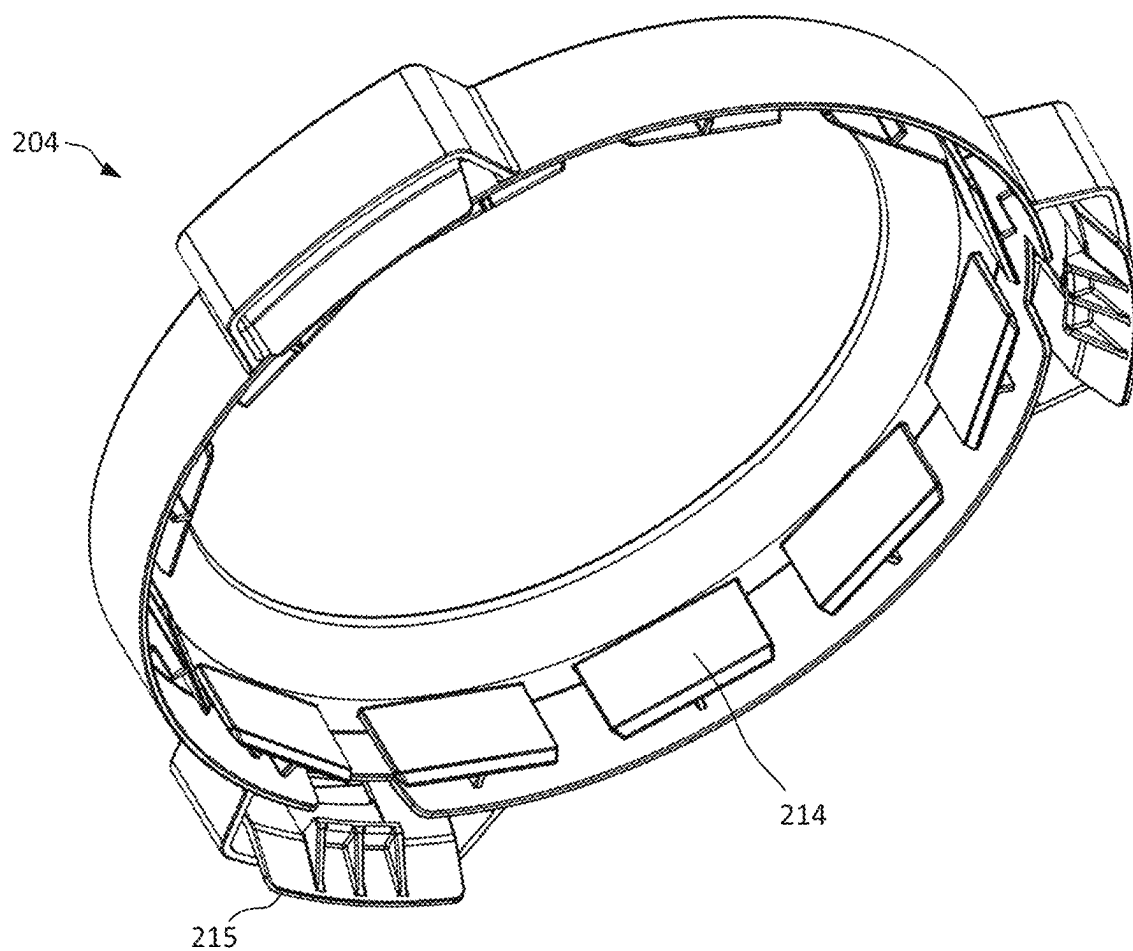


FIG. 83

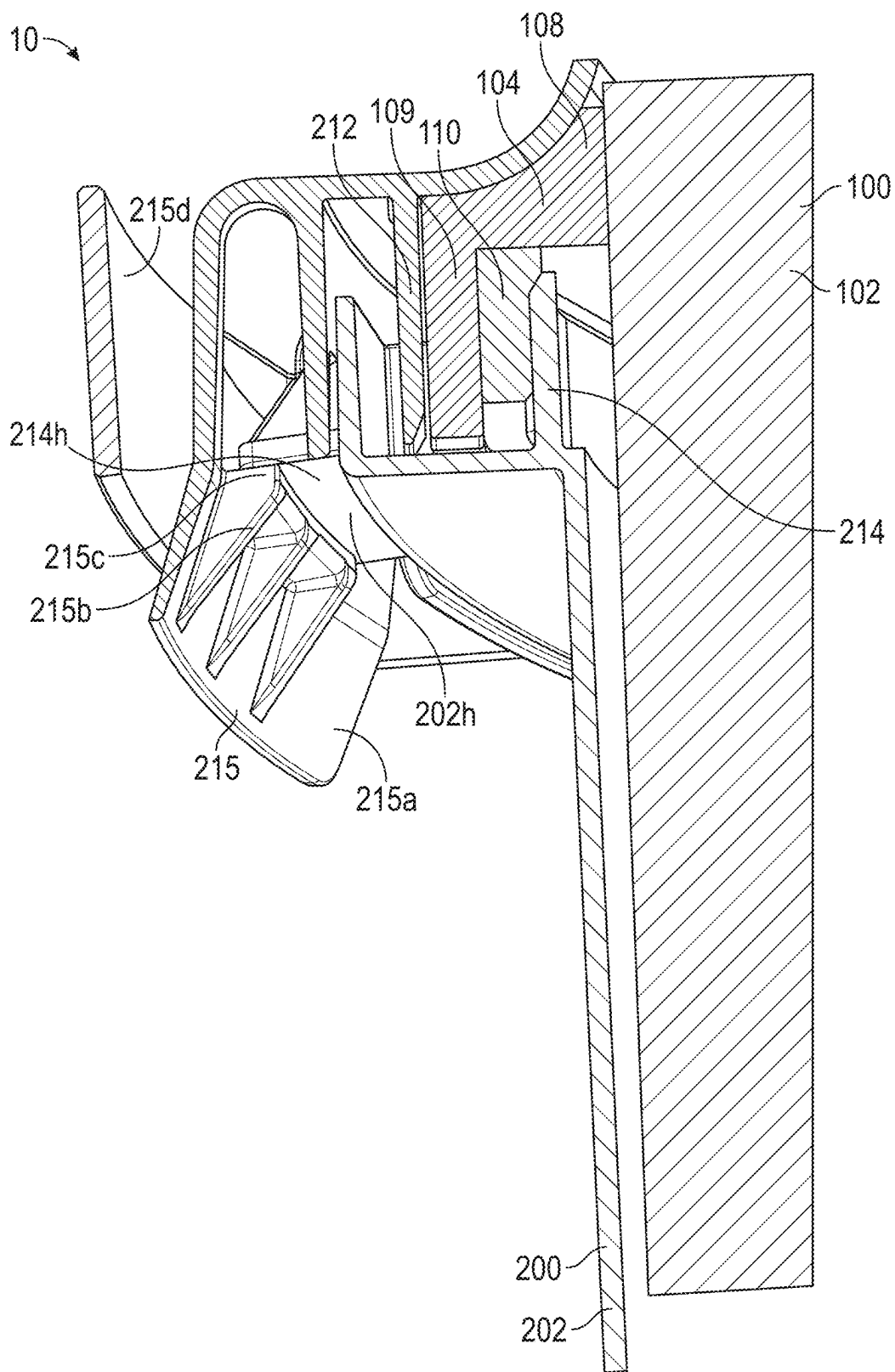


FIG. 84

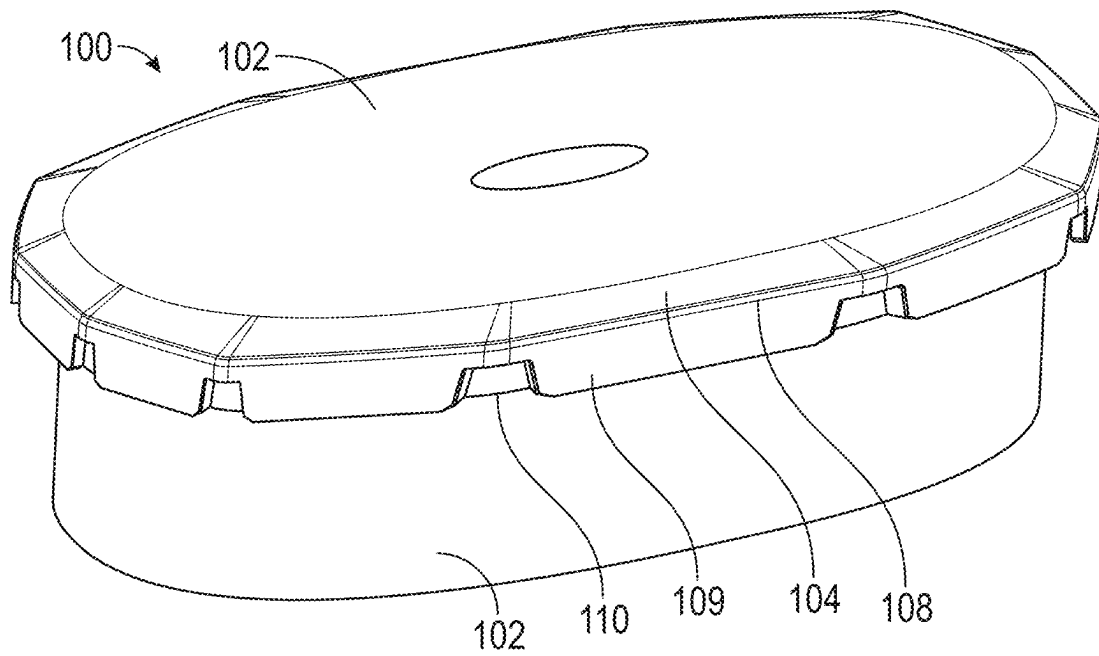


FIG. 85

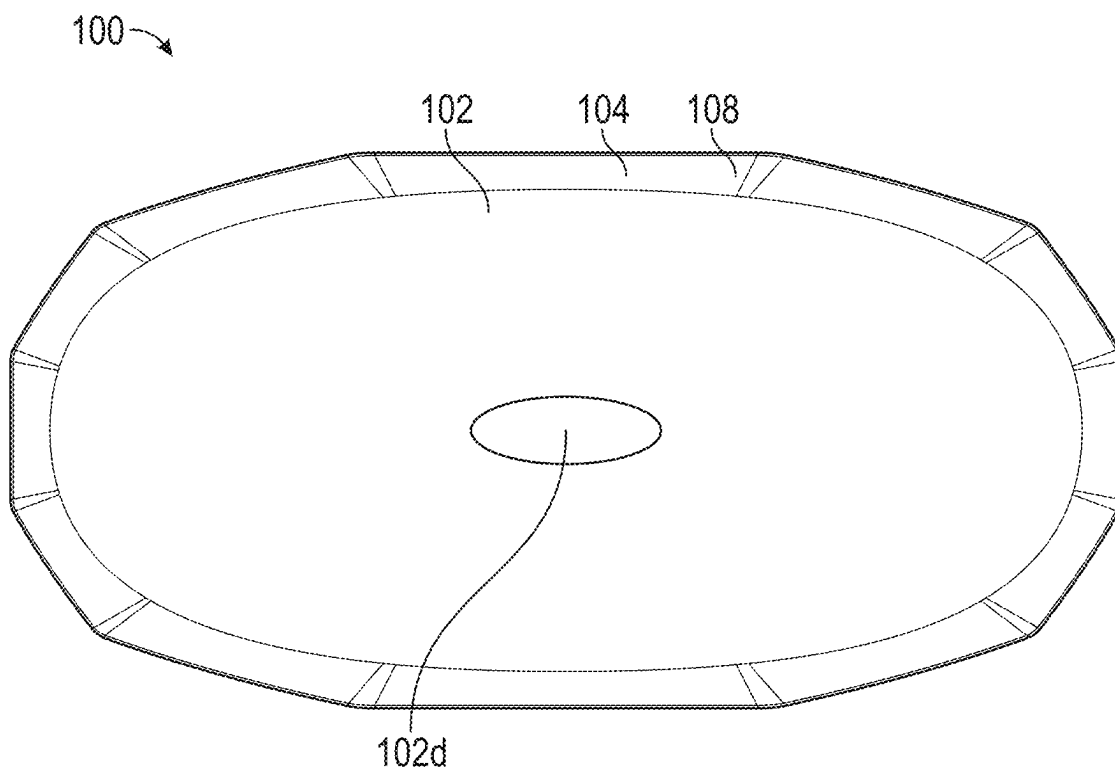


FIG. 86

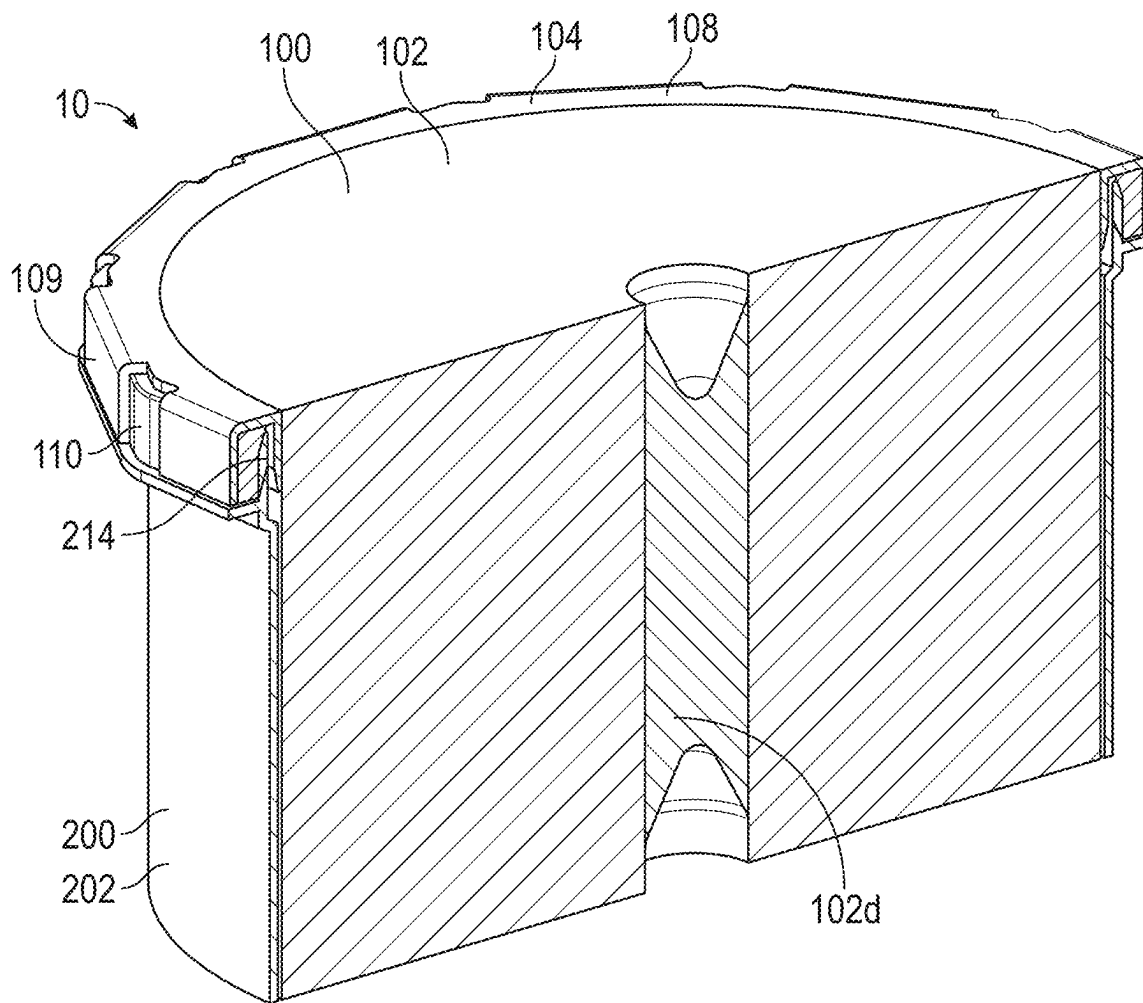


FIG. 87

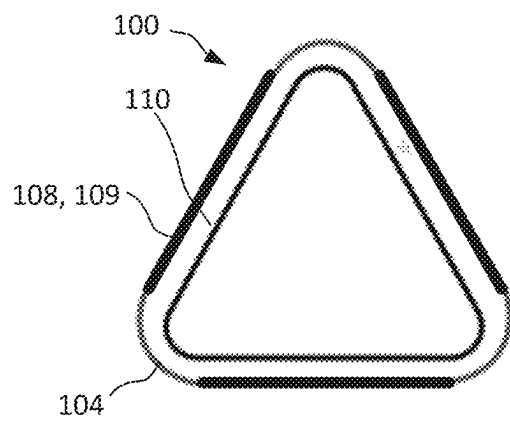


FIG. 88

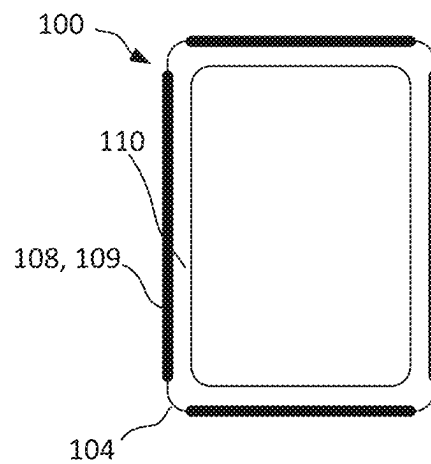


FIG. 89

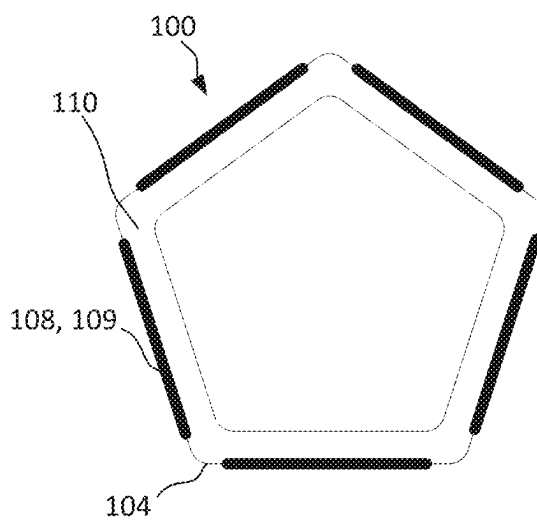


FIG. 90

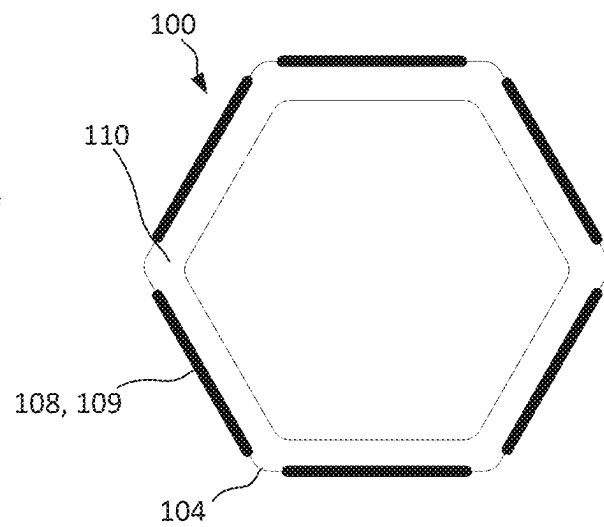


FIG. 91

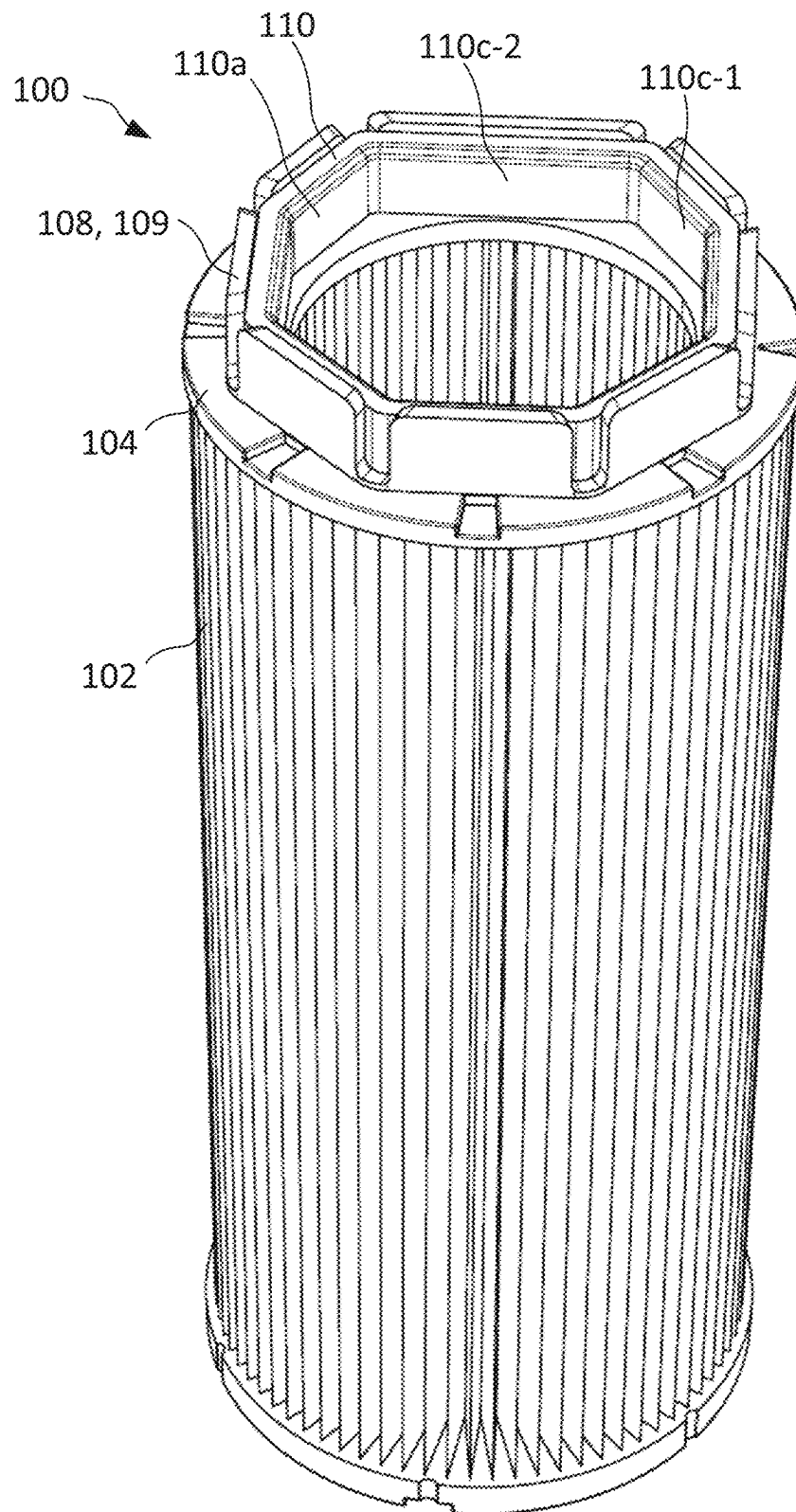


FIG. 92

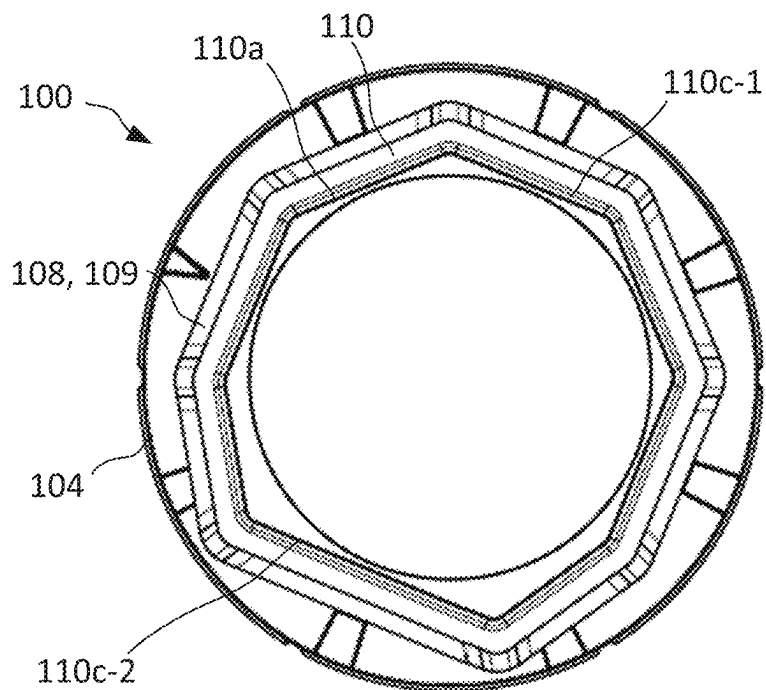


FIG. 93

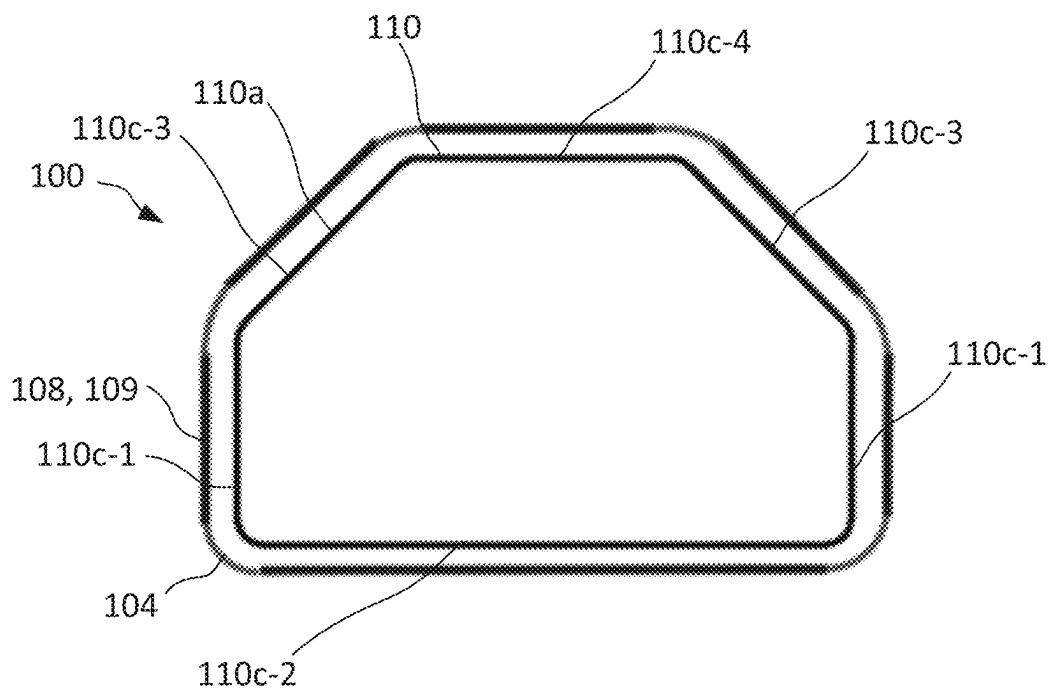


FIG. 94

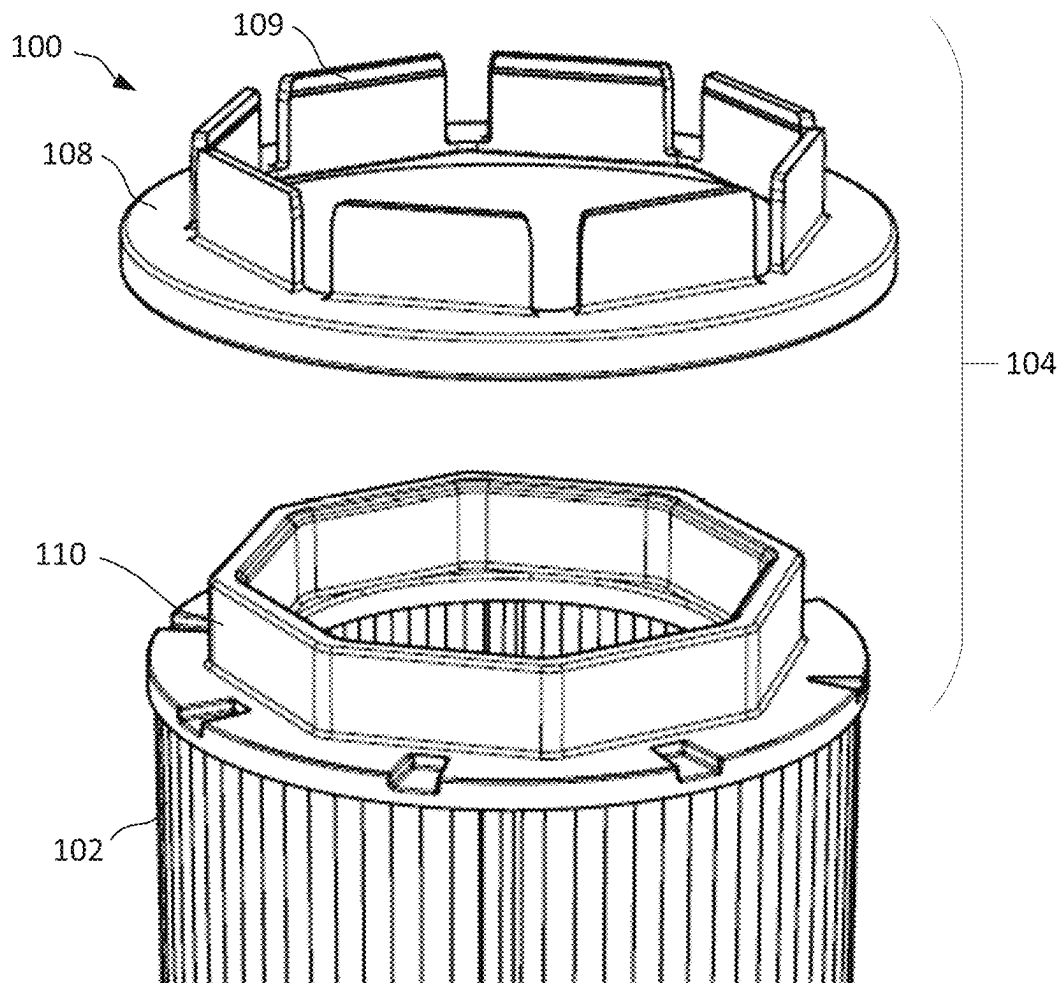


FIG. 95

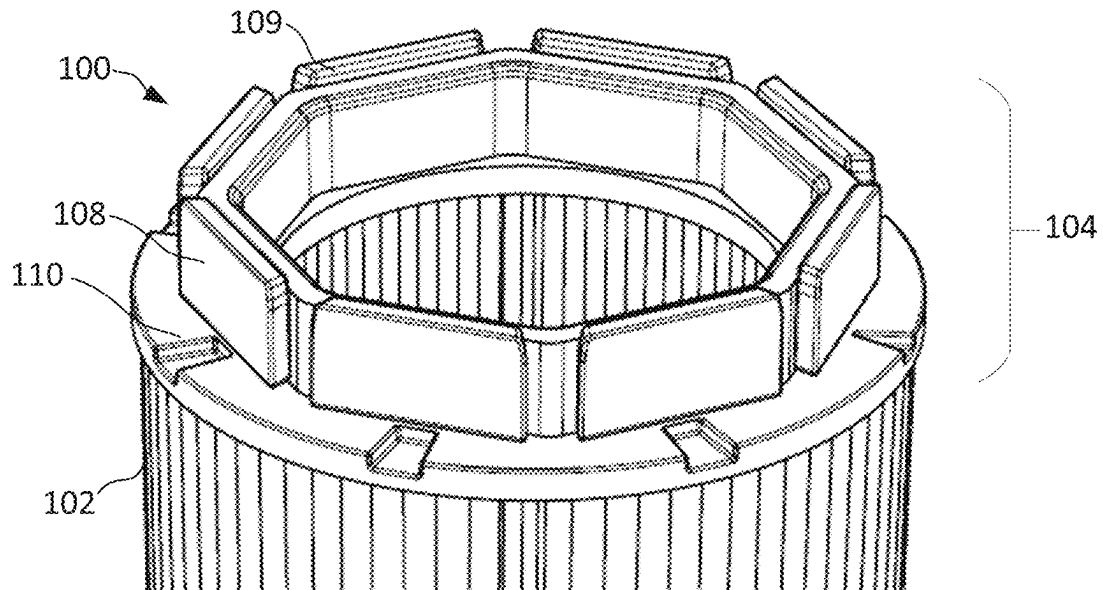


FIG. 96

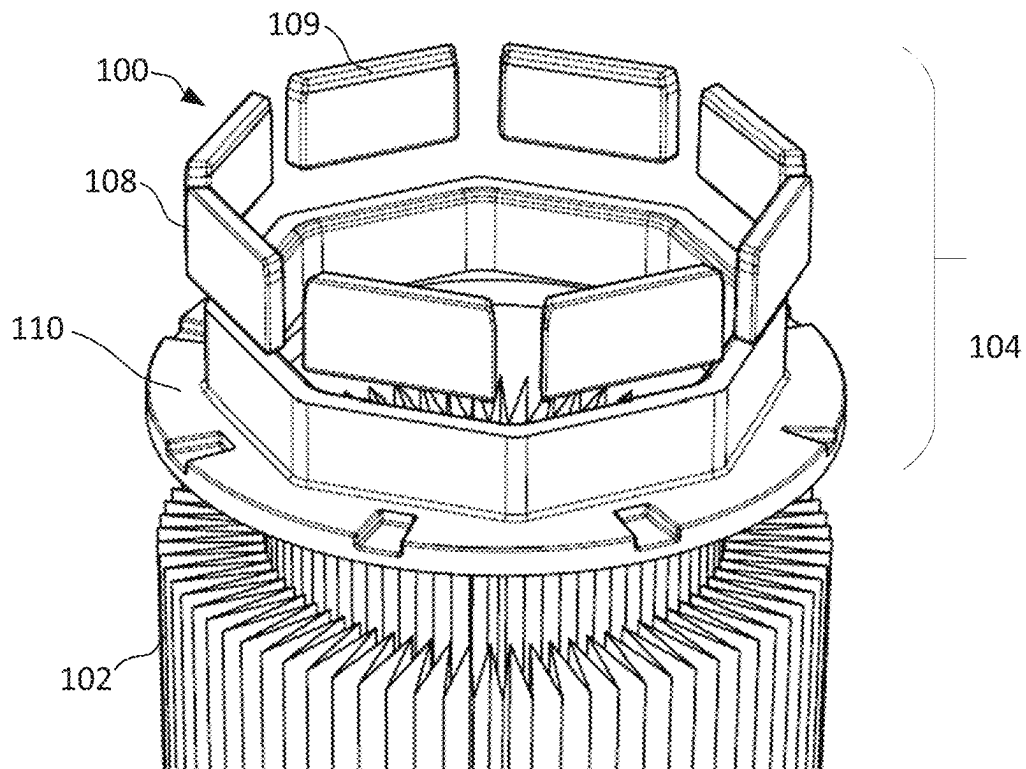


FIG. 97

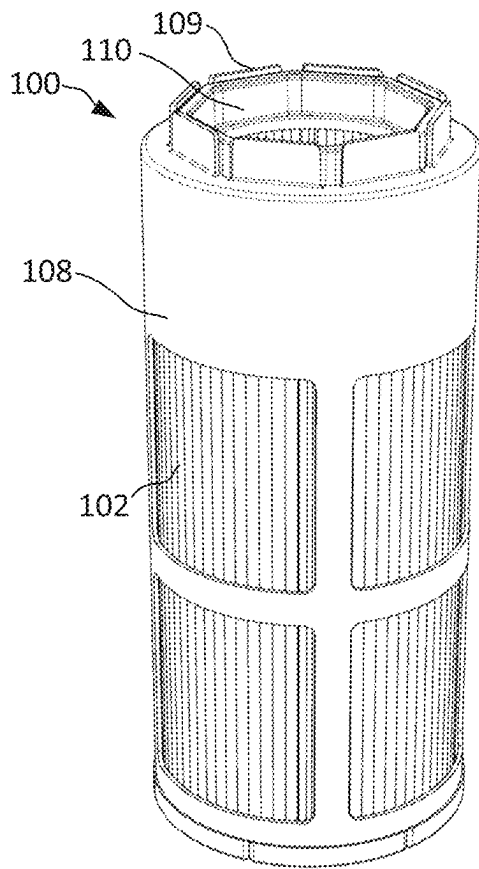


FIG. 98

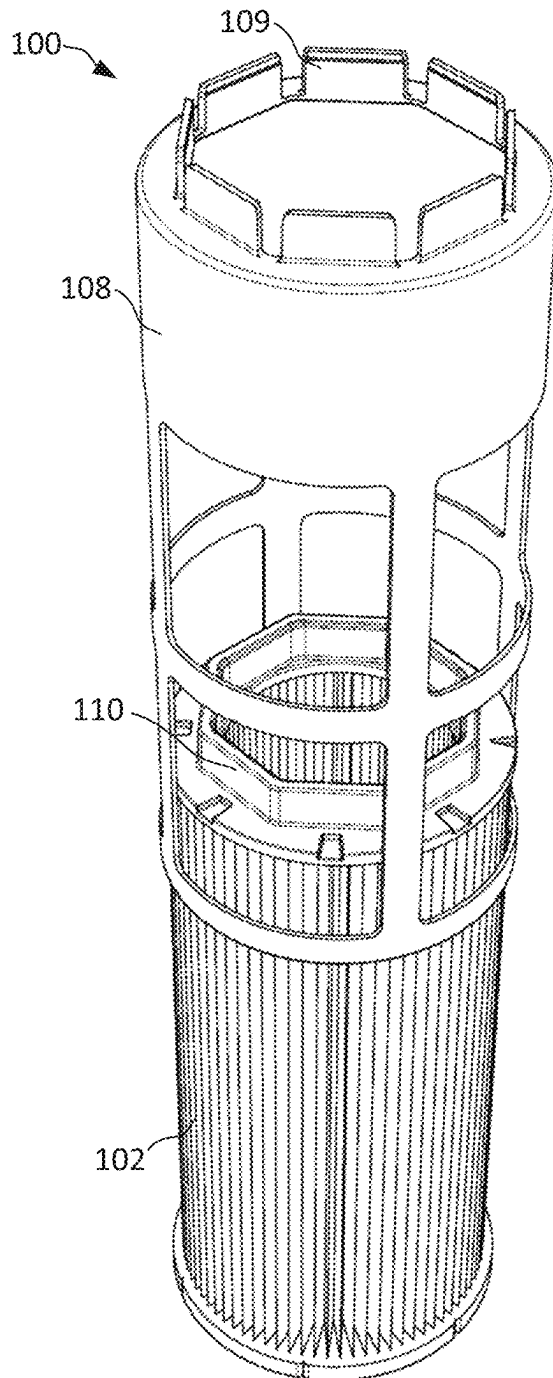


FIG. 99

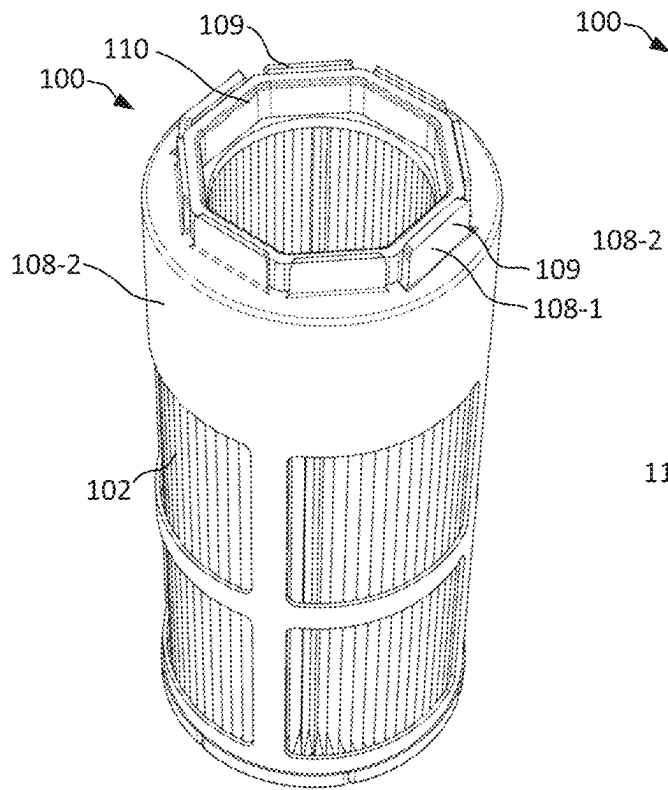


FIG. 100

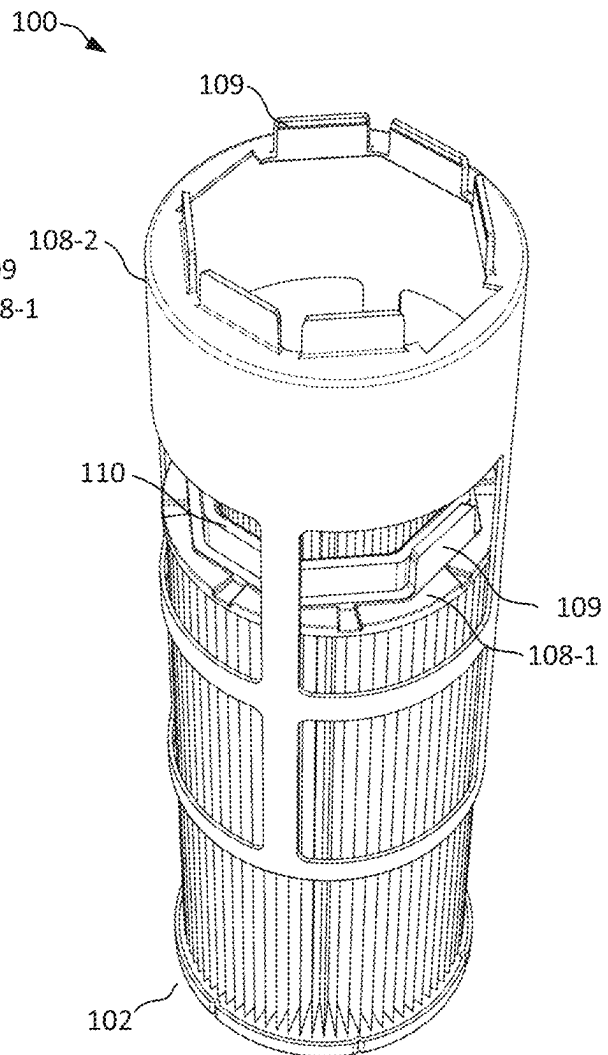


FIG. 101

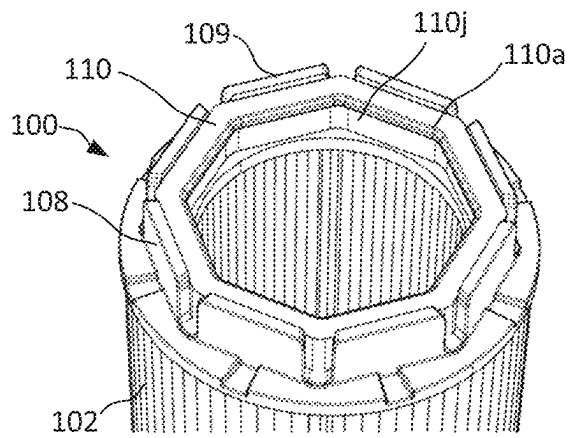


FIG. 102

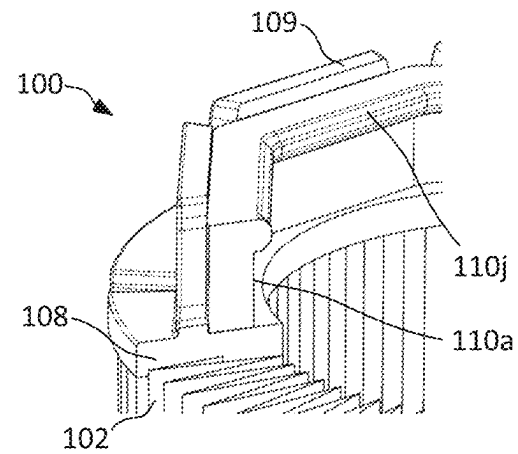


FIG. 103

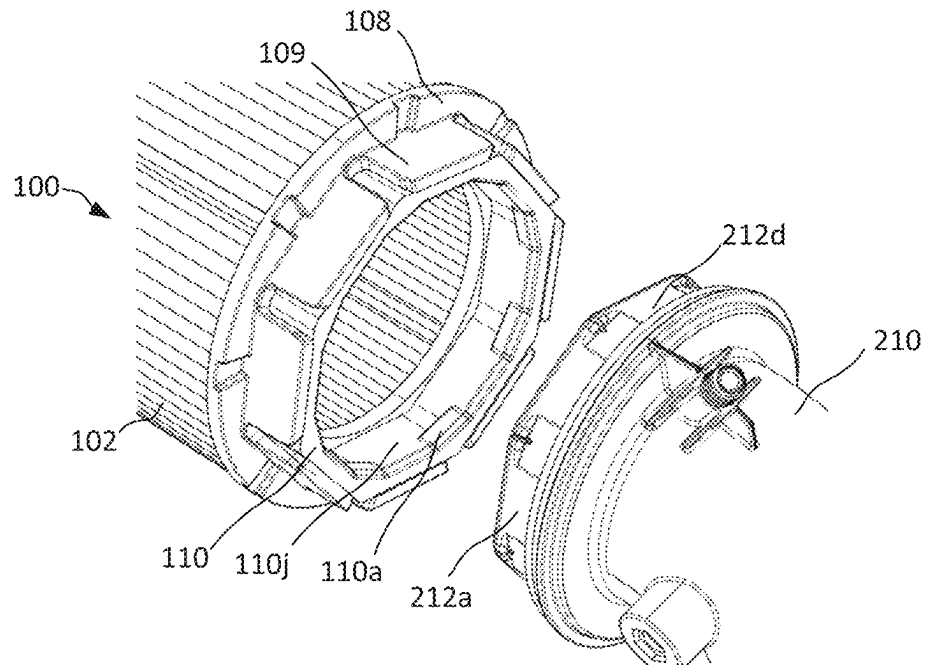


FIG. 104

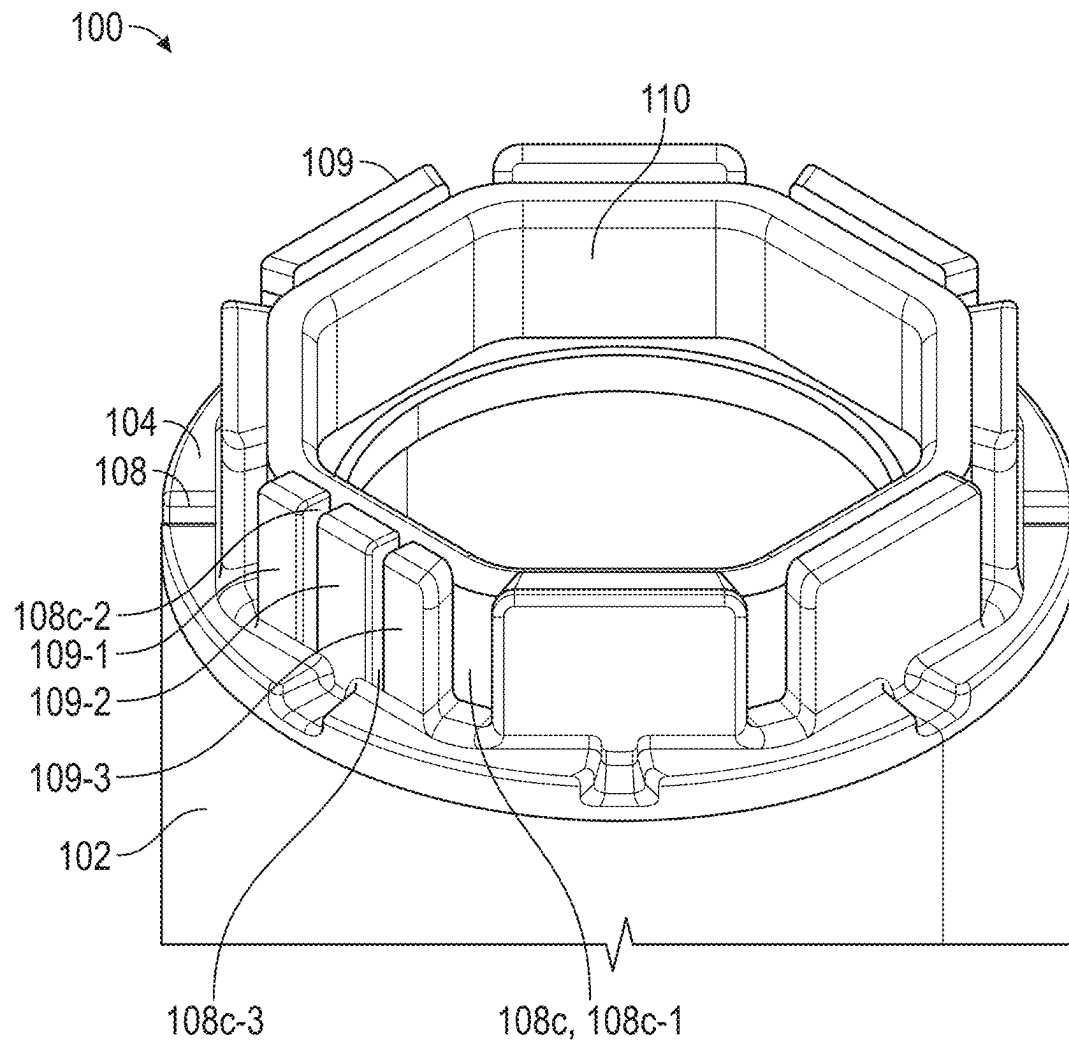


FIG. 105

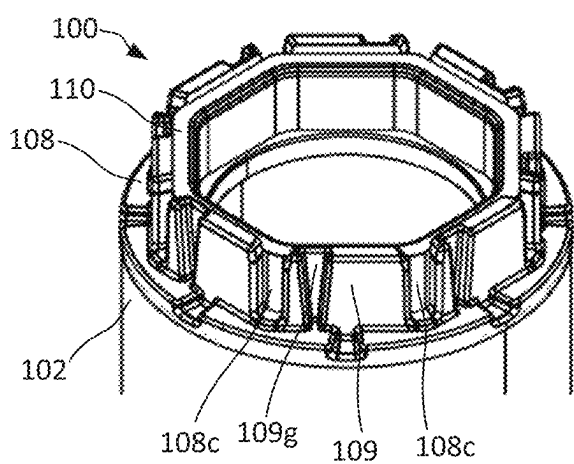


FIG. 106

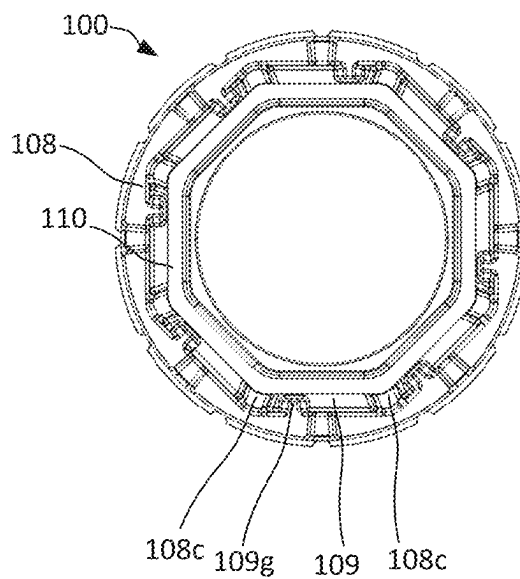


FIG. 107

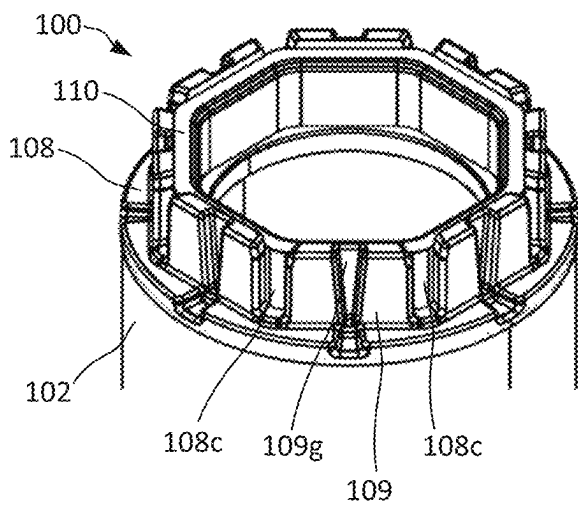


FIG. 108

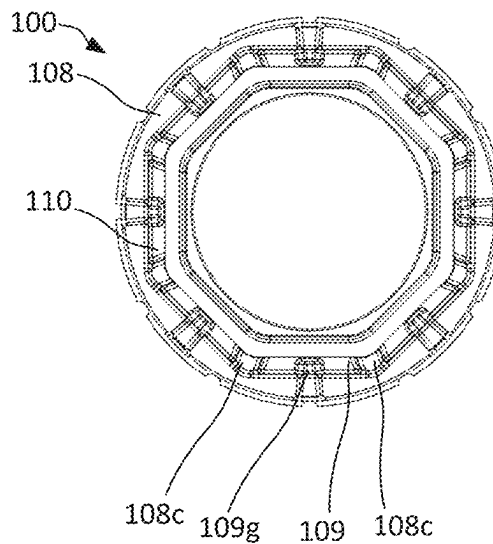


FIG. 109

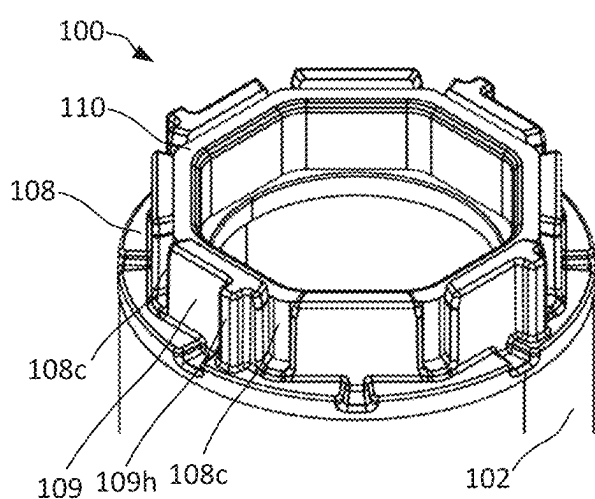


FIG. 110

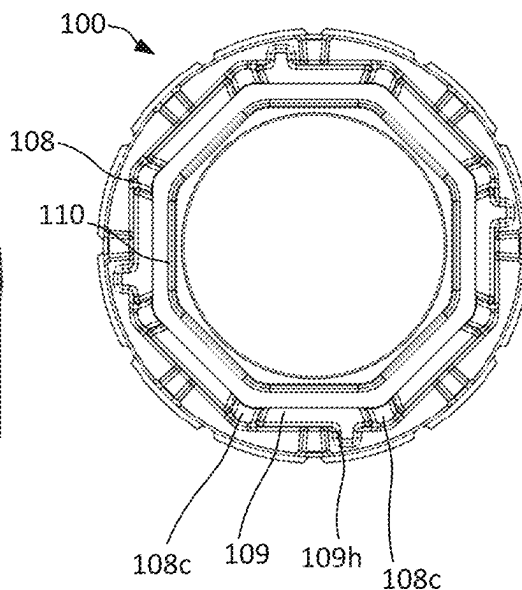


FIG. 111

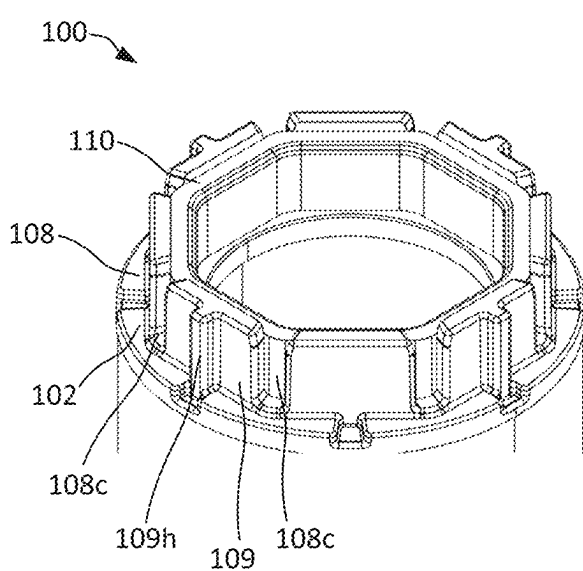


FIG. 112

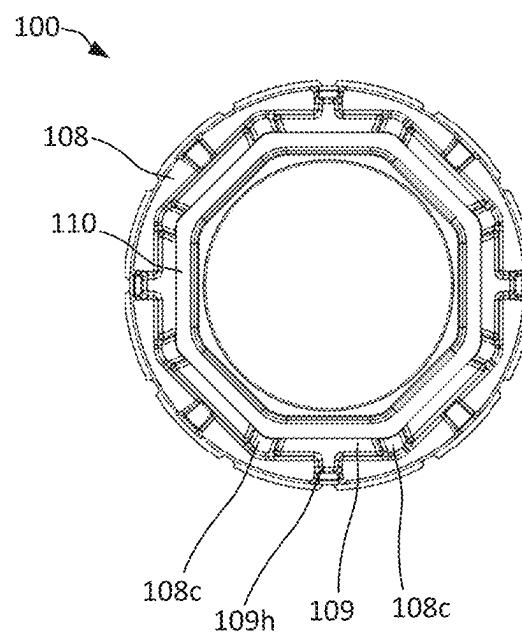


FIG. 113

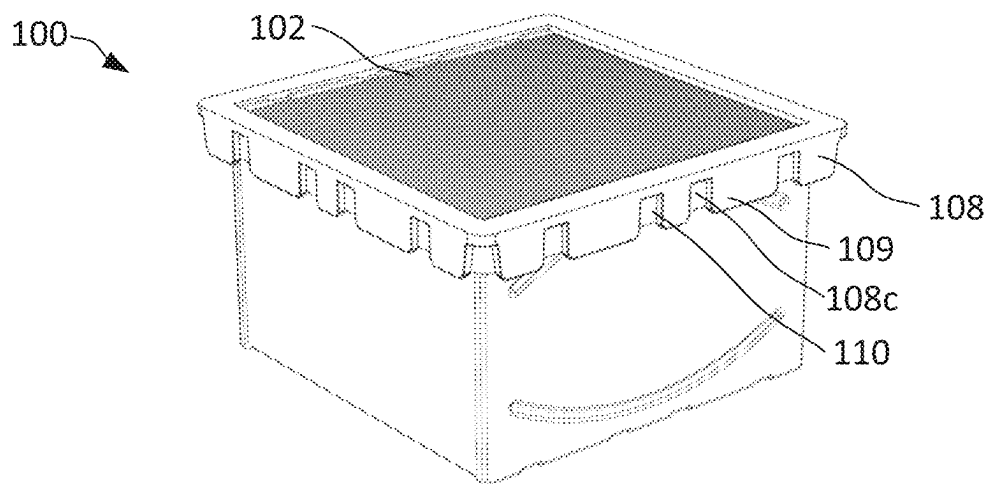


FIG. 114

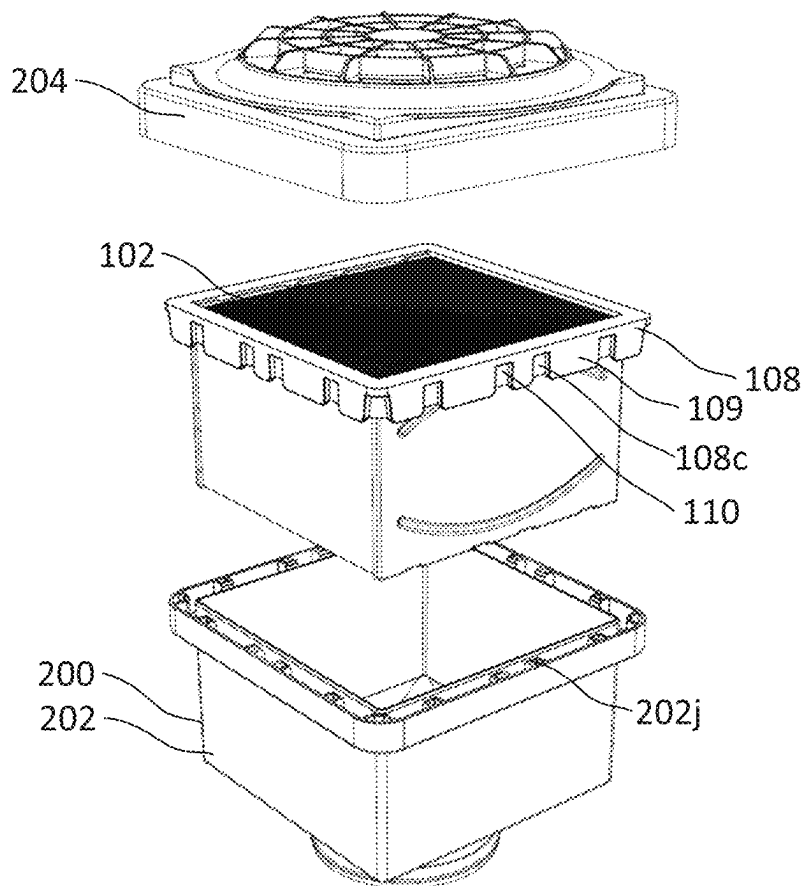


FIG. 115

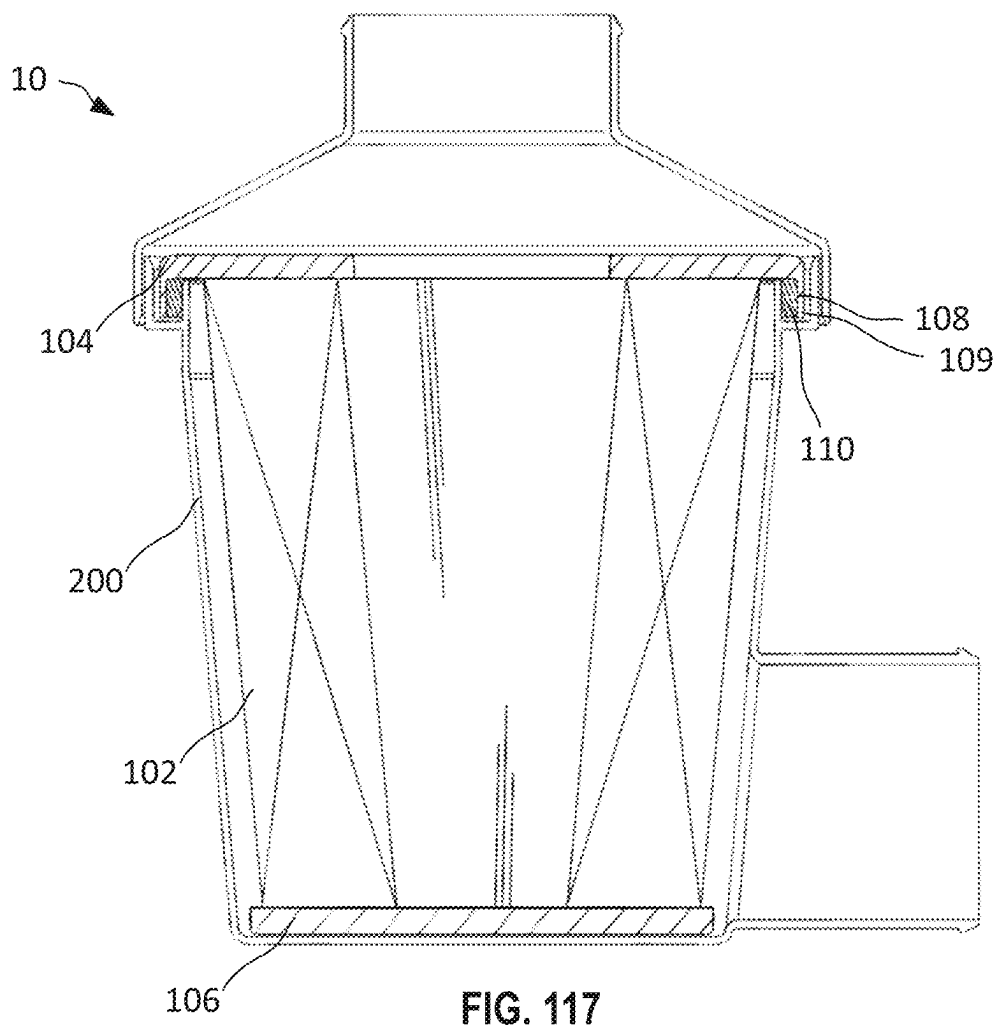
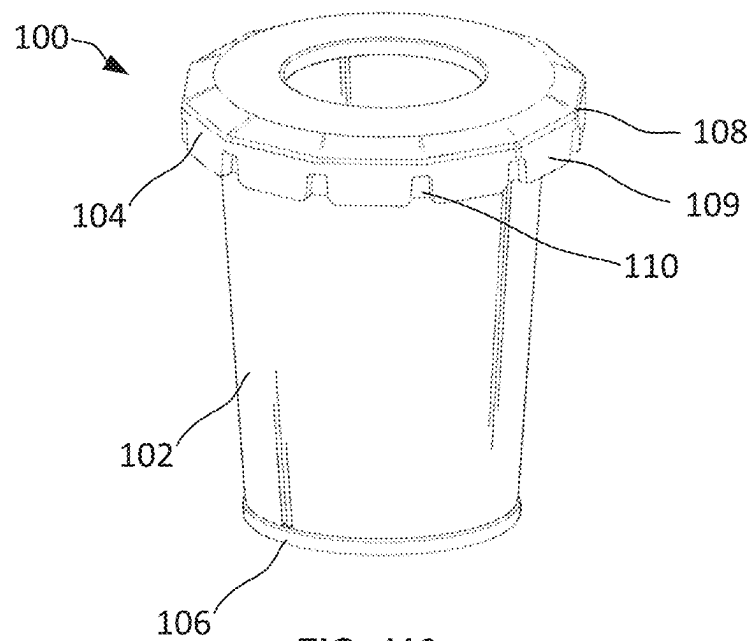


FIG. 118

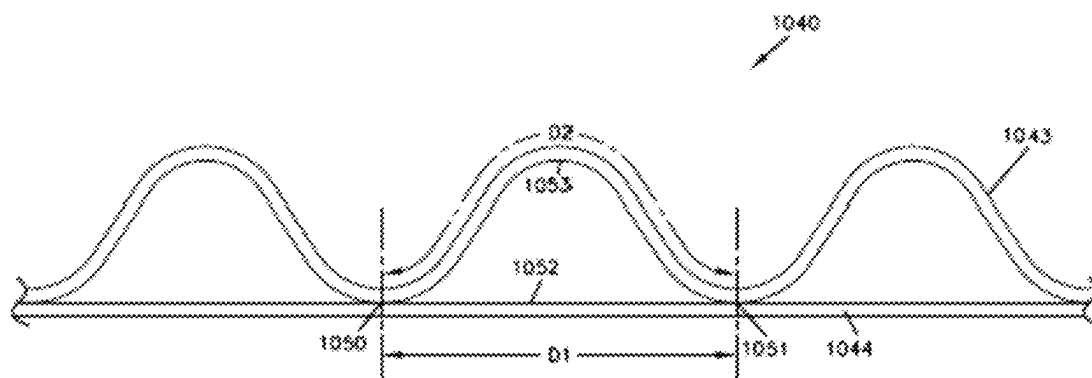


FIG. 119

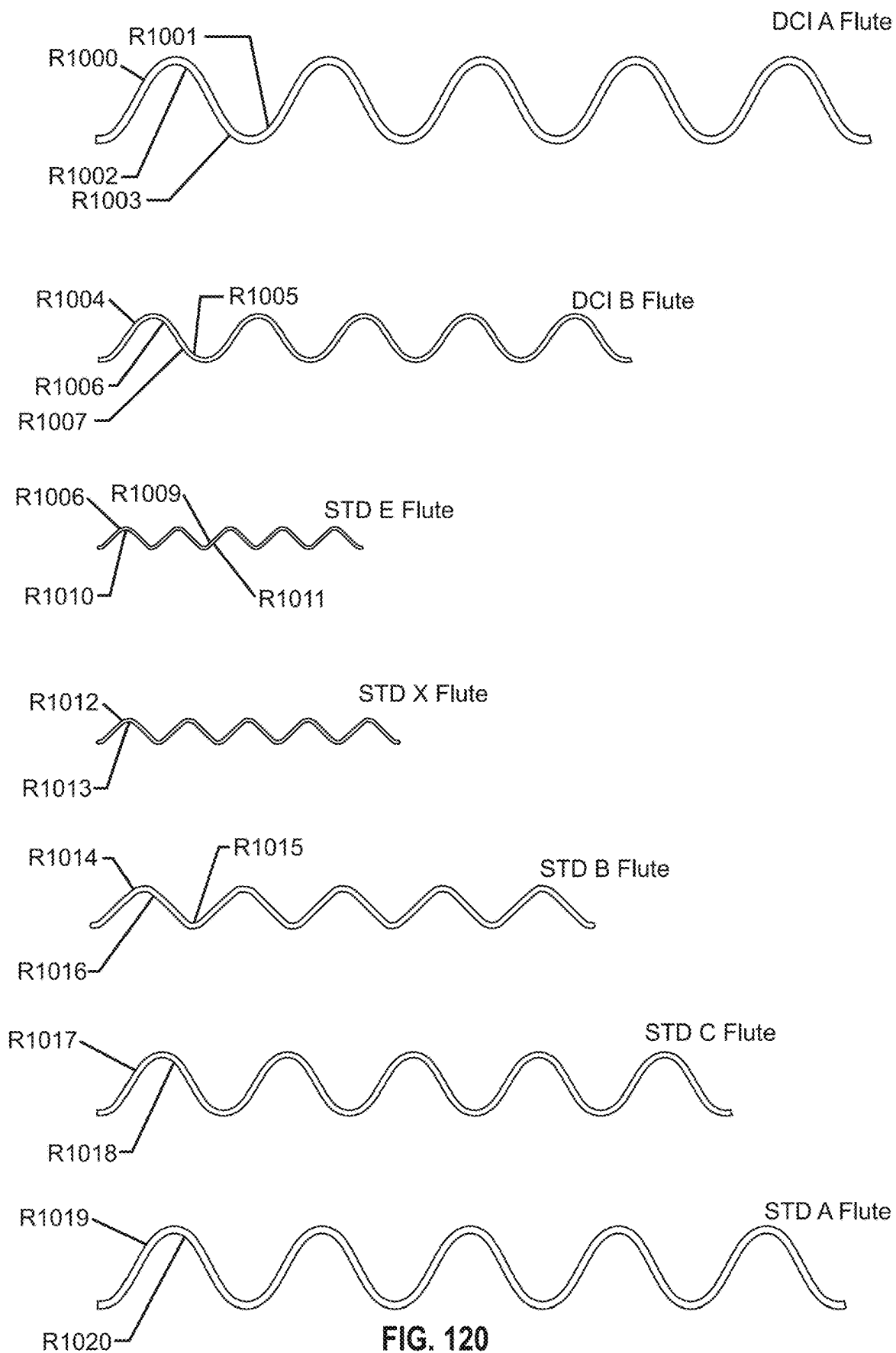


FIG. 120

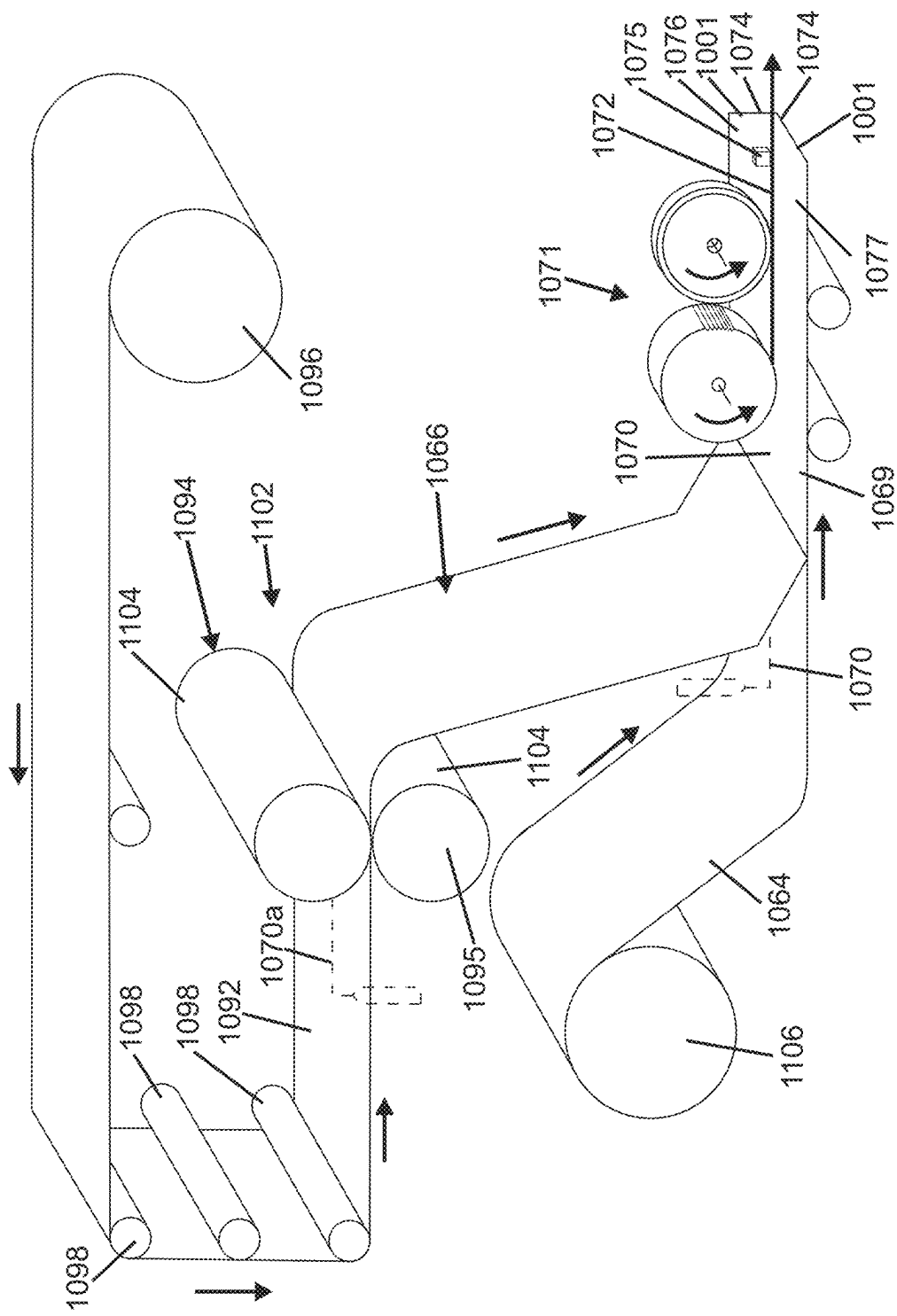


FIG. 121

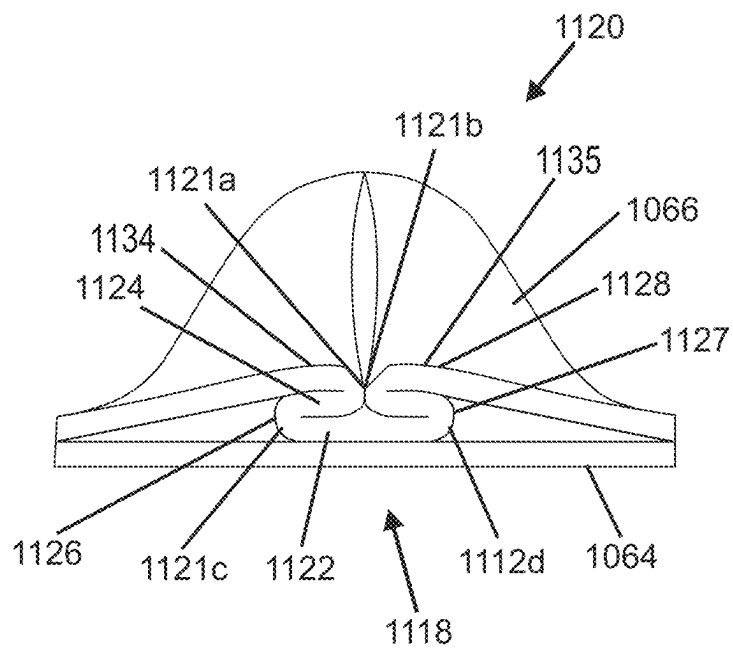


FIG. 122

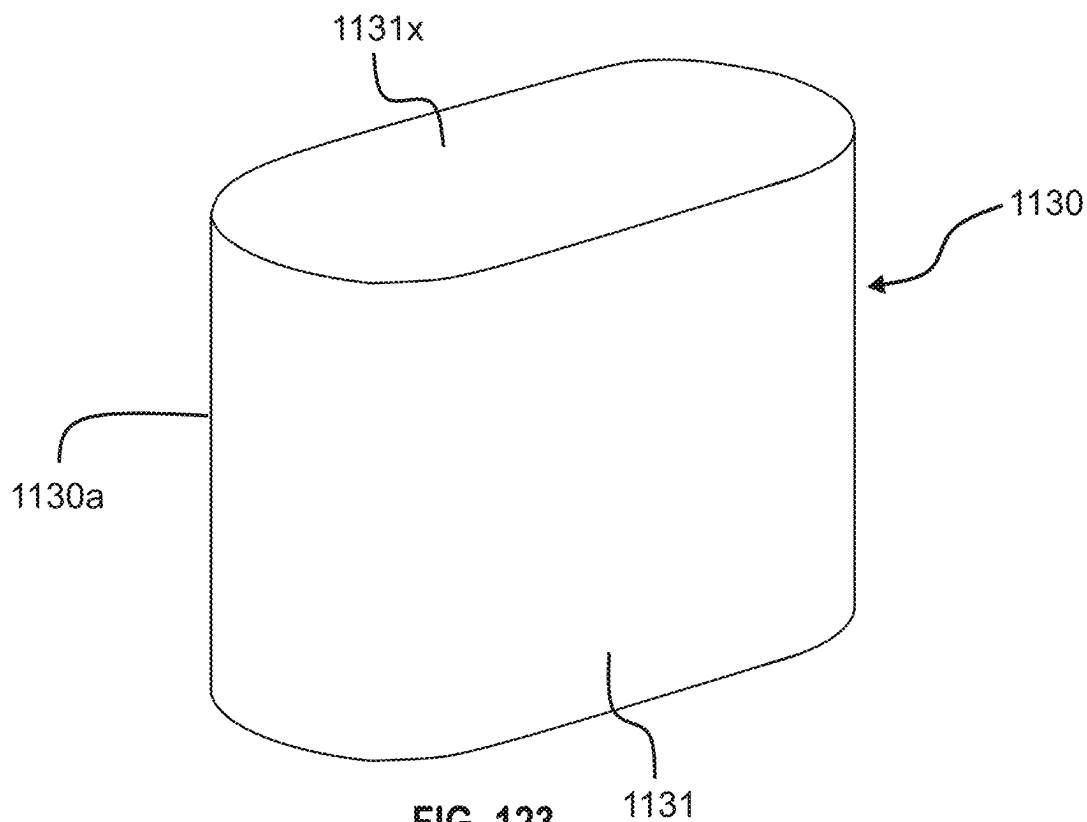


FIG. 123

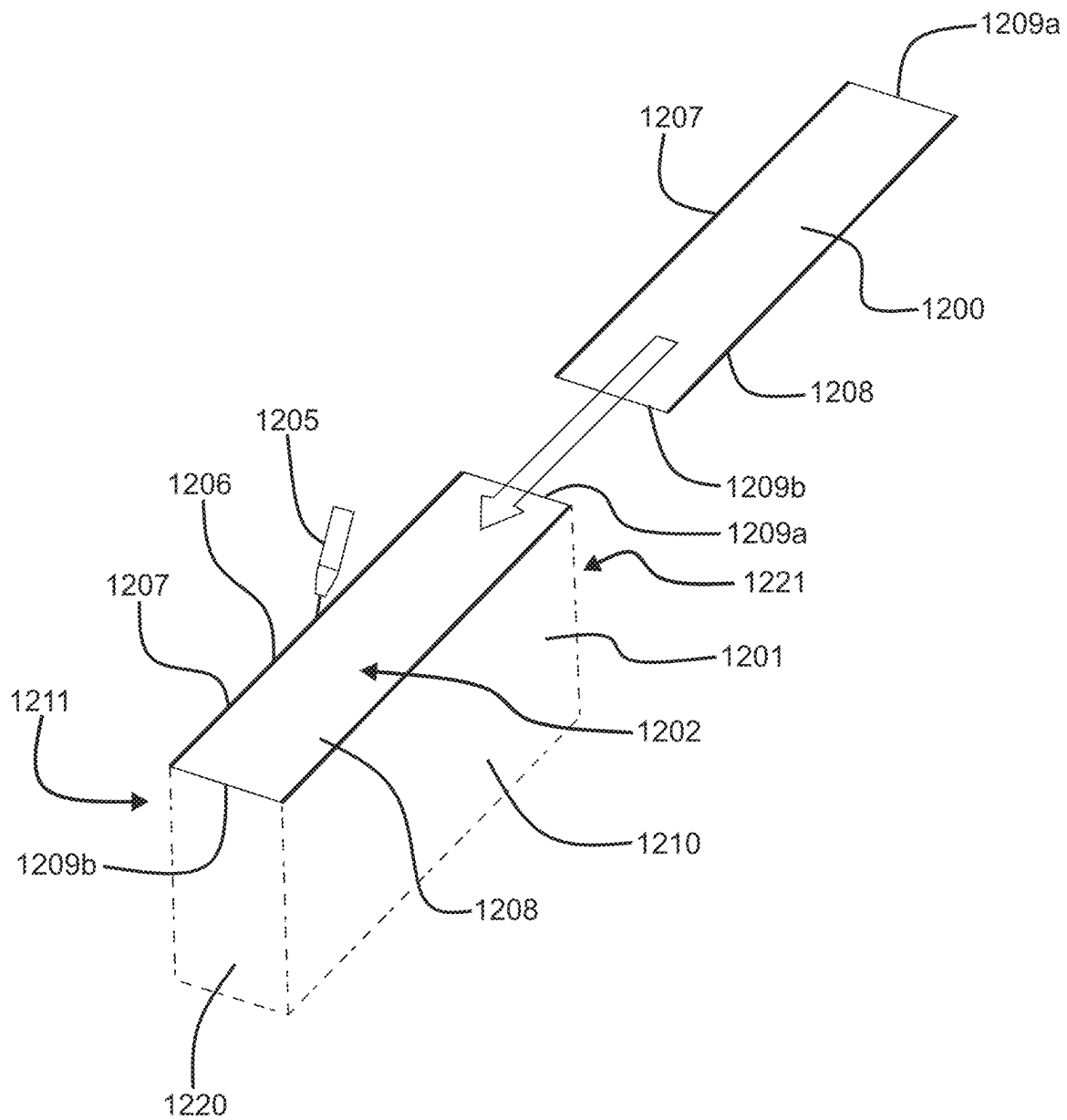


FIG. 124

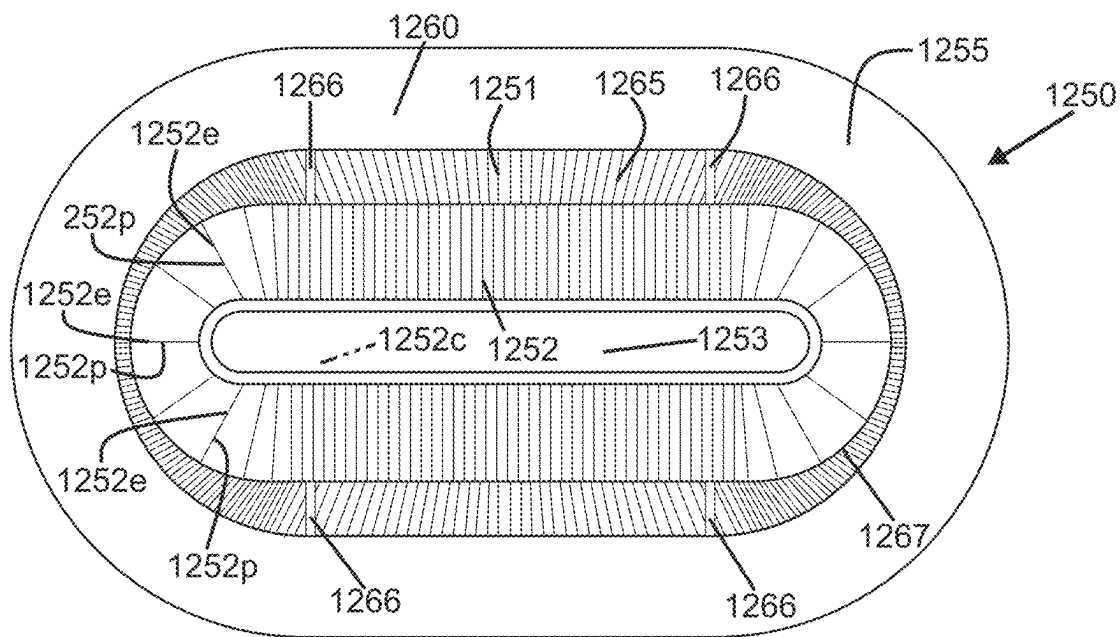


FIG. 125

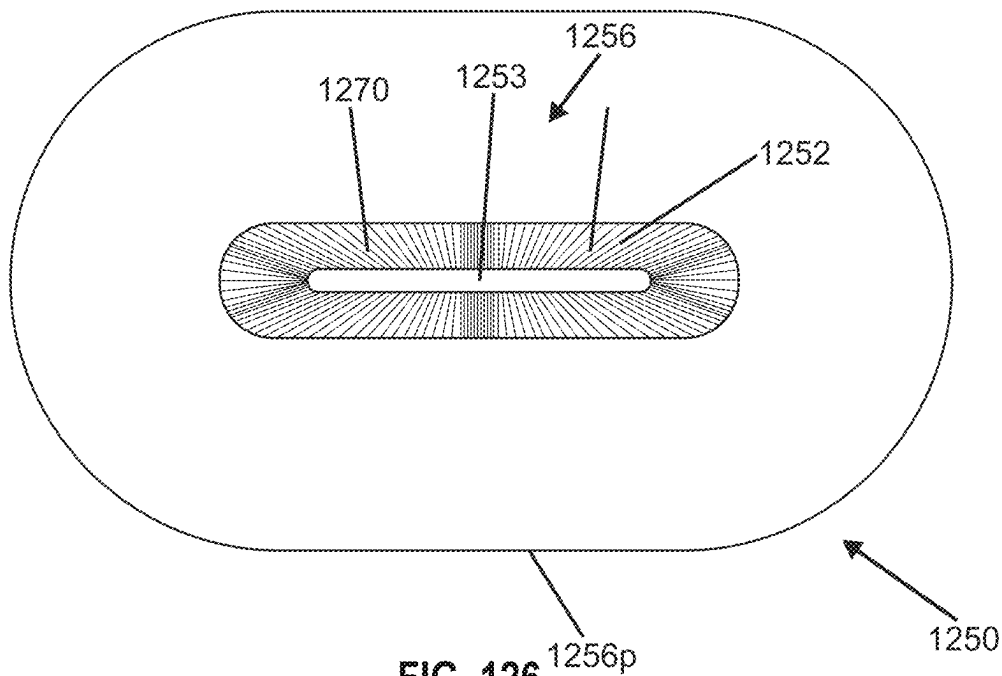


FIG. 126

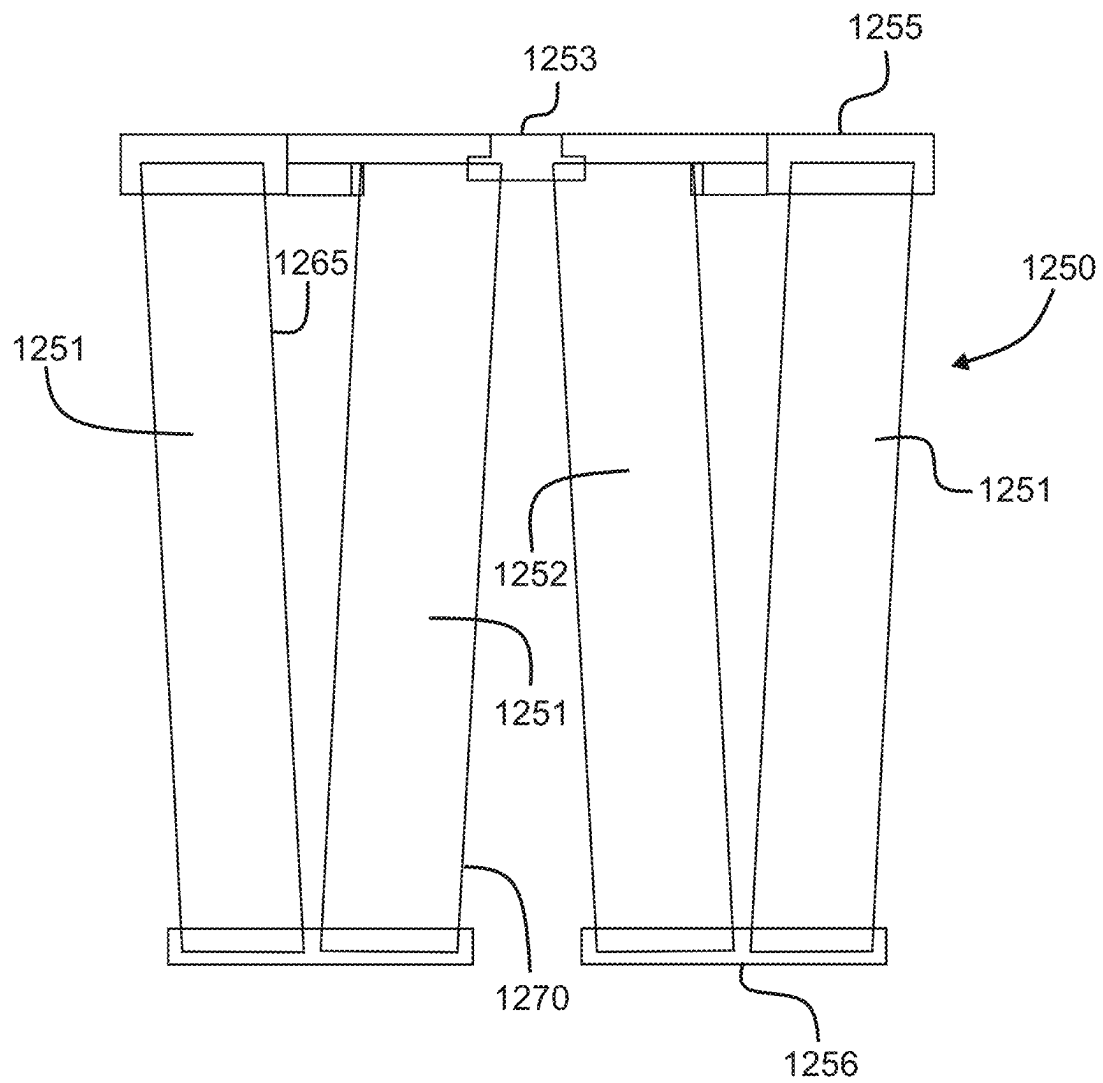


FIG. 127

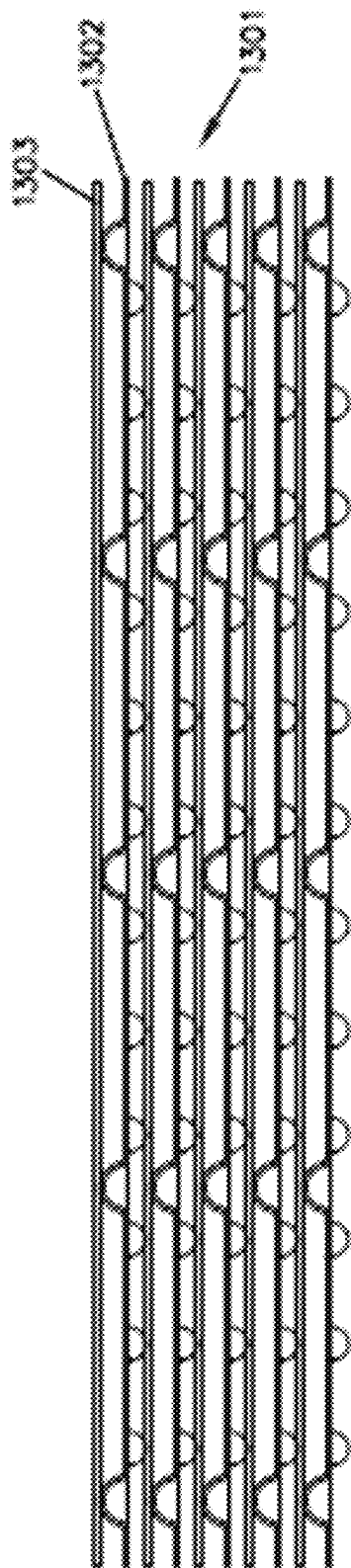


FIG. 128

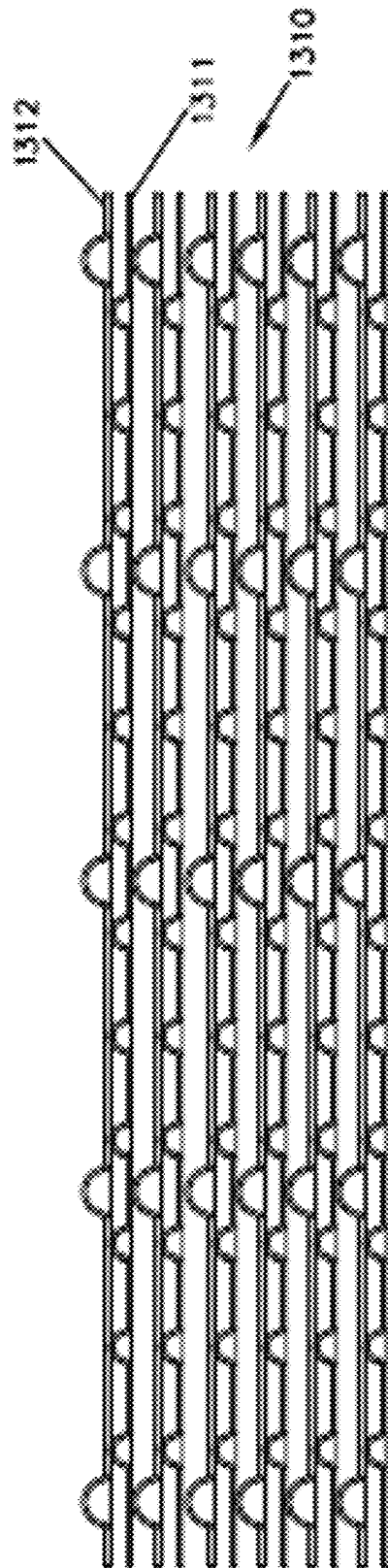


FIG. 129

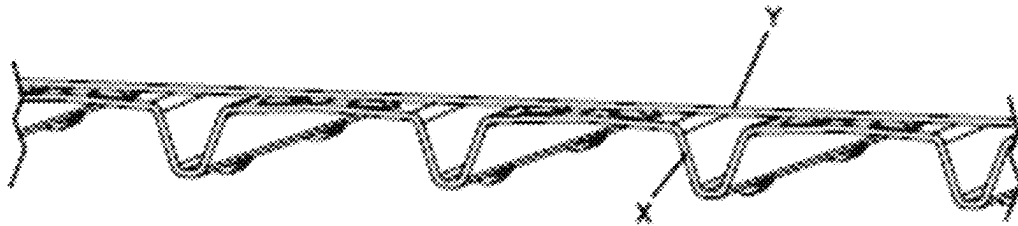


FIG. 130

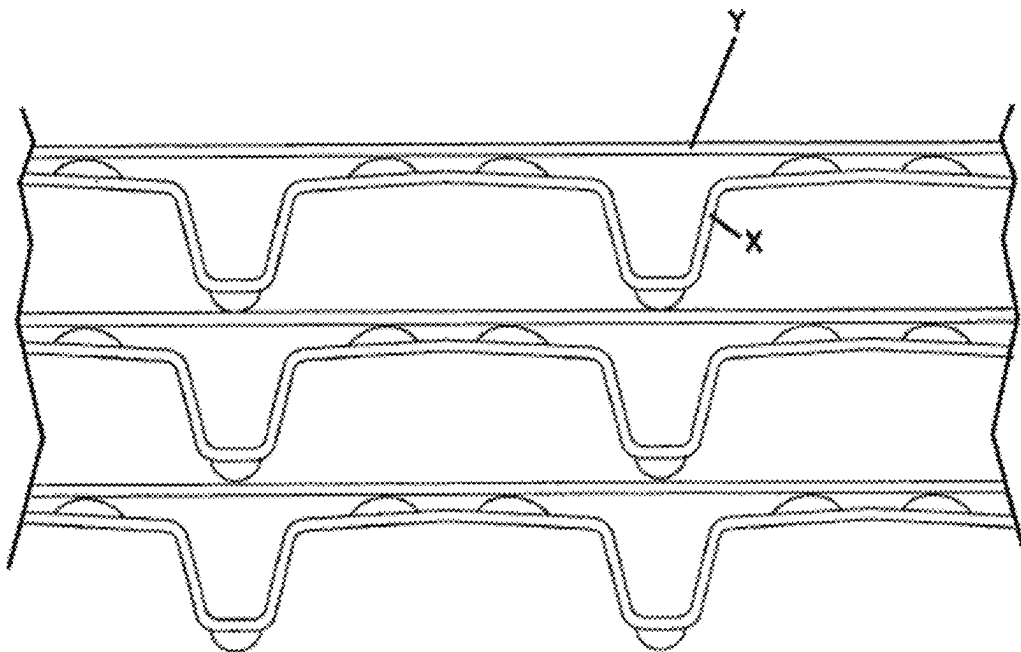


FIG. 131

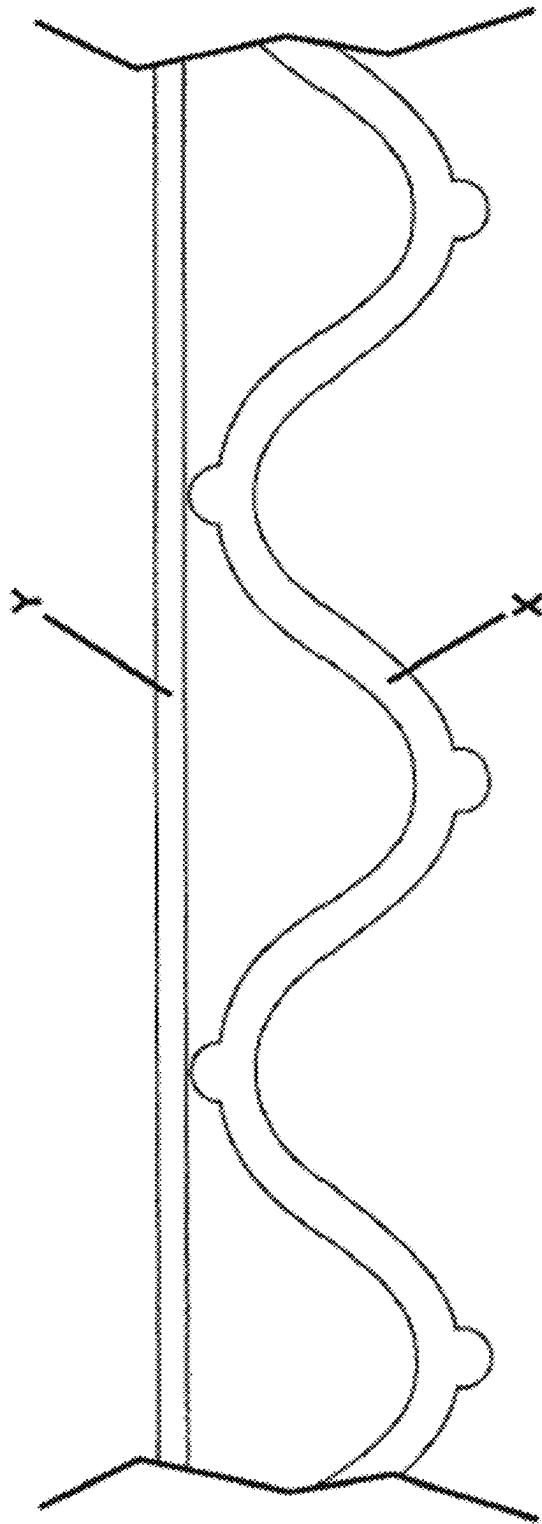


FIG. 132

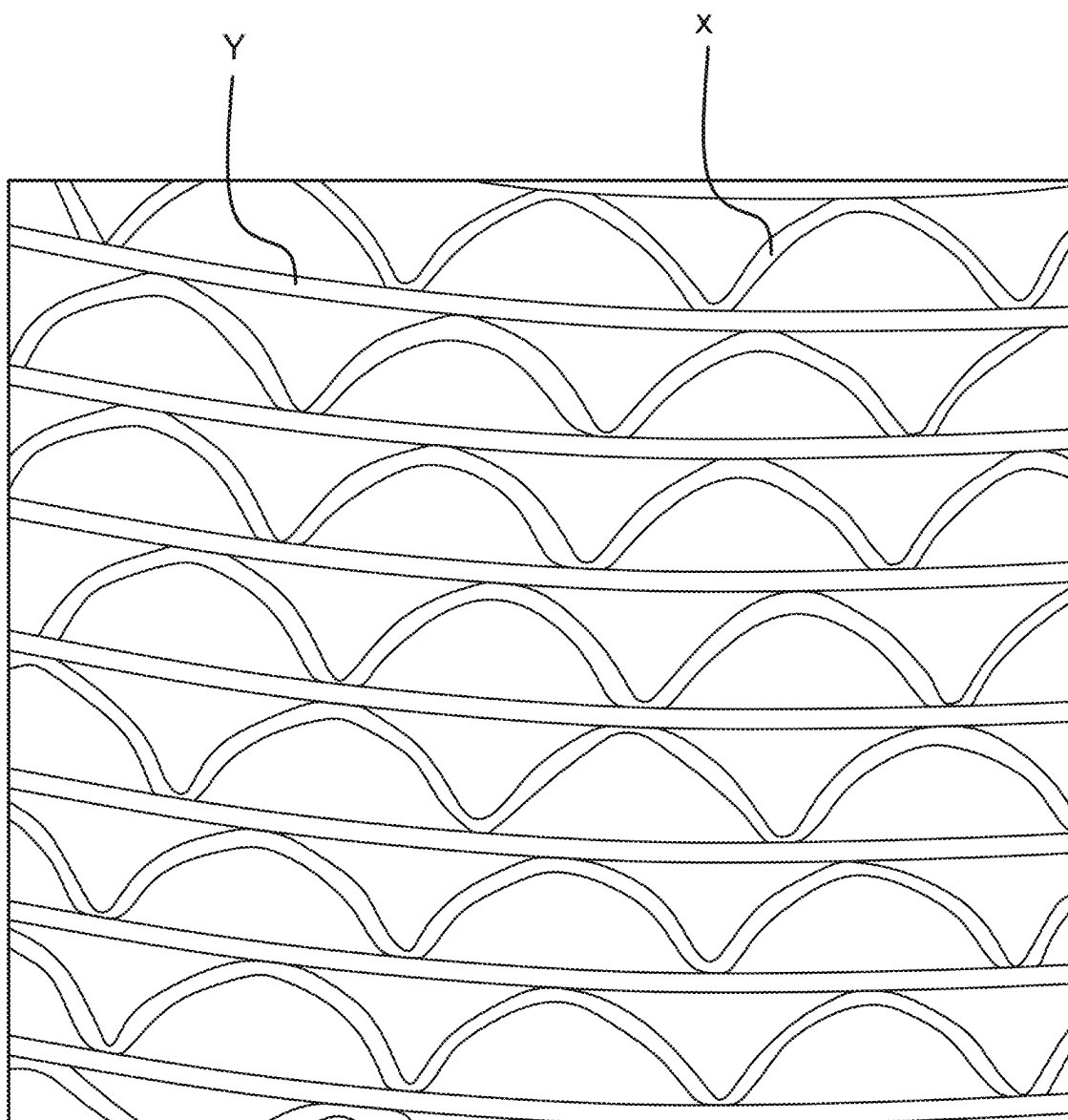


FIG. 133

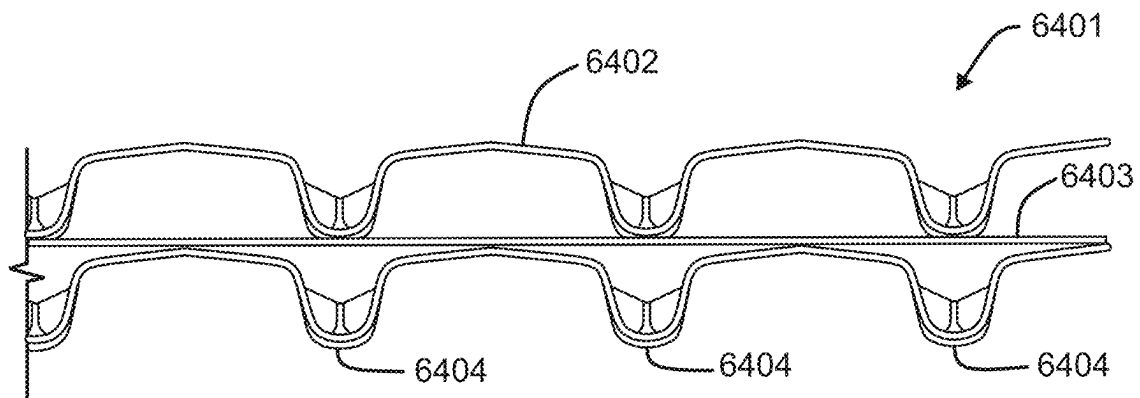


FIG. 134

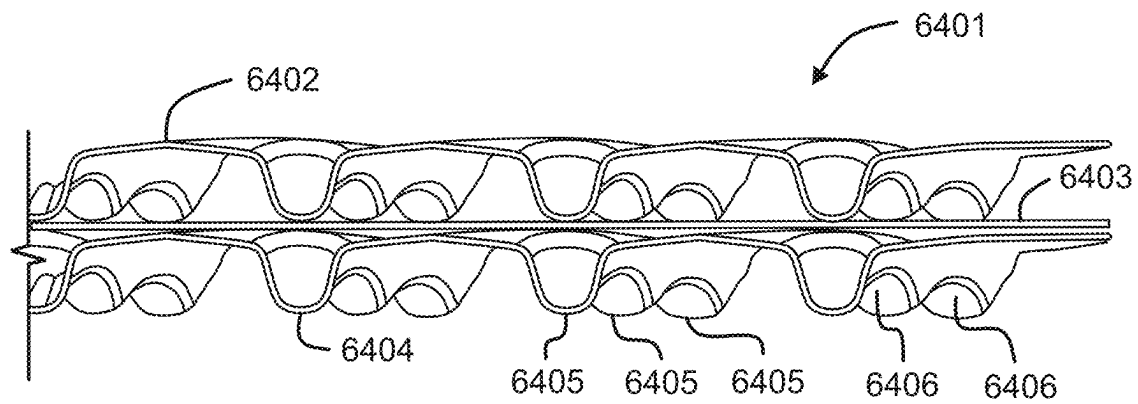


FIG. 135

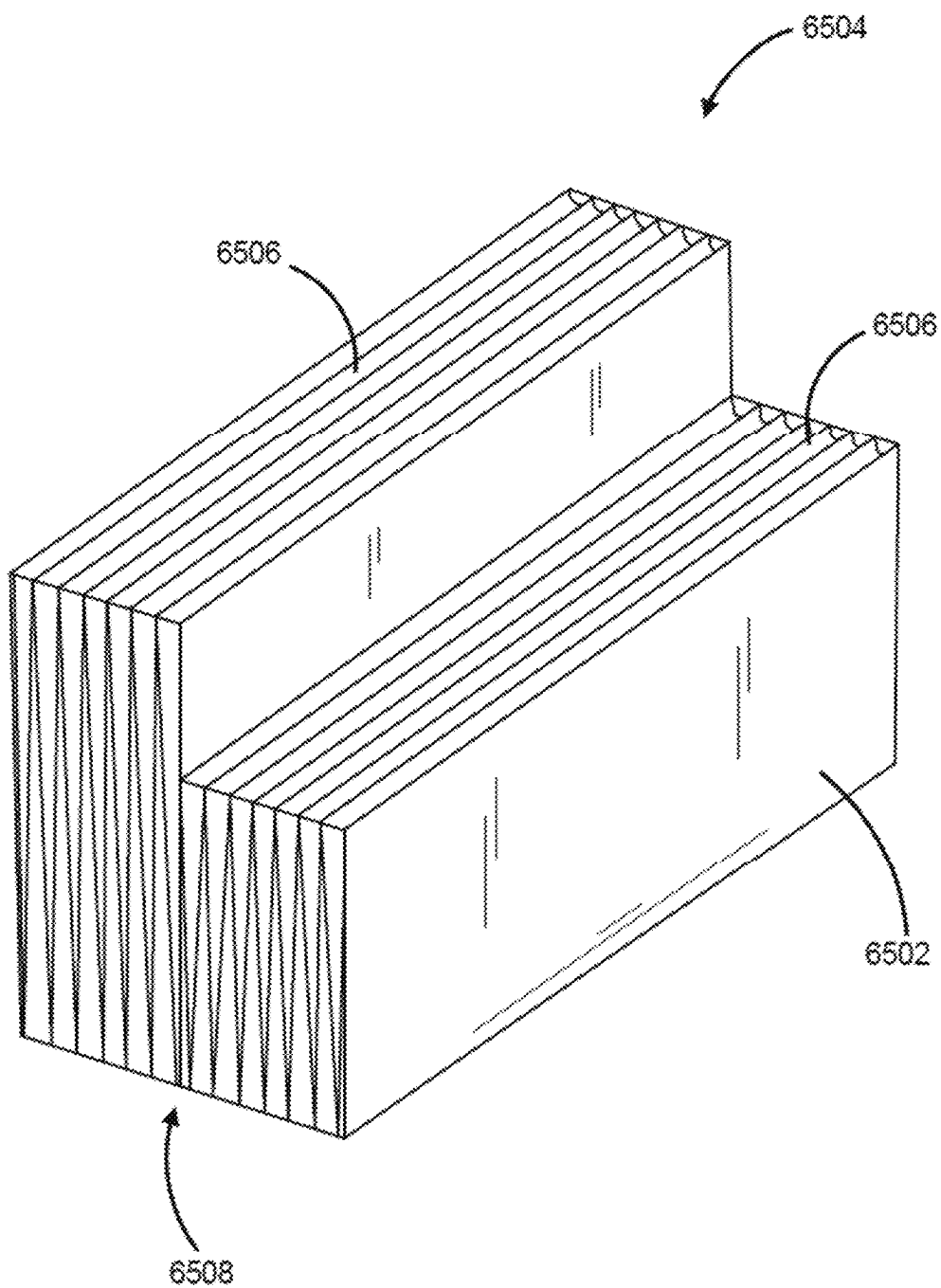


FIG. 136

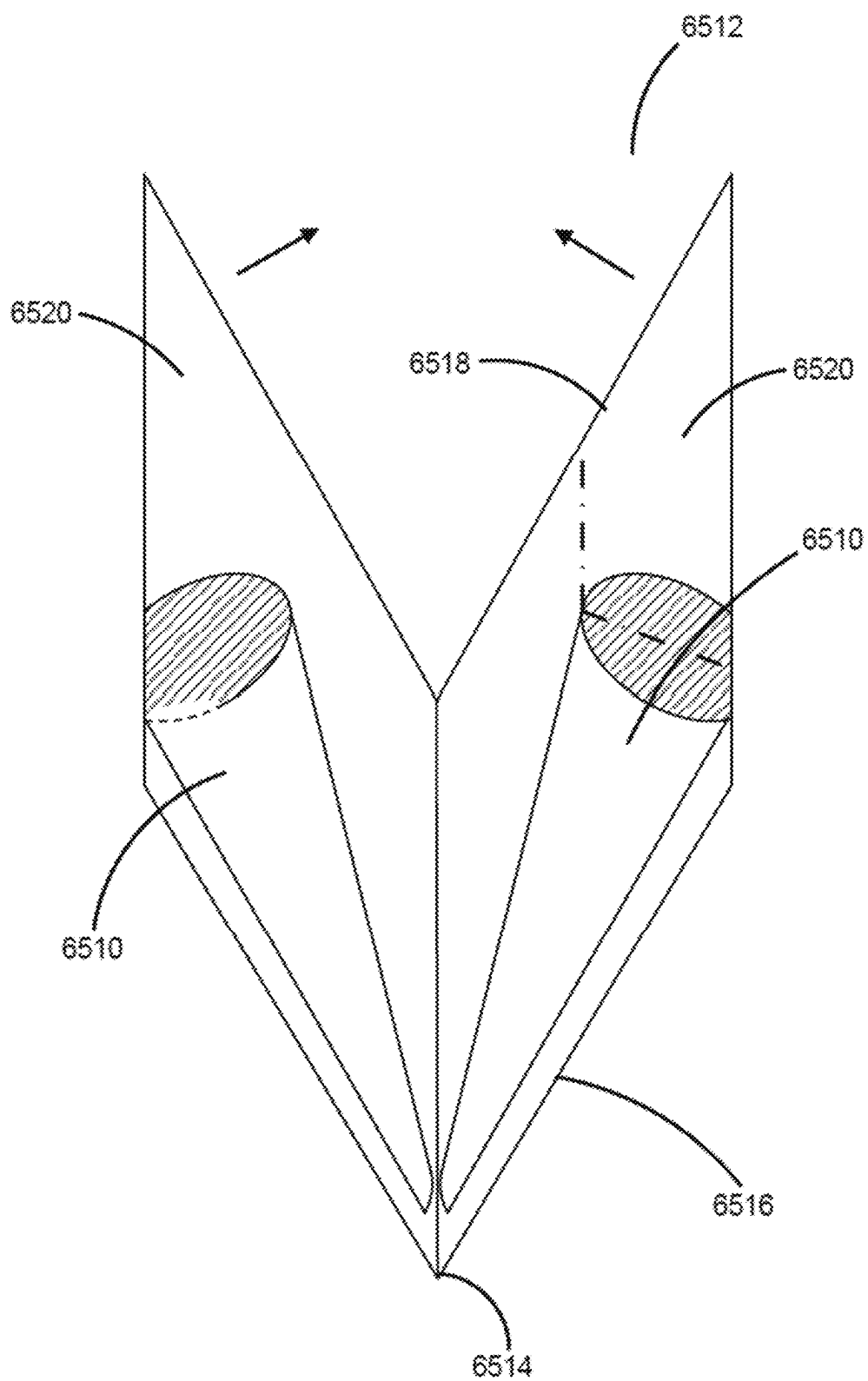


FIG. 137

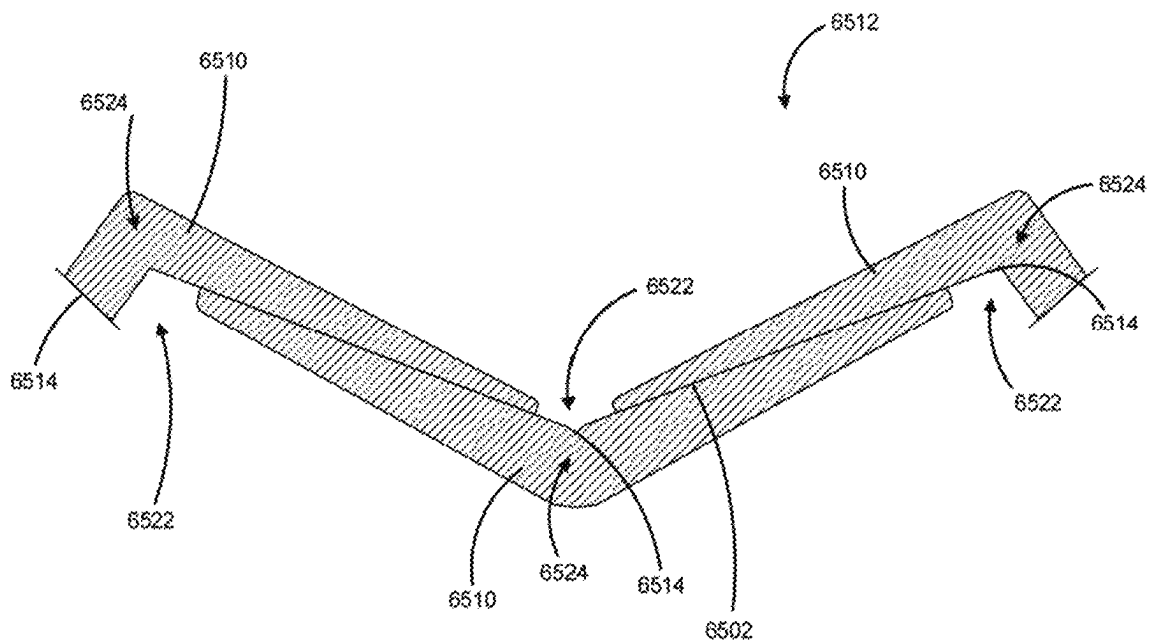


FIG. 138

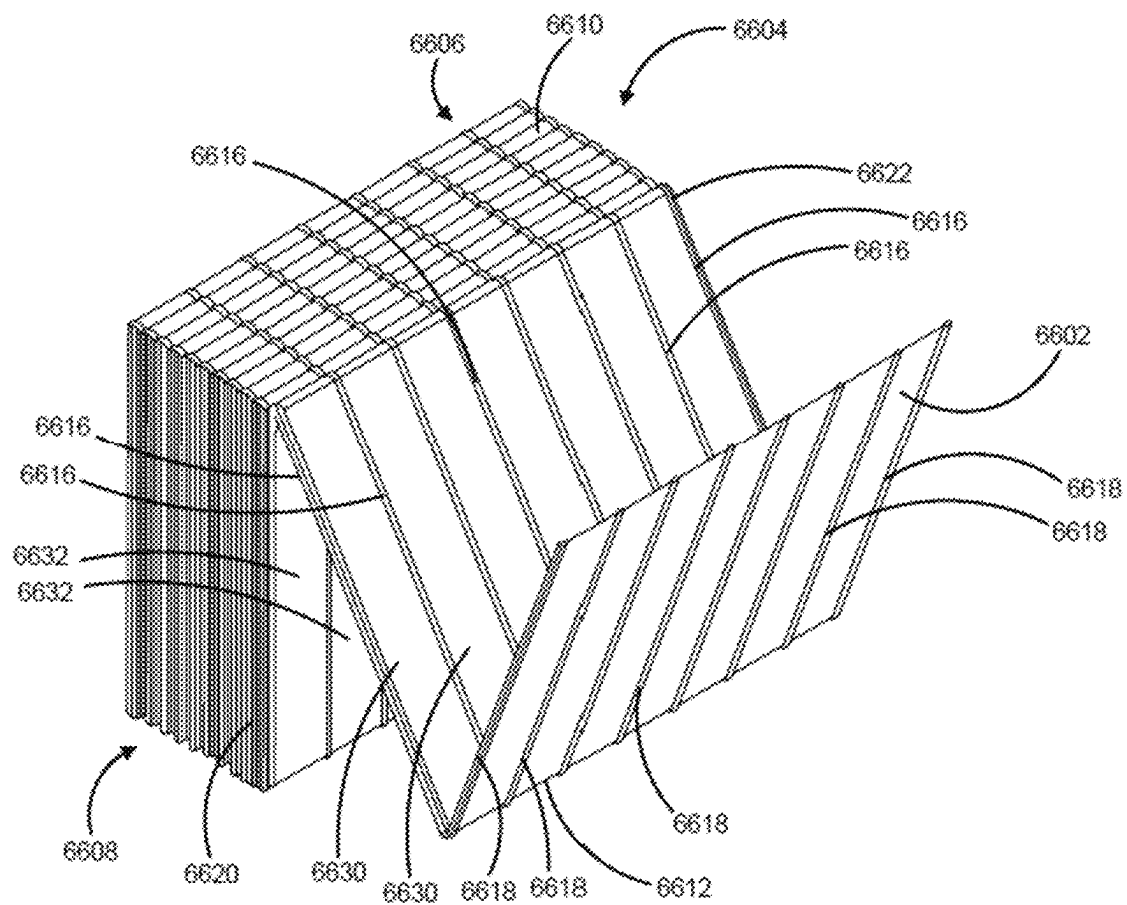


FIG. 139

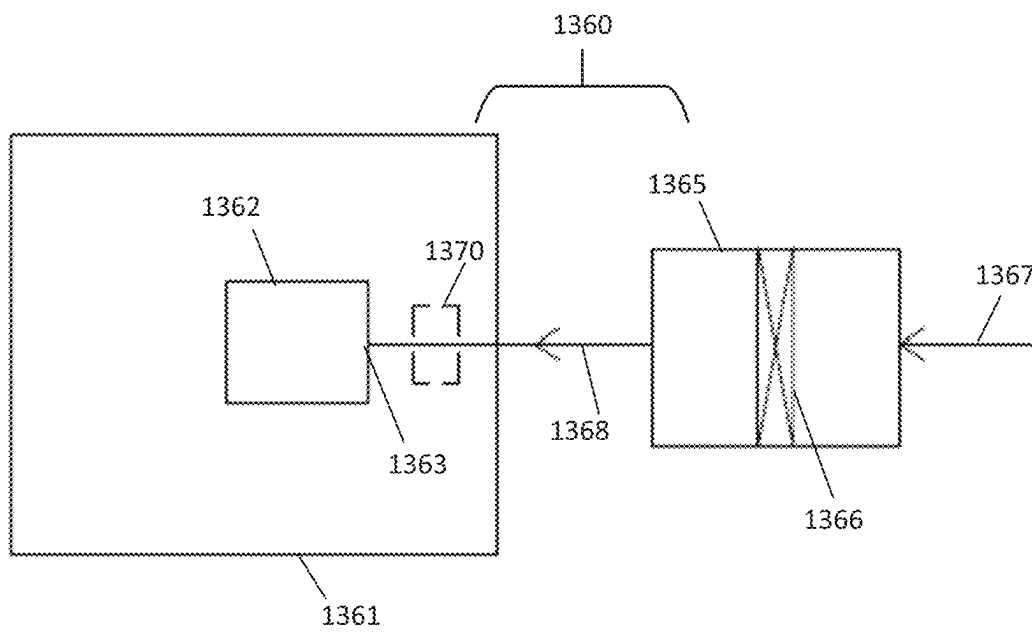


FIG. 140

FILTER ELEMENTS AND ASSEMBLIES**RELATED APPLICATIONS**

This application claims the benefit of U.S. Provisional Application Ser. No. 63/768,562, filed on Mar. 7, 2025; U.S. Provisional Application Ser. No. 63/715,456, filed on Nov. 1, 2024; and U.S. Provisional Application Ser. No. 63/649,173, filed on May 17, 2024, the disclosures of which are hereby incorporated by reference in their entireties.

FIELD OF THE DISCLOSURE

The present disclosure relates to the field of gas filtration, such as, for instance air filtration. It relates to the field of filter assemblies comprising a filter element which is arranged in a filter housing in a sealing manner, such that air passing from an inlet of the housing to an outlet of the housing is filtered by the filter element. Preferably the assemblies are used for filtering air that is afterwards directed to an engine of, for instance, a vehicle.

BACKGROUND

Filter elements, also named filter cartridges, are used for a wide variety of filtering applications and the fluid to be filtered can be a liquid or a gas, e.g., air.

Indeed, in many instances, it is desired to filter contaminant material from a fluid stream. For example, airflow streams to engines for motorized vehicles or for power generation equipment, construction equipment or other equipment, gas streams to gas turbine systems and air streams to various combustion furnaces, carry particulate contaminant therein. It is preferred for such systems that contaminant materials be removed from the fluid or at least be reduced.

Filter elements comprise filter media for removing contaminant materials when the fluid flows through the filter media. Commonly used and commercially available filter media are, for example, pleated media or fluted media. When the filter media is loaded with dust and/or particles above a certain threshold load, a filter element needs to be replaced. Filter elements are typically installed in the housing in a replaceable manner.

There is a need in industry for new solutions which allow an easy installation of a correct filter element in a filter housing, especially complex filter housing designs, and which would not allow the installation of a wrong filter element. Complex filter housing designs can, for instance, be a result of boundary conditions imposed by the geometry and positioning of other vehicle/engine components.

SUMMARY

An air filter cartridge for use within a housing assembly of an air cleaner can include a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; a first end cap operably arranged at the media pack first end, the first end cap including: a seal member presenting an inner radial side defining a seal surface and presenting an outer radial side, the inner radial side and the outer radial side each having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and a seal support arrangement including a plurality of support members located adjacent the flat sides of the seal member outer radial

side such that the corner portions of the outer radial side are unsupported by the plurality of support members.

In some examples, each of the plurality of support members extend in an axial direction parallel with the seal surface.

In some examples, the plurality of support members equals the number of the plurality of flat sides of the seal member outer radial side.

In some examples, the plurality of support members define an axial distal end of the filter cartridge.

In some examples, the seal member is formed from a first material and the plurality of support members are formed from a second material different from the first material.

In some examples, the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

An air filter cartridge for use within a housing assembly of an air cleaner can include a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; a first end cap operably arranged at the media pack first end, the first end cap including: a seal member presenting an inner radial side and an outer radial side, the inner radial side defining a seal surface having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and a seal support arrangement including a plurality of support members located radially outward from the seal surface and being radially aligned with the flat sides such that the seal surface corner portions are radially aligned with gaps formed between adjacent support members.

In some examples, the number of gaps between the plurality of support members equals the number of seal surface corner portions such that each of the gaps is radially aligned with one of the seal surface corner portions.

In some examples, the seal member outer radial side defines a generally polygonal shape having a second plurality of flat sides and adjoining corner portions, where the plurality of support members are located adjacent the second plurality of flat sides.

In some examples, an exposed portion of the plurality of support members defines an axial distal end of the filter cartridge.

In some examples, the seal member is formed from a first material and the plurality of support members are formed from a second material that is different from the first material.

In some examples, the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

In some examples, each of the plurality of support members extend in an axial direction that is generally parallel with the seal surface.

An air filter cartridge can include a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and a seal member operably arranged on the media pack, the seal member presenting a seal surface having a generally polygonal shape with flat sides and adjoining corner portions, where the seal surface is at least partially located between the media pack first and second axial ends, is spaced from the media pack outer perimeter, and faces in a radially inward direction towards the media pack outer perimeter.

In some examples, the end cap further includes a support structure including a base part and a plurality of spaced apart reinforcement members.

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In some examples, the plurality of reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

In some examples, the support structure is formed from a first material that is different from a second material associated with the seal member.

In some examples, the plurality of flat sides of the seal surface have a first length and the plurality of support members have a second length that is a majority of the first length.

In some examples, each of the plurality of support members extends in an axial direction parallel with the seal surface.

In some examples, the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

An air filter cartridge for use within a housing assembly of an air cleaner can include a media pack including filter media extending from a first end to a second end; and a first end cap arranged at the first end of the media pack, the first end cap presenting an inner radial side defining a seal surface having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions, the first end cap having a first cross-sectional thickness extending through the flat sides that is greater than a second cross-sectional thickness extending through the corner portions.

In some examples, the first end cap is formed as a single, monolithic component.

In some examples, the first end cap includes a plurality of support members at least partially defining the first cross-sectional thickness.

In some examples, a distal axial end of the filter cartridge is defined by the end cap at a location that is radially aligned with the seal surface flat sides.

In some examples, the first end cap includes a seal member defining the inner radial side and defining an outer radial side, wherein the seal member is unsupported on the outer radial side at locations radially aligned with the seal surface corner portions.

In some examples, the first end cap includes a seal member defining the inner radial side and the seal member has a constant cross-sectional thickness.

An air cleaner can include a housing assembly including a housing body defining an interior volume, an air inlet, and an air outlet, and defining a first seal surface having a generally polygonal shape with flat sides and adjoining corner portions; a filter cartridge positioned within the interior volume, the filter cartridge including: a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and a first end cap operably arranged at the media pack first end, the first end cap including: a seal member presenting an inner radial side defining a seal surface and presenting an outer radial side, the inner radial side and the outer radial side each having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions, wherein the seal surface of the seal member forms a seal against the first seal surface of the housing assembly; and a seal support arrangement including a plurality of support members located adjacent the flat sides of the seal member outer radial side such that the corner portions of the outer radial side are unsupported by the plurality of support members.

In some examples, the housing assembly includes a wall structure defining the first seal surface and that further

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defines an oppositely facing second seal surface having a generally polygonal shape with flat sides and adjoining corner portions.

In some examples, the air cleaner further includes a safety filter cartridge including filter media and having a seal member that forms a seal with the second seal surface.

In some examples, the first seal surface is formed as part of an outlet tube assembly that defines the air outlet.

In some examples, the outlet tube assembly includes a first part secured to a second part that includes a wall structure defining the first seal surface.

In some examples, the first and second parts are joined together on opposite sides of an end wall of the housing assembly.

An outlet tube assembly for use with an air cleaner housing can include a first part defining an air inlet or air outlet; and a second part joined to the first part such that a portion of the air cleaner housing is sandwiched between the first and second parts, wherein the second part includes a first sidewall having a radially outward facing seal surface with flat sides and adjoining corner portions forming a generally polygonal shape, and a second sidewall having a radially inward facing surface with a plurality of spaced apart flat sections interrupted by inwardly extending protrusions.

In some examples, the inwardly extending protrusions of the second sidewall include side edges that together with the flat sections define recess areas.

In some examples, the recess areas are configured to receive reinforcement members of a filter cartridge such that the protrusions are received within gaps between the reinforcement members.

In some examples, the flat sections and protrusions of the second sidewall are configured to guide rotational alignment of a filter cartridge during installation.

In some examples, at least one of the flat sections has a different width compared to the other flat sections.

In some examples, at least one of the protrusions has a different width compared to the other protrusions.

In some examples, the first sidewall defines a radially inward facing seal surface with flat sides and adjoining corner portions forming a generally polygonal shape.

A filter cartridge can include a media pack including media extending between a first end and a second end along a longitudinal axis; an end cap secured to the media pack and including: a seal member presenting a radially inward or outward facing seal surface having a generally polygonal shape with flat sides and adjoining corner portions, the seal member being formed from a first material, the seal member having a second surface opposite the seal surface; and a support structure including a base part and a plurality of spaced apart reinforcement members, the support structure being formed from a second material different than the first material, wherein the reinforcement members are located adjacent the seal member second surface and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

In some examples, the support structure and seal member are each formed from a polyurethane material.

In some examples, the seal member has at least three flat sides and three corner portions.

In some examples, the seal member has eight flat sides and eight corner portions.

In some examples, each of the flat sides of the seal member have an equal length.

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In some examples, the adjoining corner portions have a radius or curved shape.

In some examples, the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

In some examples, the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

In some examples, the seal member is secured to the support structure.

In some examples, the support structure is molded onto the seal member.

In some examples, the seal member is located axially beyond a distal end of the media pack.

In some examples, the seal member circumscribes the media pack.

In some examples, the media pack has one of a circular and a racetrack cross-sectional shape.

In some examples, the media pack includes pleated media.

In some examples, the media pack includes fluted media.

A filter cartridge can include a media pack including media extending between a first end and a second end along a longitudinal axis; an end cap secured to the media pack and including: a seal member presenting a radially inward or outward facing seal surface, the seal member being formed from a first material; and a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different than the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material and that defines an axial end of the cartridge and is coincident with a radial innermost portion of the end cap and/or at least a portion of the radially inward or outward facing seal surface; wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

In some examples, the support structure and seal member are each formed from a polyurethane material.

In some examples, the seal member has at least three flat sides and three corner portions.

In some examples, the seal member has eight flat sides and eight corner portions.

In some examples, each of the flat sides of the seal member have an equal length.

In some examples, the adjoining corner portions have a radius or curved shape.

In some examples, the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

In some examples, the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

In some examples, the seal member is secured to the support structure.

In some examples, the support structure is molded onto the seal member.

In some examples, the media pack has one of a circular and a racetrack cross-sectional shape.

In some examples, the media pack includes pleated media.

In some examples, the media pack includes fluted media.

A filter cartridge can include a media pack including media extending between a first end and a second end along

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a longitudinal axis; an end cap secured to the media pack and including: a seal member presenting a radially inward or outward facing seal surface, the seal member being formed from a first material; and a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different than the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material and that defines an axial end of the cartridge; wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

In some examples, the support structure and seal member are each formed from a polyurethane material.

In some examples, the seal member has at least three flat sides and three corner portions.

In some examples, the seal member has eight flat sides and eight corner portions.

In some examples, each of the flat sides of the seal member have an equal length.

In some examples, the adjoining corner portions have a radius or curved shape.

In some examples, the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

In some examples, the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

In some examples, the seal member is secured to the support structure.

In some examples, the support structure is molded onto the seal member.

In some examples, the seal member is located axially beyond a distal end of the media pack.

In some examples, the seal member circumscribes the media pack.

In some examples, the media pack has one of a circular and a racetrack cross-sectional shape.

In some examples, the media pack includes pleated media.

In some examples, the media pack includes fluted media.

A filter cartridge comprising: a media pack including media extending between a first end and a second end along a longitudinal axis; an end cap secured to the media pack and including: a seal member presenting a radially inward or outward facing seal surface and an undercut portion radially recessed from the seal surface, the seal member being formed from a first material; and a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different from the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material; wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

In some examples, the support structure and seal member are each formed from a polyurethane material.

In some examples, the seal member has at least three flat sides and three corner portions.

In some examples, the seal member has eight flat sides and eight corner portions.

In some examples, each of the flat sides of the seal member have an equal length.

In some examples, the adjoining corner portions have a radius or curved shape.

In some examples, the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

In some examples, the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

In some examples, the seal member is secured to the support structure.

In some examples, the support structure is molded onto the seal member.

In some examples, the seal member is located axially beyond a distal end of the media pack.

In some examples, the seal member circumscribes the media pack.

In some examples, the media pack has one of a circular and a racetrack cross-sectional shape.

In some examples, the media pack includes pleated media.

In some examples, the media pack includes fluted media.

A filter cartridge can include a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and an end cap arranged about the media pack outer perimeter, the end cap including: a seal member presenting a seal surface that is spaced from the media pack outer perimeter and that faces in a radial inward direction towards the media pack outer perimeter, the seal surface having a generally polygonal shape with flat sides and adjoining corner portions.

In some examples, the end cap further includes a support structure including a base part and a plurality of spaced apart reinforcement members, the support structure being formed from a material that is different than a material associated with the seal member, wherein the reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

In some examples, the support structure and seal member are each formed from a polyurethane material.

In some examples, the seal member has at least three flat sides and three corner portions.

In some examples, the seal member has eight flat sides and eight corner portions.

In some examples, each of the flat sides of the seal member have an equal length.

In some examples, the adjoining corner portions have a radius or curved shape.

In some examples, the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

In some examples, the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

In some examples, the seal member is secured to the support structure.

In some examples, the support structure is molded onto the seal member.

In some examples, the seal member is located axially beyond a distal end of the media pack.

In some examples, the seal member circumscribes the media pack.

In some examples, the media pack has one of a circular and a racetrack cross-sectional shape.

In some examples, the media pack includes pleated media.

In some examples, the media pack includes fluted media.

In some examples, the end cap is located proximate the first axial end of the media pack.

In some examples, a distal end of the end cap is generally flush with the media pack first axial end.

In some examples, the end cap is located proximate the first axial end of the media pack.

In some examples, the plurality of reinforcement members extend in a direction from the first axial end and towards the second axial end of the media pack.

In some examples, an axial end of the seal member is at least partially exposed and uncovered by the support structure.

In some examples, at least a portion of the plurality of reinforcement members circumscribe the media pack.

In some examples, the support structure further includes an axially extending portion spaced from the plurality of reinforcement members, the axially extending portion being secured to the media pack.

In some examples, the seal member is located within an annular space defined between the axially extending portion and the plurality of reinforcement members.

In some examples, an axial end of the plurality of reinforcement members is generally flush with an axial end of the seal member.

In some examples, the seal surface has at least eight flat sides.

In some examples, the seal surface has twelve flat sides.

In some examples, a length of the flat sides of the seal surface is greater than a length of the reinforcement members.

In some examples, the number of flat sides of the seal surface is equal to the number of reinforcement members.

In any of the above-described filter cartridges, the media pack can include fluted filter media, pleated filter media, or another type of media.

The present invention is defined, at least in part, in the appended independent claims. The dependent claims define advantageous embodiments.

BRIEF DESCRIPTION OF THE DRAWINGS

These and further aspects of the present disclosure will be explained in greater detail by way of example and with reference to the accompanying drawings in which:

FIG. 1 is a schematic view of a first example of an air cleaner assembly having features in accordance with the present disclosure.

FIG. 2 is a perspective exploded view of the air cleaner assembly shown in FIG. 1.

FIG. 3 is a cross-sectional side view of the air cleaner assembly shown in FIG. 1.

FIG. 4 is a perspective view of an outlet tube associated with the air cleaner assembly shown in FIG. 1.

FIG. 5 is a cross-sectional side view of the outlet tube shown in FIG. 4.

FIG. 6 is a perspective view of a collar associated with the air cleaner assembly shown in FIG. 1.

FIG. 7 is a bottom view of the collar shown in FIG. 6.

FIG. 8 is a perspective view of a filter cartridge associated with the air cleaner assembly shown in FIG. 1.

FIG. 8A is a perspective view of an alternative construction of the filter cartridge shown in FIG. 8.

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FIG. 9 is a top view of the filter cartridge shown in FIG. 8.

FIG. 10 is a side view of the filter cartridge shown in FIG. 8.

FIG. 11 is a perspective exploded view of an end cap associated with the filter cartridge shown in FIG. 8.

FIG. 12 is a side view of a portion of the filter cartridge shown in FIG. 8.

FIG. 13 is a cross-sectional side view of the filter cartridge portion shown in FIG. 12.

FIG. 14 is a perspective view of the outlet tube and filter cartridge of the air cleaner assembly shown in FIG. 1 with the filter cartridge partially installed onto the outlet tube.

FIG. 15 is a perspective view of the outlet tube and filter cartridge of the air cleaner assembly shown in FIG. 1 with the filter cartridge fully installed onto the outlet tube.

FIG. 16 is a schematic representation of the outlet tube and filter cartridge in a first sealing configuration corresponding to the configuration shown for the air cleaner assembly shown in FIG. 1.

FIG. 17 is a schematic representation of the outlet tube and filter cartridge in an alternative second configuration that is usable with the air cleaner assembly shown in FIG. 1.

FIG. 18 is a perspective view of the filter cartridge shown in FIG. 8 shown as being mounted on an alternative outlet tube that can be used with the air cleaner assembly shown in FIG. 1.

FIG. 19 is a side view of the outlet tube shown in FIG. 18.

FIG. 20 is a bottom view of the outlet tube shown in FIG. 18.

FIG. 21 is a partial cross-sectional view of an end cap arrangement usable with the filter cartridge shown in FIG. 4.

FIG. 22 is a perspective view of a portion of a filter cartridge with an end cap arrangement usable with the filter cartridge shown in FIG. 4.

FIG. 23 is a partial cross-sectional view of an end cap associated with the filter cartridge shown in FIG. 22.

FIG. 24 is a partial cross-sectional view of an end cap arrangement usable with the filter cartridge shown in FIG. 4.

FIG. 25 is a partial cross-sectional view of the filter cartridge shown in FIG. 21 shown as being mounted on an alternative outlet tube that can be used with the air cleaner assembly shown in FIG. 1.

FIG. 26 is a schematic illustration of a process for forming any of the filter cartridge end cap arrangements disclosed herein.

FIG. 27 is a perspective view of an example filter cartridge and housing assembly portion usable with an air cleaner assembly of the type shown in FIG. 1.

FIG. 28 is a lateral cross-sectional view of the assembly shown in FIG. 27.

FIG. 29 is a partial longitudinal cross-sectional view of the assembly shown in FIG. 27.

FIG. 30 is a perspective exploded view of the assembly shown in FIG. 27.

FIG. 31 is a perspective view of the filter cartridge shown in FIG. 27.

FIG. 32 is an enlarged perspective view of a portion of the filter cartridge shown in FIG. 31.

FIG. 33 is a side view of the filter cartridge shown in FIG. 31.

FIG. 34 is an enlarged portion of the filter cartridge view shown at FIG. 33.

FIG. 35 is a topview of the filter cartridge shown in FIG. 31.

FIG. 36 is a cross-sectional side view of the filter cartridge shown in FIG. 31.

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FIG. 37 is an enlarged cross-sectional side view of a portion of the filter cartridge shown in FIG. 36.

FIG. 38 is an enlarged cross-sectional side view of a portion of the filter cartridge shown in FIG. 36, taken along a line extending between support members of the end cap.

FIG. 39 is an exploded perspective view of the filter cartridge shown in FIG. 31.

FIG. 40 is a perspective view of a seal member of the filter cartridge shown in FIG. 31.

FIG. 41 is an end view of the seal member shown in FIG. 40.

FIG. 42 is a perspective view of a support structure of the filter cartridge shown in FIG. 31.

FIG. 43 is an end view of the support structure shown in FIG. 42.

FIG. 44 is a first perspective view of the housing assembly portion shown in FIG. 27.

FIG. 45 is a second perspective view of the housing assembly portion shown in FIG. 27.

FIG. 46 is an end view of the housing assembly portion shown in FIG. 27.

FIG. 47 is a lateral cross-sectional view of the housing assembly portion shown in FIG. 27 and depicting a first part and a second part of an outlet tube assembly.

FIG. 48 is an end view of a second portion of the outlet tube assembly shown in FIG. 47.

FIG. 49 is a perspective view of an air cleaner assembly within which the filter cartridge of FIGS. 30 to 41 is disposed.

FIG. 50 is an exploded perspective view of the air cleaner assembly shown in FIG. 49.

FIG. 51 is a longitudinal cross-sectional view of the air cleaner assembly shown in FIG. 49.

FIG. 51A is an enlarged portion of the cross-sectional view presented in FIG. 51.

FIG. 51B is a lateral cross-sectional view of the air cleaner assembly shown in FIG. 49.

FIG. 52 is a perspective view of a second outlet part of the housing assembly associated with the air cleaner assembly shown in FIG. 49.

FIG. 53 is a perspective view of a safety filter cartridge associated with the air cleaner assembly shown in FIG. 49.

FIG. 54 is a perspective exploded view of an end cap of the safety filter cartridge shown in FIG. 53.

FIG. 55A is a cross-sectional view of the end cap shown in FIG. 54, taken through opposite corner portions of the end cap.

FIG. 55B is a cross-sectional view of the end cap shown in FIG. 54, taken through a centerline of opposite flat sides of the end cap.

FIG. 56A is a flat schematic of the seal member and support structure of the filter cartridge shown in FIG. 31.

FIG. 56B is a flat schematic illustrating alternate arrangement options of the seal member and support structure of the filter cartridge shown in FIG. 31.

FIG. 56C is a flat schematic of a portion of a sidewall of the second part of the outlet tube assembly shown in FIGS. 47 and 48.

FIG. 56D is a flat schematic illustrating alternative arrangement options of the sidewall of the second part of the outlet tube assembly shown in FIGS. 47 and 48.

FIG. 57 is a schematic top view of the seal member and support structure of the filter cartridge shown in FIG. 31 in an alternative arrangement.

FIGS. 58A-58E are schematic top views of alternative arrangements of the second part of the outlet tube assembly shown at FIGS. 47 and 48.

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FIGS. 59-80 schematically show views of an example air cleaner assembly having a filter cartridge with an end cap mounted to an outer circumferential side of the filter cartridge.

FIGS. 81-84 schematically show views of an example air cleaner assembly generally similar to that shown in FIGS. 59-80, but with an alternative cover and latching arrangement.

FIGS. 85-86 schematically show views of an example filter cartridge in accordance with FIGS. 59-84 in which the media pack is provided with a non-round cross-sectional shape.

FIG. 87 schematically shows a further example of an air cleaner having a filter cartridge with an end cap mounted to an outer circumferential side of the filter cartridge.

FIG. 88 schematically shows an example seal and support arrangement suitable for use with any of the filter cartridges disclosed herein, wherein the seal member has three flat sides and corner portions and the support structure has three reinforcement members.

FIG. 89 schematically shows an example seal and support arrangement suitable for use with any of the filter cartridges disclosed herein, wherein the seal member has four flat sides and corner portions and the support structure has four reinforcement members.

FIG. 90 schematically shows an example seal and support arrangement suitable for use with any of the filter cartridges disclosed herein, wherein the seal member has five flat sides and corner portions and the support structure has five reinforcement members.

FIG. 91 schematically shows an example seal and support suitable for use with any of the filter cartridges disclosed herein, wherein the seal member has six flat sides and corner portions and the support structure having six reinforcement members.

FIG. 92 is a perspective view an example filter cartridge in which the seal member has one flat side that is longer than the remaining flat sides and in which the support structure has one reinforcement member that is longer than the remaining reinforcement members.

FIG. 93 is a top view of the filter cartridge of FIG. 92.

FIG. 94 is a schematic top view of a seal member and support structure in which multiple different lengths of flat sides and reinforcement members are provided.

FIG. 95 is a perspective exploded view of an example end cap construction usable with any of the filter cartridges disclosed herein.

FIG. 96 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 97 is an exploded perspective exploded view the filter cartridge construction shown in FIG. 96.

FIG. 98 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 99 is an exploded perspective exploded view the filter cartridge construction shown in FIG. 98.

FIG. 100 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 101 is an exploded perspective exploded view the filter cartridge construction shown in FIG. 100.

FIG. 102 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 103 is a cross-sectional perspective view of the filter cartridge construction shown in FIG. 102.

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FIG. 104 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 105 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 106 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 107 is a top view of the filter cartridge construction shown in FIG. 106.

FIG. 108 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 109 is a top view of the filter cartridge construction shown in FIG. 108.

FIG. 110 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 111 is a top view of the filter cartridge construction shown in FIG. 110.

FIG. 112 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 113 is a top view of the filter cartridge construction shown in FIG. 112.

FIG. 114 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 115 is a perspective exploded view of an air cleaner assembly including the filter cartridge of FIG. 114.

FIG. 116 is a perspective view of an example filter cartridge construction usable with any of the filter cartridges disclosed herein.

FIG. 117 is a cross-sectional view of an air cleaner assembly including the filter cartridge of FIG. 116.

FIG. 118 is a fragmentary, schematic, perspective view of a first example media type usable in arrangements according to the present disclosure.

FIG. 119 is an enlarged, schematic, cross-sectional view of a portion of the media type depicted in FIG. 118.

FIG. 120 includes schematic views of examples of various fluted media definitions, for media of the type of FIGS. 118 and 119.

FIG. 121 is a schematic view of an example process for manufacturing media of the type of FIGS. 118-120.

FIG. 122 is a schematic cross-sectional view of an optional end dart for media flutes of the type of FIGS. 118-121.

FIG. 123 is a schematic perspective view of a coiled filter arrangement usable in a filter cartridge having features in accordance with the present disclosure, and made with a strip of media, for example, in accordance with FIG. 118.

FIG. 124 is a schematic perspective view of a stacked media pack arrangement usable in a filter arrangement having selected features in accordance with the present disclosure and made from a strip of media for example in accordance with FIG. 118.

FIG. 125 is a schematic flow end view of a filter media pack using an alternate media to the media of FIG. 118, and alternately usable in selected filter cartridges in accordance with the present disclosure.

FIG. 126 is a schematic opposite flow end view to the view of FIG. 125.

FIG. 127 is a schematic cross-sectional view of the media pack of FIGS. 125 and 126.

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FIG. 128 is a schematic, fragmentary, cross-sectional view of a further alternate media type usable in a media pack of a filter cartridge having features in accordance with the present disclosure.

FIG. 129 is a schematic, fragmentary cross-sectional view of a first variation of the media type of FIG. 128.

FIG. 130 is a schematic, fragmentary depiction of another usable fluted sheet/facing sheet combination in accordance with the present disclosure.

FIG. 131 is a fragmentary, second schematic view of the type of media in FIG. 130 shown in a media pack.

FIG. 132 is a schematic, fragmentary, plan view of still another media variation usable in arrangements according to the present disclosure.

FIG. 133 is a schematic view of another variation of usable media in accordance with the present disclosure.

FIG. 134 is a schematic depiction of another usable fluted sheet/facing sheet combination in accordance with the present disclosure.

FIG. 135 is a perspective view of a portion of the usable fluted sheet/facing sheet combination depicted in FIG. 134.

FIG. 136 is a perspective view of another media variation usable in arrangements according to the present disclosure.

FIG. 137 is a schematic, perspective view of a portion of a support section of the filter media of FIG. 136, illustrated in a folded configuration but expanded or separated for illustrative purposes.

FIG. 138 is a schematic, cross-sectional view of a portion of the support section of the filter media of FIG. 136, illustrated in a folded configuration but expanded or separated for illustrative purposes.

FIG. 139 is a perspective view of another media variation useable in arrangements according to the present disclosure.

FIG. 140 is a schematic depiction of an equipment assembly including an air cleaner assembly in accordance with any of the air cleaner assemblies disclosed herein.

The drawings of the figures are neither drawn to scale nor proportioned. Generally, identical components are denoted by the same reference numerals in the figures. Further, it is noted that many of the drawings presented herein are generated from computer-aided design (CAD) models that show parts in their nominal, undeformed states. In particular, for assembly views showing a filter cartridge installed within the housing, the seal member is depicted in its relaxed condition even though it would actually be compressed and/or stretched when installed against the outlet tube or other seal surfaces associated with the air cleaner assembly. For example, in some views the seal member may appear to overlap with or intersect the outlet tube or seal plate geometry since the seal is shown in its pre-installation state rather than its deformed, installed state. A person having ordinary skill in the art would readily understand that such visual overlaps or intersections in the drawings represent the undeformed components and would interpret the drawings taking into account the compressible and/or stretchable nature of the seal member materials during actual installation and use.

DETAILED DESCRIPTION

The present disclosure will be described in terms of specific embodiments, which are illustrative of the disclosure and not to be construed as limiting. It will be appreciated by persons skilled in the art that the present disclosure is not limited by what has been particularly shown and/or described and that alternatives or modified embodiments

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could be developed in the light of the overall teaching of this disclosure. The drawings described are only schematic and are non-limiting.

Use of the verb “to comprise”, as well as the respective conjugations, does not exclude the presence of elements other than those stated. Use of the article “a”, “an” or “the” preceding an element does not exclude the presence of a plurality of such elements.

Furthermore, the terms first, second, and the like in the description and in the claims, are used for distinguishing between similar elements and not necessarily for describing a sequence, either temporally, spatially, in ranking or in any other manner. It is to be understood that the terms so used are interchangeable under appropriate circumstances and that the embodiments of the disclosure described herein are capable of operation in other sequences than described or illustrated herein.

Reference throughout this specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with the embodiments is included in one or more embodiment of the present disclosure. Thus, appearances of the phrases “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily all referring to the same embodiment, but may refer to one or more different embodiments. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner, as would be apparent to one ordinary skill in the art from this disclosure, in one or more embodiments.

As used herein, the term “axially” generally refers to a direction that is parallel to the longitudinal axis X while the term “radially” generally refers to a direction that is orthogonal to the longitudinal axis X. As also used herein, the term “radially inward” generally refers to a direction facing towards the longitudinal axis X while the term “radially outward” generally refers to a direction facing away from the longitudinal axis X.

Although a number of the drawings presented and discussed below are specifically addressing embodiments of a filter element for filtering air, the present disclosure is not limited thereto.

In FIG. 140, a schematic depiction of an engine equipment arrangement 1360 is depicted. The equipment system 1360, in the example, comprises a vehicle or other equipment 1361 having an internal combustion engine arrangement 1362 with a combustion air intake 1363. The equipment arrangement 1360 includes an air cleaner system 1365 having a filter arrangement 1366 therein, typically comprising a serviceable (i.e., removable and replaceable) filter cartridge. The air cleaner system 1365 and filter arrangement 1366 can include any of the below-described air cleaners and filter cartridges, and combinations thereof. Intake air to the system is shown at 1367 directed into the air cleaner assembly 1365 before filtering of unfiltered air through media of the filter cartridge arrangement 1366. At 1368, filtered air is shown being directed into the equipment air intake 1363. At 1370, optional equipment such as a turbo system is shown. Of course, alternate equipment systems can be represented by arrangements analogous to those of FIG. 140. The equipment system can be, for example, an industrial air filter, an air cleaner arrangement used in association with a turbine, etc. The use in association with an internal combustion engine is typical, but not specifically required for many of the principles characterized herein.

FIGS. 1-15 illustrate a first example of the present disclosure in a schematic manner. As shown, an air cleaner 10

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is presented including a filter cartridge **100** and a filter cartridge **150** installable within a housing assembly **200**.

In one aspect, the housing assembly **200** includes a main housing **202** defining an interior volume **202a** extending along a longitudinal axis **X1**. The housing assembly **200** is further shown as including a cover assembly **204** installable onto the main housing **202** to cover a service opening and enclose the interior volume **202a**. In one aspect, the main housing **202** further defines an air outlet **202c**, an air inlet **202d**, and a dust ejection port **202f**. In some configurations, the main housing **202** may be provided with an air inlet while the cover assembly **204** may be provided with an air outlet or vice versa. The housing assembly **200** is further shown as including an outlet tube assembly **205** which defines the air outlet **202c** and a collar **203** for attaching the outlet tube assembly **205** to the main housing **202**. The outlet tube assembly **205** is shown in isolation at FIGS. **4** and **5** while the collar **203** is shown in isolation at FIGS. **6** and **7**.

Referring to FIGS. **10** to **13**, aspects of the filter cartridge **100** are further illustrated. In one aspect, the filter cartridge **100** includes filter media **102** extending between a first end cap **104** and a second end cap **106**. In the particular example shown, the filter media **102** is pleated media arranged in a tubular manner to define a central opening **102c** around an axis **X**. The filter cartridge **150** is similarly so constructed and is installed within the filter cartridge **100** and seals to the outlet tube assembly **205** such that air passing through the filter cartridge **100** must also pass through the filter cartridge **150** before exiting the housing assembly via outlet **202c**. As detailed in a later section, many media types and configurations are possible for use with all of the disclosed example filter cartridges herein, for example, fluted media, z-media, depth media, non-pleated media, coiled media, and/or stacked media, etc.

In one aspect, the first end cap **104** is an open end cap and is connected to a first axial end **102a** of the filter media **102** in a fluid tight manner. In one aspect, the second end cap **106** is a closed end cap and is connected to a second axial end **102b** of the filter media **102** in a fluid tight manner. The first and second end caps **104**, **106** can, for instance, be formed and attached to the filter cartridge **100** by a polyurethane potting process. In one aspect, the end caps **104**, **106**, are generally annular or circular. The filter media **102** is also shown as being provided with an annular or circular shape with a hollow interior. Other shapes for the end caps **104**, **106** and filter media **102** are possible, for example, elliptical shapes, obround, oval or race-track shapes, and/or conical shapes.

In one aspect, the first end cap **104** has a multi-component construction including a support structure **108** and a seal member **110**, shown separately at FIG. **11**. In some examples, the support structure **108** is secured to the media **102** and the seal member **110** is secured to the support structure **108**. In some examples, the seal member **110** is secured to the media **102** and the support structure **108** is secured to the seal member **110**. In some examples, both the support structure **108** and the seal member **110** are secured to the media **102** and to each other. In some examples, the support structure **108** and seal member **110** are independently secured to the media or another structure without being directly secured together. In examples, the support structure **108** is formed from a material that has a higher hardness and/or stiffness in comparison to the seal member **110**. With respect to the materials described herein, materials with a higher hardness typically also happen to have a higher stiffness, and vice versa. In some examples, the material used for the support structure **108** has a lower coefficient of

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friction in comparison to the material used for the seal member **110**. In some examples, the material used for the support structure **108** and seal member **110** can be or include a metal material, a plastic material such as thermoplastic and thermoset materials, or elastomer/rubber type materials. Example materials are polyurethane, polyethylene, thermoplastic elastomers (TPE), natural and synthetic rubber materials, and acrylonitrile butadiene styrene (ABS) materials. In some examples, the material for the support structure **108** and the material for the seal member **110** are the same material or type of material. In some examples, the material for the support structure **108** and the seal member **110** are different types of materials. In some examples, the support structure **108** and seal member **110** are formed from different compositions or variations of the same material type. For example, the support structure **108** and seal member **110** can both be formed from polyurethane compositions such that the support structure **108** is relatively harder and/or stiffer than the seal member **110**. In another example, the seal member **110** is formed from a polyurethane material while the support structure **108** is formed from a polypropylene or ABS material. Many combinations of materials are possible, including any combination of the above-listed material types and examples.

As most easily seen at FIGS. **11** to **13**, the support structure **108** includes an annular base portion **108a** from which a plurality of radially flexible reinforcement members **108b** extend in an axial direction away from the axial end **102a** of the media **102**. As referenced at FIG. **13**, the reinforcement members **108a** extend at an angle **a1** from a line parallel to the longitudinal axis **X1**. In some examples, the angle **a1** may be between 0 and 60 degrees. In the example shown, the angle **a1** is about 30 degrees and can be characterized as being an oblique angle. In one aspect, the reinforcement members **108b** are circumferentially spaced apart so as to form gaps **108c** therebetween.

As shown, each of the reinforcement members **108b** includes an outer surface having portions **108d**, **108e**, **108f**, and **108g**. In one aspect, portion **108e** defines a distal axial end of the support structure **108** and of the entire filter cartridge **100** while portion **108g** defines a radial innermost portion of the support structure **108**. Portion **108f** tapers inwardly, from the portion **108e**, towards the longitudinal axis **X1** and portion **108g**, and also extends in a direction towards the end cap **106**. As most easily seen at FIG. **13**, the portions **108h**, **108i** and an upper surface **108j** of the base portion **108a** define a recess or cavity **108k** within which the seal member **110** is at least partially disposed. With such a configuration, the reinforcement members **108b** provide lateral support to the seal member **110**, as discussed in more detail below.

In one aspect, the reinforcement members **108b** are flexible in a radial direction such that a radial force applied to the second portions **108d** will cause the reinforcement members **108b** to deflect radially in the direction of the applied force. In one aspect, the distal end or portion **108e** of each reinforcement member **108b** is rounded from one lateral side to the opposite lateral side towards the adjacent gaps **108c**. As discussed later, this feature, along with tapered portion **108f** enable for easier insertion of the filter cartridge into the housing **200**.

In one aspect, the seal member **110** is formed with a radial seal surface **110a**. In the example shown, the radial seal surface **110a** faces in a radially inward direction. In one aspect, the radial seal surface **110a** is configured as a seal surface which forms a seal against a portion of the outlet tube assembly **205**. In one aspect, the radial seal surface

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110a is located opposite the reinforcement members 108b such that at least a portion of the radial seal surface 110a is coplanar with at least a portion of the reinforcement members 108b along a plane that is orthogonal to the longitudinal axis X1. Accordingly, any radial outward force exerted on the radial seal surface 110a will generate a radial outward force on the reinforcement members 108b. As explained below, such a force will cause the reinforcement members 108b to deflect radially outwardly. Reference is also made to FIG. 8A in which an alternatively constructed embodiment is illustrated wherein the reinforcement members 108b extend in an axial direction rather than being slanted in an inward direction. As with the example depicted in FIG. 8, the filter cartridge 100 illustrated at FIG. 8A has a seal surface that is round in the relaxed state and can be fit over and sealed to a polygonal-shaped outlet tube. Accordingly, the below description of the sealing interaction of the filter cartridge is generally applicable to the embodiment shown at FIG. 8A with the exception that the reinforcement members 108b are not deflected by the outlet tube in necessarily the same manner. In some examples, the cartridges shown at FIGS. 8 and 8A can be configured to form a seal with the polygonal shaped outlet tube without requiring direct interaction or contact between the outlet tube assembly 205 and the reinforcement members 108b. It is also noted that the reinforcement members 108b in FIGS. 8 and 8A are arranged with flat distal ends and sides that themselves form a generally polygonal shape or, characterized another way, extend along a polygonal shaped perimeter line or projection.

In one aspect, and as most easily seen at FIGS. 4 and 5, the outlet tube assembly 205 is provided with a number of structures configured for interaction with the first end cap 104 of the filter cartridge 100. For example, the outlet tube assembly 205 is provided with an outer radial seal surface 205a extending to a slanted distal end 205b. As shown, the seal surface 205a has a generally polygonal shape with a plurality of flat sides 205c joined together by radiused or curved portions 205d. By use of the term 'generally polygonal,' it is meant to include true polygonal shapes that have flat sides with vertices and similar shapes with flat sides having rounded corners, and is meant to include shapes with flat sides having the same length and shapes with flat sides having different lengths. By use of the term 'generally flat sides', it is meant to include surfaces that are entirely flat and surfaces which are generally flat with some degree of curvature, such as curvatures resulting from manufacturing processes. The term encompasses surfaces that maintain their generally flat characteristic while potentially having minor surface variations or features, provided such variations do not substantially alter the overall flat nature of the surface. Further, curved surfaces having a sagitta-to-chord ratio of an arc or curved surface of 10 percent or less would also be considered to be within the meaning of the term. Further, the term would also include bodies or members wherein a straight line can be drawn, within a thickness of the member, from one end of the member to the opposite end of the member. In the example shown, twelve flat sides 205c are provided but more or fewer may be used, as is discussed herein. The outlet tube is also shown as being provided with a plurality of slanted first projections 205f that extend radially inwardly and axially away from the distal end 205b. The slanted first projections 205f and distal end 205b slant in the same direction and slope toward the air outlet end 202c in a radial direction that is outward from the longitudinal axis X1. In the example shown, the slanted first projections 205f are equal in number to the flat sides 205c

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and are centered along the width of the flat sides 205c. Other configurations are possible. The outlet tube assembly 205 is further shown as including a plurality of second projections 205e that also project axially from the distal end 205b. In one aspect, the second projections 205e are equal in number and are radially aligned with the curved portions 205d. Other arrangements are possible. For example, the outlet tube could be provided with flat sides 205c in which one or more of the flat sides 205c have a length that is different from one or more of the other flat sides 205c. In such examples, the seal surface 205a can be rotationally symmetric with multiple folds of rotational symmetry or could be configured such that only single fold rotational symmetry exists.

Installation of the filter cartridge 100 onto the outlet tube assembly 205 will now be described. In a first step, and as depicted at FIG. 14, the filter cartridge 100 is axially displaced towards the outlet tube assembly 205 which will result in the end surface or portions 108e and/or 108f of the support structure 108 to contact either the axial ends of the first or second projections 205e, 205f, depending on the rotational orientation of the filter cartridge 100. In the former case, the rounded edges of the reinforcement members 108b will help the user to rotationally clock the filter cartridge 100 until the projections 205e are received into the gaps 108c between the reinforcement members 108b. Once this rotational position is achieved, as shown at FIG. 14, the slanted axial faces associated with portions 108f will contact the slanted axial faces associated with the projections 205f and distal end 205b. In the configuration shown, the slanted axial faces associated with the projections 205f and distal end 205b are generally parallel to the slanted axial faces associated with portions 108f. With such a configuration, and upon further displacement of the filter cartridge 100 towards the outlet tube assembly 205, these axial faces will force the reinforcement members 108b in a radially outward direction such that the seal member is eventually stretched over the outer radial seal surface 205a of the outlet tube, as schematically shown at FIG. 15. It is noted that the positions of the reinforcement members 108b in FIGS. 3 and 15 are not actually depicted as being flexed outwardly in the installed position and that a person skilled in the art would readily understand the resulting flexed out position of the reinforcement members 108b based on the description herein and the other figures in the application.

In one aspect, the seal member is sufficiently compressible such that the radial seal surface 110a, which is cylindrically shaped in the relaxed state, conforms to the generally polygonal shape of the outer radial seal surface 205a when the filter cartridge 100 is fully installed onto the outlet tube assembly 205. When so installed, the reinforcement members 108b provide a resistant spring force in a radial inward direction against the seal member 110 which advantageously aids in ensuring a seal is fully formed between the radial seal surface 110a and the seal surface 205a of the outlet tube assembly 205. With such a construction, it is possible to provide an end cap having a seal member 110 with a radial seal surface 110a that has a first shape in a natural or relaxed state and that has a second shape in the installed condition, wherein the first shape is different from the second shape. In the example shown, the first shape is a cylindrical or circular shape while the second shape is a generally polygonal shape. It is further noted that the first shape of the seal surface 110a has a dimension that is less than a corresponding dimension of the seal surface 205a while the second shape of the seal surface 110a has a dimension that is equal to a corresponding dimension of the seal surface 205a.

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Referring to FIGS. 16 and 17, schematic depictions of the outlet tube assembly 205 and seal surface 205a are depicted in which different configurations of the support structure 108 and seal member 110 are shown. FIG. 16 generally corresponds to the example shown at FIGS. 1-15 and illustrates that the reinforcement members 108b are aligned with the flat sides 205c such that the portions of the seal member 110 that are unsupported by the reinforcement members 108b are aligned with the corners or curved portions 205d. In such a configuration, the seal member 110 can generally compress and/or stretch to accommodate and seal against the corners 205d without requiring an additional support for force from a reinforcement member 108b. FIG. 17 shows an alternative arrangement in which the reinforcement members 108b extend across the corners 205d and in which the unsupported portions of the seal member 110 extend across the flat sides 205c. In some arrangements, the seal member 110 can still adequately form a seal against the flat sides 205c without the aid of any additional supporting structures. However, in some configurations, radially inward projections 202e associated with the housing can be provided to contact and exert a compressive force onto the seal member 110.

Referring to FIGS. 18 to 21, it can be seen that the same filter cartridge 100 can advantageously also be installed onto other types of outlet tubes that do not necessarily have a polygonal shape. In the example shown at FIGS. 18 to 20, an outlet tube assembly 205 is provided with a seal surface 205a that has a smooth frustoconical shape that tapers in a direction away from the end cap 104. In the example shown at FIG. 21, the outlet tube assembly 205 is shown with a seal surface that has a distal end that initially slants in a direction away from the longitudinal axis and then slants towards the longitudinal axis. In such a configuration, the seal surface 205a has a maximum diameter somewhere between the axial ends of the seal surface 205a such that the seal surface 205a has one or more undercut portions or surfaces relative to the maximum diameter. In one aspect, the oblique angle(s) presented by the outlet tube seal surface 205a can result in advantageous force directions. When the seal surface 205a is angled as such, a force to the seal surface 110a is applied at a non-orthogonal angle. Accordingly, there is not only a radial force applied to the seal member, but also an axial force applied to the element which pushes the element further onto the outlet tube assembly 205. In some applications, this arrangement can be beneficial to securing the filter cartridge end cap 104 to the outlet tube assembly 205, and/or holding it in place during vibration. Further, such an arrangement could eliminate the need for compression bumpers on the closed end and therefore result in space savings. The angled inlet tube surface 205a shown at FIGS. 18-20 can also provide for these effects. In other examples, the filter cartridge 100 could also be installed over a seal surface having a combination of the above-noted features, for example, a polygonal shape that is also tapered or provided with an undercut surface.

FIGS. 22 to 25 show variations in geometry of the end cap 104 that are usable with filter cartridge 100. FIGS. 22 and 23 show an example in which the reinforcement members 108b are provided with a more aggressive taper towards the longitudinal axis X1. FIG. 23 also shows that the radial innermost portion of the seal surface 110a and the reinforcement members 108b, in the relaxed state, extend to about the same diameter, which is also the case for the filter cartridge 100 shown at FIG. 13. FIG. 24 shows an example in which the seal member 110 is configured such that the seal surface 110a is radially inward of the innermost portion of the reinforcement members 108b. FIG. 25 shows an example in

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which a rib portion 110b is provided as part of the seal surface 110a, wherein the rib portion 110b is radially inward of the innermost portion of the reinforcement members 108b and wherein the remaining portion has a radial inward dimension corresponding to the radial inward dimension of the reinforcement members 108b. By providing additional seal material radially inward of the innermost radial portion of the reinforcement members 108b, an enhanced seal can be achieved with some outlet tube configurations. In one aspect, the surface 110a can be characterized as being undercut with respect to the rib portion 110b as the innermost diameter of the surface 110a is greater than that of the rib portion 110b.

Referring to FIG. 26, a schematic illustration of a process 1000 for forming any of the filter cartridge end cap arrangements disclosed herein is presented. As shown, at a stage 2010, the seal member 110 is formed in a first mold 300 having a radially outward facing seal forming surface 302, having a first diameter, that defines the seal surface 110a. In some examples, the seal member 110 is formed from a polyurethane material that is relatively soft in comparison to the material used for the support structure 108, which can also be a polyurethane material. Other materials and combinations of materials may be provided for the support structure 108 and seal member 110. Once the seal member 110 has been sufficiently cured, the seal member can be placed into a second mold 310 at a stage 2020. The second mold 310 has a radially outward facing surface 312 that has a second diameter that is greater than the first diameter. Accordingly, in order to place the seal member 110 within the second mold 310, the seal member 110 must be stretched over the surface 312. Once the seal member is so installed, the material for the support structure 108 can be poured into the second mold 310 to surround the seal member 110 at a stage 2030. The second mold 310 is formed with various ribs and other structures to ensure that the reinforcement members 108b are formed to have the desired shape shown in any of the previous drawings. In some examples, stage 2030 also includes placing the end of the media 102 within the mold such that the material for the support structure 108 bonds and seals to the media 102, thereby resulting in the fully formed end cap 104 being bonded to the media 102. In some examples, the end cap 104 is fully formed and then later attached to the media 102 by any means known in the art. Once the material for the support structure 108 has cured, the support structure and seal member 110 are sufficiently bonded together such that the seal surface 110a of the seal member is maintained at the diameter defined by the surface 312 after removal from the mold 312. Accordingly, the seal member 110 has a degree of resulting internal tension which advantageously aids in providing an additional sealing force against the outlet tube.

Example of FIGS. 27-48

Referring to FIGS. 27 to 48, aspects of an example filter cartridge 100 and related air cleaner assembly components are further illustrated. It is noted that FIGS. 27 to 30 show only a portion of the air cleaner housing assembly 200 for purposes of clarity and that the filter cartridge 100 would be completely enclosed within the housing assembly 200 in a similar manner to that shown at FIG. 1 in an actual installation. Further, the features of the air cleaner housing assembly 200 and an associated safety filter cartridge 150 described below in association with FIGS. 49 to 55B are likewise suitable for use with this example. The filter cartridge 100 of this example shares many features in

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common with the filter cartridge **100** shown in FIGS. **8** to **13** and differs primarily in that a differently configured open end cap **104** is provided, as discussed herein. As most easily viewed at FIGS. **27** to **34**, the filter cartridge **100** includes filter media **102** extending between a first end cap **104** and a second end cap **106**. In the particular example shown, the filter media **102** is pleated media arranged in a tubular manner to define a central opening **102c** around an axis X. Other media types and configurations are possible, for example, fluted media, depth media, non-pleated media, coiled media, and/or stacked media, etc.

In one aspect, the first end cap **104** is an open end cap and is connected to a first axial end **102a** of the filter media **102** in a fluid tight manner. In one aspect, the second end cap **106** is a closed end cap and is connected to a second axial end **102b** of the filter media **102** in a fluid tight manner. The first and second end caps **104**, **106** can, for instance, be formed and attached to the filter cartridge **100** by a polyurethane potting process. In one aspect, the end caps **104**, **106**, have generally annular or circular outer and innermost perimeters. The filter media **102** is also shown as being provided with an annular or circular shape with a hollow interior. Other shapes for the end caps **104**, **106** and filter media **102** are possible, for example, elliptical shapes, obround, oval or race-track shapes, polygonal, and/or conical or frustoconical shapes.

In one aspect, the first end cap **104** has a multi-component construction including a support structure **108** and an adjoining seal member **110**. In some examples, the support structure **108** is formed from a material that is harder and/or stiffer than a material forming the seal member **110**. In some examples, the support structure **108** is secured to the media **102** and the seal member **110** is secured to the support structure **108**. In some examples, the seal member **110** is secured to the media **102** and the support structure **108** is secured to the seal member **110**. In some examples, both the support structure **108** and the seal member **110** are secured to the media **102** and to each other. In examples, the support structure **108** is formed from a material that has a higher hardness and/or stiffness in comparison to the seal member **110**. With respect to the materials described herein, materials with a higher hardness typically also happen to have a higher stiffness, and vice versa. For reference, hardness refers to a material's resistance to permanent deformation or indentation while stiffness refers to a material's resistance to elastic deformation under load. In some examples, the material used for the support structure **108** has a lower coefficient of friction in comparison to the material used for the seal member **110**. According to preferred embodiments, the support structures **108** are made of a relatively hard material. Preferably, the support structures comprise materials which are from medium hard to hard and up to extra hard. To express and measure the hardness of a material, the known shore-A scale or shore-D scale can be used. In embodiments, the support structures have a hardness either measured on a shore-A scale between 60 and 100, preferably between 70 and 100, more preferably between 80 and 100 or, alternatively, when measured on a shore-D scale between 0 and 100, preferably between 15 and 100, more preferably between 30 and 100. In comparison, in examples, the seal member **110** can be provided with a hardness of less than 20 on the shore-A scale.

As most easily seen at FIGS. **31** to **43**, the seal member **110** extends between a base end **110h** and a distal end **110i** and defines a radially inward facing surface **110a** and an opposite radially outward facing surface **110e** that each have a generally polygonal cross-sectional shape in a relaxed state

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of the seal member **110**. In the example presented, the inward facing surface **110a** has a plurality of respective flat sides **110c** joined together by radiused or curved corner portions **110d**, while the outward facing surface **110e** has a plurality of respective flat sides **110f** joined together by radiused or curved corner portions **110g**. As shown, each of the surfaces **110a**, **110e** has eight flat sides and eight corner portions. Other numbers of flat sides and corner portions are possible, such as three, four, five, and six sides, as schematically depicted at FIGS. **88** to **91**. Additional sides are also possible, such as seven and nine or more flat sides and corner portions. In the example shown, the radially inward facing surface **110a** defines a seal surface for forming a seal against a correspondingly shaped outlet tube assembly **205**. In the example shown, the outlet tube assembly **205** includes a first part **208** and adjoining second part **210**. As shown, each of the first and second parts **208**, **210** define a central opening for allowing air to pass therethrough. The second part **210** is shown as having a sidewall **212** having a radially outward facing seal surface **212a** with flat sides **212b** and corners **212c** that form a generally polygonal cross-sectional shape corresponding to the shape of the seal surface **110a**. The outlet tube assembly **205** is further illustrated at FIGS. **44** to **48**. Other arrangements are possible. For example, the radially outward facing surface **110e** can instead or additionally define a seal surface for sealing against the outlet tube assembly **205**. It is noted that the radially inward facing side of the sidewall **212** that is opposite the seal surface **212a** may be used as a generally polygonal shaped seal surface for a secondary filter having a correspondingly shaped polygonal seal member. Such a use is shown and described in conjunction with the example shown at FIGS. **49** to **55B** which is fully applicable to this example as well.

As shown, the support structure **108** includes an annular base portion **108a** from which a plurality of circumferentially arranged reinforcement members **109** extend in an axial direction away from the axial end **102a** of the media **102** to distal ends **109c**. The particular size, shape, and number of the reinforcement members **109** can be a matter of design choice. As shown, the reinforcement members **109** present generally rectangular inner and outer radial side surfaces and have a generally trapezoidal cross-sectional shape such that the ends of adjacent reinforcement members **109** are parallel to each other. Further, as can be seen at FIG. **35**, the reinforcement members **109** can be provided with a thickness that is about that same as that of the seal member **110**, for example a thickness that is within 25 percent of the thickness of the seal member **110**. The support structure **108** may also be referred to as a seal support arrangement **108**. Reinforcement members **109** may also be characterized as being extension members **109** or support members **109**. With such an arrangement, gaps **109f** are formed between adjacent side edges or surfaces **109d**, **109e** of the reinforcement members **109** such that the end cap **104** has an overall cross-sectional thickness along the flat sides that is greater than an overall cross-sectional thickness at the corner portions. In one aspect, each of the reinforcement members **109** presents a radially inward facing surface **109a** and an oppositely disposed radially outward facing surface **109b**. In the example shown, the surfaces **109a**, **109b** are generally parallel to the longitudinal axis X of the filter cartridge **100**. Other angles are possible, as already discussed with respect to the first disclosed example filter cartridge **100**. In one aspect, each of the reinforcement members **109** has a generally trapezoidal cross-sectional shape such that the adjacent side or edge surfaces **109e**, **109f** of the reinforcement members **109** are generally parallel to each other. Other

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shapes are possible. In one aspect, the reinforcement members 109 have a degree of flexibility in the radial direction but are rigid in the axial direction and in a lateral direction extending between the edges 109d, 109e.

In one aspect, and as most easily viewed at FIG. 35, the seal member 110 and support structure 108 are arranged with respect to each other such that the seal member flat sides 110f of the outer surface 110e are immediately adjacent to and touching the radially inward facing surfaces 109a of the reinforcement members 109 and such that the gaps 109f defined between the reinforcement members 109 are radially aligned with the curved corner portions 110d, 110g of the seal member 110. In some examples, the support structure 108 is overmolded onto the seal member 110 in a manner similar to that described with respect to FIG. 26 and process 1000, which results in the reinforcement members 109 being bonded or otherwise adhered to the flat sides 110f of the seal member. With such an arrangement, the reinforcement members 109 can provide radial support to the seal member 110 to provide enhanced sealing between the flat sides 110c and the flat sides 112b of the outlet tube assembly 205. As the corner portions 110d, 110g of the seal member 110 are not supported by the reinforcement members 109, the corner portions 110d, 110g of the seal member 110 can be characterized as being unsupported by and/or unattached to the reinforcement members. This configuration provides for a particular advantage in that the seal member 110, at the locations of the corner portions 110d, 110g, is allowed to stretch in a circumferential direction between the reinforcement members 109 and over the corner portions 112c of the outlet tube assembly 205 which enhances sealing at the corner portions. Further, the bonding or adherence along the flat sides provides for anchor points between which the seal member 110 can stretch, for example at the corner portions where there are gaps between the reinforcement members 109. Accordingly, the seal member 110 can be characterized as being constrained from stretching in a circumferential direction along the flat sides by the bonded reinforcement members 109. In some examples, the reinforcement members 109 can be fully or partially embedded within the material of the seal member 110, along with the base portion 108a, at a location radially outward or inward of the seal surface. In some examples, the reinforcement members 109 can be covered by a thin layer of the seal member 110, such as an ultrathin urethane such that the reinforcement members 109 are generally external to the main body of the seal member 110 but are not exposed. In some examples, the reinforcement members 109 can be spaced radially from the surface of the seal member 110 such that a radial gap exists between the reinforcement members 109 and the seal member 110. In such an arrangement, the reinforcement members 109 could still radially support the seal member 110 once the seal member 110 has expanded over the outlet tube of the air cleaner housing.

To aid in orientation of the filter cartridge 100 with respect to the second part 210 of the outlet tube assembly 205, the second part 210 can be provided with a sidewall 214 having a complementary shape for receiving the reinforcement members 109. In one aspect, the second part 210 may be referred to as a receiver plate 210. As detailed at FIG. 48, the sidewall 214 defines a radially inward facing radial surface having a plurality of spaced apart flat sections 214a interrupted by inwardly extending protrusions 214b having side edges 214c, 214d. Taken together, the surfaces 214a, 214c, and 214d form a recess areas 216 that can receive the reinforcement members 109 such that surface 109b faces surface 214a and edge surfaces 109d, 109e face surfaces

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214c, 214d, and such that the protrusions 214b are received within gaps 109f between the reinforcement members 109. As distal ends 109c of the reinforcement members 109 define the distal end of the support structure 108 and of the filter cartridge 100 itself, the protrusions 214b can act as guide surfaces during insertion of the filter cartridge 100. For example, if the filter cartridge 100 is initially rotated such that the polygonal shapes of the seal member 110 and sidewall 212 are misaligned, the ends 109c will contact the top surfaces of the protrusion 214b. As the reinforcement members 109 and sidewall 214 are formed of relatively hard material (e.g., ABS plastic), relatively little friction exists between these components. Accordingly, once contact is made between these components, an operator can easily rotate the cartridge 100 until the gaps 109f are aligned with the protrusions 214b, at which point the filter cartridge can drop fully into place. Once this position is achieved, the protrusions 214b are adjacent the corner portions 110d of the seal member 110 but do not prevent the seal material from stretching or otherwise extending about the corners 112c of the sidewall 212.

With reference to FIG. 47, it can be seen that the first part 208 of the outlet tube assembly 205 is provided with a U-shaped, annular wall structure 209 defining a cavity 209a, while the second part 210 is provided with a single annular wall structure 217 that is received within the cavity 209a such that the housing wall 202 is sandwiched between the first and second parts 208, 210. In one example, the wall structures 209, 217 form a tongue and groove weld joint formed by a spin welding process. Spin welding is a friction welding process used to join two plastic parts. It involves rotating one part against another at high speed, generating heat through friction. The heat softens the plastic surfaces, allowing them to fuse together as they cool, forming a strong bond. Such a process allows for modularity of parts in that different second parts 210 may be used with the same air cleaner housing and first part 208 without requiring modification thereof. Such a construction also better supports the second part 210 and related seal surfaces against moment forces exerted on the outlet tube first part 208. Where the outlet tube assembly 205 is constructed as a single component, as is shown with the first disclosed embodiment, forces exerted on the exterior of the outlet tube can cause the seal surfaces to rotate and compromise the seal between the outlet tube and the filter cartridge. As the parts 208 and 210 are joined together with a fused or welded attachment, the potential for undesirable leakage is minimized. Further, providing the wall structures 208 and 210 in separate parts allows for greater ease of manufacturing, such as molding, while also advantageously providing for a degree of modularity to accommodate differently sized and/or shaped housings and outlets.

Example of FIGS. 49 to 55B

Referring to FIGS. 49 to 55B an air cleaner assembly 10 is disclosed that is a more detailed version of that presented at FIGS. 27 to 48 and that includes an alternative configuration for the first and second parts 208, 210. The air cleaner assembly 10 associated with FIGS. 49 to 55B also presents a safety or secondary filter cartridge 150. The features associated with the example presented in FIGS. 49 to 55B should be considered overlapping with and compatible with those presented at FIGS. 27 to 48. Accordingly, where components or features are in common, the above-provided description for FIGS. 27 to 48 is fully applicable for the example shown at FIGS. 49 to 55B and need not be repeated

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herein. Rather, this section will focus on the additional or different features presented at FIGS. 49 to 55B. Further, the description for the example of FIGS. 49 to 55B is also fully applicable to the example presented at FIGS. 27 to 48.

In one aspect, and as shown at FIGS. 49 and 50, the air cleaner assembly 10 includes the above-described filter cartridge 100 and a safety filter cartridge 150 that are housed within an interior volume of air cleaner housing 200. The air cleaner housing 200 is shown as being provided with a main housing body 202 and a cover 204. A mounting structure 201 for mounting the main housing body 202 to another structure is also shown. The air cleaner housing 200 also includes an outlet tube assembly 205 having joined together first and second parts 208, 210. In the example shown, the outlet tube assembly 205 is rotatable with respect to the main housing body 202. The air cleaner assembly 10 is also provided with an air inlet 202d. In some examples, assembly 205 can be configured as an air inlet while structure 202d can be configured as an air outlet.

As most easily viewed at FIGS. 53 to 55B, the safety filter cartridge 150 is provided with filter media 152 extending between a first end cap 153 and a second end cap 154. In the particular example shown, the filter media 152 is non-pleated media arranged in a tubular manner to define a central opening. As detailed in a later section, many media types and configurations are possible for use with all of the disclosed example filter cartridges herein, for example, fluted media, z-media, depth media, non-pleated media, coiled media, and/or stacked media, etc. In one aspect, the second end cap 154 is an open end cap and presents a radially outward facing seal surface defined by a plurality of flat sides 156a and adjoining rounded corner portions 156b. Accordingly, the seal surface of the second end cap 154 may be characterized as having a generally polygonal shape. In the example shown, eight flat sides 156a and eight corner portions 156b are provided that are aligned with the flat sides 212e and corner portions 212f of the second part 210. As most easily seen at FIG. 54, the end cap 154 is provided with a seal member 156 and a support member 158. As shown, the seal member 156 is provided with a plurality of slots 156c located opposite corner portions 156b and a plurality of slots 156d located opposite flat sides 156a. The support member 158 is provided with corresponding tabs or protrusions 158c, 158d that are received into the slots 156c, 156d. It is noted that the seal member 156 can be overmolded onto the support member 158 such that the slots 156c, 156d are simply the areas at which the tabs or protrusions 158c, 158d are present within the material of the seal member 156. The seal member 156 can also be a separately formed member and later joined to the support member 158. The support member 158 is also shown as having a cylindrically shaped sidewall 158a and top flange member 158b which support the seal member 156 on those sides. As can be seen at FIGS. 55A and 55b, the tabs or protrusions 158c aligned with the corner portions 156b are larger than the tabs or protrusions 158d at the flat sides 156a. With such a construction, the tabs or protrusions 158c, 158d provide radial support to the flat sides 156a and corner portions 156b to aid in ensuring that an adequate seal is formed against corresponding flat sides 212d and corner portions 212f of a seal surface 212d of the wall structure. The cross-sectional view shown at FIG. 51B provides an illustration showing these features in an installed condition with respect to the second part 210 and air cleaner housing 202.

Examples of FIGS. 56A to 56D and 57

With reference to FIGS. 56A to 56B, flat schematic illustrations of the seal member 110 and support structure

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108 are shown to illustrate further concepts associated with the disclosure. As shown at FIG. 56A, it can be seen that the reinforcement members 109 may all be provided with the same length L1 such that the resulting gaps therebetween all have the same length G1 and such that the distal ends of the reinforcement members 109 extend beyond the distal end of the seal member. FIG. 56B shows that alternatives to this configuration are possible. For example, FIG. 56B illustrates that the reinforcement members 109 may be provided with different or dissimilar lengths L1, L2, may form gaps with different lengths G1, G2, and may be provided with different or dissimilar heights, for example a height H2 that is greater than a height H1 of the seal member, a height H3 that is the same as height H1, and a height H4 that is less than the height H1. In examples, the lengths of the support members 109 are a majority of the length of the corresponding flat sides 110f. In some examples, the combined lengths of all of the support members 109 are more than 75 percent of the combined lengths of all of the flat sides 110f. FIG. 56C illustrates that the wall structure 212 may be provided with flat surfaces 212b having the same width W1 and corner portions 212c having the same width C1. FIG. 56D illustrates that alternatives for the wall structure 212 are possible. For example, the wall structure 212 may be provided with flat surfaces 212b having different or dissimilar widths W1, W2 and may be provided with corner portions 212c having different or dissimilar widths C1, C2. In one advantageous aspect of the disclosed design, a wall structure 212 having a radial surface 212a with dissimilarly sized corner portions 212c can be readily accommodated by the seal member 110 as the corner portions of the seal member are unconstrained by the reinforcement members 109 and can therefore stretch over variously sized corner portions 212c of the sidewall 212. In some examples, and as noted elsewhere, the seal surface 110a can also be provided with different or dissimilar widths which may correspond to widths C1, C2. It is noted that all of the examples depicted in FIGS. 56A to 56D are usable with any of the disclosed arrangements herein, for example, with the configurations shown at FIGS. 23 to 48 and FIGS. 49 to 55.

Referring to FIG. 57, it is illustrated that the reinforcement member 109 may alternatively be provided with an arc-shaped cross-sectional profile such that at least a center section of the reinforcement members 109 is in contact with the seal member 110. Such a configuration can be advantageous in creating a spring effect onto the seal member to effectuate a seal of the seal member at the central location of the flat sides where sealing may be relatively difficult to ensure. In such arrangements, the outside corner portions of the reinforcement members 109 can use the wall structure 214 as support for creating the spring force against the seal member 110. In some examples, the reinforcement members 109 are curved such that the side edges are out of contact with the seal member 110. In some examples, the seal member 110 is adhered with the reinforcement members 109 to a degree such that the seal member 110 remains in contact with the curved length of the reinforcement members 109. As the depicted curvature is relatively minor (e.g., a straight line can extend within the thickness of each member 109 between its ends), the reinforcement members 109 may be characterized as being generally flat, in accordance with the above, or as having a curved or slightly curved surface.

Examples of FIGS. 58A to 58E

Referring to FIGS. 58A to 58E, alternative arrangement configurations for the sidewall 214 of the second part 210 are presented, as described below.

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FIG. 58A presents an example showing that the recesses 216 defined by the sidewall 214 can have different shapes and sizes instead of being identical or matching, as is the case with the example shown at FIG. 48. As shown, the sidewall 214 defines eight recesses 216 in which two oppositely located recesses 216b have a larger or deeper shape in comparison to the remaining recesses 216a. The recesses 216b are also defined by a curved surface 214a rather than the flat surface 214a defining the recesses 216a. In the disclosed example, the sidewall 216 and second part 210 have two-fold or mirror symmetry. However, other configurations are possible. For example, the sidewall 214 could be configured with recesses 216b such that only a single fold symmetry results. For example, the sidewall 214 could be provided with a single different recess 216b or multiple recesses 216b arranged in an asymmetrical shape. Further, more than two recesses 216b may be provided in either a symmetrical or asymmetrical arrangement. It is also noted that the sidewall 214 could be formed to define only recesses 216b without any recesses 216a being present. In one aspect, by providing differently shaped recesses 216, an indexing feature is provided which ensures that only an appropriate filter cartridge 100 having a similarly or complementarily shaped support structure 108 and/or seal member 110 can be received by the second part 210 in a desired rotational orientation.

FIG. 58B presents an example similar to that shown in FIG. 58A but in which recesses 216b are each provided with a plurality of radial inward extensions 214e extending from surface 214a. In the example shown, three extensions 214e are provided for each recess 216b. More or fewer extensions 214e may be provided. In one aspect, the center extension 214e has a longer length in comparison to the two outermost extensions 214e. Other arrangements are possible such as: all of the extensions 214e having the same length, none of the extensions having an equal length, extensions 214e being provided in differing arrangements on different recesses, and extensions 214e being provided on all of the recesses. Further, the recesses 216a may also be provided with such extensions 214e. With the inclusion of such extensions 214e, further indexing features are provided which ensure that only an appropriate filter cartridge 100 having a similarly or complementarily shaped support structure 108 and/or seal member 110 can be received by the second part 210 in a desired rotational orientation.

FIG. 58C shows an example in which the surface 214a is provided with a gap rather than being continuous. Such an arrangement can allow for receiving a protrusion associated with the support structure 108 to be received into the gap to provide a further indexing feature.

FIG. 58D shows an example in which the sidewall 214 is provided without surfaces 214b and 214c such that no recess 216 is formed with the sidewall 214a. With such an arrangement, the surfaces 214a of the sidewall 214 simply abut the outward facing surfaces 109b of the reinforcement members 109 without any securement of the side edges of the reinforcement members. As a result, the indexing function of the sidewall 214 is provided solely by the surfaces 214a as the filter cartridge 100 must still be rotated to sufficiently align the surfaces 214a, 109b before the filter cartridge 100 will fully drop into place.

FIG. 58E shows a further example that is generally similar to that shown in FIG. 57, but in which two of the surfaces 214a are each provided with a single extension 214e extending in a radial inward direction to provide for a further indexing function. Other arrangements for the extension 214e are possible, as described above.

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It should be understood that the various individual features disclosed in FIG. 54 to FIG. 59 may be provided in different combinations than what is specifically shown in the drawings. For example, any of the disclosed designs may be provided with similar or differently shaped recesses 216, with or without radial extensions 214e, with or without gaps in the surface 214a, and with or without one or both of surfaces 214b, 214c, and any combination thereof.

Additional Examples of FIGS. 59 to 117

Referring now to FIGS. 59 to 87, additional air cleaner embodiments are presented illustrating the concept that the end cap 104 can be mounted to the outer circumferential side filter cartridge 100 rather than to the axial face of the media pack, as is shown in previous embodiments. In some examples, and as shown, the end cap 104 is mounted directly to the media pack outer perimeter 102e. Other arrangements are possible. For example, the filter cartridge 100 can be provided with an outer shell surrounding the media pack 102 and the end cap 104 can be mounted or formed as part of the shell.

The examples illustrated at FIGS. 59 to 87 also differ from the previously illustrated embodiments in that the filter media pack 102 is shown as being formed from fluted or z-media. This type of filter media can be constructed by combining a corrugated or fluted sheet with a flat facing sheet to create a series of parallel channels. In some examples, the manufacturing process involves forming a corrugated or fluted pattern in a first sheet of filter material through the application of heat and pressure between corrugating rollers. Adhesive beads are then applied along the peaks of the flutes on one side, after which a flat facing sheet is bonded to the fluted sheet to create enclosed channels. The combined sheets are then rolled or coiled into the desired shape, such as cylindrical or oval configurations. Many other processes are also possible for forming a media pack with fluted channels, as is described in PCT Publication WO2020/163756, the entirety of which is incorporated by reference herein. While the examples shown in FIGS. 60 to 87 illustrate a fluted media pack, it should be understood that the end cap 104 of these figures may be used with other types of media such as any of the media types shown and described in the WO '756 publication, including tubular pleated media of the type shown in association with FIGS. 1 to 59. It should also be understood that the end caps disclosed in association with FIGS. 1 to 59 may also be used with any of the media types shown and described in the WO '756 publication, including fluted filter media of the type shown in association with FIGS. 60 to 87. In one aspect, the media pack 102 shown in FIGS. 59-87 may be provided with a hollow space that is filled by a central core 102d, as is illustrated most clearly at FIG. 64. Media packs without a hollow space or central core can also be utilized without departing from the concepts presented herein.

Referring now to FIGS. 59-80, an air cleaner 10 is shown including a filter cartridge 100 and a housing assembly 200. The filter cartridge 100 comprises a filter media 102 formed of rolled fluted media and an end cap 104 mounted to the circumferential outer surface of the filter media 102. The filter cartridge 100 is provided with an end cap 104 that circumscribes at least a portion of the outer perimeter 102e of the media pack 102. A seal member 110, having a polygonal shape, is carried by the end cap 104 and is configured to form a seal between the filter cartridge 100 and the housing assembly 200 when installed. The end cap further includes a support structure 108 having circumfer-

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entially arranged reinforcement members 109 that support the seal member 110 in the same manner as previously described. However, in contrast to previously disclosed embodiments and as most easily seen at FIG. 64, the reinforcement members 109 circumscribe the media pack 102 such that they are located between the media pack axial ends 102a, 102b and extend in a direction towards the axial end 102b. In the example shown, the seal member 110 has twelve flat sides and associated corner portions while the support structure 108 includes twelve corresponding reinforcement members 109. Other numbers of flat sides and reinforcement members 109 are possible, as discussed herein.

With continued reference to FIG. 64, it can be seen that the configurations of the sidewalls 212 and 214 are also modified, in comparison to previous embodiments, in that the sidewall 212 is located on the main housing 202 while the sidewall 214 remains associated with the cover 204. With such an arrangement, during installation, the filter cartridge 100 is received into the interior volume 202a of the main housing 202 such that the seal member 110 engages with and forms a seal against the outer surface of the sidewall 212 with the seal member stretching around the corners of the sidewall 212. Subsequently, once the cover 204 is installed, the sidewall 214 engages against the outer surfaces of the reinforcement members 109 to result in an enhanced seal, as previously described for the earlier-disclosed embodiments. As is most easily seen at FIGS. 75 and 76, the sidewall 214 is configured with flat portions separated by gaps in an arrangement similar to the example shown at FIG. 57. However, any of the arrangements disclosed at FIGS. 48 and 54 to 58 may be utilized for the sidewall 214. To additionally support the seal member 110 at the exposed corner portions or segments, the housing 202 can be provided with radially inward extending protrusions 213 that extend to and contact the seal member 110, as is illustrated at FIG. 64B. The protrusions 213 are further illustrated at FIGS. 77 to 80 and function in the same manner as already described for projections 202e illustrated at FIG. 17. As illustrated at FIG. 64A, it can be seen that the filter cartridge 100 is clamped or secured between the cover 204 and the main body 202 by virtue of contact between a first bottom facing side 204a of the cover 204 and an upward facing side 104a of the end cap 104 and by virtue of contact between a second bottom facing side 204b of the cover 204 and an upward facing side 202g of the main housing 202.

FIGS. 81-84 illustrate a variation of the filter cartridge design shown in FIGS. 59-80 that is generally similar but in which an additional latch arrangement 215 is provided for securing the cover 204 to the housing main body 202. The latch arrangement 215 allows for tool-free removal and installation of the cover 204. As shown, the cover 204 is provided with a plurality of radially deflectable latch members or arms 215a that carry a ramped latch member 215b presenting an upward facing surface 215c. The housing 202 is provided with an plurality of corresponding radial extensions or protrusions 202h against which the ramped latch members 215b can act against to expand the radial arms 215a outward during installation. Once the cover 204 is sufficiently depressed onto the housing 202, the ramped latch members 215b extend past the extensions 202h and the deflectable arms 215a can return towards their natural positions. In this position, downward facing surfaces 214h of the extensions 202h engage against the upward facing surfaces 215c of the ramped latch members 215b to retain the cover 204 onto the main housing 202. To remove the cover 204, an operator need only to deflect the arms 215a in

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a radial outward direction to disengage the surfaces 215c, 214h from each other. In the example shown, the latch arrangement is further provided with surrounding portions 215d on the cover 204 that surround the deflectable arms 215a and can act as a guard for the arms 215a and/or as handles.

With reference to FIGS. 85 to 87 and 116 to 117, further variations are illustrated. For example, FIGS. 85 to 86 depict another variation of the filter cartridge 100 having an oval or racetrack-shaped cross-section rather than circular. The oval configuration maintains the same basic components and mounting arrangement as described above, while providing an alternative form factor that may be advantageous for certain installation spaces. FIG. 87 shows yet another embodiment wherein the filter cartridge 100 is designed for use without a removable cover 204 and in which the support structure only partially covers an axial face of the seal member 110. FIGS. 116 to 117 show an example similar to that shown in FIGS. 60 to 80 but in which pleated media 102 with a hollow center is utilized instead of fluted type media.

FIGS. 88 to 91, as noted above, illustrate example arrangements in which the seal arrangement is provided with different numbers of flat sides and corner portions than previously depicted. As shown, FIG. 88 shows a three-sided seal member 110 with reinforcement members 109 being provided along each of the flat sides between adjacent the corner portions. As shown, FIG. 89 shows a four-sided seal member 110 with reinforcement members 109 being provided along each of the flat sides between adjacent the corner portions. As shown, FIG. 90 shows a five-sided seal member 110 with reinforcement members 109 being provided along each of the flat sides between adjacent the corner portions. As shown, FIG. 91 shows a six-sided seal member 110 with reinforcement members 109 being provided along each of the flat sides between adjacent the corner portions. The number of flat sides and/or corner portions can be a matter of design choice.

FIGS. 92 to 93 disclose filter cartridge examples and aspects usable with any of the filter cartridges disclosed herein, wherein the seal member 110 has at least one flat side that is longer than at least one other side. Such arrangements may be referred to as forming generally irregular polygonal shapes. In the particular example shown, the seal member 110 has eight flat sides in which seven of the flat sides 110c-1 have the same length and in which one of the flat sides 110c-2 has a length that is longer than that of the flat sides 110c-1. The reinforcement member 109 extending along the flat side 110c-2 is also longer in comparison to the remaining reinforcement members 109. With such an arrangement, single fold symmetry results in which the filter cartridge 100 can be received by the housing in only a single rotational orientation. Many other combinations are possible. For example, every other flat side 110c could have the same length while the remaining flat sides 110c are provided with a longer or shorter length such that half of the flat sides have one length and the other half have a second length that is different than the first length. Such an arrangement would provide for rotational symmetry on an order that is half the number of flat sides. In some example configurations, none of the sides have an equal length to any other side. FIG. 94 shows yet another example in which a six-sided arrangement is provided in which two flat sides 110c-1 have a first length, one flat side 110c-2 has a second length, two flat sides 110c-3 have a third length, and one flat side 110c-4 has a fourth length, wherein each of the first, second, third, and fourth lengths are different from each other. With the above-described arrangements, in which varying lengths are pres-

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ent, an advantage can result in ensuring that the filter cartridge **100** is oriented in a desired rotational position before being allowed to drop fully into place and into sealing engagement with the seal part **212** of the housing.

FIGS. **95** to **113** show further examples illustrating different construction and arrangement approaches that may be used with any of the filter cartridges disclosed herein. FIG. **95** shows an example end cap **104** for a filter cartridge **100** and illustrates that the seal member **110** may be molded directly onto the media pack **102**. FIG. **95** further shows that the support structure **108** may be either overmolded onto the seal member **110** and media pack **102** or may be a separately formed component that is placed over the seal member **110**. In the latter case, the support arrangement **108** can be secured to the seal member **110** via various means, such as with an adhesive. Alternatively, the support arrangement **108** can be attached with a friction or press fit onto the seal member **110** such that the support arrangement **108** could be re-used on another filter cartridge **100**.

FIGS. **96** and **97**, similar to the example shown in FIG. **95**, show an end cap **104** for a filter cartridge **100** in which the seal member **100** is molded directly onto the media pack **102**. In contrast to the example of FIG. **95**, the support structure **108** in FIGS. **96** and **97** is shown as being formed without a base part such that the support structure **108** is formed only by the reinforcement members **109**. The reinforcement members **109** also have a length that is a majority of the length of the flat sides **110f**. In such an arrangement, the reinforcement members can be either molded or otherwise formed and adhered onto the seal member **100**.

FIGS. **98** and **99** also show a seal member **110** that is molded directly onto the media pack and having a separately formed reinforcement structure **108**. However, in contrast to the examples shown at FIGS. **95** to **97**, the support structure **108** in FIGS. **98** and **99** is configured as an apertured outer support tube that extends the length of the media pack **102**, wherein the support tube has an end wall that supports the reinforcement member **109**. The reinforcement structure **108** may be permanently secured to the media pack **102** and/or seal member **110** or may be removable and reusable. Although usable with any of the embodiments disclosed herein, it is specifically noted that a similar construction could be applied to the filter cartridge **100** depicted in FIGS. **8** to **10** such that the radially flexible reinforcement members **108b** are formed as part of an outer support tube for the media pack **102**.

FIGS. **100** and **101** present yet another construction alternative. In the example shown, the support structure **108** is provided in two separate part **108-1** and **108-2**. Support structure part **108-1** is shown as being molded onto the media pack **102** and seal member **110** and forming two of the eight total reinforcement members **109**. Support structure part **108-2** is shown as being generally similar to the support tube configuration shown in FIGS. **98** and **99** but with apertures for receiving the two reinforcement members **109** formed by the support structure part **108-1**. Other combinations are possible in which the reinforcement members **109** are provided on the support structure parts **108-1** and **108-2** in different combinations. It is noted that FIGS. **100** and **101** show that reinforcement members **109** associated with the filter cartridge **100** have a larger size in comparison to those provided on the support structure **108**. Accordingly, the resulting construction has only two orders of rotational symmetry. In some examples, all reinforcement members **109** are the same such that rotational symmetry equals the number of polygonal sides that are provided. In some

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examples, one of the reinforcement members **109** is different from the others such that single fold symmetry results.

FIGS. **102** to **104** show embodiments in which the filter cartridge **100** is provided with an undercut or recessed portion **110j** below the seal surface **110a** for receiving a correspondingly shaped radial protrusion **212g** formed by the outlet tube assembly **205**. With such a construction, seal member **110** must initially expand around the radial protrusion **212g**. After sufficient axial displacement of the filter cartridge **100**, the seal surface **110a** moves past the radial protrusion **212g** and comes into sealing contact with seal surface **212a**. To accommodate this compression, the reinforcement members **109** can be configured to deflect radially outward during the insertion process. In the example shown at FIGS. **102** and **103**, the undercut portion **110j** extends continuously while in the example shown at FIG. **104**, the undercut portion **110j** is provided at the locations of the corner portions as the radial protrusions on the outlet tube assembly **205** are provided at only those locations.

FIGS. **105** to **113** illustrate across several examples that the number of reinforcement members **109** and associated gaps **108c** can be different than the number of flat sides **110c**, **110f** and corner portions **110d**, **110g** of the seal member **110**. In the example shown at FIG. **105**, one of the flat sides is supported by three separate, spaced apart reinforcement members **109-1**, **109-2**, **109-3** creating gaps **108c-2** and **108c-3** in addition to the already provided and described gaps **108c-1** at the corner portions. Accordingly, while all of the corner portions **110d**, **110g** are radially aligned with a gap **108c-1** between adjacent reinforcement members **109**, not all gaps, such as **108c-2**, **108c-3** are necessarily aligned with a corner portion **110d**, **110g**. In some examples, the gaps **108c-2** and/or **108c-3** can be configured to accept radial protrusions extending radially inwardly from the sidewall **214**. When so configured, the additionally-provided gaps **108c-2**, **108c-3** and corresponding housing protrusions can operate as a verification or safety feature such that only a filter cartridge **100** designed for use with the housing assembly **200** can be successfully installed into the housing as the housing protrusions would otherwise block the cartridge from being fully inserted into the housing assembly **200**.

FIGS. **106** and **107** show a variation of such an approach but in which each of the reinforcement members **109** is provided with a partial thickness groove **109g** for receiving a corresponding protrusion associated with the housing assembly **200**. As shown, the grooves **109g** are tapered or narrow from the distal end of the filter cartridge **100** in a direction towards the closed end cap **106** and are located on one side of the reinforcement members **109** such that the grooves **109g** are proximate one of the gaps **108c**. FIGS. **108** and **109** show a variation in which the grooves **109g** are centered within the reinforcement members **109**. In some examples, the grooves **109g** can be the full thickness of the reinforcement members **109** such that each reinforcement member **109** is divided into two separate members. In some examples, the grooves have axial or straight sidewalls, rather than being tapered. However, it is noted that the wider initial opening provided by the tapered grooves **109g** can help to serve as a lead-in surface during installation. In some examples, only some of the reinforcement members **109** are provided with a groove **109g**. In some examples, more than one groove **109g** can be provided for each reinforcement member **109**. In some examples, the grooves **109g** are provided at different locations on different reinforcement members **109**. In some examples, not all reinforcement members **109** are provided with a groove **109g**. However, it is noted that by providing a similarly shaped and located

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groove 109g on each reinforcement member 109, a particular rotational position of the filter cartridge 100 beyond what is already required by the shape of the polygonal seal surface is not required.

FIGS. 110 to 113 show additional examples similar to that shown in FIGS. 106 to 109, but in which radial protrusions 109h are provided instead of a groove 109g. With such a construction, the housing assembly 200 is configured with recesses or gaps to receive the radial protrusions 109h rather than being provided with protrusions. As shown at FIGS. 110 and 111, the radial protrusion 109h is provided at an offset location along the reinforcement member 109 such that each radial protrusion 109h is located proximate a gap 108c. FIGS. 112 and 113 show the radial protrusion 109h located in a centered location along the reinforcement member 109. It is noted that in the examples shown at FIGS. 110 to 113, the protrusions 109h are provided on every other reinforcement member rather than on every single reinforcement member 109. Accordingly, during installation, the filter cartridge 100 must be rotated into one of four acceptable radial positions rather than the eight that would be dictated by the octagonal shaped seal member 110 alone. It is noted that in some examples, one or more radial protrusions 109h and one or more grooves 109g may be provided for one or more reinforcement members 109. In such cases, the housing assembly 200 would also be provided with a combination of corresponding protrusions and recesses. In one aspect, the above-described grooves 109g and protrusions 109h may be more generically described as keying features, alignment features, and/or guide features.

FIGS. 114 to 115 show an additional illustrative example in which a filter cartridge 100 having a four-sided inwardly directed radial seal spaced from the media pack outer perimeter is provided. As shown, the seal member 110 is carried by a shell that forms the support structure 108 and reinforcement members 109, wherein the multiple spaced-apart reinforcement members 109 are provided along each side. In one aspect, the reinforcement members 109 have different lengths from each other such that the gaps 108c formed between them are provided at different distances from each other. As can be seen at FIG. 115, the housing assembly 200 is shown as including radially inward extending protrusions 202j which extend into the gaps 108c between the reinforcement members 108. Accordingly, and as explained above, only a filter cartridge 100 having a matching construction can be successfully installed into the housing body 202. In the example shown, the media 102 is fluted media.

Example Media Configurations, Generally

Any type of filter media can be used as the media pack 102, 152 for the above-disclosed filter cartridges 100, 150 as further described herein with relation to FIGS. 118-140. Further, the media type for filter cartridge 100 may be the same type or a different type of media than that for an associated filter cartridge 150 that may also be present in the air cleaner housing. For example, the filter cartridge 100 may have pleated or fluted type media while another filter cartridge 150 may be provided with fluted or pleated type media, respectively.

The media can be of a variety of types and configurations, and can be made from using a variety of materials. For example, pleated media arrangements can be used in cartridges according to the principles of the present disclosure, as discussed below.

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The principles are particularly well adapted for use in situations in which the media is quite deep in extension between the inlet and outlet ends of the cartridge, but alternatives are possible. Also, the principles are often used in cartridges having relatively large cross-dimension sizes. With such arrangements, alternate media types to pleated media will often be desired.

In this section, examples of some media arrangements that are usable with the techniques described herein are provided. It will be understood, however, that a variety of alternate media types can be used. The choice of media type is generally one of preference for: availability, function in a given situation of application, ease of manufacturability, etc., and the choice is not necessarily specifically related to the overall function of selected ones of various filter cartridge/air cleaner interaction features characterized herein.

a. Media Pack Arrangements Using Filter Media Having Media Ridges (Flutes) Secured to Facing Media

Fluted filter media (media having media ridges) can be used to provide fluid filter constructions in a variety of manners. One well known manner is characterized herein as a z-filter construction. The term “z-filter construction” as used herein, is meant to include (but not be limited) a type of filter construction in which individual ones of corrugated, folded or otherwise formed filter flutes are used to define (typically in combination with facing media) sets of longitudinal, typically parallel, inlet and outlet filter flutes for fluid flow through the media. Some examples of z-filter media are provided in U.S. Pat. Nos. 5,820,646; 5,772,883; 5,902,364; 5,792,247; 5,895,574; 6,210,469; 6,190,432; 6,350,291; 6,179,890; 6,235,195; Des. 399,944; Des. 428,128; Des. 396,098; Des. 398,046; and Des. 437,401; each of these cited references being incorporated herein by reference.

One type of z-filter media, utilizes two specific media components joined together, to form the media construction. The two components are: (1) a fluted (typically corrugated) media sheet or sheet section, and (2) a facing media sheet or sheet section. The facing media sheet is typically non-corrugated, however it can be corrugated, for example perpendicularly to the flute direction as described in PCT Publication WO 2005/077487, incorporated herein by reference.

The fluted media section and facing media section can comprise separate materials between one another. However, they can also be sections of the single media sheet folded to bring the facing media material into appropriate juxtaposition with the fluted media portion of the media. For example, a single continuous sheet of media formed with alternating fluted and flat sections along the length of the media can be folded upon itself in zig-zag fashion to form a fluted media configuration.

The fluted (typically corrugated) media sheet and the facing media sheet or sheet section together, are typically used to define media having parallel flutes. In some instances, the fluted sheet and facing sheet are separate and then secured together and are then coiled, as a media strip, to form a z-filter media construction. Such arrangements are described, for example, in U.S. Pat. Nos. 6,235,195 and 6,179,890, each of which is incorporated herein by reference. In certain other arrangements, some non-coiled sections or strips of fluted (typically corrugated) media secured to facing media, are stacked with one another, to create a

filter construction. An example of this is described in FIG. 11 of U.S. Pat. No. 5,820,646, incorporated herein by reference.

Herein, strips of material comprising fluted sheet (sheet of media with ridges) secured to corrugated sheet, which are then assembled into stacks to form media packs, are sometimes referred to as “single facer strips,” “single faced strips,” or as “single facer” or “single faced” media. The terms and variants thereof, are meant to refer to a fact that one face, i.e., a single face, of the fluted (typically corrugated) sheet is faced by the facing sheet, in each strip.

Typically, coiling of a strip of the fluted sheet/facing sheet (i.e., single facer) combination around itself, to create a coiled media pack, is conducted with the facing sheet directed outwardly. Some techniques for coiling are described in PCT Publication WO 2004/082795, which is incorporated herein by reference. The resulting coiled arrangement generally has, as the outer surface of the media pack, a portion of the facing sheet, as a result.

The term “corrugated” used herein to refer to structure in media, is often used to refer to a flute structure resulting from passing the media between two corrugation rollers, i.e., into a nip or bite between two rollers, each of which has surface features appropriate to cause corrugations in the resulting media. The term “corrugation” is however, not meant to be limited to such flutes, unless it is stated that they result from flutes that are by techniques involving passage of media into a bite between corrugation rollers. The term “corrugated” is meant to apply even if the media is further modified or deformed after corrugation, for example by the folding techniques described in PCT Publication WO 2004/007054 incorporated herein by reference.

Corrugated media is a specific form of fluted media. Fluted media is media which has individual flutes or ridges (for example formed by corrugating or folding) extending thereacross.

Serviceable filter element or filter cartridge configurations utilizing z-filter media are sometimes referred to as “straight through flow configurations” or by variants thereof. In general, in this context what is meant is that the serviceable filter elements or cartridges generally have an inlet flow end (or face) and an opposite exit flow end (or face), with flow entering and exiting the filter cartridge in generally the same straight through direction. The term “serviceable” in this context is meant to refer to a media containing filter cartridge that is periodically removed and replaced from a corresponding fluid (e.g. air) cleaner. In some instances, each of the inlet flow end (or face) and outlet flow end (or face) will be generally flat or planar, with the two parallel to one another. However, variations from this, for example non-planar faces, are possible.

A straight through flow configuration (especially for a coiled or stacked media pack) is, for example, in contrast to serviceable filter cartridges such as cylindrical pleated filter cartridges of the type shown in U.S. Pat. No. 6,039,778, incorporated herein by reference, in which the flow generally makes a substantial turn as it passes into and out of the media. That is, in a U.S. Pat. No. 6,039,778 filter, the flow enters the cylindrical filter cartridge through a cylindrical side, and then turns to exit through an open end of the media (in forward-flow systems). In a typical reverse-flow system, the flow enters the serviceable cylindrical cartridge through an open end of the media and then turns to exit through a side of the cylindrical filter media. An example of such a reverse-flow system is shown in U.S. Pat. No. 5,613,992, incorporated by reference herein.

The term “z-filter media construction” and variants thereof as used herein, without more, is meant to include, but not necessarily be limited to, any or all of: a web of corrugated or otherwise fluted media (media having media ridges) secured adjacent to (facing) media, whether the sheets are separate or part of a single web, with appropriate sealing (closure) to allow for definition of inlet and outlet flutes; and/or a media pack constructed or formed from such media into a three dimensional network of inlet and outlet flutes; and/or, a filter cartridge or construction including such a media pack.

In FIG. 118, an example of media 1001 usable in z-filter media construction is shown. The media 1001 is formed from a fluted, in this instance corrugated, sheet 1003 and a facing sheet 1004. A construction such as media 1001 is referred to herein as a single facer or single faced strip.

Sometimes, the corrugated fluted or ridged sheet 1003, shown in FIG. 118, is of a type generally characterized herein as having a regular, curved, wave pattern of flutes, ridges, or corrugations 1007. The term “wave pattern” in this context, is meant to refer to a flute, ridge or corrugated pattern of alternating troughs 1007b and ridges 1007a. The term “regular” in this context is meant to refer to the fact that the pairs of troughs and ridges (1007b, 1007a) alternate with generally the same repeating corrugation (flute or ridge) shape and size. (Also, typically in a regular configuration each trough 1007b is substantially an inverse ridge for each ridge 1007a.) The term “regular” is thus meant to indicate that the corrugation (or flute) pattern comprises troughs (inverted ridges) and ridges with each pair (comprising an adjacent trough and ridge) repeating, without substantial modification in size and shape of the corrugations along at least 70% of the length of the flutes. The term “substantial” in this context, refers to a modification resulting from a change in the process or form used to create the corrugated or fluted sheet, as opposed to minor variations from the fact that the media sheet 1003 is flexible. With respect to the characterization of a repeating pattern, it is not meant that in any given filter construction, an equal number of ridges and troughs is necessarily present. The media 1001 could be terminated, for example, between a pair comprising a ridge and a trough, or partially along a pair comprising a ridge and a trough. (For example, in FIG. 118, the media 1001 depicted in fragmentary has eight complete ridges 1007a and seven complete troughs 1007b.) Also, the opposite flute ends (ends of the troughs and ridges) may vary from one another. Such variations in ends are disregarded in these definitions, unless specifically stated. That is, variations in the ends of flutes are intended to be covered by the above definitions.

In the context of the characterization of a “curved” wave pattern of corrugations, in certain instances the corrugation pattern is not the result of a folded or creased shape provided to the media, but rather the apex of each ridge 1007a and the bottom of each trough 1007b is formed along a radiused curve. A typical radius for such z-filter media would be at least 0.25 mm and typically would be not more than 3 mm.

An additional characteristic of the particular regular, curved, wave pattern depicted in FIG. 118, for the corrugated sheet 1003, is that at approximately a midpoint region 1030 between each trough and each adjacent ridge, along most of the length of the flutes 1007, is located a transition region where the curvature inverts. For example, viewing a back side or face 1003a, in FIG. 118, trough 1007b is a concave region, and ridge 1007a is a convex region. Of course, when viewed toward front side or face 1003b, trough 1007b of side 1003a forms a ridge; and ridge 1007a of face

1003a, forms a trough. (In some instances, region **1030** can be a straight segment, instead of a point, with curvature inverting at ends of the region **1030**.)

A characteristic of the particular regular, wave pattern fluted (in this instance corrugated) sheet **1003** shown in FIG. **118**, is that the individual corrugations, ridges or flutes are generally straight, although alternatives are possible. By “straight” in this context, it is meant that through at least 70%, typically at least 80% of the length, the ridges **1007a** and troughs (or inverted ridges) **1007b** do not change substantially in cross-section. The term “straight” in reference to corrugation pattern shown in FIG. **118**, in part distinguishes the pattern from the tapered flutes of corrugated media described in FIG. 1 of PCT Publication WO 97/40918 and PCT Publication 2003/47722, each of which is incorporated herein by reference. The tapered flutes of FIG. 1 of WO 97/40918, for example, would be a curved wave pattern, but not a “regular” pattern, or a pattern of straight flutes, as the terms are used herein.

Referring to FIG. **118** and as referenced above, the media **1001** has first and second opposite edges **1008** and **1009**. When the media **1001** is formed into a media pack, in general edge **1009** will form an inlet end or face for the media pack and edge **1008** will form an outlet end or face, although an opposite orientation is possible.

In the example depicted, the various flutes **1007** extend completely between the opposite edges **1008**, **1009**, but alternatives are possible. For example, they can extend to a location adjacent or near the edges, but not completely therethrough. Also, they can be stopped and started partway through the media, as for example in the media of U.S. Publication 2014/0208705, incorporated herein by reference.

When the media is as depicted in FIG. **118**, adjacent edge **1008** can provided a sealant bead **1010**, sealing the corrugated sheet **3** and the facing sheet **1004** together. Bead **1010** will sometimes be referred to as a “single facer” or “single face” bead, or by variants, since it is a bead between the corrugated sheet **1003** and facing sheet **1004**, which forms the single facer (single faced) media strip **1001**. Sealant bead **1010** seals closed individual flutes **1011** adjacent edge **1008**, to passage of air therefrom (or thereto in an opposite flow).

In the media depicted in FIG. **118**, adjacent edge **1009** is provided seal bead **1014**. Seal bead **1014** generally closes flutes **1015** to passage of unfiltered fluid therefrom (or flow therein in an opposite flow), adjacent edge **1009**. Bead **1014** would typically be applied as media **1001** is configured into a media pack. If the media pack is made from a stack of strips **1001**, bead **1014** will form a seal between a backside **1017** of facing sheet **1004**, and side **1018** of the next adjacent corrugated sheet **1003**. When the media **1001** is cut in strips and stacked, instead of coiled, bead **1014** is referenced as a “stacking bead.” (When bead **1014** is used in a coiled arrangement formed from a long strip of media **1001**, it may be referenced as a “winding bead.”).

In alternate types of through-flow media, seal material can be located differently, and added sealant or adhesive can even be avoided. For example, in some instances, the media can be folded to form an end or edge seam, or the media can be sealed closed by alternate techniques such as ultrasound application, etc. Further, even when sealant material is used, it need not be adjacent opposite ends.

Referring to FIG. **118**, once the filter media **1001** is incorporated into a media pack, for example by stacking or coiling, it can be operated as follows. First, air in the direction of arrows **1012**, would enter open flutes **1011** adjacent end **1009**. Due to the closure at end **1008**, by bead

1010, the air would pass through the filter media **1001**, for example as shown by arrows **1013**. It could then exit the media or media pack, by passage through open ends **1015a** of the flutes **1015**, adjacent end **1008** of the media pack. Of course, operation could be conducted with air flow in the opposite direction.

For the particular arrangement shown herein in FIG. **118**, the parallel corrugations **1007a**, **1007b** are generally straight completely across the media, from edge **1008** to edge **1009**. Straight flutes, ridges or corrugations can be deformed or folded at selected locations, especially at ends. Modifications at flute ends for closure are generally disregarded in the above definitions of “regular,” “curved” and “wave pattern.”

Z-filter constructions which do not utilize straight, regular curved wave pattern corrugation shapes are known. For example, in Yamada et al., U.S. Pat. No. 5,562,825, corrugation patterns which utilize somewhat semicircular (in cross section) inlet flutes adjacent narrow V-shaped (with curved sides) exit flutes are shown (see FIGS. 1 and 3, of Yamada '825). In Matsumoto, et al., U.S. Pat. No. 5,049,326, circular (in cross-section) or tubular flutes defined by one sheet having half tubes attached to another sheet having half tubes, with flat regions between the resulting parallel, straight, flutes are shown, see FIG. 2 of Matsumoto '326. In Ishii, et al. U.S., 4,925,561, (FIG. 1) flutes folded to have a rectangular cross section are shown, in which the flutes taper along their lengths. In WO 97/40918 (FIG. 1), flutes or parallel corrugations which have a curved, wave patterns (from adjacent curved convex and concave troughs) but which taper along their lengths (and thus are not straight) are shown. Also, in WO 97/40918, flutes which have curved wave patterns, but with different sized ridges and troughs, are shown. Yamada '825, Matsumoto '326, Ishii '561, and WO 97/40918 are incorporated herein by reference. Also, flutes, which are modified in shape to include various ridges, are known.

In general, the filter media is a relatively flexible material, typically a non-woven fibrous material (of cellulose fibers, synthetic fibers or both) often including a resin therein, sometimes treated with additional materials. Thus, it can be conformed or configured into the various corrugated patterns, without unacceptable media damage. Also, it can be readily coiled or otherwise configured for use, again without unacceptable media damage. Of course, it must be of a nature such that it will maintain the required corrugated configuration, during use.

Typically, in the corrugation process, an inelastic deformation is caused to the media. This prevents the media from returning to its original shape. However, once the tension is released, the flute or corrugations will tend to spring back, recovering only a portion of the stretch and bending that has occurred. The facing media sheet is sometimes tacked to the fluted media sheet, to inhibit this spring back in the corrugated sheet. Such tacking is shown at **1020**.

Also, typically, the media contains a resin. During the corrugation process, the media can be heated to above the glass transition point of the resin. When the resin then cools, it will help to maintain the fluted shapes.

The media of the corrugated (fluted) sheet **1003**, facing sheet **1004**, or both, can be provided with a fine fiber material on one or both sides thereof, for example in accordance with U.S. Pat. No. 6,673,136, incorporated herein by reference. In some instances, when such fine fiber material is used, it may be desirable to provide the fine fiber on the upstream side of the material and inside the flutes. When this occurs, air flow, during filtering, will typically be into the edge comprising the stacking bead.

An issue with respect to z-filter constructions relates to closing of the individual flute ends. Although alternatives are possible, typically a sealant or adhesive is provided, to accomplish the closure. As is apparent from the discussion above, in typical z-filter media especially that which uses straight flutes as opposed to tapered flutes and sealant for flute seals, large sealant surface areas (and volume) at both the upstream end and the downstream end are needed. High quality seals at these locations are important to proper operation of the media structure that results. The high sealant volume and area, creates issues with respect to this. Attention is now directed to FIG. 119, in which z-filter media; i.e., a z-filter media construction 1040, utilizing a regular, curved, wave pattern corrugated sheet 1043, and a non-corrugated flat sheet 1044, i.e., a single facer strip is schematically depicted. A distance D1, between points 1050 and 1051, defines the extension of flat sheet 1044 in region 1052 underneath a given corrugated flute 1053. A length or distance D2 of the arcuate media for the corrugated flute 1053, over the same distance D1 is of course larger than D1, due to the shape of the corrugated flute 1053. For a typical regular-shaped media used in fluted filter applications, the linear length D2 of the media 1053 between points 1050 and 1051 will often be at least 1.2 times D1. Typically, D2 would be within a range of 1.2 to 2.0 times D1, inclusive. One particularly convenient arrangement for air filters has a configuration in which D2 is about 1.25 to 1.35 times D1. Such media has, for example, been used commercially in Donaldson Powercore™ Z-filter arrangements. Another potentially convenient size would be one in which D2 is about 1.4 to 1.6 times D1. Herein the ratio D2/D1 will sometimes be characterized as the flute/flat ratio or media draw for the corrugated media.

In the corrugated cardboard industry, various standard flutes have been defined. For example, the standard E flute, standard X flute, standard B flute, standard C flute and standard A flute. FIG. 120 in combination with Table A below provides definitions of these flutes.

Donaldson Company, Inc., (DCI) the assignee of the present disclosure, has used variations of the standard A and standard B flutes, in a variety of z-filter arrangements.

TABLE A

(Flute definitions for FIG. 120)

DCI A Flute: Flute/flat = 1.52:1; The Radii (R) are as follows:
 R1000 = .0675 inch (1.715 mm); R1001 = .0581 inch (1.476 mm);
 R1002 = .0575 inch (1.461 mm); R1003 = .0681 inch (1.730 mm);
 DCI B Flute: Flute/flat = 1.32:1; The Radii (R) are as follows:
 R1004 = .0600 inch (1.524 mm); R1005 = .0520 inch (1.321 mm);
 R1006 = .0500 inch (1.270 mm); R1007 = .0620 inch (1.575 mm);
 Std. E Flute: Flute/flat = 1.24:1; The Radii (R) are as follows:
 R1008 = .0200 inch (.508 mm); R1009 = .0300 inch (.762 mm);
 R1010 = .0100 inch (.254 mm); R1011 = .0400 inch (1.016 mm);
 Std. X Flute: Flute/flat = 1.29:1; The Radii (R) are as follows:
 R1012 = .0250 inch (.635 mm); R1013 = .0150 inch (.381 mm);
 Std. B Flute: Flute/flat = 1.29:1; The Radii (R) are as follows:
 R1014 = .0410 inch (1.041 mm); R1015 = .0310 inch (.7874 mm);
 R1016 = .0310 inch (.7874 mm);
 Std. C Flute: Flute/flat = 1.46:1; The Radii (R) are as follows:
 R1017 = .0720 inch (1.829 mm); R1018 = .0620 inch (1.575 mm);
 Std. A Flute: Flute/flat = 1.53:1; The Radii (R) are as follows:
 R1019 = .0720 inch (1.829 mm); R1020 = .0620 inch (1.575 mm).

Of course, other, standard, flutes definitions from the corrugated box industry are known.

In general, standard flute configurations from the corrugated box industry can be used to define corrugation shapes or approximate corrugation shapes for corrugated media.

Comparisons above between the DCI A flute and DCI B flute, and the corrugation industry standard A and standard B flutes, indicate some convenient variations.

It is noted that alternative flute definitions such as those characterized in U.S. Publication 2009/0127211; U.S. Publication 2008/0282890; and/or U.S. Publication 2010/0032365 can be used, with air cleaner features as characterized herein below. The complete disclosures of each of U.S. 2009/0127211, U.S. 2008/0282890, and U.S. 2010/0032365 are incorporated herein by reference.

Another media variation comprising fluted media with facing media secured thereto, can be used in arrangements according to the present disclosure, in either a stacked or coiled form, is described in U.S. Publication 2014/0208705, owned by Baldwin Filters, Inc. and incorporated herein by reference.

B. Manufacture of Media Pack Configurations Including the Media of FIGS. 118-120, see FIGS.

121-124

In FIG. 121, one example of a manufacturing process for making a media strip (single facer) corresponding to strip 1001 of FIG. 118 is shown. In general, a facing sheet 1064 and a fluted (corrugated) sheet 1066 having flutes 1068 are brought together to form a media web 1069, with an adhesive bead located therebetween at 1070. The adhesive bead 1070 will form a single facer bead 1010, see FIG. 118. An optional darting process occurs at station 1071 to form center darted section 1072 located mid-web. The z-filter media or Z-media strip 1074 can be cut or slit at 1075 along the bead 1070 to create two pieces or strips 1076, 1077 of z-filter media 1074, each of which has an edge with a strip of sealant (single facer bead) extending between the corrugating and facing sheet. Of course, if the optional darting process is used, the edge with a strip of sealant (single facer bead) would also have a set of flutes darted at this location.

Techniques for conducting a process as characterized with respect to FIG. 121 are described in PCT Publication WO 2004/007054, incorporated herein by reference.

Still in reference to FIG. 121, before the z-filter media 1074 is put through the darting station 1071 and eventually slit at 1075, it must be formed. In the schematic shown in FIG. 121, this is done by passing a sheet of filter media 1092 through a pair of corrugation rollers 1094, 1095. In the schematic shown in FIG. 121, the sheet of filter media 1092 is unrolled from a roll 1096, wound around tension rollers 1098, and then passed through a nip or bite 1102 between the corrugation rollers 1094, 1095. The corrugation rollers 1094, 1095 have teeth 1104 that will give the general desired shape of the corrugations after the flat sheet 1092 passes through the nip 1102. After passing through the nip 1102, the sheet 1092 becomes corrugated across the machine direction and is referenced at 1066 as the corrugated sheet. The corrugated sheet 1066 is then secured to facing sheet 1064. (The corrugation process may involve heating the media, in some instances.)

Still in reference to FIG. 121, the process also shows the facing sheet 1064 being routed to the darting process station 1071. The facing sheet 1064 is depicted as being stored on a roll 1106 and then directed to the corrugated sheet 1066 to form the Z-media 1074. The corrugated sheet 1066 and the facing sheet 1064 would typically be secured together by adhesive or by other means (for example, by sonic welding).

Referring to FIG. 121, an adhesive line 1070 is shown used to secure corrugated sheet 1066 and facing sheet 1064 together, as the sealant bead. Alternatively, the sealant bead

for forming the facing bead could be applied as shown as **1070a**. If the sealant is applied at **1070a**, it may be desirable to put a gap in the corrugation roller **1095**, and possibly in both corrugation rollers **1094**, **1095**, to accommodate the bead **1070a**.

Of course, the equipment of FIG. **121** can be modified to provide for the tack beads **1020** of FIG. **118**, if desired.

The type of corrugation provided to the corrugated media is a matter of choice, and will be dictated by the corrugation or corrugation teeth of the corrugation rollers **1094**, **1095**. One useful corrugation pattern will be a regular curved wave pattern corrugation, of straight flutes or ridges, as defined herein above. A typical regular curved wave pattern used, would be one in which the distance **D2**, as defined above, in a corrugated pattern is at least 1.2 times the distance **D1** as defined above. In example applications, typically **D2**=1.25 to 1.35 times **D1**, although alternatives are possible. In some instances, the techniques may be applied with curved wave patterns that are not "regular," including, for example, ones that do not use straight flutes. Also, variations from the curved wave patterns shown, are possible.

As described, the process shown in FIG. **121** can be used to create the center darted section **1072**. FIG. **122** shows, in cross-section, one of the flutes **1068** after darting and slitting.

A fold arrangement **1118** can be seen to form a darted flute **1120** with four creases **1121a**, **1121b**, **1121c**, **1121d**. The fold arrangement **1118** includes a flat first layer or portion **1122** that is secured to the facing sheet **1064**. A second layer or portion **1124** is shown pressed against the first layer or portion **1122**. The second layer or portion **1124** is preferably formed from folding opposite outer ends **1126**, **1127** of the first layer or portion **1122**.

Still referring to FIG. **122**, two of the folds or creases **1121a**, **1121b** will generally be referred to herein as "upper, inwardly directed" folds or creases. The term "upper" in this context is meant to indicate that the creases lie on an upper portion of the entire fold **1120**, when the fold **1120** is viewed in the orientation of FIG. **122**. The term "inwardly directed" is meant to refer to the fact that the fold line or crease line of each crease **1121a**, **1121b**, is directed toward the other.

In FIG. **122**, creases **1121c**, **1121d**, will generally be referred to herein as "lower, outwardly directed" creases. The term "lower" in this context refers to the fact that the creases **1121c**, **1121d** are not located on the top as are creases **1121a**, **1121b**, in the orientation of FIG. **122**. The term "outwardly directed" is meant to indicate that the fold lines of the creases **1121c**, **1121d** are directed away from one another.

The terms "upper" and "lower" as used in this context are meant specifically to refer to the fold **1120**, when viewed from the orientation of FIG. **122**. That is, they are not meant to be otherwise indicative of direction when the fold **1120** is oriented in an actual product for use.

Based upon these characterizations and review of FIG. **122**, it can be seen that a regular fold arrangement **1118** according to FIG. **122** in this disclosure is one which includes at least two "upper, inwardly directed, creases." These inwardly directed creases are unique and help provide an overall arrangement in which the folding does not cause a significant encroachment on adjacent flutes.

A third layer or portion **1128** can also be seen pressed against the second layer or portion **1124**. The third layer or portion **1128** is formed by folding from opposite inner ends **1134**, **1135** of the third layer **1128**.

Another way of viewing the fold arrangement **1118** is in reference to the geometry of alternating ridges and troughs

of the corrugated sheet **1066**. The first layer or portion **1122** is formed from an inverted ridge. The second layer or portion **1124** corresponds to a double peak (after inverting the ridge) that is folded toward, and in preferred arrangements, folded against the inverted ridge.

Techniques for providing the optional dart described in connection with FIG. **122**, in a preferred manner, are described in PCT Publication WO 2004/007054, incorporated herein by reference. Techniques for coiling the media, with application of the winding bead, are described in PCT Publication WO 2004/082795, and incorporated herein by reference.

Alternate approaches to darting the fluted ends closed are possible. Such approaches can involve, for example: darting which is not centered in each flute; and rolling, pressing or folding over the various flutes. In general, darting involves folding or otherwise manipulating media adjacent to fluted end, to accomplish a compressed, closed, state.

Techniques described herein are particularly well adapted for use in media packs that result from a step of coiling a single sheet comprising a corrugated sheet/facing sheet combination, i.e., a "single facer" strip. However, they can also be made into stacked arrangements.

Coiled media or media pack arrangements can be provided with a variety of peripheral perimeter definitions. In this context the term "peripheral, perimeter definition" and variants thereof, is meant to refer to the outside perimeter shape defined, looking at either the inlet end or the outlet end of the media or media pack. Typical shapes are circular as described in PCT Publication WO 2004/007054. Other usable shapes are obround, some examples of obround being oval shape. In general, oval shapes have opposite curved ends attached by a pair of opposite sides. In some oval shapes, the opposite sides are also curved. In other oval shapes, sometimes called racetrack shapes, the opposite sides are generally straight. Racetrack shapes are described, for example, in PCT Publications WO 2004/007054 and WO 2004/082795, each of which is incorporated herein by reference.

Another way of describing the peripheral or perimeter shape is by defining the perimeter resulting from taking a cross-section through the media pack in a direction orthogonal to the winding access of the coil.

Opposite flow ends or flow faces of the media or media pack can be provided with a variety of different definitions. In many arrangements, the ends or end faces are generally flat (planar) and perpendicular to one another. In other arrangements, one or both of the end faces include tapered, for example, stepped, portions which can either be defined to project axially outwardly from an axial end of the side wall of the media pack; or, to project axially inwardly from an end of the side wall of the media pack.

The flute seals (for example from the single facer bead, winding bead or stacking bead) can be formed from a variety of materials. In various ones of the cited and incorporated references, hot melt or polyurethane seals are described as possible for various applications.

In FIG. **123**, a coiled media pack (or coiled media) **1130** constructed by coiling a single strip of single faced media is depicted, generally. The particular coiled media pack depicted is an oval media pack **1130a**, specifically a race-track shaped media pack **1131**. The tail end of the media, at the outside of the media pack **1130** is shown at **1131x**. It will be typical to terminate that tail end along straight section of the media pack **1130** for convenience and sealing. Typically, a hot melt seal bead or seal bead is positioned along that tail end to ensure sealing. In the media pack **130**, the opposite

flow (end) faces are designated at **1132**, **1133**. One would be an inlet flow end or face, the other an outlet flow end or face.

In FIG. **124**, there is (schematically) shown a step of forming stacked z-filter media (or media pack) from strips of z-filter media, each strip being a fluted sheet secured to a facing sheet. Referring to FIG. **124**, single facer strip **1200** is being shown added to a stack **1201** of strips **1202** analogous to strip **1200**. Strip **1200** can be cut from either of strips **1076**, **1077**, FIG. **121**. At **1205** in FIG. **124**, application of a stacking bead **1206** is shown, between each layer corresponding to a strip **1200**, **1202** at an opposite edge from the single facer bead or seal. (Stacking can also be done with each layer being added to the bottom of the stack, as opposed to the top.)

Referring to FIG. **124**, each strip **1200**, **1202** has front and rear edges **1207**, **1208** and opposite side edges **1209a**, **1209b**. Inlet and outlet flutes of the corrugated sheet/facing sheet combination comprising each strip **1200**, **1202** generally extend between the front and rear edges **1207**, **1208**, and parallel to side edges **1209a**, **1209b**.

Still referring to FIG. **124**, in the media or media pack **1201** being formed, opposite flow faces are indicated at **1210**, **1211**. The selection of which one of faces **1210**, **1211** is the inlet end face and which is the outlet end face, during filtering, is a matter of choice. In some instances, the stacking bead **1206** is positioned adjacent the upstream or inlet face **1211**; in others, the opposite is true. The flow faces **1210**, **1211**, extend between opposite side faces **1220**, **1221**.

The stacked media configuration or pack **1201** shown being formed in FIG. **124**, is sometimes referred to herein as a "blocked" stacked media pack. The term "blocked" in this context, is an indication that the arrangement is formed to a rectangular block in which all faces are 90° relative to all adjoining wall faces. For example, in some instances, the stack can be created with each strip **1200** being slightly offset from alignment with an adjacent strip, to create a parallelogram or slanted block shape, with the inlet face and outlet face parallel to one another, but not perpendicular to upper and bottom surfaces.

In some instances, the media or media pack will be referenced as having a parallelogram shape in any cross-section, meaning that any two opposite side faces extend generally parallel to one another.

It is noted that a blocked, stacked arrangement corresponding to FIG. **124** is described in the prior art of U.S. Pat. No. 5,820,646, incorporated herein by reference. It is also noted that stacked arrangements are described in U.S. Pat. Nos. 5,772,883; 5,792,247; 7,351,270; and U.S. Publication 2004/0187689. Each of these latter references is incorporated herein by reference. It is noted that a stacked arrangement shown in U.S. Publication 2005/0130508 is a slanted stacked arrangement.

It is also noted that, in some instances, more than one stack can be incorporated into a single media pack. Also, in some instances, the stack can be generated with one or more flow faces that have a recess therein, for example, as shown in U.S. Pat. No. 7,625,419 incorporated herein by reference.

C. Selected Media or Media Pack Arrangements Comprising Multiple Spaced Coils of Fluted Media; FIGS. **125-127**

Alternate types of media arrangements or packs that involve flutes between opposite ends extending between can be used with selected principles according to the present disclosure. An example of such alternate media arrangement or pack is depicted in FIGS. **125-127**. The media of FIGS.

125-127 is analogous to one depicted and described in DE 202008017059; and as can sometimes found in arrangements available under the mark "IQORON" from Mann & Hummel.

Referring to FIG. **125**, the media or media pack is indicated generally at **1250**. The media or media pack **1250** comprises a first outer pleated (ridged) media loop **1251** and a second, inner, pleated (ridged) media loop **1252**, each with pleat tips (or ridges) extending between opposite flow ends. The view of FIG. **125** is toward a media pack (flow) end **1255**. The end **1255** depicted, can be an inlet (flow) end or an outlet (flow) end, depending on selected flow direction. For many arrangements, using principles characterized having the media pack **1250** would be configured in a filter cartridge such that end **1255** is an inlet flow end.

Still referring to FIG. **125**, the outer pleated (ridged) media loop **1251** is configured in an oval shape, though alternatives are possible. At **1260**, a pleat end closure, for example molded in place, is depicted closing ends of the pleats or ridges **1251** at media pack end **1255**.

Pleats or ridges **1252** (and the related pleat tips) are positioned surrounded by and spaced from loop **1251**, and thus pleated media loop **1252** is also depicted in a somewhat oval configuration. In this instance, ends **1252e** of individual pleats or ridges **1252p** in a loop **1252** are sealed closed. Also, loop **1252** surrounds a center **1252c** that is closed by a center strip **1253** of material, typically molded-in-place.

During filtering, when end **1255** is an inlet flow end, air enters gap **1265** between the two loops of media **1251**, **1252**. The air then flows either through loop **1251** or loop **1252**, as it moves through the media pack **1250**, with filtering.

In the example depicted, loop **1251** is configured slanting inwardly toward loop **1252**, in extension away from end **1255**. Also, spacers **1266** are shown supporting a centering ring **1267** that surrounds an end of the loop **1252**, for structural integrity.

In FIG. **126**, an end **1256** of the cartridge **1250**, opposite end **1255** is viewable. Here, an interior of loop **1252** can be seen, surrounding an open gas flow region **1270**. When air is directed through cartridge **1250** in a general direction toward end **1256** and away from end **1255**, the portion of the air that passes through loop **1252** will enter central region **1270** and exit therefrom at end **1256**. Of course, air that has entered media loop **1251**, as shown in FIG. **125**, during filtering would generally pass around (over) an outer perimeter **1256p** of end **1256**.

In FIG. **127**, a schematic cross-sectional view of cartridge **1250** is provided. Selected identified and described features are indicated by like reference numerals.

It will be understood from a review of FIGS. **124-127**, the above description, that the cartridge **1250** described, is generally a cartridge which has media tips extending in a longitudinal direction between opposite flow ends **1255**, **1256**.

In the arrangement of FIGS. **125-127**, the media pack **1250** is depicted with an oval, in particular racetrack, shaped perimeter. It is depicted in this manner since the air filter cartridges, in many examples below, also have an oval or racetrack shaped configuration. However, the principles can be embodied in a variety of alternate peripheral shapes.

D. Other Media Variations, FIGS. **128-133**

Herein, in FIGS. **128-133**, some schematic, fragmentary, cross-sectional views are provided of still further alternate variations of media types that can be used in selected applications of the principles characterized herein. Certain

examples are described in PCT Publication WO 2016/077377, owned by the Assignee of the present disclosure, Donaldson Company, Inc. In general, each of the arrangements of FIGS. 128-133 represents a media type that can be stacked or coiled into an arrangement that has opposite inlet and outlet flow ends (or faces), with straight through flow.

In FIG. 138, an example media arrangement 1301 from PCT Publication WO 2016/077377 is depicted, in which an embossed sheet 1302 is secured to a non-embossed sheet 1303, then stacked and coiled into a media pack, with seals along opposite edges of the type previously described for FIG. 118 herein.

In FIG. 129, an alternate example media pack 1310 from PCT Publication WO 2016/077377 is depicted, in which a first embossed sheet 1311 is secured to a second embossed sheet 1312 and then formed into a stacked or coiled media pack arrangement, having edge seals generally in accordance with FIG. 118 herein.

Edge seals can be conducted in either the upstream end or the downstream end, or in some instances, both. Especially when the media is likely to encounter chemical material during filtering, it may be desirable to avoid a typical adhesive or sealant.

In FIG. 130, a cross-section is depicted in which the fluted sheet X has various embossments on it for engagement with the facing sheet Y. Again, these can be separate, or sections of the same media sheet.

In FIG. 131, a schematic depiction of such an arrangement between the fluted sheet X and facing sheet Y is also shown.

In FIG. 132, a still further variation of such a principle is shown between a fluted sheet X and a facing sheet Y. These are meant to help understand how a wide variety of approaches are possible.

In FIG. 133, still another possible variation in fluted sheet X and facing sheet Y is shown.

In FIGS. 134 and 135, an example media arrangement 6401 is depicted, in which a fluted sheet 6402 is secured to a facing sheet 6403. The facing sheet 6403 may be a flat sheet. The media arrangement 6401 can then be stacked or coiled into a media pack, with seals along opposite edges of the type previously described for FIG. 118 herein. In the embodiment shown, flutes 6404 of fluted sheet 6402 have an undulating ridgeline including a series of peaks 6405 and saddles 6406. The peaks 6405 of adjacent flutes 6404 can be either aligned as shown in FIGS. 134 and 135, or offset. Further the peak height and/or density can increase, decrease, or remain constant along the length of the flutes 6404. The ratio of the peak flute height to saddle flute height can vary from about 1.5, typically from 1.1 to about 1.

It is noted that there is no specific requirement that the same media be used for the fluted sheet section and the facing sheet section. A different media can be desirable in each, to obtain different effects. For example, one may be a cellulose media, while the other is a media containing some non-cellulose fiber. They may be provided with different porosity or different structural characteristics, to achieve desired results.

A variety of materials can be used. For example, the fluted sheet section or the facing sheet section can include a cellulose material, synthetic material, or a mixture thereof. In some embodiments, one of the fluted sheet section and the facing sheet section includes a cellulose material and the other of the fluted sheet section and facing sheet section includes a synthetic material.

Synthetic material(s) can include polymeric fibers, such as polyolefin, polyamide, polyester, polyvinyl chloride,

polyvinyl alcohol (of various degrees of hydrolysis), and polyvinyl acetate fibers. Suitable synthetic fibers include, for example, polyethylene terephthalate, polyethylene, polypropylene, nylon, and rayon fibers. Other suitable synthetic fibers include those made from thermoplastic polymers, cellulosic and other fibers coated with thermoplastic polymers, and multi-component fibers in which at least one of the components includes a thermoplastic polymer. Single and multi-component fibers can be manufactured from polyester, polyethylene, polypropylene, and other conventional thermoplastic fibrous materials.

The examples of FIGS. 118-134, are meant to indicate generally that a variety alternate media packs can be used in accordance with the principles herein. Attention is also directed to PCT Publication WO 2016/077377, incorporated herein by reference, with respect to the general principles of construction and application of some alternates media types.

E. Additional Media Pack Arrangements Including Pleated Media with Flutes; FIGS. 136-139

Additional examples of alternative types of media arrangements or packs that involve filtration media having flutes extending between opposite ends or flow faces in a straight through flow configuration are depicted in FIGS. 136-139. The flutes can be considered inlet flutes when they are arranged to receive dirty air via an inlet flow face, and they can be considered outlet flutes when they are arranged to permit filtered air to flow out via an outlet flow face.

The filtration media 6502 depicted in FIGS. 136-138, which is analogous to ones depicted in U.S. Pat. Nos. 8,479,924 and 9,919,256 assigned to Mann+Hummel GmbH, is illustrated in an arrangement that shows how the filtration media 6502 can be formed into a media pack arrangement 6504.

The media pack arrangement 6504 can be considered as having relatively long or deep pleats from an inlet flow face 6506 to an outlet flow face 6508, and can also have varying pleat depths as illustrated. As the depth of pleats of a media pack increases, there is a tendency of the filtration media to collapse on each other thereby causing masking. Masking is undesirable because masked filtration media tends to no longer be available for filtration thereby decreasing dust holding capacity and flow through the media pack, and also potentially increasing pressure drop across the media pack. In order to reduce masking and to help the filtration media retain its shape, support structures are known to be applied to pleated media. In FIGS. 137 and 138, support sections or spacers 6510 are provided. It should be appreciated that FIGS. 137 and 138 are illustrated in a folded configuration 6512 having pleat folds 6514, but are expanded or separated to show how the filtration media 6502 and the support sections or spacers 6510 can be arranged.

As illustrated in FIGS. 137 and 138, the filtration media 6502 extends between a first side 6516 and a second side 6518. Although only one support section 6510 is shown on each pleat face 6520, it should be appreciated that multiple support sections 6510 can be arranged along each pleat face 6520 so that when the filtration media 6502 is arranged into a media pack, as illustrated in FIG. 139 as media pack 6604, the volume between each of the support sections 6510 can be considered flutes extending between the inlet flow face 6506 and the outlet flow face 6508. The support sections 6510 can be arranged on each flow face 6520 so that opposite support sections 6510 contact or engage each other to help maintain the media pack shape while also limiting the amount of filtration media that would be contacted by the

support sections **6510**, as illustrated in FIG. **138**. Furthermore, by providing that the support sections **6510** have adhesive properties, the support sections **6510** can be provided so that opposing support sections **6510** can adhere to each other when the filtration media **6502** is arranged into the media pack **6504**.

The support sections **6510** can be arranged in a tapered configuration where support sections **6510** have a cross section at an interior fold **6522** and wherein the cross section increases toward an exterior fold **6524**. In this context, the phrase “interior fold” refers to the side of the media that forms an acute angle, and the phrase “exterior fold” refers to the side of the media that forms an obtuse angle when the media is arranged into a media pack. Furthermore, the reference to changing the cross section of the support sections **6510** can refer to one or both of the height that the support section extends away from the media to which it is adhered and also to the width along the media to which it is adhered to in a direction toward or away from other support sections across adjacent flutes. Changing the shape of the support sections **6510** can help maintain the shape of the media pack and the resulting flutes, and can help reduce the amount of media that would otherwise be contacted by the support sections **6510** if they were not arranged in a tapered configuration. In addition, the support sections **6510** can be arranged in a non-tapered configuration. As illustrated in FIG. **138**, the support sections **6510** can be provided so that they extend over the exterior folds **6524**, although it is not necessary for the support sections **6510** to extend over the exterior folds. In addition, it is not necessary for the support sections **6510** to extend into the interior folds **6522**, although, if desired, the support sections **6510** can be provided so that they extend into the interior folds **6522**.

The support sections **6510** can be applied to the filtration media **6502** as adhesive extruded onto the filtration media **6502** where the adhesive forms the support sections **6510**. Before the adhesive has a chance to fully cure, the filtration media **6502** can be folded into the media pack arrangement **6504**, which may or may not have varying pleat depths. By forming the media pack arrangement **6504** before the adhesive has fully cured, the opposing support sections **6510** can become bonded or adhered to each other thereby forming flutes extending between the inlet flow face **6506** and the outlet flow face **6508**.

It should be appreciated that the filtration media **6502** can be provided with deformation, such as corrugations, extending across the media. The direction of deformation, such as corrugation, can be parallel or perpendicular to the pleat fold direction.

The filtration media **6602** depicted in FIG. **139** is analogous to filtration media depicted in U.S. Publication 2018/0207566, assigned to Champion Laboratories, Inc., as another example of a media pack arrangement **6604** having inlet and outlet flutes in a straight through flow arrangement.

The filtration media pack arrangement **6604** can be formed by folding the filtration media **6602** to form an inlet flow face **6606** and an outlet flow face **6608**. Pleat tips **6610** form the inlet flow face **6606**, and pleat tips **6612** form the outlet flow face **6608**. Adhesive beads **6616** and **6618**, which may be continuous or discontinuous, extend along the filtration media **6602** in multiple lines across the filtration media **6602** from a media first side **6620** to a media second side **6622**. The adhesive beads **6616** and **6618** along the media first side **6620** and along the media second side **6620** can be thickened, if desired, and can be arranged to provide an edge seal along the media first side **6620** and the media second side **6622**. By providing that the adhesive beads **6616**

and **6618** adhere to each other as the filtration media **6602** is folded, inlet flutes **6630** and outlet flutes **6632** can be formed in the straight through media pack arrangement **6604**.

A similar type of filtration media pack arrangement is commercially available under the name Enduracube from Baldwin Filters, Inc. The filtration media pack available under the name Enduracube from Baldwin Filters, Inc. is arranged in a pleated configuration forming inlet flutes and outlet flutes extending between an inlet flow face and an outlet flow face.

F. Still Further Media Types

Many of the techniques characterized herein will preferably be applied when the media is oriented for filtering between opposite flow ends of the cartridge is media having flutes or pleat tips that extend in a direction between those opposite ends. However, alternatives are possible. The techniques characterized herein with respect to seal arrangement definition can be applied in filter cartridges that have opposite flow ends, with media positioned to filter fluid flow between those ends, even when the media does not include flutes or pleat tips extending in a direction between those ends. The media, for example, can be depth media, can be pleated in an alternate direction, or it can be a non-pleated material.

It is indeed the case, however, that the techniques characterized herein are particularly advantageous for use with cartridges that are relatively deep in extension between flow ends, usually at least 100 mm, typically at least 150 mm, often at least 200 mm, sometimes at least 250 mm, and in some instances 300 mm or more, and are configured for large loading volume during use. These types of systems will typically be ones in which the media is configured with pleat tips or flutes extending in a direction between opposite flow ends.

It is also noted that while the techniques described herein were typically developed for advantageous application and arrangements involving media packs with straight through flow configurations, the techniques can be applied to advantage in other systems. For example, the techniques can be applied when the cartridge comprises media surrounding a central interior, in which the cartridge has an open end. Such arrangements can involve “forward flow” in which air to be filtered enters the central open interior by passage through the media, and the exits through the open end; or, with reverse flow in which air to be filtered enters the open end and then turns and passes through the media. A variety of such arrangements are possible, including pleated media and alternate types of media. Configurations usable would include cylindrical and conical, among others.

Aspects

The present disclosure includes various aspects that may be claimed in the future. The following aspects are intended to highlight certain features without limiting the scope of protection. It should be understood that any of the following aspects may be claimed in a patent application claiming priority to the present disclosure, either alone or in combination with other aspects. Further, the following aspects may be modified or combined in any suitable manner apparent to one skilled in the art in light of the teachings herein. Features which are described in the context of separate aspects and embodiments of the disclosure may be used together and/or be interchangeable. Similarly, features described in the

context of a single embodiment may also be provided separately or in any suitable subcombination. The aspects are numbered for convenience only and should not be construed as requiring a particular order or limiting the scope of what may be claimed. In various aspects, an air cleaner housing and related filter cartridges are provided that include features and components as described in one or more of the following aspects. The aspects may be combined or modified in ways apparent to those skilled in the art based on the teachings herein. While specific materials, dimensions, and configurations are described for certain aspects, these are exemplary only and other materials, dimensions and configurations may be used within the scope of the disclosure.

Aspect 1: A filter cartridge comprising: a) a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and b) an end cap arranged about the media pack outer perimeter, the end cap including: i) a seal member presenting a seal surface that is spaced from the media pack outer perimeter and that faces in a radial inward direction towards the media pack outer perimeter, the seal surface having a generally polygonal shape with flat sides and adjoining corner portions.

Aspect 2: The filter cartridge of Aspect 1, or any of Aspects 3 to 155, wherein the end cap further includes a support structure including a base part and a plurality of spaced apart reinforcement members, the support structure being formed from a material that is different than a material associated with the seal member, wherein the reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

Aspect 3: The filter cartridge of Aspect 1, or any of Aspects 2 and 4 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 4: The filter cartridge of Aspect 1, or any of Aspects 2 to 3 and 5 to 155, wherein the seal member has at least three flat sides and three corner portions.

Aspect 5: The filter cartridge of Aspect 1, or any of Aspects 2 to 4 and 6 to 155, wherein the seal member has eight flat sides and eight corner portions.

Aspect 6: The filter cartridge of Aspect 1, or any of Aspects 2 to 5 and 7 to 155, wherein each of the flat sides of the seal member have an equal length.

Aspect 7: The filter cartridge of Aspect 1, or any of Aspects 2 to 6 and 8 to 155, wherein the adjoining corner portions have a radius or curved shape.

Aspect 8: The filter cartridge of Aspect 2, or any of Aspects 1, 3 to 7, and 9 to 155, wherein the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 9: The filter cartridge of Aspect 2, or any of Aspects 1, 3 to 8, and 10 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 10: The filter cartridge of Aspect 2, or any of Aspects 1, 3 to 9, and 11 to 155, wherein the seal member is secured to the support structure.

Aspect 11: The filter cartridge of Aspect 10, or any of Aspects 1 to 9 and 12 to 155, wherein the support structure is molded onto the seal member.

Aspect 12: The filter cartridge of Aspect 1, or any of Aspects 2 to 11 and 13 to 155, wherein the seal member is located axially beyond a distal end of the media pack.

Aspect 13: The filter cartridge of Aspect 1, or any of Aspects 2 to 12 and 14 to 155, wherein the seal member circumscribes the media pack.

Aspect 14: The filter cartridge of Aspect 1, or any of Aspects 2 to 13 and 15 to 155, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 15: The filter cartridge of Aspect 1, or any of Aspects 2 to 14 and 16 to 155, wherein the media pack includes pleated media.

Aspect 16: The filter cartridge of Aspect 1, or any of Aspects 2 to 15 and 17 to 155, wherein the media pack includes fluted media.

Aspect 17: The filter cartridge of Aspect 1, or any of Aspects 2 to 16 and 18 to 155, wherein the end cap is located proximate the first axial end of the media pack.

Aspect 18: The filter cartridge of Aspect 1, or any of Aspects 2 to 17 and 19 to 155, wherein a distal end of the end cap is generally flush with the media pack first axial end.

Aspect 19: The filter cartridge of Aspect 1, or any of Aspects 2 to 18 and 20 to 155, wherein the end cap is located proximate the first axial end of the media pack.

Aspect 20: The filter cartridge of Aspect 2, or any of Aspects 1, 3 to 19 and 21 to 155, wherein the plurality of reinforcement members extend in a direction from the first axial end and towards the second axial end of the media pack or extend in a direction from the second axial end towards the first axial end of the media pack.

Aspect 21: The filter cartridge of Aspect 20, or any of Aspects 1 to 19 and 22 to 155, wherein an axial end of the seal member is at least partially exposed and uncovered by the support structure.

Aspect 22: The filter cartridge of Aspect 20, or any of Aspects 1 to 19, 21, and 23 to 155, wherein at least a portion of the plurality of reinforcement members circumscribe the media pack.

Aspect 23: The filter cartridge of Aspect 20, or any of Aspects 1 to 19, 21, 22, and 24 to 155, wherein the support structure further includes an axially extending portion spaced from the plurality of reinforcement members, the axially extending portion being secured to the media pack.

Aspect 24: The filter cartridge of Aspect 23, or any of Aspects 1 to 22 and 25 to 155, wherein the seal member is located within an annular space defined between the axially extending portion and the plurality of reinforcement members.

Aspect 25: The filter cartridge of Aspect 20, or any of Aspects 1 to 19, 21 to 24, and 26 to 155, wherein an axial end of the plurality of reinforcement members is generally flush with an axial end of the seal member.

Aspect 26: The filter cartridge of Aspect 1, or any of Aspects 1 to 25 and 27 to 155, wherein the seal surface has at least eight flat sides.

Aspect 27: The filter cartridge of Aspect 26, or any of Aspects 1 to 25 and 28 to 155, wherein the seal surface has twelve flat sides.

Aspect 28: The filter cartridge of any preceding aspect, wherein a length of the flat sides of the seal surface is greater than a length of the reinforcement members.

Aspect 29: The filter cartridge of any preceding aspect, wherein the number of flat sides of the seal surface is equal to the number of reinforcement members.

Aspect 30: An air cleaner comprising: a) a housing assembly including a housing body defining an interior volume, an air inlet, and an air outlet, and defining a first seal surface having a generally polygonal shape with flat sides and adjoining corner portions; b) a filter cartridge positioned within the interior volume, the filter cartridge including: i) a

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media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and ii) a first end cap operably arranged at the media pack first end, the first end cap including a seal member presenting an inner radial side defining a seal surface and presenting an outer radial side, the inner radial side and the outer radial side each having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions, wherein the seal surface of the seal member forms a seal against the first seal surface of the housing assembly; and a seal support arrangement including a plurality of support members located adjacent the flat sides of the seal member outer radial side such that the corner portions of the outer radial side are unsupported by the plurality of support members.

Aspect 31: The air cleaner of Aspect 30, or any of Aspects 1 to 29 and 32 to 155, wherein the housing assembly includes a wall structure defining the first seal surface and that further defines an oppositely facing second seal surface having a generally polygonal shape with flat sides and adjoining corner portions.

Aspect 32: The air cleaner of Aspect 30, or any of Aspects 1 to 29, 31, and 33 to 155, further comprising: a) a safety filter cartridge including filter media and having a seal member that forms a seal with the second seal surface.

Aspect 33: The air cleaner of Aspect 32, or any of Aspects 1 to 31 and 33A to 155, wherein the first seal surface is formed as part of an outlet tube assembly that defines the air outlet.

Aspect 33A: The air cleaner of Aspect 32, or any of Aspects 1 to 31, 33, and 34 to 155, wherein the first seal surface is defined by an outlet tube assembly having a plurality of spaced apart protrusions at a distal end and arranged such that the first seal surface is an undercut seal surface.

Aspect 34: The air cleaner of Aspect 33, or any of Aspects 1 to 32, 33A, and 35 to 155, wherein the outlet tube assembly includes a first part secured to a second part that includes a wall structure defining the first seal surface.

Aspect 35: The air cleaner of Aspect 34, or any of Aspects 1 to 33 and 35 to 155, wherein the first and second parts are joined together on opposite sides of an end wall of the housing assembly.

Aspect 36: An outlet tube assembly for use with an air cleaner housing, the outlet tube assembly comprising: a) a first part defining an air inlet or air outlet; and b) a second part joined to the first part such that a portion of the air cleaner housing is sandwiched between the first and second parts, wherein the second part includes a first sidewall having a radially outward facing seal surface with flat sides and adjoining corner portions forming a generally polygonal shape, and a second sidewall having a radially inward facing surface with a plurality of spaced apart flat sections interrupted by inwardly extending protrusions.

Aspect 37: The outlet tube assembly of Aspect 36, or any of Aspects 1 to 35 and 38 to 155, wherein the inwardly extending protrusions of the second sidewall include side edges that together with the flat sections define recess areas.

Aspect 38: The outlet tube assembly of Aspect 37, or any of Aspects 1 to 36 and 39 to 155, wherein the recess areas are configured to receive reinforcement members of a filter cartridge such that the protrusions are received within gaps between the reinforcement members.

Aspect 39: The outlet tube assembly of Aspect 36, or any of Aspects 1 to 35, 37 to 38, and 40 to 155, wherein the flat

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sections and protrusions of the second sidewall are configured to guide rotational alignment of a filter cartridge during installation.

Aspect 40: The outlet tube assembly of Aspect 36, or any of Aspects 1 to 35, 37 to 39, and 41 to 155, wherein at least one of the flat sections has a different width compared to the other flat sections.

Aspect 41: The outlet tube assembly of Aspect 36, or any of Aspects 1 to 3, 37 to 40, and 42 to 155, wherein at least one of the protrusions has a different width compared to the other protrusions.

Aspect 42: The outlet tube assembly of Aspect 36, or any of Aspects 1 to 35, 37 to 41, and 43 to 155, wherein the first sidewall defines a radially inward facing seal surface with flat sides and adjoining corner portions forming a generally polygonal shape.

Aspect 43: An air filter cartridge for use within a housing assembly of an air cleaner, the filter cartridge comprising: a) a media pack including filter media extending from a first end to a second end; and b) a first end cap arranged at the first end of the media pack, the first end cap presenting an inner radial side defining a seal surface having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions, the first end cap having a first cross-sectional thickness extending through the flat sides that is greater than a second cross-sectional thickness extending through the corner portions.

Aspect 44: The air filter cartridge of Aspect 43, or any of Aspects 1 to 42 and 45 to 155, wherein the first end cap is formed as a single, monolithic component.

Aspect 45: The air filter cartridge of Aspect 43, or any of Aspects 1 to 42, 44 to 44, and 46 to 155, wherein the first end cap includes a plurality of support members at least partially defining the first cross-sectional thickness.

Aspect 46: The air filter cartridge of Aspect 43, or any of Aspects 1 to 42, 44 to 45, and 47 to 155, wherein a distal axial end of the filter cartridge is defined by the end cap at a location that is radially aligned with the seal surface flat sides.

Aspect 47: The air filter cartridge of Aspect 43, or any of Aspects 1 to 42, 44 to 46, and 48 to 155, wherein the first end cap includes a seal member defining the inner radial side and defining an outer radial side, wherein the seal member is unsupported on the outer radial side at locations radially aligned with the seal surface corner portions.

Aspect 48: The air filter cartridge of Aspect 43, or any of Aspects 1 to 42, 44 to 47, and 49 to 155, wherein the seal member has a constant cross-sectional thickness.

Aspect 49: An air filter cartridge for use within a housing assembly of an air cleaner, the filter cartridge comprising: a) a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and b) a first end cap operably arranged at the media pack first end, the first end cap including: i) a seal member presenting an inner radial side defining a seal surface and presenting an outer radial side, the inner radial side and the outer radial side each having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and ii) a seal support arrangement including a plurality of support members located adjacent the flat sides of the seal member outer radial side such that the corner portions of the outer radial side are unsupported by the plurality of support members.

Aspect 50: The air filter cartridge of Aspect 49, or any of Aspects 1 to 48 and 51 to 155, wherein each of the plurality of support members extends in an axial direction parallel with the seal surface.

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Aspect 51: The air filter cartridge of Aspect 49, or any of Aspects 1 to 48, 50, and 52 to 155, wherein the plurality of support members equals the number of the plurality of flat sides of the seal member outer radial side.

Aspect 52: The air filter cartridge of Aspect 49, or any of Aspects 1 to 48, 50 to 51, and 53 to 155, wherein the plurality of support members define an axial distal end of the filter cartridge.

Aspect 53: The air filter cartridge of Aspect 49, or any of Aspects 1 to 48, 50 to 52, and 54 to 155, wherein the seal member is formed from a first material and the plurality of support members are formed from a second material different from the first material.

Aspect 54: The air filter cartridge of Aspect 49, or any of Aspects 1 to 48, 50 to 53, and 55 to 155, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

Aspect 55: An air filter cartridge for use within a housing assembly of an air cleaner, the filter cartridge comprising: a) a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and b) a first end cap operably arranged at the media pack first end, the first end cap including: i) a seal member presenting an inner radial side and an outer radial side, the inner radial side defining a seal surface having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and ii) a seal support arrangement including a plurality of support members located radially outward from the seal surface and being radially aligned with the flat sides such that the seal surface corner portions are radially aligned with gaps formed between adjacent support members.

Aspect 56: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54 and 56 to 155, wherein the number of gaps between the plurality of support members equals the number of seal surface corner portions such that each of the gaps is radially aligned with one of the seal surface corner portions.

Aspect 57: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54, 56, and 58 to 155, wherein the seal member outer radial side defines a generally polygonal shape having a second plurality of flat sides and adjoining corner portions, and wherein the plurality of support members are located adjacent the second plurality of flat sides.

Aspect 58: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54, 56 to 58, and 59 to 155, wherein an exposed portion of the plurality of support members defines an axial distal end of the filter cartridge.

Aspect 59: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54, 56 to 58, and 60 to 155, wherein the seal member is formed from a first material and the plurality of support members are formed from a second material that is different from the first material.

Aspect 60: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54, 56 to 59, and 61 to 155, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

Aspect 61: The air filter cartridge of Aspect 55, or any of Aspects 1 to 54, 56 to 60, and 62 to 155, wherein each of the plurality of support members extend in an axial direction that is generally parallel with the seal surface.

Aspect 62: An air filter cartridge comprising: a) a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and b) a seal member operably arranged on the media pack, the seal member presenting a seal surface

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having a generally polygonal shape with flat sides and adjoining corner portions, wherein the seal surface is at least partially located between the media pack first and second axial ends, is spaced from the media pack outer perimeter, and faces in a radially inward direction towards the media pack outer perimeter.

Aspect 63: The air filter cartridge of Aspect 62, or any of Aspects 1 to 61 and 64 to 155, further including a support structure including a base part and a plurality of spaced apart reinforcement members.

Aspect 64: The air filter cartridge of Aspect 63, or any of Aspects 1 to 62 and 65 to 155, wherein the plurality of reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

Aspect 65: The air filter cartridge of Aspect 63, or any of Aspects 1 to 62, 64, and 66 to 155, wherein the support structure is formed from a first material that is different from a second material associated with the seal member.

Aspect 66: The air filter cartridge of Aspect 65, or any of Aspects 1 to 64 and 67 to 155, wherein the plurality of flat sides of the seal surface have a first length and the plurality of support members have a second length that is a majority of the first length.

Aspect 67: The air filter cartridge of Aspect 63, or any of Aspects 1 to 62, 64 to 66, and 68 to 155, wherein each of the plurality of support members extends in an axial direction parallel with the seal surface.

Aspect 68: The air filter cartridge of Aspect 64, or any of Aspects 1 to 63, 65 to 67, and 69 to 155, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

Aspect 69: A filter cartridge comprising: a) a media pack including media extending between a first end and a second end along a longitudinal axis; b) an end cap secured to the media pack and including: i) a seal member presenting a radially inward or outward facing seal surface having a generally polygonal shape with flat sides and adjoining corner portions, the seal member being formed from a first material, the seal member having a second surface opposite the seal surface; and ii) a support structure including a base part and a plurality of spaced apart reinforcement members, the support structure being formed from a second material different than the first material, wherein the reinforcement members are located adjacent the seal member second surface and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

Aspect 70: The filter cartridge of Aspect 69, or any of Aspects 1 to 68 and 71 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 71: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70, and 72 to 155, wherein the seal member has at least three flat sides and three corner portions.

Aspect 72: The filter cartridge of Aspect 71, or any of Aspects 1 to 70, 71, and 73 to 155, wherein the seal member has eight flat sides and eight corner portions.

Aspect 73: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 72, and 74 to 155, wherein each of the flat sides of the seal member have an equal length.

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Aspect 74: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 73, and 75 to 155, wherein the adjoining corner portions have a radius or curved shape.

Aspect 75: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 74, and 76 to 155, wherein the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 76: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 75, and 77 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 77: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 76, and 78 to 155, wherein the seal member is secured to the support structure.

Aspect 78: The filter cartridge of Aspect 77, or any of Aspects 1 to 76 and 79 to 155, wherein the support structure is molded onto the seal member.

Aspect 79: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 78, and 80 to 155, wherein the seal member is located axially beyond a distal end of the media pack.

Aspect 80: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 79, and 81 to 155, wherein the seal member circumscribes the media pack.

Aspect 81: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 80, and 82 to 155, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 82: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 81, and 83 to 155, wherein the media pack includes pleated media.

Aspect 83: The filter cartridge of Aspect 69, or any of Aspects 1 to 68, 70 to 82, and 84 to 155, wherein the media pack includes fluted media.

Aspect 84: A filter cartridge comprising: a) a media pack including media extending between a first end and a second end along a longitudinal axis; b) an end cap secured to the media pack and including: i) a seal member presenting a radially inward or outward facing seal surface, the seal member being formed from a first material; and ii) a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different than the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material and that defines an axial end of the cartridge and is coincident with a radial innermost portion of the end cap and/or at least a portion of the radially inward or outward facing seal surface; iii) wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

Aspect 85: The filter cartridge of Aspect 84, or any of Aspects 1 to 83 and 86 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 86: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85, and 86 to 155 wherein the seal member has at least three flat sides and three corner portions.

Aspect 87: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 84 to 86, and 88 to 155 wherein the seal member has eight flat sides and eight corner portions.

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Aspect 88: The filter cartridge of Aspect 87, or any of Aspects 1 to 86 and 89 to 155, wherein each of the flat sides of the seal member have an equal length.

Aspect 89: The filter cartridge of Aspect 87, or any of Aspects 1 to 86, 88, and 89 to 155, wherein the adjoining corner portions have a radius or curved shape.

Aspect 90: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 89, and 91 to 155, wherein the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 91: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 90, and 91 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 92: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 91, and 93 to 155, wherein the seal member is secured to the support structure.

Aspect 93: The filter cartridge of Aspect 92, or any of Aspects 1 to 91 and 94 to 155, wherein the support structure is molded onto the seal member.

Aspect 94: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 93, and 95 to 155, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 95: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 94, and 96 to 155, wherein the media pack includes pleated media.

Aspect 96: The filter cartridge of Aspect 84, or any of Aspects 1 to 83, 85 to 95, and 97 to 155, wherein the media pack includes fluted media.

Aspect 97: A filter cartridge comprising: a) a media pack including media extending between a first end and a second end along a longitudinal axis; b) an end cap secured to the media pack and including: i) a seal member presenting a radially inward or outward facing seal surface, the seal member being formed from a first material; and ii) a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different than the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material and that defines an axial end of the cartridge; iii) wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

Aspect 98: The filter cartridge of Aspect 97, or any of Aspects 1 to 96 and 99 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 99: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98, and 100 to 155, wherein the seal member has at least three flat sides and three corner portions.

Aspect 100: The filter cartridge of Aspect 99, or any of Aspects 1 to 98 and 101 to 155, wherein the seal member has eight flat sides and eight corner portions.

Aspect 101: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 100, and 102 to 155, wherein each of the flat sides of the seal member have an equal length.

Aspect 102: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 101, and 103 to 155 wherein the adjoining corner portions have a radius or curved shape.

Aspect 103: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 102, and 104 to 155, wherein the

plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 104: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 103, and 105 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 105: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 104, and 106 to 155, wherein the seal member is secured to the support structure.

Aspect 106: The filter cartridge of Aspect 105, or any of Aspects 1 to 104 and 102 to 155, wherein the support structure is molded onto the seal member.

Aspect 107: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 105, and 108 to 155, wherein the seal member is located axially beyond a distal end of the media pack.

Aspect 108: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 108, and 109 to 155, wherein the seal member circumscribes the media pack.

Aspect 109: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 108, and 110 to 155, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 110: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 109, and 111 to 155, wherein the media pack includes pleated media.

Aspect 111: The filter cartridge of Aspect 97, or any of Aspects 1 to 96, 98 to 110, and 112 to 155, wherein the media pack includes fluted media.

Aspect 112: A filter cartridge comprising: a) a media pack including media extending between a first end and a second end along a longitudinal axis; b) an end cap secured to the media pack and including: i) a seal member presenting a radially inward or outward facing seal surface and an undercut portion radially recessed from the seal surface, the seal member being formed from a first material; and ii) a support structure including a base part and a plurality of spaced apart, radially flexible extensions, the support structure being formed from a second material different from the first material, wherein at least a portion of the base part is covered by and/or embedded within the first material, the plurality of flexible extensions including an exposed portion that is uncovered by and/or embedded within the first material; iii) wherein the seal surface and the exposed portion of the flexible extensions are axially aligned such that a plane orthogonal to the longitudinal axis can pass through the seal member and through the exposed portion of the plurality of flexible extensions.

Aspect 113: The filter cartridge of Aspect 112, or any of Aspects 1 to 111 and 113 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 114: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113, and 115 to 155, wherein the seal member has at least three flat sides and three corner portions.

Aspect 115: The filter cartridge of Aspect 114, or any of Aspects 1 to 113 and 116 to 155, wherein the seal member has eight flat sides and eight corner portions.

Aspect 116: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 115, and 117 to 155, wherein each of the flat sides of the seal member have an equal length.

Aspect 117: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 116, and 118 to 155, wherein the adjoining corner portions have a radius or curved shape.

Aspect 118: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 117, and 119 to 155, wherein the

plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 119: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 118, and 120 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 120: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 119, and 121 to 155, wherein the seal member is secured to the support structure.

Aspect 121: The filter cartridge of Aspect 120, or any of Aspects 1 to 119 and 122 to 155, wherein the support structure is molded onto the seal member.

Aspect 122: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 121, and 123 to 155, wherein the seal member is located axially beyond a distal end of the media pack.

Aspect 123: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 122, and 123 to 155, wherein the seal member circumscribes the media pack.

Aspect 124: The filter cartridge of Aspect 112, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 125: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 124, and 125 to 155, wherein the media pack includes pleated media.

Aspect 126: The filter cartridge of Aspect 112, or any of Aspects 1 to 111, 113 to 125, and 127 to 155, wherein the media pack includes fluted media.

Aspect 127: A filter cartridge comprising: a) a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and b) a seal member presenting a seal surface that faces in a radial inward direction, the seal surface having a generally polygonal shape with flat sides and adjoining corner portions, wherein the seal member is constrained from stretching in a circumferential direction along at least a portion of the flat sides and is unconstrained from stretching in a circumferential direction along the corner portions.

Aspect 128: The filter cartridge of Aspect 127, further comprising: a) a support structure including a base part and a plurality of spaced apart reinforcement members, the support structure being formed from a material that is different than a material associated with the seal member, wherein the reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

Aspect 129: The filter cartridge of Aspect 128, or any of Aspects 1 to 127 and 128 to 155, wherein the support structure and seal member are each formed from a polyurethane material.

Aspect 130: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 129 and 131 to 155, wherein the seal member circumscribes and is spaced from the media pack outer perimeter.

Aspect 131: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 130 and 132 to 155, wherein the seal surface has between three and eight flat sides and between three and eight corner portions.

Aspect 132: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 131 and 133 to 155, wherein each of the flat sides of the seal member have an equal length.

Aspect 133: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 132 and 134 to 155, wherein the adjoining corner portions have a radius or curved shape.

Aspect 134: The filter cartridge of Aspect 128, or any of Aspects 1 to 127, 129 to 133 and 135 to 155, wherein the plurality of spaced apart reinforcement members define a distal end of the filter cartridge.

Aspect 135: The filter cartridge of Aspect 128, or any of Aspects 1 to 127, 129 to 134 and 136 to 155, wherein the plurality of spaced apart reinforcement members have a height that is greater than a height of the seal member.

Aspect 136: The filter cartridge of Aspect 128, or any of Aspects 1 to 127, 129 to 135 and 137 to 155, wherein the seal member is secured to the support structure.

Aspect 137: The filter cartridge of Aspect 136, or any of Aspects 1 to 135 and 138 to 155, wherein the support structure is molded onto the seal member.

Aspect 138: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 137 and 139 to 155, wherein the seal member is located axially beyond a distal end of the media pack.

Aspect 139: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 138 and 140 to 155, wherein the seal member circumscribes the media pack.

Aspect 140: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 139 and 141 to 155, wherein the media pack has one of a circular and a racetrack cross-sectional shape.

Aspect 141: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 140 and 142 to 155, wherein the media pack includes pleated media.

Aspect 142: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 141 and 143 to 155, wherein the media pack includes fluted media.

Aspect 143: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 142 and 144 to 155, wherein the seal member is provided on an end cap located proximate the first axial end of the media pack.

Aspect 144: The filter cartridge of Aspect 143, or any of Aspects 1 to 142 and 145 to 155, wherein a distal end of the end cap is generally flush with the media pack first axial end.

Aspect 145: The filter cartridge of Aspect 143, or any of Aspects 1 to 142, 144, and 146 to 155, wherein the end cap is located proximate the first axial end of the media pack.

Aspect 146: The filter cartridge of Aspect 128, or any of Aspects 1 to 127, 129 to 145 and 147 to 155, wherein the plurality of reinforcement members extend in a direction from the first axial end and towards the second axial end of the media pack.

Aspect 147: The filter cartridge of Aspect 146, or any of Aspects 1 to 145 and 148 to 155, wherein an axial end of the seal member is at least partially exposed and uncovered by the support structure.

Aspect 148: The filter cartridge of Aspect 146, or any of Aspects 1 to 145, 147, and 148 to 155, wherein at least a portion of the plurality of reinforcement members circumscribe the media pack.

Aspect 149: The filter cartridge of Aspect 146, or any of Aspects 1 to 145, 146 to 148, and 148 to 155, wherein the support structure further includes an axially extending portion spaced from the plurality of reinforcement members, the axially extending portion being secured to the media pack.

Aspect 150: The filter cartridge of Aspect 149, or any of Aspects 1 to 148 and 151 to 155, wherein the seal member is located within an annular space defined between the axially extending portion and the plurality of reinforcement members.

Aspect 151: The filter cartridge of Aspect 146, or any of Aspects 1 to 145, 146 to 150, and 152 to 155, wherein an

axial end of the plurality of reinforcement members is generally flush with an axial end of the seal member.

Aspect 152: The filter cartridge of Aspect 127, or any of Aspects 1 to 126, 128 to 151, and 153 to 155, wherein the corner portions are rounded corner portions.

Aspect 153: The filter cartridge of Aspect 148, or any of Aspects 1 to 147, 148 to 152, and 154 to 155, wherein the plurality of reinforcement members extend in a direction from the second axial end and towards the first axial end of the media pack.

Aspect 154: The filter cartridge of any preceding Aspect 1-153, wherein a length of the flat sides of the seal surface is greater than a length of the reinforcement members.

Aspect 155: The filter cartridge of any preceding aspect 11-154, wherein the number of flat sides of the seal surface is equal to the number of reinforcement members.

SUMMARY

The principles described herein can be applied in a variety of filter assemblies. Examples described in which the principles applied to (air) gas filter assemblies. Examples are described include air filters, for example, air filters used for treating engine intake airflows. The principles can be applied to a variety of alternate gas filtration arrangements, in some instances even with liquid filter assemblies. Again, the principles, techniques, and features described herein can be applied in a variety of systems, and there is no requirement that all of the advantageous features identified be incorporated in an assembly, system or component to obtain some benefit according to the present disclosure.

What is claimed is:

1. An air filter cartridge for use within a housing assembly of an air cleaner, the filter cartridge comprising:

- a) a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and
- b) a first end cap operably arranged at the media pack first end, the first end cap including:
 - i) a seal member presenting an inner radial side defining a seal surface and presenting an outer radial side, the inner radial side and the outer radial side each having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and
 - ii) a seal support arrangement including a plurality of support members located adjacent the flat sides of the seal member outer radial side such that the corner portions of the outer radial side are unsupported by the plurality of support members.

2. The air filter cartridge of claim 1, wherein each of the plurality of support members extend in an axial direction parallel with the seal surface.

3. The air filter cartridge of claim 1, wherein the plurality of support members equals the number of the plurality of flat sides of the seal member outer radial side.

4. The air filter cartridge of claim 1, wherein the plurality of support members define an axial distal end of the filter cartridge.

5. The air filter cartridge of claim 1, wherein the seal member is formed from a first material and the plurality of support members are formed from a second material different from the first material.

6. The air filter cartridge of claim 1, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

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7. An air filter cartridge for use within a housing assembly of an air cleaner, the filter cartridge comprising:

- a) a media pack including filter media extending from a first end to a second end and forming a tubular shape having a hollow center; and
- b) a first end cap operably arranged at the media pack first end, the first end cap including:
 - i) a seal member presenting an inner radial side and an outer radial side, the inner radial side defining a seal surface having a generally polygonal shape formed by a plurality of flat sides and adjoining corner portions; and
 - ii) a seal support arrangement including a plurality of support members located radially outward from the seal surface and being radially aligned with the flat sides such that the seal surface corner portions are radially aligned with gaps formed between adjacent support members.

8. The air filter cartridge of claim 7, wherein the number of gaps between the plurality of support members equals the number of seal surface corner portions such that each of the gaps is radially aligned with one of the seal surface corner portions.

9. The air filter cartridge of claim 7, wherein the seal member outer radial side defines a generally polygonal shape having a second plurality of flat sides and adjoining corner portions, and wherein the plurality of support members are located adjacent the second plurality of flat sides.

10. The air filter cartridge of claim 7, wherein an exposed portion of the plurality of support members defines an axial distal end of the filter cartridge.

11. The air filter cartridge of claim 7, wherein the seal member is formed from a first material and the plurality of support members are formed from a second material that is different from the first material.

12. The air filter cartridge of claim 7, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

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13. The air filter cartridge of claim 7, wherein each of the plurality of support members extends in an axial direction that is generally parallel with the seal surface.

14. An air filter cartridge comprising:

- a) a media pack including media defining an outer perimeter extending along a longitudinal axis between a first axial end and a second axial end; and
- b) a seal member operably arranged on the media pack, the seal member presenting a seal surface having a generally polygonal shape with flat sides and adjoining corner portions, wherein the seal surface is at least partially located between the media pack first and second axial ends, is spaced from the media pack outer perimeter, and faces in a radially inward direction towards the media pack outer perimeter.

15. The air filter cartridge of claim 14, further including a support structure including a base part and a plurality of spaced apart reinforcement members.

16. The air filter cartridge of claim 15, wherein the plurality of reinforcement members are located adjacent a radially outward facing side of the seal member and are radially aligned with the flat sides of the seal member such that gaps between adjacent reinforcement members are radially aligned with the seal surface corner portions.

17. The air filter cartridge of claim 15, wherein the support structure is formed from a first material that is different from a second material associated with the seal member.

18. The air filter cartridge of claim 15, wherein the plurality of flat sides of the seal surface have a first length and the plurality of support members have a second length that is a majority of the first length.

19. The air filter cartridge of claim 15, wherein each of the plurality of support members extend in an axial direction parallel with the seal surface.

20. The air filter cartridge of claim 14, wherein the plurality of flat sides includes one flat side having a different length in comparison to at least one other of the plurality of flat sides.

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